Axiom

AJ Cason

Contents

1 Preface **1**

I The Foundation **3**

2 Introduction **5**

3 Logic **7**

3.1 Introduction 7

3.2 Defining an Axiom 8

3.3 ***** 9

3.3.0.1 Propositional logic 9

3.3.0.2 Axiom System for the Propositional Calculus 9

3.3.0.3 Defining Tautologies 10

3.3.0.4 Truth Tables 10

4 Philosophy of Mathematics **13**

5 Fundamental Metaphysics **15**

5.1 Scope 15

5.2 Reality 15
5.2.1 Existence Exists 15
5.2.2 The Entity and Observable reality 16
5.2.3 Beyond Observable 17
5.2.4 Independence of Reality 18
5.3 The Entity and Observations 18
5.3.1 Introduction 18
5.3.2 Properties 18
5.3.3 Abstraction: Math, sets, and Relations . . 18
5.3.4 Matter 19
5.3.5 SpaceTime 19
5.3.6 Nous 19
5.3.7 Non-material 20
5.4 Assumptions(Axioms) 20
5.5 Platoism and Realism 20

6 Epistemology **23**
6.1 Introduction 23
6.2 Ontology 23
6.2.1 Statements 23
6.2.2 Events 24
6.2.3 Relations 25
6.3 Verifying Statements 25
6.3.1 Deduction 25
6.3.2 Induction 26
6.3.3 Abduction 26
6.3.4 Statement Functions 26
6.3.5 Probability Space 27

6.3.6 Probability Rules 27
6.3.7 Bayes 28
6.4 Information 28
6.4.1 Information of Distinction 29
6.4.2 Formulisms of Information 30
6.4.2.1 Shannon Information 30
6.4.2.2 Algorithmic Information 30
6.4.2.3 Semantic Information 31
6.4.2.4 Programme 31
6.5 Justification 31
6.6 Belief 31

7 Philosophy of Science 33

8 Philosophy of Physics 35

9 Foundations of Physics 37

10 Foundations of Mathematics 39

11 Foundation of Computations 41

12 Foundations of Mathematics 43

13 Theology 45

14 Ethics 47
14.1 The Primacy of God 47

15 Social Philosophy 49

16 Personal Philosophy **51**

II Notes **53**

17 Introduction **55**

18 Physics **57**

18.1 Introduction 57
18.2 Theoretical Physics by Georg Joos 57
18.3 On Sympathetic Reduction in Classical Mechanics 60
18.4 Magnetic Fields and Magnetic Diagnostics for Tokamak Plasmas by Alan Wooton 60
18.5 Advanced MHD with Applications to Laboratory and Astrophysical Plasmas by Cambridge 60
18.6 Hamiltonian description of the ideal Fuid by P.J.Morrison 61
18.7 Classical Dynamics a Modern Perspective 61
18.7.1 Chapter 6 61
18.7.2 Chapter 7 62
18.8 Differential Geometry - Robert Davie 62
18.8.1 1-forms 62
18.8.2 2-forms 63
18.8.3 3-forms 63
18.8.4 k-form 63
18.8.5 Introduction to Exterior calculus 63
18.8.6 Exterior calculus-2 64
18.8.7 Wedge Product-2 65
18.8.8 Introduction to the Hodge Star Operation 65
18.8.9 Hodge star 2 66

18.8.10 The push forward of vectors on manifolds 66
18.8.11 Pull-Back in k-forms 67
18.8.12 Pull-Back of volume forms 69
18.8.13 Integration on Manifolds Using the Pull-back of Volume Forms 70
18.8.14 Line Integrals on Manifolds 71
18.8.15 Classical Stokes Thereom in Manifolds . . 72
18.9 Symplectic geometry & classical mechanics . . . 72
18.9.1 Lecture 1 72
18.9.2 Lecture 2 72

19 Calculus of Variations by Gelfand & Fomin 73
19.0.1 Elements of the Theory 73
19.1 MIT 8.323 Relativistic Quantum Field Theory 1 . 74
19.1.1 Lecture 1: Classical Field Theories and Principle of Locality 74
19.1.1.1 Equations of motion 75
19.1.2 Lecture 2: Symmetries and Conservation Laws 76
19.1.3 Lecture 3: Why Quantum Field Theory . . 77
19.1.4 Lecture 4: Canonical Quantization of a Free Scalar Field Theory 78

20 Numerical Hamiltonian Problems 79
20.0.1 Chapter 1: Hamiltonian Systems 79
20.0.2 Chapter 2: Symplecticness 80
20.1 Chapter 2: Numerical Methods 80
20.1.1 Chapter 4: Order Condition 80
20.1.2 Chapter 5: Implementation 81

20.1.3 Chapter 6: Symplectic Integration 81
20.1.4 Chapter 7: Symplectic Order Conditions . 81
20.1.5 Chapter 8: Numerical Experimentations . 81
20.1.6 Chapter 10: Properties of Symplectic Integrators 81
20.1.6.1 Backward error analysis 81
20.1.6.2 Alternative Approach 82
20.1.6.3 Conserved Energy and Quantities 82
20.1.6.4 KAM 82
20.1.7 Chapter 11: Generating Functions 82
20.1.7.1 First kind 83
20.1.7.2 Third kind 83
20.1.7.3 All kinds 83
20.1.7.4 Hamiltonian-Jacobi 84
20.1.7.5 Symplectic Integrators on Generating Functions 84
20.1.8 Chapter 12: Lie Formulism 84
20.1.8.1 Lie Operators & series 85
20.1.8.2 Multiplication of exponentials . . 85
20.1.8.3 Baker-Campbell-Hausdorff formula 85
20.1.8.4 Fractional-Step Method 86
20.1.9 Chapter 13: Higher Order Methods 86

21 General Relativity by Robert M. Wald 87
21.0.1 Manifolds and Tensor Fields 87
21.0.1.1 Manifolds 87
21.0.1.2 Vectors 88
21.0.2 Tensors; the Metric Tensor 88
21.0.2.1 Abstract Index Notation 91

21.0.3 Curvature 91
21.0.3.1 Derivative Operators and Parallel Transport 91
21.0.3.2 Curvature 93
21.0.3.3 Geodesics 95
21.0.3.4 Methods of Computing Curvature 98
21.0.3.5 Orthonormal Basis (Tetrad) Methods 99
21.0.3.6 Einstein's Equations 100
21.0.3.6 Special 100
21.0.3.7 General 101
21.0.3.8 Linearized Gravity 101
21.1 Homogeneous, Isotropic Cosmology 102
21.2 The Schwarzschild Solution 102
21.2.1 Derivation 102
21.2.2 Interior Solutions 103
21.2.3 Geodesics 104
21.2.4 Kruskal Extention 104
21.3 Methods for Solving Einstein's Equation 104
21.3.1 Stationary, Axisymmetric Solutions 104

22 The Geometry of Kerr Black Holes by Barrett O'Neill 105
22.1 Beginning Kerr Spacetime* 105
22.2 Maximal Extention 105

23 Assumptions of Physics 107
23.0.1 Overview 107
23.0.2 Reverse Physics 107
23.0.3 Physical Mathematics 108

24 A Concise Introduction to Geometric Numerical Integration 109
24.1 Variational Integrator 109
24.2 Volume-Preserving Methods 110
24.3 Lie Group Method 110
24.4 Time-Splitting PDE 111

25 Geometric Numerical Integration 113
25.1 Symplectic Integrators of Hamiltonian systems . 113
25.1.1 Variational Integrators 113
25.1.1.1 Hamilton's Principle 113
25.1.1.2 Discretization of Hamilton's principle 114
25.1.1.3 Noether's theorem 114
25.1.2 Characterization of Symplectic Methods . 114
25.2 Non-Canonical Hamiltonian 114
25.2.1 Constrained Mechanical Systems 114
25.2.2 Poisson System 114
25.2.2.1 Poisson Integrators 114

26 Introduction to Mechanics and Symmetry 115
26.1 Hamiltonian Systems on Linear Symplectic Spaces 115
26.1.1 Introduction 115

27 The Relativistic Boltzmann Equation 117
27.1 Special Relativity 117
27.1.1 Minkowski Force 118
27.1.2 Electrodynamics in free space 118
27.1.2.1 Maxwell Equations 118

27.2 Relativistic Boltzmann Equation 119
27.2.1 Single Non-degenerate Gas 119
27.3 Vlasov 120
27.3.1 The Vlasov-Maxwell system 120
27.3.2 The Threshold of Black hole formation . . 121

28 Introduction to 3+1 Numerical Relativity 123
28.1 3+1 formulism 123
28.1.1 3+1 split 123
28.1.2 Extrinsic Curvature 124
28.1.3 The ADM evolution Equation 125
28.1.4 BSSNOK formulation 125
28.2 Gravitational Waves 127
28.2.1 Gauge invariant Perturbations of Schwarzchild 127
28.2.2 Gravitational Radiation in the TT Gauge . 128
28.2.3 Energy and Momentum of GRW 128

29 Applications of Lie Groups to Difference Equations 129
29.1 Finite Differences and Transformation Groups in Space of Discrete Variables 129

30 Topological Methods in Hydrodynamics 131
30.1 Group and Hamiltonian Structures 131

31 Chaos and Structure in Nonlinear Plasma 133
31.1 Nonlinear Oscillations 133
31.1.1 Harmonic Generations in Small Amplitude Oscillations 133
31.2 Hamiltonian Dynamics 133
31.2.1 Poincare Surface of Section 134

31.2.2 Fixed Points and Invariant Curves 134
31.2.3 KAM Theory 134
31.2.4 KAM Torii 134
31.2.4.1 Ergodic Theorem 134
31.2.5 KAM theory in Labrotory Plasmas 135
31.2.6 Visualization of the Magnetic Flux surfaces by Electron beam mapping 135
31.2.7 Magnetic Islands and the Unstable-Chaotic Trajectories of Field Lines in Tokamaks . . 136

32 Papers 137

32.1 Operator Splitting Methods: Numerical Solutions of Ordinary Differential Equations via Separation of Variables 137
32.1.1 General Idea 137
32.1.2 Notes On Introduction 137
32.1.3 Strongly Continuous and bi-continuous semi-groups 138
32.2 Splitting methods for differential equations . . . 138
32.2.1 Notes on introduction and Abstract 138
32.2.2 Higher Order Splitting and Composition . 139
32.3 Core Collapse Supernova Modeling: The Next Ten Years 139
32.4 Physical, numerical, and computational challenges of modeling neutrino transport in core-collapse supernovae 139
32.4.1 Structure Preserving 139
32.5 Discontinuous Galerkin Methods 139

32.6 Energy conserving and well-balanced discontinuous Galerkin methods for the Euler–Poisson equations in spherical symmetry 139
32.6.1 Numerical Methods 139

33 Colloquia: UTK 141
33.1 August 24: Controlling Topological Quantum Phases through Epitaxy 141
33.1.1 Main notes 141
33.1.1.1 Phase and strain control 141
33.1.1.2 Strain-controlled phase stabilization 142
33.1.1.3 Quantum Oscillation in strained a-Sn 142
33.1.1.4 Dimensioality Control 142
33.1.1.5 Geometry and Interfaces 142
33.1.1.6 Emerging 142
33.1.2 Connections 143
33.1.3 Summary 143
33.1.4 Questions 143
33.2 August 31 - Electron Pairing and Fractionalization in age of AI and Quantum Computing 143
33.2.1 Main notes 143
33.2.1.1 Introduction 143
33.2.2 Concepts 144
33.2.3 Learning Hamiltonians 144
33.2.4 Kinetic Magnetism 145
33.2.5 Quantum Computing 145
33.2.6 Connections 145

33.2.7 Summary 145
33.2.8 Questions 146
33.3 Curiosity and Perseverance: Martian Surface . . 146
33.3.1 Main notes 146
33.3.2 Connections 147
33.3.3 Summary 147
33.3.4 Questions 147
33.4 Weighing the Earth: The Quest to Measure the Gravitational Constant 147
33.4.1 Main notes 147
33.4.2 Connections 147
33.4.3 Summary 148
33.4.4 Questions 148
33.5 Direct Observation of the Superallowed α-Decay of 104 Te 148
33.5.1 Main notes 148
33.5.2 Connections 149
33.5.3 Summary 149
33.5.4 Questions 149

34 Coding **151**
34.1 *Linux Terminal 151
34.2 *Github 151
34.3 *FORTRAN 151
34.4 *Python 151
34.5 *Emacs 151
34.6 *LaTeX 151

35 Bible Reading **153**

35.1 2026-16-02 153
35.1.1 Genesis 41:17-42:17 153
35.1.2 Matthew 13:24-46 153
35.1.3 Psalm 18:1-15 154
35.1.4 Proverbs 4:1-6 154
35.2 2026-17-02 154
35.2.1 Old testament 154
35.2.2 New Testament 154
35.2.3 Psalm 154
35.2.4 Proverbs 154
35.3 2026-24-03 155
35.3.1 Genesis 46:1-47:31 155
35.3.2 Matthew 15:1-28 155
35.3.3 Psalm 19:1-14 155
35.3.4 Proverbs 4:14-19 155
35.4 03/25/26 156
35.4.1 Genesis 48:1-49:33 156
35.4.2 Matthew 15:29-16:12 156
35.4.3 Psalm 20:1-9 157
35.4.4 Proverbs 4:20-22 157
35.4.5 Take-away 157
35.5 4/24/26 158
35.5.1 Exodus 158
35.5.2 Matthew 158
35.5.3 Psalm 158
35.5.4 Proverbs 159
35.6 4/25/26 159
35.6.1 Exodus 159

35.6.2 Matthew 160
35.6.3 Psalm 160
35.6.4 Proverbs 160
35.7 4/29/26 160
35.7.1 Exodus 160
35.7.2 Matthew 161
35.7.3 Psalms 162
35.7.4 Proverbs 162
35.8 5/24/26 162
35.8.1 Exodus 162
35.8.2 Matthew 163
35.8.3 Psalm 163
35.8.4 Proverbs 163
35.9 5/26/26 163
35.9.1 Matthew 163
35.10 6/8/26 164
35.10.1 Exodus 164
35.10.2 Matthew 164
35.10.3 Proverbs 164
35.11 6/16/26 164
35.11.1 Exodus 164
35.11.2 Matthew 165
35.11.3 Psalms 165
35.11.4 Proverbs 165
35.12 6/23/26 165
35.12.1 Exodus 165
35.12.2 Matthew 166
35.12.3 Psalms 166

35.12.4 Proverbs 166
35.13 Meeting 167
35.13.1 Order of events 167
35.13.2 Things I have noticed 167
35.13.3 Discussion Questions 168
35.14 6/24/26 169
35.14.1 Exodus 169
35.14.2 Matthew 170
35.14.3 Psalms 170
35.14.4 Proverbs 170
35.15 7/27/26 170
35.15.1 Exodus 170
35.15.2 Matthew 171
35.15.2.1 Psalm 171
35.15.2.2 Proverbs 171

36 Philosophy **173**
36.1 Scope 173
36.2 Nicomachean Ethics 173
36.2.1 3/26/26 173

37 Random readings **175**
37.1 Plan 175
37.2 Doing it 176
37.2.1 Sept 23, 2026 176

III Projects **177**

38 Introduction **179**

39 Previous Research **181**

39.1 3D Modeling of Non-Equilibrium Dynamics in Compressed Plasma Using Lattice Boltzmann Method 181

39.1.1 3D Modeling of Non-Equilibrium Dynamics in Compressed Plasma Using Lattice Boltzmann Method Introduction 182

39.1.2 Physical Model 183

39.1.3 Density Distribution Function 184

39.1.4 Potential Energy Distribution 184

39.1.5 Hyperbolic Maxwell Equations 184

39.1.6 Kinetic Non-Equilibrium Method 185

39.1.7 Discussion 185

39.2 A Hamiltonian Framework on ICF Implosions Rocket Equation Based on Rayleigh–Taylor Instabilities 186

39.2.1 Background 186

39.2.2 Problem Statement 187

39.2.3 Assumptions 187

39.2.4 Framework 188

39.2.5 Shell Kinematics 188

39.2.6 Shell Areal Density 189

39.2.7 Compressed-Fuel Pressure 189

39.2.8 Variational Principles 189

39.2.8.1 Phase-Space Lagrangian 189

39.2.8.2 Conservation Laws 190

39.2.9 Conclusion 190

39.3 Basic MHD 191
39.3.1 MHD Module 191
39.3.2 Hamiltonain MHD 192
39.4 A Metriplectic Formulation of Reduced Magneto-hydrodynamics with Resistivity 192
39.4.1 Introduction 192
39.4.2 Defining the Hamiltonian 193
39.4.3 Entropy and Dissipation 193
39.4.4 Metric Bracket and Metriplectic Structure 193
39.4.5 Combined Dynamics 194
39.5 Death Star 194
39.6 Interstellar travel 195
39.6.1 Speeds and forces 195
39.6.2 Magnetic Fusion Plasma Drive 198
39.6.2.1 Theoretical Basis 198
39.6.3 *Optimization 199
39.6.4 *Special Circumstances 199
39.6.4.1 *Strong Magnetic Fields 199
39.6.4.2 *Gravitational Fields 199
39.7 Kerr Black hole stuff 199
39.7.1 Energy (finish) 199
39.7.2 *Multiversal Travel 201
39.7.3 *Other effects 201
39.8 *Extreme Theoretical Work 201
39.9 Astroid Game 201
39.9.1 Introduction 201
39.9.2 Classical 201
39.9.3 Newtonian Gravity 201

39.9.4 Dark Matter 201
39.9.5 Relativistic 201

40 Omega-X **203**

40.1 Pre-planing(part 1) 203
40.1.1 Ideas & Overarching Plan 203
40.1.2 Metriplectic 4-bracket algorithm for constructing thermodynamically consistent dynamical systems 204
40.1.2.1 Math 204
40.1.3 A thermodynamically consistent discretizations of 1D thermal-fluid models using their metriplectic 4-bracket structure 205
40.1.3.1 Math 205
40.1.3.2 Coding 208
40.1.3.3 Future Work 208
40.1.4 Metriplectic Framework for Dissipative Magneto-Hydrodynamics 209
40.1.4.1 My Introduction 209
40.1.4.2 Notes on introduction 209
40.1.5 METRIPLECTIC FORMULATION OF VISCO-RESISTIVE MHD 211
40.1.6 A least Action principle for visco-resistive Hall Magnetohydrodynamic with metriplectic reformulation 212
40.1.6.1 Abstract 212
40.1.6.2 Metriplectic bracket, Hamiltonian and Entropy for Viscoresistive Hall MHD 213

40.1.7 Variational discretizations of viscous and resistive magnetohydrodynamics using structure-preserving finite elements 213
40.1.7.1 Abstract 213
40.2 part 2 214
40.2.1 Coding(1D fluid) 214
40.2.1.0.1 Introduction 214
40.2.1.0.2 mesh_1d 214
40.2.1.0.3 states_1d 216
40.2.1.0.4 projection_matrix_1d . . . 218
40.2.1.0.5 mass_1d 219
40.2.1.0.6 momentum_1d 220
40.2.1.0.7 entropy_1d 222
40.2.1.0.8 velocity_1d 223
40.2.1.0.9 EOS_1d_thermal 224
40.2.1.0.10 time_integrator_1d 226
40.2.1.0.11 io_1d 227
40.2.1.0.12 2D_Thermal_driver 228
40.2.1.1 Plans 230
40.3 part 3 230
40.3.1 Notes 230
40.3.1.1 Notes on module FORTRAN 230
40.3.1.2 Diminsionaless programing 231
40.3.1.3 Generalize 231
40.3.2 Plan 232
40.3.3 New Structure 233
40.4 Part 4 233
40.4.1 Summary On the Past 234

40.4.2 AI Attempt 234
40.4.2.1 AI Analysis 234
40.4.2.2 Code 234
40.4.2.3 My Analysis 234

41 Axiom **235**
41.1 *Structure 235
41.2 LaTeX 235
41.2.1 Division of Structure 235
41.2.2 Commands 236

42 Linux & Customizing my Computer **239**
42.1 Introductory Plans 239
42.2 Package Manegment 239
42.3 Terminal Emulator 240
42.4 Editor 240
42.5 AI summary 240
42.5.1 Core System Setup 240
42.5.1.1 Installation 240
42.5.1.2 Purpose 240
42.5.2 Compilers 241
42.5.2.1 Installation 241
42.5.2.2 Why Multiple Compilers? 241
42.5.2.3 Usage Example 241
42.5.3 Scientific Libraries 241
42.5.3.1 Installation 241
42.5.3.2 Mathematical Role 242
42.5.3.3 MPI Example 242

42.5.4 Python Scientific Environment 242
42.5.4.1 Installation (Conda) 242
42.5.4.2 Role in Workflow 242
42.5.4.3 Example 243
42.5.5 Fortran Development 243
42.5.5.1 Tools 243
42.5.5.2 Why Fortran? 243
42.5.6 Visualization 243
42.5.6.1 Installation 243
42.5.6.2 Purpose 243
42.5.7 LaTeX Environment 244
42.5.7.1 Installation 244
42.5.7.2 Purpose 244
42.5.8 Editors 244
42.5.8.1 Installation 244
42.5.8.2 Usage Philosophy 244
42.5.9 Containers and Reproducibility 244
42.5.9.1 Installation 244
42.5.9.2 Purpose 245
42.5.10 Workflow Architecture 245
42.5.10.1 Full Pipeline 245
42.5.11 Directory Structure 245
42.5.12 Advanced Topics 246
42.5.13 Performance Optimization 246
42.5.13.1 Parallel Scaling 246
42.5.13.2 Reproducibility Strategy 246
42.6 3/20/26 246

43 Emacs **247**

44 manim **249**
44.1 Notes from making WHCKL 249

45 Website **251**
45.1 Brainstorm 251
45.2 2/16/2026 251

46 Category-Theoretic Metriplectic **253**
46.1 Plans 253
46.2 Papers 253
46.3 Review of Metriplectic systems 254
46.4 Understanding of Category-Theoretic Systems . . 254
46.5 My work 254

47 Gravitational Memory study **255**
47.1 First meeting 255
47.1.1 First go: Detecting Gravitational Wave Memory in the Next Galactic Core-Collapse Supernova 256
47.1.1.1 Action Notes 256
47.1.1.2 Code 256
47.1.1.3 General Readings/Videos 256
47.2 September 10 257
47.2.1 Code 257
47.2.2 Quadrupoles 257
47.3 September 24 258
47.3.1 Meeting overview 258
47.4 October 1 258

47.5 October 15 258

48 Website **259**
48.1 Brainstorm 259
48.2 2/16/2026 259

49 YouTube **261**
49.1 Scope and Goals 261
49.2 Getting Started 261
49.2.1 Ideas 261
49.3 Introduction to Symplectic Integrators 264
49.3.1 General overview of direction 264
49.3.2 Opening 265
49.3.3 Motivation 265
49.3.4 Hamiltonian Mechanics Review 265
49.3.5 Why standard models fail 267
49.3.6 Symplectic Integrators 268
49.3.7 Symplectic Euler's method 269
49.3.8 The LeapFrog/Verlet Method 270
49.3.9 Splitting Method 271
49.3.10 Beyond the Basics 272
49.4 Physical Interpretations of Hamiltonian Dynamics and Byond 273
49.4.1 Planning 273
49.4.2 Opening 273
49.4.3 What is the Hamiltonian? 274
49.4.4 What is Phase Space? 275
49.4.5 Is Momentum more Fundamental than Velocity 276

49.4.6 Hamiltonian Mechanics and Teleology . . 277
49.4.7 Noether's Theorem 277
49.4.8 Canonical Transformations 277
49.4.9 The Question of Dualism 277
49.4.10 Hamiltonian Branch offs 277
49.4.11 Hamiltonian and Dissipation 277
49.4.11.1 Metriplectic and Others 277
49.4.12 Structural Realism 277
49.5 Variational Integrator for RMHD 277
49.5.1 Planning 277
49.5.2 Opening 278
49.5.3 Introduction 278
49.5.4 Variational Integrators 278
49.5.5 RMHD 280
49.5.6 Formal Lagrandian 281
49.5.7 Conservation Laws 282
49.5.8 Electron Inertia 282
49.5.9 Symmetrisation 283
49.5.10 Discrete Action Principle 284
49.5.11 Discrete Conservation Laws 286
49.5.12 Numerical Methods 287
49.5.13 Validation 287
49.5.14 Random Test 288
49.6 Why: Developing Physics 288
49.6.1 Planing 288
49.6.2 Introduction 288
49.7 Can you Destroy a Universe with a Force? 289
49.7.1 Introduction 289

49.7.2 Planck Force 289
49.7.3 Striking the Earth with a Planck Force . . 290
49.7.4 Assumptions about a Universe 291
49.7.5 Modeling a Universe 292
49.7.6 Results 294
49.7.7 Interprating the Results 294
49.7.8 General Questions 295
49.8 Over Engineering Celestial Mechanics: WHCKL model 296
49.8.1 Planning 296
49.8.2 Introduction 297
49.8.3 Scope 297
49.8.4 Deriving WH 297
49.8.5 WH Maping 300
49.8.6 Symplectic Correctors 302
49.8.7 Building the Code 302
49.8.8 Results 303
49.8.9 Watching it 303
49.9 Lie Operator Splitting: One of the ten Pillars . . . 304
49.9.1 Introduction 304
49.9.2 Scope 304
49.9.3 Motivating Operator Splitting 305
49.9.4 Deriving Strang Splitting 306
49.9.5 Adjoint formulism 308
49.9.6 Higher Order Splitting and Composition . 308
49.10 Discontinuous Galerkin Method: Self Gravitating Euler Equations 309
49.10.1 Planning 309

49.10.2 Introduction 310
49.10.3 Scope 310
49.10.4 Introduction to DG 310
49.10.5 Physical Model 313
49.10.6 Mesh and Projection 313
49.10.7 Flux 314
49.10.8 Scheme 315
49.10.8.1 Why local DG, specifically 316
49.10.9 Standard LDG 316
49.10.9.1 Fluxes for the elliptic part 318
49.10.10 Structure preserving DG 320
49.10.10.1 Fix one: splitting the potential for well-balancedness 322
49.10.10.1.1 The modified HLLC flux. . 322
49.10.10.1.2 The well-balanced source term. 323
49.10.10.2 Fix two: a conservative form for total energy 325
49.10.11 Time Discretization 328
49.10.12 Dissilation Elimination 330
49.10.13 Structure Preserving Properties 332
49.10.13.1 Theorem: well-balancedness . . . 332
49.10.13.2 Theorem: total energy conservation 333
49.10.14 The Code itself (Flowchart) 335
49.10.15 Results 336
49.11 What is Velocity(joke video) 336
49.11.1 Introduction 338
49.11.2 Basic 341

49.11.3 Newtonian 341
49.11.4 Special Relativity 342
49.11.5 GR 343
49.11.6 Dynamics 344
49.11.7 Other 345
49.12 Developing Numerical General Relativity: Part 1, Vacuum Kerr Black Whole 346
49.12.1 Planning 346
49.12.2 Introduction 347
49.12.3 Scope 347
49.12.4 Kerr SpaceTime 348
49.12.5 3+1 Split 350
49.12.6 ADM Formulism 351
49.12.7 BSSN formalism 353
49.12.8 Guage Conditions 354
49.12.9 Initial Data 355
49.12.10 Cartoon Method 356
49.12.11 Excision 357
49.12.12 Method of lines + RK4 359
49.12.13 Boundary Conditions 360
49.12.14 Code 361
49.12.15 Validation 361
49.12.16 Next Steps 361
49.12.17 Results 361
49.13 Metriplectic Part 1: Why Can't Ordinary Hamiltonians be Dissipative 361
49.13.1 Introduction 362
49.13.2 Example 362

49.13.3 Why? 363
49.13.4 What a Hamiltonian System Actually Is . . 364
49.13.5 Why This Disallows Dissipation 365
49.13.5.1 Loophole 1: Hide the Dissipation in Time 366
49.13.5.2 Loophole 2: Give Up the Canonical Structure 366
49.13.6 Reversibility 367
49.13.7 Entropy 368
49.13.8 Metriplectic Dynamics 369
49.13.9 Conclusion and a Look Ahead 371
49.14 Numerically Simulating Navier-Stokes Singularity 372
49.15 Developing Numerical Relativity: Simulating a Star with TOV 373

50 GHMDH: Kerr black hole inside of Star 375
50.1 Scope 375
50.2 Stage 1: Vacuum Kerr Black Hole 375
50.2.1 Initializing Values 375
50.2.2 Hamiltonian Constraints 376
50.2.3 Integration 376
50.2.3.1 RK4 376
50.2.3.2 RHS 376
50.3 Theorizing about Hydrodynamics 377
50.3.1 Initialization 377
50.4 Orientation 378
50.4.1 What this code is 378
50.4.2 Why the design is shaped this way 378
50.4.3 Quick start 379

50.4.4 Repository map 380
50.4.4.1 Core evolution path 380
50.4.4.2 Not on the production path 381
50.4.4.3 Tests and diagnostics 382
50.5 The Physics 383
50.5.1 From ADM to BSSN 383
50.5.1.1 ADM 383
50.5.1.2 The conformal split 383
50.5.2 Constraints 384
50.5.3 Gauge 385
50.6 The Discretisation 386
50.6.1 The grid 386
50.6.2 The Cartoon method in full 387
50.6.2.1 What is stored 387
50.6.2.2 The radial interpolation 387
50.6.2.3 Rotation 388
50.6.3 Finite differences and the fallback cascade 389
50.6.3.1 Every stencil in the code 389
50.6.3.2 The fallback logic 389
50.6.3.3 The variable mask contract 390
50.7 Boundaries 391
50.7.1 Inner: excision 391
50.7.1.1 The mask 391
50.7.1.2 The ghost fill 391
50.7.1.3 Onion-peeling and the frozen snapshot 392
50.7.1.4 Positivity enforcement 393
50.7.1.5 The characteristic SAT 393

50.7.2 Outer: Sommerfeld 394
50.7.3 Time integration 395
50.7.3.1 The timestep 395
50.7.3.2 The algebraic projection 395
50.8 Initial Data 396
50.8.0.1 The conformal connection 396
50.9 Constraints and Diagnostics 397
50.9.1 The margin trap 397
50.9.2 The near-mask band 398
50.9.3 The diagnostic programs 399
50.9.3.1 diagnose-ghost-accuracy 399
50.9.3.2 diagnose-momentum-point 400
50.9.3.3 diagnose-inner-boundary 400
50.9.3.4 A methodological warning 400
50.10 Build and Test 401
50.10.1 The parity gate 401
50.10.2 Sources must be listed in dependency order 402
50.10.3 Environment overrides 402
50.11 A Tour of Every File 403
50.11.0.1 The first-order track 405
50.11.1 SBP-SAT: why it is not finished 406
50.12 Findings 407
50.12.1 The failure 407
50.12.2 What was established 407
50.12.2.1 Negative results, all measured . . 407
50.12.2.2 The live picture 408
50.12.3 Excision order: a trade-off, not a win . . . 409

50.12.4 How this investigation went wrong, repeatedly 409
50.13 Writing Your Own 410
50.13.1 Where to start 410
50.13.2 Design lessons worth stealing 412
50.13.3 Where the physics is genuinely unfinished 412
50.14 Deriving BSSN, and Why It Exists 413
50.14.1 The failure of ADM in practice 413
50.14.2 What the constraint propagation mechanism needs 414
50.14.3 Where each BSSN piece comes from 415
50.14.4 Why the code writes $\tilde{R}$ in connection form 416
50.15 A Worked Example: the Cartoon Method 417
50.15.1 The problem 417
50.15.2 Step 1: establish the symmetry correctly . 418
50.15.3 Step 2: reduce to radius and interpolate . 418
50.15.4 Step 3: rotate by rank 419
50.15.5 Step 4: validate, in this order 419
50.16 Debugging Numerical Relativity Code 420
50.16.1 Rule 1: verify the instrument before interpreting it 420
50.16.2 Rule 2: cross-check against an independent truth 421
50.16.3 Rule 3: check the order of accuracy you claim 421
50.16.4 Rule 4: know which knobs are inert 422
50.16.5 Rule 5: gates must be able to fail 422
50.16.6 Rule 6: keep a dated findings log, including retractions 422

50.16.7 Rule 7: restate the baseline whenever anything moves . . . 423
50.17 Where the Numbers Come From . . . 423
50.17.1 The baseline of record . . . 423
50.17.2 Lifetimes . . . 424
50.17.3 Near-mask measurements . . . 424
50.17.4 Driver worst-point locations, corrected filter 425
.1 Traps, Collected . . . 425
.2 Environment Overrides and Reproduction . . . 426
.2.0.0.1 Environment. . . . 427

A Different Numerical Methods for Metriplectic Simulations 429

B DG CCSN 431
B.1 Scope and goal . . . 431
B.2 Mathematical Model . . . 431
B.3 General DG . . . 432
B.4 My DG . . . 432
B.4.1 AI overview . . . 433
B.5 General Notes . . . 434
B.6 Results . . . 434

C WarpDrive Collision 435
C.1 Scope . . . 435

D Projects for YouTube 437
D.1 Planck Force on Earth . . . 437
D.2 Collapsing Universe . . . 437
D.3 Post-Newtonian Simulations . . . 437

D.3.1 General idea 437
D.3.2 Notes on Brans-Dicke 438
D.4 CCSN via DG 438

E Hamiltonian, Boltzmann, to Navier stokes 439
E.1 Introduction 440
E.2 Scope 440
E.3 Brief Overview 440
E.4 From Liouville to BBGKY 440
E.4.1 The BBGKY Hierarchy 443
E.5 Motivating the Boltzmann Equation 444

F The Eudaimonia Society 445
F.1 Initial Exploration 445
F.1.1 General Thoughts & Scope 445
F.1.1.1 Moral Development 446
F.1.1.2 Intellectual Development 446
F.1.1.3 Social Cohesion 447
F.1.1.4 Social Impact 447
F.1.2 First layer of Refinement 447
F.1.3 Mission Statement 447
F.1.4 Structure 447
F.1.5 Meetings 448

G NEP 449
G.1 Preliminary plans 449
G.1.1 Second week 449
G.1.1.1 Goals for next week 449
G.1.1.2 Goal 450

G.1.1.2.1 Scope, Assumptions, and Decision Criteria 451
G.1.1.2.1.1 The actual decision 451
G.1.1.2.1.2 Assumed propulsion architecture 452
G.1.1.2.1.3 Performance relations453
G.1.1.2.1.4 What constitutes a good mission for this project? 454
G.1.1.2.2 Mission A: Atomic Clock in Mars Orbit 454
G.1.1.2.2.1 Concept and potential purpose 454
G.1.1.2.2.2 Arguments in favor 455
G.1.1.2.2.3 Arguments against 459
G.1.1.2.2.4 Assessment 462
G.1.1.2.3 Mission B: Atomic Clock near Mars–Sun L1 462
G.1.1.2.3.1 Clarifying the location 462
G.1.1.2.3.2 Arguments in favor 463
G.1.1.2.3.3 Arguments against 465
G.1.1.2.3.4 Assessment 468
G.1.1.2.4 Mars Orbit versus Mars–Sun L1 468
G.1.1.2.4.1 The fundamental difference 468
G.1.1.2.4.2 A quantitative way to compare them . . . 469

G.1.1.2.4.3 Recommended comparison experiment 470
G.1.1.2.4.4 Decision 471
G.1.1.2.5 Mission C: Mars Flyby . . 471
G.1.1.2.5.1 Concept and mission objective 471
G.1.1.2.5.2 Arguments in favor 472
G.1.1.2.5.3 Arguments against 474
G.1.1.2.5.4 Assessment 476
G.1.1.2.6 Mission D: Mars Rover . . 477
G.1.1.2.6.1 Concept and mission objective 477
G.1.1.2.6.2 Arguments in favor 477
G.1.1.2.6.3 Arguments against 480
G.1.2 Assessment 483
G.1.2.0.1 Cross-Mission Comparison 483
G.1.2.0.1.1 Qualitative comparison 483
G.1.2.0.1.2 The four missions answer different questions 485
G.1.2.0.2 Recommended Quantitative Evaluation 486
G.1.2.0.2.1 Step 1: Define the mission requirements486
G.1.2.0.2.2 Step 2: Build a baseline mass budget . 487
G.1.2.0.2.3 Step 3: Compute first-order propulsion requirements 487

G.1.2.0.2.4 Step 4: Model the trajectory at an appropriate fidelity 488
G.1.2.0.2.5 Step 5: Compare against a credible alternative 489
G.1.2.0.2.6 Step 6: Tie the mission to the thruster design 490
G.1.2.0.3 Final Recommendation . 491
G.1.2.0.3.1 If the priority is a clear mission justification 491
G.1.2.0.3.2 If the priority is a more novel navigation study491
G.1.2.0.3.3 If the priority is a focused trajectory project491
G.1.2.0.3.4 If the priority is an exploration mission 492
G.1.2.0.3.5 Overall conclusion . 492
G.2 Design 496
G.3 Optimization 496
G.4 Teachings 496
G.4.1 Numerics 496
G.4.2 Plasma Physics 497

H Story 499
H.1 Scope and Concept 499
H.2 Characters 499
H.2.1 POV Characters 499
H.2.1.1 Idea for POV characters 499
H.2.1.2 Dr. John Jackson, 32 499

H.2.1.3 Dr. Marcus Venn, 46 500
H.2.1.4 Commander John Cram 501
H.2.1.5 Secretary 501
H.3 Plot 501

I Random Small stuff **503**
I.1 Singularity Formulation in 3D Euler and Nonuniqueness in unforced 3D Navier-Stokes 503

IV Essays **505**

J Introduction **507**

K Post-Surgery **509**
K.1 Overview 509
K.2 1/9/2026 Essay: Cultivation in Degregation . . 509
K.3 3/../2026: New Beginnings and Failures 511

L Metaphysics **513**
L.1 Scientific Materialism and God 513
L.2 Scientific Realism vs Anti-realism 515
L.3 Causality 515
L.4 Divine Law Theory of Causation 521
L.5 Material, Immaterial, and Reality itself 521
L.6 Identity 522
L.7 SpaceTime 523
L.8 Interpretation of Quantum Mechanics 524
L.9 *Other Questions in the MetaPhysics of Quantum Mechanics 527

L.10 Quantum Consciousness, Arminianism, Super-Gödel thinking, OR Orch, why I even decided to write this book, and how long can I make the title for a section within a chapter 528
L.11 Compatibilist Free Will 531
L.12 *Super-determinism, semi-determinism, Retro-causality, and Chaos theory 531
L.13 Role of God 531
L.14 Post-Axiomatic Nature of God 532
L.15 Metaphysics of cosmology 533
L.16 Ontology of Information and Entropy 534
L.17 Dualism 535
L.18 The Problem of the one-way speed of light 535
L.18.1 *Phase-space and the Euclidean Mind . . 537
L.19 *Ontology of Numbers 537
L.20 Reality and the Abstract 537
L.21 Dialects' Critique of Relativity 537
L.22 Structure as Fundamental 540

M Epistemology 541
M.1 Limited Objectivism 541
M.2 Consciousness and Reality 542
M.3 Nature and Validation of Axioms 544
M.4 Non-infinite Definition of words and Axiomatic Semantics 545
M.5 Induction & Deduction 545
M.6 The Problem of Induction 546

N Ethics 547

N.1 Introduction to ethics 547
N.2 Virtue Ethics 548
N.3 Stewardship 550
N.4 Categorical Good 551
N.5 Cardinal Virtues 552
N.6 *Salient Virtues 553
N.7 The Philotimic Virtue 553
N.8 *The Internal Promise and Dignity 555
N.9 Good Life and Eternal Struggle 555
N.10*Personal Virtues, Values, and God's Morality . . 556
N.11Intentions 556
N.12*Morality and Religion 557
N.13*The Ethics of Knowledge 557
N.14Recursive Connection of Civilization & Virtue . . 557
N.14.1*Social Rules and Morality 558
N.14.2*Culture as Moral Education 558
N.14.3Intensity of Duty* 558
N.15Morality in the End of the World 559

O Personal Philosophy 561
O.1 *God as Foundation 561
O.2 Will as Identity 561
O.3 *Fanatical Ethics 563
O.4 The Duty as an Actor 563
O.5 *The Aesthetic Pursuit of Truth 563

P Personal-Analysis 565
P.1 Introduction 565

P.2 Disgrace and Pride 565
P.3 On Ayn Rand 569
P.4 On the Pursuit of Thought 571
P.5 Nietzsche's Sovereign Man and Morality 572
P.6 Half-Growth and Half-Death 574
P.7 Lack of Aesthetic 578
P.8 Boredom 578
P.9 Why Physics 580
P.10 Lack of Resistance 581
P.11 Am I Misanthropic?* 582
P.12 Depth of Feeling 582
P.13 Language Usage* 583
P.14 Euphesus* 585

Q Self-Authorship 589
Q.1 Scope and Understanding 589
Q.2 Ideals of Becoming 590
Q.2.1 Overview of Virtues 590
Q.2.1.0.1 Intellectual Virtues 590

R Philotimic 593
R.1 Intro 593
R.2 Overview 593
R.2.0.0.1 Self-Perfection 593
R.2.0.0.2 Outward image 594
R.2.0.0.3 Speech, Vocabulary, Diction, and Rhetoric 594
R.2.0.0.4 Academic Communication 594

R.2.0.0.5 Assumptions and Cognative Habits 595
R.2.0.0.6 Bayes Reasoning 595
R.2.0.0.7 Meta-analysis 595
R.2.0.0.8 Attention 595
R.2.0.0.9 Observational awarness . 596
R.2.0.0.10 Leadership 596
R.2.0.0.11 Dealines 596
R.2.0.0.12 Body Language 596
R.2.0.0.13 Memory 596
R.2.0.0.14 Desicion making 596
R.2.0.0.15 Teaching 596
R.2.0.0.16 Schedule 596
R.2.0.0.17 Humor 597
R.2.0.0.18 Writing 597
R.2.0.0.19 Socratic Dialog 597
R.3 Doing it 597
R.3.1 9/21/26 597
R.3.1.1 Morning Outlook 597
R.3.1.2 Notes 598
R.3.1.3 Review of the Day 598

S The Structure of Motion: A Philosophical Journey Through Hamiltonian Mechanics 599
S.1 Introduction 599
S.1.1 Physical Reality and Frameworks 600
S.1.2 Philosophical Context* 601
S.1.3 Notion of Time, States, and Trajectories . . 602
S.1.4 Introduction to Phase Space 603

S.1.5 Teleology vs Causality 603
S.1.6 Role of Variational Principles in Mechanics 604
S.2 Axiomatic Mechanics 604
S.2.1 Axiomatic Derivation of 604
S.2.1.1 Introduction 604
S.2.1.2 Axiom of Real numbers 605
S.2.1.3 Axiom of Euclidean Geometry . . 605
S.2.1.4 Axiom of Set Theory (Basic) 606
S.2.1.5 Axiom of Calculus (Differential and Integral Calculus) 606
S.2.1.6 Axioms of Variational Calculus . . 607
S.2.1.7 Axioms of Linear Algebra (Basic) . 607
S.2.1.8 Axioms of Variational Calculus . . 608
S.2.1.9 Physical Assumptions (Not Strictly Mathematical Axioms) 608
S.2.2 Axiomatic Derivation of Hamiltonian Mechanics 609
S.2.3 *Hamiltonian Mechanics ad a Geometric Theory 611
S.2.4 *Poisson Brackets and the Algebra of Dynamics 611
S.2.5 *Canonical Transformations: Symmetry, Simplicity, an Structure 611
S.3 Philosophical Implications* 611
S.3.1 *Time in Essence 611
S.3.2 *Hamiltonian Constraints and the Problem with Time 611
S.3.3 Noether's Theorem 611
S.3.3.1 *Ontolological Priority of Symmetry 611

S.3.3.2 Laws as Autonomous Structures . 612
S.3.3.3 The Continual Definition of Energy 612
S.3.3.4 Nature of Time 612
S.4 Modern Hamiltonian 612
S.4.1 Modern Research 612
S.4.1.1 *Hamiltonian Formulation in Plasma Physics 612
S.4.1.2 *Hamiltonian and Metriplectic Mechanics 612
S.4.1.3 *Symplectic Integrators: Preserving the Structure of Nature 612
S.4.1.4 *HNNN(Hamiltonian Neural Networks)612
S.4.2 Fun Mess Around 612
S.4.2.1 Non-Canonical Poisson Brackets . 612

T Dialog 615
T.1 Introduction and Scope 615
T.2 Politics 615
T.2.0.0.1 AI is a Net Positive* 615
T.3 Personal 615
T.3.0.0.1 Who Should I Become(part 1, Broad) 615
T.3.0.0.2 My use of AI 619
T.4 Philosophy 619
T.4.0.0.1 The Paradox of Absolute Altruism 619
T.5 Physics 620
T.5.0.0.1 The Laws of Physics are the same everywhere 620

T.5.0.0.2 Computational Models and their Accuracy* 624
T.5.0.0.3 Further Development in Structural preserving methods 624
T.5.0.0.4 Explaining Gravitational Waves624
T.6 Mathematics 626
T.7 Research Culture 626
T.8 Broader Culture 626
T.9 General Conversing 626

U Clarification **627**
U.1 Scope and Purpose 627
U.2 Physics 627
U.3 Math 627
U.3.0.0.1 Multilinear 627

V Random Stuff **629**
V.1 Euphesus, Continuation 629
V.1.1 Anaysis of the Person 629

V Journaling **633**

W *Introduction **635**

X Plans **637**
X.1 Now 637
X.1.1 Self-Education 637
X.1.2 Research 641
X.1.3 Other activities 643

X.2 Pre-University 644
X.2.1 Independent Learning 644
X.3 Undergrad 644
X.3.1 Classes 644
X.3.1.1 Requirments 644
X.3.1.1.1 General 644
X.3.1.1.2 Academic Physics Major 645
X.3.1.1.3 Applied Maths Major 646
X.3.1.1.4 Computer Science Minor 647
X.3.1.1.5 Electives 647
X.3.1.1.6 Total 648
X.3.1.2 Schedule 650
X.3.1.3 First Year 651
X.3.1.4 Second Year 651
X.3.1.5 Third year 652
X.3.1.6 Fourth year 653
X.3.1.7 Others 653
X.3.1.8 Possible summer 654
X.3.2 Goals for the first three weeks 654
X.3.3 First Week 656
X.4 Second Semester 656
X.4.1 Classes 656
X.5 *Grad 657
X.6 *Early career 657
X.7 *Late Career 657

Y Schedule Templates **659**
Y.1 Preliminary Semester 1 659

Z Journal **665**
Z.1 2025-11-27 665
Z.2 Emacs 665
Z.3 2025-11-29 666
Z.4 Axiom 666
Z.5 Emacs 666
Z.6 Media 667
Z.6.1 Zettelkasten 667
Z.7 2025-12-03 667
Z.8 Links to Book Chapters 667
Z.9 Emacs 667
Z.9.1 Emacs updates 667
Z.9.2 Emacs vs Vim 667
Z.9.3 Org vs Tex 668
Z.9.4 Future 668
Z.10 Omega-X 669
Z.10.1 Arguments 669
Z.10.2 Updates 669
Z.11 Future Ideas 669
Z.11.1 Categorical Metriplectic 670
Z.11.2 Cason Brackets Dynamics(name pending) 670
Z.11.3 Metriplectic PINN 670
Z.12 Media 670
Z.12.1 Star Trek 670
Z.13 Self-analysis 671
Z.13.1 Stress & TV 671
Z.14 2025-12-04 671
Z.15 Links to books 671

Z.16 Projects 671
Z.16.1 Formal projects 671
Z.16.2 Skills 672
Z.16.3 Plan 672
Z.17 2025-12-05 674
Z.18 Links to Book Chapters 674
Z.19 Plans for Today: 674
Z.19.1 Omega-X Notes 675
Z.19.2 Emacs Build 676
Z.19.3 Categoric Metreplectic 678
Z.20 2026-01-03 678
Z.21 Links to Book Chapters 678
Z.22 Emacs 678
Z.22.1 TABS 678
Z.23 2026-02-05 679
Z.24 Links to Book Chapters 679
Z.25 Plans 679
Z.26 Schedule 679
Z.26.1 9:00 - 11:00 679
Z.26.2 11:00 - 1:00 679
Z.26.3 1:00 - 4:00 679
Z.26.4 4:00 - 5:00 680
Z.26.5 5:00 - 5:45 680
Z.26.6 5:45 - 6:00 680
Z.26.7 6:00 - 7:00 680
Z.26.8 7:30 - 8:30 680
Z.26.9 8:30 - 10:00 680
Z.27 2026-02-16 681

Z.28 Bible Reading 681
Z.28.1 Genesis 41:17-42:17 681
Z.28.2 Matthew 13:24-46 681
Z.28.3 Psalm 18:1-15 681
Z.28.4 Proverbs 4:1-6 681
Z.29 Today 682
Z.30 2026-02-17 682
Z.31 Bible reading 682
Z.31.1 Old testament 682
Z.31.2 New Testament 682
Z.31.3 Psalm 682
Z.31.4 Proverbs 683
Z.32 Ideas for projects 683
Z.33 Schedule 683
Z.33.1 Coding 683
Z.33.2 Writing 683
Z.33.3 Math/Physics 684
Z.34 April 14 684
Z.35 7/24/26 685

Chapter 1

Preface

"If you start with the right axioms, the rest will follow."— David Hilbert

This book "Axiom" is a continuation of my previous project Lógos. Which was mostly a collection of essays on several topics; physics, philosophy, religion, civilization, psychology, personal plans, etc.

It was previously meant to work through my ideas, using the fact I was writing it out. It allowed me to keep track of the dozens of ideas and how they all interacted with each other. Developing on the complexity without forgetting and creating a full framework with defined axioms.

Creating this new project is for multiple reasons.

1. To create a more organized book
    - What I mean is have an order, like create a fully formalized epistemology and basic metaphysics and then using that to create a fully organized Ontology. Rather than a chaotic coll
2. To mark the end of my high school career and beginning

of University

3. Lógos had become too big and had too many errors
    - Too big to run on Overleaf, too many errors to run directly in the terminal.
        - This will start smaller, and once it is bigger will be made without errors so it can be run on the terminal once it gets bigger. Also more modular
4. Be able to rewrite with my new knowledge to update it.
5. Write it with a better understanding of the packages, LaTeX, and even create my own custom packages

In the end, it serves the same purpose; to help me better understand things through writing, develop thing iteratively over time using writing, and write out notes to remember things for the future.

As of writing this section, the text will be broken into three parts. Part one being "The Foundation." It would be a axiomatic approach to creating the foundations needed for future systems. Two, being "Notes" on things that I have learned. Three, being writings on my "Projects"; mostly physics research projects but also projects in further developing philosophy or other similar things. Finally, "Life in Practice" being more practical things; plans for the future, self-analysis, ideals on society, etc.

Part I

The Foundation

Chapter 2

Introduction

As mentioned in the preface, this section is made to axiomatically develop the foundations needed.

To illustrate, in order to develop physical theories, mathematics must be developed, a philosophy of science must be made, metaphysics, epistemology, and formal logic. Thus, a complete theory of all of these things and more should be created. One that has fundamental, expressible axioms.

Also, most of these theories rely on one another, so a logical chronology must be defined. At the moment of writing this it is defined to be logic $\rightarrow$ metaphysics $\rightarrow$ Epistemology$\rightarrow$ philosophy of science $\rightarrow$ philosophy of physics $\rightarrow$ mathematics $\rightarrow$ physics $\rightarrow$ theology $\rightarrow$ ethics $\rightarrow$ social philosophy $\rightarrow$ personal philosophy.

One additional thing to add, you may notice the separation of metaphysics and the philosophy of physics. These are both meant to signify a difference where metaphysics represents extremely fundamental questions like does existence exist. Things that must be answered before epistemology can be fully introduced. The next section, philosophy of physics, are slightly less fundamental, like the ontology of entropy. Ques-

tions that require a theory of epistemology to be answered.

Chapter 3

Logic

"One must be able to say at all times—instead of points, lines, and planes—tables, chairs, and beer mugs." — David Hilbert

3.1 Introduction

Logic is an important tool for the creation of all formal systems. It is the foundation of mathematics, which is for the most part the foundation of everything else. It is also simply important to anything else by clarify concepts, giving them formal definitions, and insures consistency. Allowing for a bedrock behind all analytical thinking.

Though, before I can get into it too much, I should define what the study of logic is. It is the branch of philosophy that critically examines the fundamental nature, scope, and principles of logic itself.

It has a couple fo subfields as well.

- Philosophy of Logic
  - The study of logic at a high level, analyzing its nature

and scope.

- Formal Logic
  - A creation and usage of symbolic logic symbols to create systems; mainly mathematics.
- Metalogic
  - The technical, mathematical study of the properties of formal logical systems. The philosophy of logic often uses the results of metalogic (like Gödel's incompleteness theorems) to address its conceptual questions, such as the limits of a formal axiomatic system

3.2 Defining an Axiom

This entire text is named Axiom, for it is the foundation. To define systems axiomatically and then build them out into their true complexity is not only one of my favorite things to do, but as an extension of that it is what this book is trying to do.(Obviously, I wouldn't do something like this if I didn't enjoy it.)

So let's define it. As I mention extensively in my last book, Lógos, we must define things extremely rigorously. By testing against edge cases and keep it with consistent.

A good way to start is with the dictionary definition. "a statement accepted as true as the basis for argument or inference." For instance, our understanding of scientific realism is the understanding of the fact that our observations of reality are accurate.

A more defined definition can then be defined to say that an axiom is a statement that either cannot be proved or cannot be proved currently that is used within a greater statement

that can be proved upon its edifice. These axioms, while don't require 'proof' generally desire some type of reasoning even if it is not objective or purely analytical in nature. For instance, the above arguments for the objective nature of reality aren't true objective proof, they still exist as semi-arguments. While we cannot prove the objectivity of reality with true and objective reasoning, some reasoning can be applied to get some kind of 'proof' of the statement so that further studies can be applied based upon that said axiom.

An astute observer can find that these mean the same thing, my is more of an explanation so, the dictionary definition does work:

Definition 3.1. *Axiom: a statement accepted as true as the basis for argument or inference.*

3.3 *****

I am currently a little stuck on how to write this chapter, so I will return at a later point.

3.3.0.1 Propositional logic

The most basic form of classical logic, but instead of telling you now, I will develop it.

3.3.0.2 Axiom System for the Propositional Calculus

For this simple system a number of axioms must be assumed:

- $A \Longrightarrow (B \Longrightarrow A)$
- $(A \Longrightarrow (B \Longrightarrow C)) \Longrightarrow ((A \Longrightarrow B) \Longrightarrow (A \Longrightarrow C))$

- $(\neg B \Longrightarrow \neg A) \Longrightarrow (A \Longrightarrow B)$

Then one rule of inference is needed. Modus Ponens(MP)

From A and $A \Longrightarrow B$ infer B

This defines all tautologies and thus the larger system as a whole.

3.3.0.3 Defining Tautologies

A tautology is simply a statement form that is always true, regardless of the truth values of the statement letters.

An example would be the law of the excluded middle. $(A \vee (\neg A))$. This is a statement that is true by definition and there is no case where it can be false. Axioms in logic are generally tautologies(such is the way are the once mentioned in the previous subsubsection).

3.3.0.4 Truth Tables

Negation is the first an simplest form. The symbol is $\neg$, so $\neg A$ is the negation of A.

so the table can be constructed

A	$\neg A$
T	F
F	T

Where T, represents true, and F represents false.

A further complication is the conjunction operator $\wedge$ or "and."

It combines to functions into a greater set.(a union of function)

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

$A \wedge B$ is only true when both "A" and "B" are true.

$\vee$ is the inverse, only one of the functions must be true. It is called a disjunction.

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

$A \Longrightarrow B$ means "if A, then B" meaning that A needs to happen for B or that A causes B.

	A	B	$A \Longrightarrow B$
T	T		T
T	F		T
F	T		F
F	F		T

There are a couple of other operators used in prepositional logic, but for not I will move on and if needed bring those back up at a later point.

Chapter 4

Philosophy of Mathematics

Chapter 5

Fundamental Metaphysics

5.1 Scope

This chapter will cover fundamental metaphysics, what I mean by this is for my progression of metaphysics I need a theory of epistemology, philosophy of science, inductive and deductive research on physics, etc; however, for all of these things I need some concepts of metaphysics. Things like the existence, reality, consciousness, questions of God, and so much more. Thus, this will serve that purpose.

5.2 Reality

5.2.1 Existence Exists

To truly come to a thorough understanding of a system, one must create it from first principle slowly and carefully. To question every implicit assumption, this for the most part is the quest of philosophy; especially metaphysics.

Thus let us begin our creation of reality.

Existence Exists, this is by definition an irrefutable point. To negate an argument an antithesis must be supplanted, non-existing, but non-existence by definition implies a existence. This is because you cannot have a non-existence without an existence. $\neg E(x) \Longrightarrow E(x) \Longleftrightarrow ((\forall P(Prop(P) \Longrightarrow E(p))) \wedge (\exists Prop(E(\neg E(x)))))$. By the simple definition of existence to be everything. Another more concrete argument is that to deny the existence is a claim, a claim is an act, an act requires and actor, and thus an actor must exist, an actor exists within the subsect of existence, so if it exists so must existence.

Though, this doesn't say anything about the existence, in fact there is a special case in which existence is the absent of all things. Using the second argument, we create a conscious entity. *Cognito, ergo sum*. I think therefore I am. Now upon inspection of this entity it is quite simple, the only capacity it necessarily has is the ability to make a claim, such a claim needs no basis in reality, sense, logic, or anything of such. It could be the dream of a butterfly, random interactions of particles in the dead of space, or anything of the such.

One constraint does exist, that entity is me. While it could still(at least for now) be some random assortment of particles or what ever machinations you may dream of, there is some knowledge we do know. It has the ability to observe, reason, and has innate concepts like space, time, similarities, and differences. One thing to be clear, at this point their is no justification that any of these things have any connection to reality, but they are apart of the entities observations.

5.2.2 The Entity and Observable reality

Now, it is technically true that the entity has no way to truly and faithfully validate his observations and reasoning beyond self-referential, due the fact that these are the entities' only capacity to make claims.

Thus, taking these as true is a fair axiom, unprovable but logical in the necessity of the argument. I do not mean taking all observations as true(in fact always questioning them is of great importance as I will address later). I mean regarding that our observations are a consequence of reality. Taking the reality that existed around us as the "truth." While observations can be complete, wrong, they are not false by the nature of being observations by the entity.

Now this creates a dichotomy between the observed reality and "true reality"(whatever that might be). There is a spectrum of how much these two overlap.

Ranging from:

A: Scientific Materialism; that all of observed reality and true reality overlap because these observations are directly true, and nothing exists beyond what is within the capacity of theoretical observation.

B: Idealism; there is no overlap at all. Such as the Boltzmann brain.

Now, this observable reality, is still apart of existence, and is thus no metaphysically distinct if some greater "true reality" and is thus given the same importance and study. While other sub-existences beyond our observable are lesser for reasons, these are more ethical, aesthetic, and theological rather than truly metaphysical.

5.2.3 Beyond Observable

Given this spectrum, it is entirely possible for there to exist unobservable reality that is true. This reality has unknown capacity and can theoretically interact and cause observabes.

5.2.4 Independence of Reality

Given that this entity is me, I observe the continuation of reality beyond my direct observation. Once again this is not direct proof that it is, but makes it indistinguishable from if it was.

5.3 The Entity and Observations

5.3.1 Introduction

Now this entity observes a great many things; matter, other entities, change, relations, space, time, identity, theory, mathematics, difference, categories, individuation, emotions, structure, aesthetics, fundamentalism, layers, evidence, argumentation, possibility, necessity, abstracta, causation, and much more. Categorizing these and understanding them is the foundation of metaphysics.

5.3.2 Properties

All observables are made up of properties, and many of the observables themselves are properties.

5.3.3 Abstraction: Math, sets, and Relations

Given this nature, these properties can act as sets of 1 and can make up object which are thus observables. They thus follow the same mathematical rules of sets.

- Relations to other sets(and themselves)
- Shared Properties with other sets
- can be apart of larger sets(categories)

- and so much more

Now I won't examine all of this because it is mentioned in the earlier chapter so we will move on.

These sets can be whatever, such as the other properties or something new all together.

5.3.4 Matter

One of the most important sets is matter, within this set is a near infinite amount of subsets. These subsets can "theoretically" have about any property. The main one that define it is that it must have the capacity to affect the properties of other matter(and thus be effected too). It must also have the properties of existing within space and time. Direct space or time is not needed, but having the property is required. Finally, it must be independent of a thinker.

5.3.5 SpaceTime

Once again, given the simplicity of this model this examination will be quite simple. Essentially all that can be taken at this moment is that space and time are properties of many things. Extremely important ones. Dimensions to be considered, later an address the geometry and importance of these fundemental dimensions.

5.3.6 Nous

A thinker is an entity like object of matter. These entities are capable of creating, holding, using, being effected by, etc by non-material objects. These thinkers do not need to be human, or even alive for that matter. For future refrences I will refer to it as nous; based upon the greek word.

5.3.7 Non-material

Already defined to be objects that exist(observable have effects) dependent of a thinker.

5.4 Assumptions(Axioms)

Get it, I said the name of the book, Axiom.

Anyways, here I define and summerize the assumptions I have made.

- Principle of Causal Order
- Time and Spacial symmetry of the laws of nature
- Reality Obeys Fundemental Logic and Tautologies

Many of the other commonly noted axioms like that the unieverse follows mathematic or that nature is orderly follow naturally from these assumptions.

5.5 Platoism and Realism

With this view of reality, to somewhat conflicting views occur, Platoism and Realism.

Given the view of the surronding reality as 'real,' all other principles of Realsim become natural and logical.

From the fact that this view of reality is simply our visiable abstraction of reality based upon logic and mathematics, Platoism becomes natural. Further analysis of these argumentations have already been presented and will be presented in Essays and this section may latter be updated.

For the purposes of this book, the the equavlency of these expressions will generally be used. With, given the mathematical and abstract nature of this exersice, Platoism will be used for the majority of the book with equivlent materialistic expressions being used as a translation.

Chapter 6

Epistemology

"Epistemology without contact with science becomes an empty scheme. Science without epistemology is... primitive and muddled." — Albert Einstein

6.1 Introduction

Here we will derive the basic principles of Epistemology that will later be expanded upon by the next chapter on the philosophy of science.

6.2 Ontology

6.2.1 Statements

There are so many possible statements one can make *"an electron is 511 ± 0.5 keV"* or *"I am a person"* or *brutalism is the best form of artwork*. Given that these statements will be at the corners stone of the structure and development of knowledge; clear definitions, rules, and such must be derived.

One assumption one can take is that these statements exist within the Boolean domain thus that $S \in \mathbb{B} = FALSE, TRUE$. Making it so that not only can truth values be assigned to a statement, a statement can only be true or false. This is clear from any logical analysis that for their to be a logically consistent truth system, that their only be true or false. Nuance, while useful outside of strict epistemology, doesn't actually change the fact that the statement is either true or false.

From here, the rules developed in Formal logic ring true. Addition, negation, or, etc. They are the rules that govern these statements and allow for knowledge to be constructed in the first place.

Another rule to add is equivalence, but there are different types of equivalence. Semantic equivalence(saying the same thing in a different form), Logical equivalence(One proves another, or both prove each other), and finally material equivalence(they both have the same truth value). These tools of equivalence will be useful in the future.

An addition to traditional logic are adding relations like narrower, broader, compatible, or incompatible.

With these in mind we satisfy De Morgan Laws and Boolean algebra infinite distribution with all of the rules we need for the construction of statements. Other similar methods can be used, but this minimal set traditionally holds.

6.2.2 Events

As we have already defined an agent observing objects with observable properties. A new axiom would be as the property of time tends to move forwards, properties tend to change. The position in time state space in which said changes happens is defined to be the *event*

6.2.3 Relations

As we have already defined causal order, a simple extrapolation of simple logic

$$P \to Q$$

$$\therefore \forall P \exists Q$$

If P causes Q, than if P happens then Q happens. A simple relation between the two.

There are other simple derivations to get other relations; biconditional, common causal variable, and coincidental.

6.3 Verifying Statements

Now we must talk about how to verify these statements for the development of knowledge.

6.3.1 Deduction

Using the principles of formal logic, one may derive statements from one another. These statements will be sound by definition but not technically true by definition. For they rely on the statements, Axioms, taken to be true.

These rules must be furfiled by how we defined a statement to be. Being Boolean and thus susceptible to the relations of formal logic.

this can be trivially proved from our earlier axiom that reality obeys such logical rules.

One can use these for several examples. Say an understanding of the laws of gravitation, one can derive the tragectories of objects flying on the moon with only information of the mass, radius, and air resitince on the moon. Or derive

General Relativity from special relativity and the principle of equivalence.

6.3.2 Induction

This brings us to our next form of developing logic, induction. From our earlier Axiom of UoN(uniformity of Nature), induction arises. Allowing us to use past observations to inform natural laws and models. Finding that predictive power is evidence for the validity of a model.

Essentially, past actions are valid evidence for future ones.

There is also abduction which is closely combined with induction, this will also be addressed in the next sub-section.

6.3.3 Abduction

While these both give us information. It doesn't give the why, only the fact it happened(or could). The development of theories, causal order is beyond their scope unless it is completely obvious from earlier theories(which is technically still abduction in most cases, unless solely deductive with no inductive reasoning).

Abduction is the use of models, models being human made ideas that are meant to represent the world. Whether they are meant to be ontologically true or just predecive, at the end they are simply predictive. One creates models and evaluates them based upon a set of criteria to influence where they are on a hierarchy of models.

6.3.4 Statement Functions

Statement functions can be identified as variation within a statement, such as the mass of the ball is x, where $5kg < x <$

$6kg$ in this, there are infinitely many individual statements. When attempting to validate it through induction, one cannot truly validate or invalidate all possible statemements as all experiments(acting on an object in a way to measure a property.)

6.3.5 Probability Space

We take a manifold of a metric space. This manifold satisfies

$$P(\Omega) = 1$$

$$(\Omega, F, P)$$

Where Ω is defined as sample space, F is events, and P is Probability measure.

From here the capacity of probability space is quite clear

6.3.6 Probability Rules

From here many things are proved

$$P(\emptyset) = 0$$

$$P(A^c) = 1 - P(A)$$

and

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Further, variables become a metric function

$$X : \Omega \to \mathbb{R}$$

Once these variables are defined:

$$P(X \in B) = P(X^{-1}(B))$$

This makes it so the pushforward measure is the probability distribution. Elementry probability distributions are all simply particular pushforward methods.

After that, the expectation can be clearly defined through the Lebesgue integral

$$E(X) = \int_{\Omega} X dP$$

Where variance follows as

$$Var(X) = E((X - E(X))^2)$$

6.3.7 Bayes

Simple derivation brings

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

This simple theorem holds much of modern science. As P(A), where how strongly evidence should be treated is based upon the likelyhood of said thing happening with or without A being true.

6.4 Information

Information is a defined non-material existence, being as it exists in conclusion of a person. Properties of either a material object or non-material object can both be described as information

6.4.1 Information of Distinction

Information is simply a way to distinguish between states, possible and existant.

Formally,

Suppose

$$S = s_1, s_2, \ldots$$

is the set of possible states.

Information requires

$$\exists s_i \neq s_j.$$

Without distinguishable alternatives, no information exists. If every possible observation produces the same result, the system conveys nothing.

Suppose a nous initially considers

$$\Omega = s_1, s_2, \ldots, s_n.$$

An observation transforms

$$\Omega \longrightarrow \Omega'$$

where

$$\Omega' \subseteq \Omega.$$

Information is therefore the reduction of admissible possibilities. Knowledge will later be defined as appropriately justified reductions that correspond to reality.

6.4.2 Formulisms of Information

6.4.2.1 Shannon Information

Claude Shannon deliberately removed meaning from information.

Suppose an event occurs with probability

$$p.$$

Its self-information is

$$I = -\log p.$$

Rare events contain more information because they eliminate more uncertainty.

A probability of 1 contains no information.

Nothing new has been learned.

Entropy becomes

$$H(X) = -\sum_i p_i \log p_i.$$

Entropy measures expected uncertainty before observation.

Information is the decrease in uncertainty after observation.

6.4.2.2 Algorithmic Information

Algorithmic information defines information by the length of the most compressible form.

6.4.2.3 Semantic Information

The traditional view includes the syntax and semantics of a statement, what properties are transported through activity; ie talking, painting, mesurement, etc.

6.4.2.4 Programme

While all of these formulisms work in their respective fields and are all incredibly important; for the purposes of this programme, using Shannon is most efficient. As it works best for physics, scientific studies, and mathematical repsersentation.

6.5 Justification

6.6 Belief

Chapter 7

Philosophy of Science

Chapter 8

Philosophy of Physics

Chapter 9

Foundations of Physics

Chapter 10

Foundations of Mathematics

Chapter 11

Foundation of Computations

Chapter 12

Foundations of Mathematics

Chapter 13

Theology

Chapter 14

Ethics

14.1 The Primacy of God

Chapter 15

Social Philosophy

Chapter 16

Personal Philosophy

Part II

Notes

Chapter 17

Introduction

This part is for the purpose of writing my notes on other people's work, rather than my own. Whether it be notes from a class, a textbook, a regular book, or anything of that nature. Pretty simple.

Chapter 18

Physics

18.1 Introduction

Here I will write my notes related to physics

18.2 Theoretical Physics by Georg Joos

The vector analysis portion is intresting due to the fact that it takes something as simple as vector analysis and brings rigorous ideas to it like that $\oint ds = 0$ for close surfaces.

I also never really thought about using vector analysis instead of tensors.

The rest of curl, gauss, and such is very simple. Though, now it is getting into tensors though vectors.

$$dv_x = ds \nabla v_x$$

$$dv_y = ds \nabla v_y$$

$$dv_z = ds \nabla v_z$$

So therefore $dv = ds\nabla v$

To calculate v, three vectors or nine scaler must be known.

Another interesting addition is the fact that in physics, symmetric tensors can be represented as a surface to the second degree.

One thing I have noticed is I need a more intuitive grasp into the relationships between curl, div, laplance, and grad. They use it a lot to simplify the calculations.

$$\nabla^2 f = \nabla \cdot (\nabla f)$$

The Laplace operator is equal to the divergence of the gradient.

$$\nabla \times (\nabla f) = 0$$

For smooth scaler fields. The curl of the gradient is zero.

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0$$

The same is true for the divergence of the curl. $\nabla^2\mathbf{F} = \nabla(\nabla\cdot\mathbf{F}) - \nabla \times (\nabla \times \mathbf{F})$ For vector fields, the Laplace operator has different relations.

Next, onto calculus of variations. I will derive Euler-Lagrange differential equation.

Let: $\tilde{y}$ be a neighboring function to y. Where $\in$ be a small quantity and $\eta(x)$ be a arbitrary function of x. so if $\tilde{y} = y + \epsilon\eta$ then $\tilde{y'} = y' + \epsilon\eta'$. Here we stipulate that the two functions $\tilde{y}$ and y converge at the beginning and end. Thus, η must vanish at the ends. So if we substitute an integral I„ we find that it becomes a function of ϵ. Then we require that $I(\epsilon)$ must have an extreme value of $\epsilon = 0$. Here it is in mathematical terms:

$$I(\epsilon) = \int_{x_0}^{x_1} F(x, y + \epsilon\eta, y' + \epsilon\eta')dx = extremum\ for\ \epsilon = 0$$

This gives us a simple way of determining the extreme value for a given integral. The condition is:

$$(\frac{dI}{D\epsilon})_{\epsilon=0} = 0$$

We can then expand the integrad function F in Taylor's series, according to the powers of ϵ.

The differentiate with respect to ϵ.

This expression then vanishes for $\epsilon = 0$. Thus then simply remains the condition for the extremum.

Integrate this to get the Euler-Lagrange differential equation:

$$\frac{d}{dx}\frac{\partial F(x,y,y')}{\partial y'} - \frac{\partial F(x,y,y')}{\partial y} = 0$$

For writing constraining forces, we find it to be

$$Z = \lambda \; grad \; G$$

Where $G(x,y,z)$ is the equation of the surface

18.3 On Sympathetic Reduction in Classical Mechanics

18.4 Magnetic Fields and Magnetic Diagnostics for Tokamak Plasmas by Alan Wooton

18.5 Advanced MHD with Applications to Laboratory and Astrophysical Plasmas by Cambridge

For wide variety of MHD instabilities operating in tokamaks, represented by normal modes of the form(assuming that cylindrical approximation and the toroidal representation may be ignored):

$$f(\psi, \vartheta, \varphi, t) = \sum\nolimits_m \tilde{f}(\psi) e^{i(m\vartheta + \eta\varphi - \omega t)}$$

Is only unstable for perpendicular wave vectors.

The reason is the enormous field line bending energy of the Alfven waves

18.6 Hamiltonian description of the ideal Fuid by P.J.Morrison

18.7 Classical Dynamics a Modern Perspective

18.7.1 Chapter 6

A lot more importation stuff I forgot to write down, well I will start in chapter 6: If multiple canonical transformations, directly transforming would be canonical. $w \to w' \to w''$ then $w \to w''$. This is the group Composition Law.

Next is the Associativity of Composition Law: if $w' = \phi(w)$, $w" = \Phi(w')$, and $w''' = \xi(w")$ so no matter what order of transformation they will be the same, whether ($T_3(T_2T_2)$ or $(T_3T_2)T_1$ or anything of the such.

Identity: The identity transformation w' = w obviously obeys the conditions for being canonical.

Inverses: If $\omega \to \omega'$ is canonical, then so is the transformation id $\omega' \to \omega$.

"Thus all the properties needed to define a group are trivially obeyed. (In Chapter 12 we discuss group structure in a little more detail). "

The canonical group is characterized by the dimension, 2k, of the phase space.

The subgroups of such a phasespace is very large, though three can be easily defined.

Another subgroup is the contact transformations.(when a poission bracket can be formed)

Note: changing coordinate systems can satisfy phasespace

rules, but isn't generally considered to be a transformation

"Intuitively speaking, a finite canonical transformation that can be connected continuously to the identity should be built up as a succession of infinitesimal transformations."

To express the relation between Poission Brackets and canonical transformations through differential equations:

$$\frac{d\omega}{d\theta} = \{\omega^\mu, \phi(\omega)\}_\omega = \epsilon^{\mu v}\frac{\partial\phi(\omega)}{\partial\omega^v}$$

Let us first clarify the meaning of this equation. We look upon the w^μ as functions of θ; ϕ is a function with a fixed functional form, and w^μ are the unknowns in the differential equation. Thus on the right-hand side of this equation, the arguments of ϕ are just the quantities we are trying to solve for. From the theory of differential equations, we know that there is a unique solution for w^μ if we are given the values of w^μ at $\theta = 0$. Calling these boundary values of w^μ as w_0^μ, the solution of the equation can be written as:

$$\omega^\mu = \varphi^\mu(\omega_0;\theta), \varphi^\mu(\omega_0;0) = \omega_0^\mu$$

18.7.2 Chapter 7

...

18.8 Differential Geometry - Robert Davie

18.8.1 1-forms

These are linear functionals that map tangent vectors at a point on a manifold to real numbers. In local coordinates on a manifold M, a 1-form can be written as $\omega = \omega_i dx^i$, where ω_i

are smooth functions and dx^i are basis 1-forms (dual to coordinate basis vectors). For example, the differential of a function $df = \frac{\partial f}{\partial x^i} dx^i$ is a 1-form. They are used to measure vectors, like gradients or line integrals.

18.8.2 2-forms

These are antisymmetric bilinear maps that take two tangent vectors and produce a real number. In coordinates, a 2-form looks like $\omega = \omega_{ij} dx^i \wedge dx^j$, where $\wedge$ denotes the wedge product (ensuring antisymmetry: $dx^i \wedge dx^j = -dx^j \wedge dx^i$). 2-forms are used to measure oriented areas, such as in surface integrals or the electromagnetic field tensor in physics.

18.8.3 3-forms

These are antisymmetric trilinear maps, taking three tangent vectors to a real number. In coordinates, a 3-form is $\omega = \omega_{ijk} dx^i \wedge dx^j \wedge dx^k$, with antisymmetry in all indices. They measure oriented volumes and are used in integrals over 3-dimensional submanifolds, like in fluid dynamics or general relativity.

18.8.4 k-form

This can then be made general as

$$\omega = \Sigma_{i_1..i_k} \omega_{i_1...i_k} dx^{i_1} ... \wedge dx^{i_k}$$

18.8.5 Introduction to Exterior calculus

An exterior derivative maps k-forms into k+1-forms.

$$d : \wedge^k(M) \to \wedge^{k+1}(M)$$

and must follow these rules Linearity: $d(\alpha+\beta)=d\alpha+d\beta$ Leibniz(product rule): $\wedge(\alpha+\beta)=d\alpha\wedge\beta+(-1)^k\alpha\wedge d\beta$ Nilpotency: $d(d\alpha)=0$

$$\therefore d\omega=\Sigma_{i<j}(\frac{\partial\omega_i}{\partial x_j}-\frac{\partial\omega_j}{\partial x_i})dx^i\wedge dx^j$$

In 3D, the exterior derivative of the 2-form correspondence to the divergence in vector field.

Allows for integration of manifolds. Shows that the integral of the exterior derivative of a differential form over a manifold of the integral of the form itself over the boundary of the manifold:

$$\int_M d\omega=\int_{\partial M}\omega$$

This unifies vector calculus in areas like divergence or green's theorem.

It also show vector flow

Lie derivatives measure the change of forms

18.8.6 Exterior calculus-2

The exterior for a k-form is

$$d\omega=\sum(\sum_{n=1}\frac{\partial x_{i.....k}}{\partial x^{i_n}}dx^{i_n})\wedge dx^{i_1}\wedge....\wedge dx^{i_k}$$

4-form is the highest form

Back to to integration of manifolds with stokes theorem. While $\int_M d\omega=\int_{\partial M}\omega$ is true $\int dx\wedge dy$ and $\int dxdy$ represent slightly different mathematical concepts. In diff forms, it is the integral of an oriental area. In Multi-variable calculus it is the integral of over a region in the xy-plane

18.8.7 Wedge Product-2

Properties of the wedge product: Anti-symmetry: $\omega \wedge \eta = (-1)^{pq}\eta \wedge \omega$. Where p & q are the dimensionality of ω, η

Symmetric group: S_{p+q} is the group of all possible permutations.

Permutation and sign of permutations: σ is essentially a specific way of ordering a set of elements. The sign of permutation $sign(\sigma)$ tells us if the permutation can be achieved in an even or odd number of swaps. Where odd is negative and even is positive

Role of these two things: The sign tells us the sign of a wedge product given a permutation. $dx^1 \wedge dx^2 \rightarrow -dx^2 \wedge dx^1$. The symmetric groups tells us all possible permutations.

$$\therefore (\omega \wedge \eta)(X_1..., X_{p+q}) = \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} sign(\sigma)\omega(X_{\sigma(1)}..., X_{\sigma(p)}), \eta(X_{\sigma(p+1)}...X_{\sigma(p+q)})$$

This $\frac{1}{p!q!}$ ensures correct normalization.

18.8.8 Introduction to the Hodge Star Operation

This operator allows us to relate manifolds. Crucial in formulating coordinate free theories. It transforms k-forms into (n-k)-forms. Specifically it is an isomorphism between the spaces of k-forms and (n-k)-forms on an n-dimensional Riemannian(M,g). Where g is the metric tensor. It is denoted as:

$$\star : \wedge^k(T^*M) \rightarrow \wedge^{n-k}(T^*M)$$

The way it works is hard to write out, so just remember it.

To construct a hodge star

$$\omega \wedge \star\eta = \langle \omega, \eta \rangle vol_M$$

Where $\langle ., . \rangle$ is the inner product and Vol is the volume form of M. Defined as $vol_M = \sqrt{|g|} dx^1 \wedge .. \wedge dx^n$ Where $|g|$ is the determinate of the Metric Tensor.

The equation is long, so just look it up, it is simple but long.

18.8.9 Hodge star 2

Hodge star establishes duality between geometric theories. Involution: Doing it twice returns it to the original Linear transformations: You know what this means Interacts like multiplication with exterior derivative: $\star(d\omega) = d(\star\omega)$ The co-differential is defined using the Hodge Star.

18.8.10 The push forward of vectors on manifolds

A pushforward is a way to map a vecotr on a tangent space onto another manifold's tangent space.

Properties:

Linearity: The pushforward $\phi_* : T_pM \to T_{\phi(p)}N$ is a linear map. For vectors $v, w \in T_pM$ and scalars a, b,

$$\phi_*(av + bw) = a\phi_* v + b\phi_* w$$

Chain rule: If you have another smooth map $\psi : N \to P$, the pushforward satisfies the composition rule:

$$(\psi \circ \phi)_* = \psi_* \circ \phi_*$$

This follows from the chain rule for derivatives.

Action on curves: If you think of a vector $v \in T_pM$ as the tangent to a curve $\gamma : (-\epsilon, \epsilon) \to M$ with $\gamma(0) = p$, then $\phi_* v$ is the tangent vector to the curve $\phi \circ \gamma$ at $\phi(p)$.

The pushforward $\phi_* v \in T_{\phi(p)}N$ is given by:

$$\phi_* v = v^i \frac{\partial \phi^j}{\partial x^i} \frac{\partial}{\partial y^j}$$

Here, v^i are the components of v, and $\frac{\partial \phi^j}{\partial x^i}$ are the Jacobian entries. The result is a vector in $T_{\phi(p)}N$, expressed in the basis $\left\{ \frac{\partial}{\partial y^1}, \ldots, \frac{\partial}{\partial y^n} \right\}$.

18.8.11 Pull-Back in k-forms

Given a smooth map $\phi : M \to N$ between manifolds M (dimension m) and N (dimension n), and a k-form $\omega \in \Omega^k(N)$ on N, the pullback $\phi^* \omega \in \Omega^k(M)$ is a k-form on M. The pullback essentially "transfers" ω from N to M by composing it with the map ϕ. The pullback is defined pointwise. For a point $p \in M$, and k tangent vectors $v_1, \ldots, v_k \in T_pM$, the pullback $\phi^* \omega$ at p is:

$$(\phi^* \omega)_p (v_1, \ldots, v_k) = \omega_{\phi(p)} (\phi_* v_1, \ldots, \phi_* v_k)$$

Here:

$\phi_* : T_pM \to T_{\phi(p)}N$ is the pushforward (differential) of ϕ, which maps each tangent vector v_i to $\phi_* v_i$. $\omega_{\phi(p)} \in \wedge^k T^*_{\phi(p)}N$ is the k-form ω evaluated at $\phi(p)$, acting on the pushed-forward vectors.

This definition ensures that $\phi^* \omega$ is a k-form on M, as it takes k vectors in T_pM and produces a number.

Key Properties of the Pullback

Linearity: The pullback $\phi^* : \Omega^k(N) \to \Omega^k(M)$ is a linear map.

Preserves degree: The pullback of a k-form is a k-form.

Functoriality: For maps $\phi : M \to N$ and $\psi : N \to P$,

$$(\psi \circ \phi)^* = \phi^* \circ \psi^*$$

Commutes with exterior derivative: For any k-form ω,

$$d(\phi^*\omega) = \phi^*(d\omega)$$

This is a powerful property, making pullbacks compatible with the exterior calculus.

Wedge product: The pullback respects the wedge product:

$$\phi^*(\omega \wedge \eta) = (\phi^*\omega) \wedge (\phi^*\eta)$$

Computing the Pullback in Coordinates To compute $\phi^*\omega$ in practice, we typically use local coordinates. Here's the step-by-step process:

Choose coordinates:

On M, use coordinates $(x^1, \ldots, x^m)$ around p.

On N, use coordinates $(y^1, \ldots, y^n)$ around $\phi(p)$.

The map ϕ is expressed as $y^i = \phi^i(x^1, \ldots, x^m)$.

Express the k-form:

Let $\omega \in \Omega^k(N)$ be written in coordinates as:

$$\omega = \sum_I a_I(y)\, dy^{i_1} \wedge \cdots \wedge dy^{i_k}$$

where $I = (i_1, \ldots, i_k)$ is a multi-index with $1 \leq i_1 < \cdots < i_k \leq n$, and $a_I(y)$ are smooth functions on N.

Pull back the form:

The pullback is:

$$\phi^*\omega = \sum_I (a_I \circ \phi)\, \phi^*(dy^{i_1} \wedge \cdots \wedge dy^{i_k})$$

Compute $a_I \circ \phi$: Replace y with $\phi(x)$, so $a_I(y) = a_I(\phi(x))$.

Compute the pullback of the basis k-form:

$$\phi^*(dy^{i_1} \wedge \cdots \wedge dy^{i_k}) = \phi^*(dy^{i_1}) \wedge \cdots \wedge \phi^*(dy^{i_k})$$

For a 1-form dy^j, the pullback is:

$$\phi^*(dy^j) = d(y^j \circ \phi) = d(\phi^j(x)) = \frac{\partial \phi^j}{\partial x^\ell} dx^\ell$$

(using the chain rule, summing over ℓ). Thus:

$$\phi^*(dy^{i_1} \wedge \cdots \wedge dy^{i_k}) = \left(\frac{\partial \phi^{i_1}}{\partial x^{\ell_1}} dx^{\ell_1}\right) \wedge \cdots \wedge \left(\frac{\partial \phi^{i_k}}{\partial x^{\ell_k}} dx^{\ell_k}\right)$$

This is a k-form on M, expressed in the dx^ℓ basis.

Simplify:

Expand the wedge products and collect terms. The result is a sum of terms involving $dx^{\ell_1} \wedge \cdots \wedge dx^{\ell_k}$, with coefficients that depend on the Jacobian $\frac{\partial \phi^j}{\partial x^\ell}$ and $a_I(\phi(x))$.

18.8.12 Pull-Back of volume forms

The pullback of a volume form $\omega = f(y)\, dy^1 \wedge \cdots \wedge dy^n$ on N under a map $\phi : M \to N$ is an n-form on M, computed as:

$$\phi^*\omega = (f \circ \phi) \cdot \det\left(\frac{\partial \phi^i}{\partial x^{j_k}}\right) dx^{j_1} \wedge \cdots \wedge dx^{j_n}$$

If $\dim M = \dim N$, and ϕ is a local diffeomorphism, $\phi^*\omega$ is a volume form on M. If $\dim M < \dim N$, the pullback is zero. The Jacobian determinant captures the volume scaling, making this operation central to integration and geometric transformations.

18.8.13 Integration on Manifolds Using the Pullback of Volume Forms

The pullback of volume forms allows for integration of manifolds. Which is useful in the integration of curved coordinate systems.

volumes generalize "dx" or "dxdy."

The steps in integration of manifolds using volume forms is:

Step 1: M: The domain of integration (often $\mathbb{R}^n$ or a simpler manifold). N: The target manifold with a volume form ω. $\phi : M \to N$: A smooth map, typically a parametrization or diffeomorphism. $f : N \to \mathbb{R}$: The function to integrate (if $f = 1$, we're computing the volume).

Express the volume form:

Write $\omega = f_\omega(y)\, dy^1 \wedge \cdots \wedge dy^n$ in coordinates on N.

Compute the pullback:

Compute $\phi^*\omega$, which involves:

Substituting $y = \phi(x)$ into $f_\omega(y)$. Computing the Jacobian determinant of ϕ. Forming $\phi^*\omega = (f_\omega \circ \phi) \cdot \det\left(\frac{\partial \phi^i}{\partial x^j}\right) dx^1 \wedge \cdots \wedge dx^n$ (if $m = n$).

Set up the integral:

The integral becomes:

$$\int_{\phi(M)} f\,\omega = \int_M (f \circ \phi) \cdot \phi^*\omega$$

In coordinates, if $\phi^*\omega = g(x)\,dx^1 \wedge \cdots \wedge dx^n$, and M corresponds to a region $U \subset \mathbb{R}^n$:

$$\int_U (f \circ \phi)(x) \cdot g(x)\,dx^1 \dots dx^n$$

Evaluate the integral:

Perform the integral over U using standard multivariable calculus techniques, accounting for the Jacobian determinant and any orientation changes.

18.8.14 Line Integrals on Manifolds

Parameterize the curve: Define the curve $\gamma(t) = (x(t), y(t), \dots)$ for $t \in [a, b]$.

Define the one-form: Let the one-form be $\omega = f(x, y, \dots)dx + g(x, y, \dots)dy + \dots$.

Calculate the pullback: Substitute the coordinate functions $x(t), y(t), \dots$ and their differentials into the one-form. This gives the pullback one-form $\gamma^*\omega$ on the interval $[a, b]$. For example, if $\omega = f(x, y)dx + g(x, y)dy$ and $\gamma(t) = (x(t), y(t))$, the pullback is $\gamma^*\omega = (f(x(t), y(t))\frac{dx}{dt} + g(x(t), y(t))\frac{dy}{dt})dt$.

Integrate the pullback: Integrate the resulting expression with respect to t from a to b.

This gives the value of the line integral:

$$\int_\gamma \omega = \int_a^b (f(x(t), y(t))\frac{dx}{dt} + g(x(t), y(t))\frac{dy}{dt})dt$$

.

18.8.15 Classical Stokes Thereom in Manifolds

18.9 Symplectic geometry & classical mechanics

18.9.1 Lecture 1

Symplectic doesn't work well in dissipative structures.

It also goes over the definition of local eucledian space, which is pretty basic.

They also go over how you can make an object unique by maxionable defining coordinate systems.

This is given by an atlas of coordinate charts

To get a differntiable structure, an atlas is described through coordinate charts (U_α, ϕ_α) such that:

Each $\psi_\alpha : U_\alpha \to R^n$ is a homeimorphism Also, on overlaps between $U_\alpha \cap U_\beta$ it is continously differentiable.

18.9.2 Lecture 2

They go over catogories in a brief overview, and that this lecture will be going over a map of differential geometry.

So, let us start with a manifold: $f(p) \in N$

Now there isn't much we can say about this basic manifold, though let's add a bit more context. Say that manifolds seem like flat-space(euclidian) locally.

So we can take the charts in the manifold to project euclidian space upon it to differentiate.

When k is $\inf$ f is smooth.

Chapter 19

Calculus of Variations by Gelfand & Fomin

19.0.1 Elements of the Theory

A functional is defined as a function where the variable is in it of itself a function. It can also be seen as a class of functions.

While the calculus of functionals most developed and used branch is finding the extrema of variations, this is hardly its extent.

A great example is using the path length formula $L = \int_a^b \sqrt{1 + \dot{y}^2} dx$ and finding which y achieves the minimum value for L, this is by definition the straightest line, or geodesic. It can also attempt to maximize. It is most often applied to any form of optimization theory or minimizing the action for Lagrandian dynamics.

In this book they will primarily use geometric language defining a space, the functional space, to be a space whose elements are functions. The nature and dimensions of such a space are defined by the functional defined. It should be as specific and defined with the problem. Continuity is the main constraint.

To formalize the "closeness" needed for continuity the space will use normed linear space. Which is defined by the following axioms:

- $x + y = y + x$
- $(x + y) + z = x + (y + z)$
- There exists a 0 element
- $\forall x \exists (-x) : -x + x = 0$
- Multiplication of 1
- Multiplatice communication

Normed is defined to be the magnitude of the length.

As I am flipping through this book I am seeing that the subject matter is quite trivial but presented in a overtly complex way. I may return to it but that is unlikely.

19.1 MIT 8.323 Relativistic Quantum Field Theory 1

19.1.1 Lecture 1: Classical Field Theories and Principle of Locality

The first part goes over trivial procceses in field theory and general mathematics. The main note on this section would be the principle of Locality:

$$L = \int d^3x \mathcal{L} \left(\phi_a, \partial_t \phi_a, \partial_i \phi_a \right)$$

Where $\phi_a(\vec{x}, t)$: general fields. $\mathcal{L}$ is a function of ϕ and its derivatives. $\mathcal{L}$ is called the Lagrangian density.

For this form it constrains all dynamics so that all actions must be local. It is used by a single integral to enforce this. Another constraint is one will only use the first derivative in time. This is because including a second order time derivative will make an equation of motion with an initial condition with an acceleration which includes arbitrary complications without need. This rule is less important and fundamental but is used to make things easier. Lastly, in this class, being that it is relativistic one won't need additional spacial derivatives, but in classical models this in not the case.

Canonical momentum density is exactly what one would expect.

$$\Pi_a = \frac{\partial \mathcal{L}}{\partial \phi_a}$$

Hamiltonian is defined the same way.

One brief note on notation, in this x is simply a label for a degree of freedom, there can be infinite such degrees of freedom. Also, the teacher is using Einstein's notation that repeated indices will be summed.

19.1.1.1 Equations of motion

He gives an example through:

$$\delta s = 0$$

$$S = \int dt L$$

$$0 = \delta s = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \delta(\partial_\mu \phi_a \right]$$

One can simplify this through integration by parts

As for boundary conditions. One will choose $\delta\phi_a$ so that boundary conditions will vanish. Thus one will not need to worry about that.

This thus implies that $\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) = 0$.

From now on all things will be translationally invariant and Lorentz invariant.

He briefly goes over the simplest form of a quantum field, a scalar field.(Pion or higgs)

It follows a very similar model:

$$S = \int d^4x \left[-1/2, \partial_\mu \phi, \partial^\mu \phi - V(\phi) \right]$$

Transnational symmetry means that all parameters of $V(\phi)$ must be constant. V is the same as the Lagrangian density

Using this equation one can once again find the Hamiltonian Density and thus Hamiltonian. To give a simple equation of motion.

He goes through a famous Klein-Gordan:

$$\partial^2 \phi - m^2 \phi = 0$$

19.1.2 Lecture 2: Symmetries and Conservation Laws

I forgot to save my notes last time, so those are gone. It was generally pretty simple going over Noether theorem and defining types of symmetries at the beginning.

Some how it deleted again... oh well, moving on.

19.1.3 Lecture 3: Why Quantum Field Theory

Non-Relativistic Schrodinger equation

$$\partial^2\psi - m^2\psi = 0$$

Simplest scalar field theory

$$S = -\frac{1}{2}\int dx(\partial_\mu\phi\partial^\nu\phi + m^2\phi^2)$$

Implies

$$\partial^2\phi - m^2\phi = 0$$

This changes $\phi(t, x)$ where x becomes an eigenvalue rather than a coordinate.

In relativistic quantum mechanics, there are several challenges. For one, no way to define a positive probability density. Also, there exists negative energy states. You must describe a fixed number of particles without changing. As equations change as wavefunctions change. This troubles with relativity as energy can create particles, adding additional difficulty. Finally, there is a fundamental asymmetry between the parameter t and the eigenvalues of x.

Therefore, relativistic quantum mechanics one cannot give a fundamental description.(at best an approximate, no particle creation or annihilation) The solution of this is quantum field theory, simply the quantization of classical field theory.(using t and x as parameters not eigenvalues) Field theory can also arise as a limit of discrite systems.

In classical field theory, one takes the originally discrite Lagrangian and defines $\phi(x_n, t)$ to be the discrite parts. From here a simple Taylor expanse can be used to change all of the interaction terms to in the form of the new continuum. Finally one may derive the field-theoretic Euler-Lagrange equation.

Quantization of harmonic oscillator in Heisenberg picture.

$$L = \frac{1}{2}\dot{x} - \frac{\omega^2}{2}x^2$$
$$p = \dot{x}$$
$$H = \frac{p^2}{2} + \frac{1}{2}\omega^2 x^2$$
$$\ddot{x} + \omega^2 x = 0$$

The quantum solution is exactly the same, exept x becomes an operator:

$$\hat{x}(t) = \hat{A}\cos\omega t + \hat{B}\sin\omega t$$

From here one can define the hilbert space to define through $[\hat{a}, \hat{a^+}]i$

$$|h>= \frac{1}{\sqrt{n!}}(\hat{a^+})^n|0>$$

One can then use this to quantize a field theory. First, find the most general solution of the EOM. Second, promote the integration constants to constant operators to give full time evolution. Impose canonical quantization conditions$\{q, p\} = 1$ thus, $[\hat{q}, \hat{p}] = ih$. Finally, use this to develop Hilbert Space.

19.1.4 Lecture 4: Canonical Quantization of a Free Scalar Field Theory

...

Chapter 20

Numerical Hamiltonian Problems

20.0.1 Chapter 1: Hamiltonian Systems

MacKay and Meiss have compiled great papers on the subject in 1987, Marsen (1992) offers a more geometric introduction.

Usually in mechanics, q is generalized coordinates while p is conjugated generalized momenta. H being total mechanical energy.

I like how they show the nature of the Hamiltonian system to be a logical nessesity based upon how the phase space is constructed, rather than purely physical construction.

It is sometimes useful to combine dependent variables into a 2d $y = (p, q)$.

$$\frac{\mathrm{d}y}{\mathrm{d}t} = J^{-1}\nabla H$$

Where, ∇ is the gradient operator and J is skew-symmetric Matrix

20.0.2 Chapter 2: Symplecticness

As mentioned, Hamiltonian problems are exeptional. Most systems do not allow for area preservation(energy conservation, time symmetry, etc), which is a requirment of Hamiltonian systems. That is why exploration into other systems like Nambu, Metriplectic, etc is needed.

This area preservation is not simply required as a general rule but is the basis of the structure. All other properties of Hamiltonian dynamics is derived from the area property.

One can check for the preservation ogarea by checking if the Jacobian is exactly 1 or a simpler way is to find it is if $dp * \wedge dq* = dp \wedge dq$ or that the pull back of the 2-form is equal to the origional 2-form.

In multiple-diminsions. Look for multiple area preservations rather than multi-diminsional preservation.

20.1 Chapter 2: Numerical Methods

Stiff problems are defined to be where numerical stability, not accuracy, forces you to take extremely small steps.

Explicit methods requires s number of evaluations per steps while implicit provides coupled system of $s \times D$ algebraic systems and $s \times D$ components of stage vectors.

After reading this I wrote a program in Julia to compare the time it takes for both an explicit RK and Implicit to solve a ODE. It was more challenging than it should have been.

20.1.1 Chapter 4: Order Condition

More so Computer science stuff, not so important at the moment but something to look at later.

20.1.2 Chapter 5: Implementation

Really just the stuff I did earlier when writing my own implicit RK.

20.1.3 Chapter 6: Symplectic Integration

If $b_i a_{ij} + b_j aji - b_i b_j = 0$ for $i, j = 1, ..., s$. Then the method is symplectic. They have a nice proof that I can use as a reference.

Most all symplectic methods are implicit.

They go through solutions of various families of symplectic RK integration.

20.1.4 Chapter 7: Symplectic Order Conditions

I still don't get order conditions. I will end it at one point.

20.1.5 Chapter 8: Numerical Experimentations

Page 115 lists multiple symplectic numerical methods, the rest of the chapter examines them. Too much for me to write but useful and interesting information nonetheless.

One thing I will say is that they claimed that Calvo is clearly the best and very little improvement is likely possible.

20.1.6 Chapter 10: Properties of Symplectic Integrators

20.1.6.1 Backward error analysis

How much must the numerical problem be perturbed so that the numerical problem is equal to the analytical. This is espe-

cially useful in uncertain models, in which perturbation is the best we can do.

This only holds if the numerical solution of time t_n is computed by iterating n times 1 in the same symplectic maps. Thus, variable-step integrations fail.

20.1.6.2 Alternative Approach

There is another method through repeating periodically; though, this system is now non-autonomous and is discontinous.

20.1.6.3 Conserved Energy and Quantities

While there are exceptions for unique cases, generally one must choice whether to have conserved energy or sympletic structure. Obviously, choosing must be done by the researcher, but generally sympletic is superior for the fact that it does not confine the nature of the dynamics.

There is a great deal of research into methods that exactly conserve H. This is not covered in the book.

There are several ways to come with close approximations of H though.

20.1.6.4 KAM

KAM theory proves that symplectic methods can show that nonlinear effects are not destabilizing.

20.1.7 Chapter 11: Generating Functions

Mentioned earlier, S can be used to express symplectic mappings and transformations. S is a single real value function. This function it called the *generating function*.

20.1.7.1 First kind

Let $(p^*, q^*) = \phi(p, q)$ be a sympletic transformation defined in a simply connected Ω domain. For each closed path γ:

$$\int_\gamma pdq - \int_\gamma p * dq* = 0$$

Where pdq is the differential form.

Making it so that

$$dS = pdq - p^*dq^*$$

Further assume that q and $q*$ are functions independent of Ω. If this is true we can express $S(p, q)$ as function S^1 of q and q*.. Thus,

$$p = \frac{\partial S^1}{\partial q}, \tag{20.1}$$

$$p* = \frac{\partial S^1}{\partial q*} \tag{20.2}$$

20.1.7.2 Third kind

What happended to *second*, I don't know.

If q and q* are not independent. Then,if p and q* are independent you can express $S^3(p, q*)$. The rest is self-explanitory.

20.1.7.3 All kinds

One can use the Poincare generating function.

20.1.7.4 Hamiltonian-Jacobi

This is another way to generate functions by turning dynamics into single nonlinear partial differential equations throught the equation:

$$\frac{\partial S^3}{\partial t} + H(\mathbf{p}, \mathbf{q}, t) = 0$$

While in writing this I use the third kind, it can be used for any kind.

20.1.7.5 Symplectic Integrators on Generating Functions

They talk about a lot of different research, not going into detail about any of it

20.1.8 Chapter 12: Lie Formulism

This being something that I have research well in the past is relatively simple. The Poisson Bracket:

$$\{F, G\} = \sum \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right)$$

is defined by a couple of rules

- Bileaniarity
- Skew-Symmetry
- Jacobi Condition
- Leibiz' rule

By introducing 2d-dimensional vectors the equation becomes

$$\{F, G\} = \boldsymbol{\nabla} F^T J^{-1} \boldsymbol{\nabla} G$$

Where J is the skew-symmetry matrix.

From here we can see that this works well the Hamiltonain equations of motion as

$$\dot{y} = \{y, H\}$$

Also, the hamiltonain is canonical iff the Poisson Bracket of any two functions remains invariant under changes in variables.

20.1.8.1 Lie Operators & series

Powers of lie operators can be expanded to compositions of lie operators, or vise versa. This can be used as an equivalent to a derivative as a function of time for Hamiltonian systems.

Using the exponential operator on the lie operator, one can find the taylor series expansion of a flow(for integration) by choosing G to be one of the coordinate functions.

20.1.8.2 Multiplication of exponentials

Composing two exponential lie oporators repersents the composition of flows; specifically $\phi_{t-F_1}(\phi_{t-F_2})$ in that order.

20.1.8.3 Baker-Campbell-Hausdorff formula

If one wishes to find $e^x e^y = e^z$ and X and Y are non-communters,it can be found through the infinite series:

$$Z = X + Y + \frac{1}{2}\{x, y\} + \frac{1}{12}\{X, \{x, y\}\} - \frac{1}{12}\{Y, \{X, Y\}\} + ...$$

20.1.8.4 Fractional-Step Method

I don't get this...

20.1.9 Chapter 13: Higher Order Methods

After reading my book on Lie Algebra I should come back to this

Chapter 21

General Relativity by Robert M. Wald

21.0.1 Manifolds and Tensor Fields

21.0.1.1 Manifolds

Spacetime is a four-dimensional continuum and by that nature a manifold is simply a multi-dimensional continuum.

When going over this concept they first present an open ball, where all points distance from a chosen center is always strictly less than some radius. Essentially the inside of some interval, circle, sphere, etc. Then an open set in $\mathbb{R}^n$ is any set that can be expressed as a union of open balls. This make $\mathbb{R}^n$ a topological space.

Essentially a manifold is a set made up of peices that "look like" open subsets of $\mathbb{R}^n$. These peices can be "sewn together" smoothly. This can be more precisely defined a n-dimensional, C^∞, real manifold M is a set together with a collection of subsets $\{O_\alpha\}$ that satisfies the following properties: O_α covers the entire manifold, all points can be mapped onto another subset of $\mathbb{R}^n$, such subsets must be differentiable.

These maps ϕ_α are generally called charts by mathematicians and coordinate systems by physicist. They further define the maps to be homeomorphic, or that all mappings perserve topological properties.

In Euclidean space they map a sphere into a two dimensional manifold; because, pieces can be smoothly sewn together. This then satisfies the above conditions.

21.0.1.2 Vectors

Because traditional vectors do not work in geometric curvature, vectors will be instead used as directional derivatives, rather than an intrinsic structure. These directional derivatives are characterized by their linearity and "Leibnitz" rule. Letting F denote collections of smoothly differentiable functions from M to R, $v : F \to \mathbb{R}$. Also, they give a proof that dim V = n = dim M. They also define the vector transformation law which is defined as:

$$v'^{v} = \sum_{\mu=1}^{n} v^{\mu} \frac{\partial x'^{v}}{\partial x^{\mu}}$$

21.0.2 Tensors; the Metric Tensor

To define a tensor one must first define a dual vector space. A dual vector space is a space of the functional transformations from vectors to linear measurements. A double vector space takes those linear measurements and turns them into scalars. This double dual space is isomorphic to the original vector space. Reconstructing all measurements. This double dual space can be used to reconstruct the vector space using the dual vector space.

Now to define a tensor, T, of type (k,l):

$$T : V^* \times ... \times V^* \times V \times ... \times V \to \mathbb{R}$$

In other words, given k dual vectors and l ordinary vectors, T produces a number. A number that if one where to fix the values of all of the other vectors and dual vectors, it is a linear map on the remaining variable. Thus, according to this definition, a tensor of type (0,1) is a dual vector. Of type (1, 0) a double dual vector. Because double dual vectors identify with traditional vectors, they are simply vectors. From this identification one can construct higher types of tensors. (1,1) is a multilinear map from V to V or for the other side.

There are two important operations on tensors. The first being called the contraction. Defined as $C : \lambda(k,l) \to \lambda(k-1, l-1)$. Thus if T is a tensor of type (k,l), then:

$$CT = \sum_{\sigma=1}^{n} T(..., v^{\sigma^*}, ...; ..., v_\sigma, ...)$$

Where v_σ is the basis of V and v^{sigma^*} is its dual basis. These vectors are inserted into the ith and jth slot of T.(note this is specifically the contraction of tensor of type (1,1)). The next operation being the outer product if T and T'. Denoted as $T \otimes T'$, by the following rules. Given (k+k') dual vectors $v^{k+1*}, ..., v^{k+k'*}$ and the equivalent for l vectors. $T \otimes T^{*'}$ is defined as the product of $T(v^{1*}, ..., v^{k*}; w_1, ..., w_l)$ and $T(v^{k+1*}, ..., v^{k+k'*}; w_{l+1}, ..., w_{l+l'})$. Thus, one way of constructing tensors is to take the outer product of vectors and dual vectors. A tensor of this would be called simple. From this any tensor of type (k,l) can be expressed as a sum of simple tensors.

$$T = \sum_{\mu_1,...,v_l=1}^{n} T^{\mu_1 ... \mu_k}_{v_1,...,v_l} v_{\mu_1} \otimes ... \otimes v^{v_l^*}$$

The basis expansion of coefficens are called components of the tensor.

This discussion above refers to arbitrary finite-dimensional vector spaces. Now let us move onto where V is the tangent space V_p at point p of a manifold M. In this V_p^* is refereed to as the contangent space. Where at this point the vectors are cotangent vectors. Another term would be in V they are contravariant vectors and V* as covariant vectors.(remember dx^μ is a symbol that merely is defined as $dx^\mu(\frac{\partial}{\partial x^v}) = \delta_v^\mu$).

Finally, we have the defining property of all tensors, tensor transformation law.(though a multilinear map on vectors and dual vectors also works)

$$T'^{\mu_1\ldots\mu_k}{}_{\nu_1\ldots\nu_l} = \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}} \cdots \frac{\partial x'^{\mu_k}}{\partial x^{\alpha_k}} \frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}} \cdots \frac{\partial x^{\beta_l}}{\partial x'^{\nu_l}} T^{\alpha_1\ldots\alpha_k}{}_{\beta_1\ldots\beta_l}$$

(for future notice this in Einstein summation)

Now we have a tensor field. I will also add that coovariant vector field is as smooth as contravariant vector field. The tensor field's smoothness is defined the same way.

Now one introduces a metric. Intuitively the metric gives the "infinitesimal square distance" associated with "infinitesimal displacement." As mentioned earlier this displacement is defined by the tangent vector. Thus a metric g, should be a linear map from $V_p \times v_p$ into numbers, ie a tensor of type (0,2). In addition, the metric must be symmetric and nondegenerate. Meaning $g(v_1, v_2) = g(v_2, v_1)$ and $g(v, v_1) = 0$. Where $v_1 = 0$.

In a coordianate basis, we may expand a metric in terms of its components as:

$$g = sum_{\mu,v} g_{\mu\nu} dx^\mu \otimes dx^\nu$$

Sometimes ds^2 is used in place of g to represent the metric tensor. Given a metric g, we can always find an orthonormal

basis. The number of + and - signs occurring is called the signature of the metric. The Lorentzian is defined as - + + +.

21.0.2.1 Abstract Index Notation

In this indices do not represent components or numbers but rather the type of tensor. Repeated indices imply contraction(as lables not coordinate slots.)

Also, (ab) implies symmetry and [ab] implies antisymmetry.

Lastly, repeated index implies summation automatically.

21.0.3 Curvature

Generally curvature is defined as extrinsic curvature of a "surface" embedded in a higher dimensional space. There is also intrinsic curvature.

Essentially, this intrinsic curvature can be defined through parallel transport. From here a geodesic is a curve whose tangent is parallel-transported along itself, or the "straightest possible" curve. A space is curved if a parallel geodesic is no longer parallel. One can use the definition of a derivative to define parallel transport as when the derivative along a given curve is 0. Thus the definition of curvature can be defined in terms of failure of successive differentiation on tensor fields to commute. Another thing to add is that given a metric manifold there exists a unique notion of parallel transport in which perserves the inner product of all pairs of vectors.

21.0.3.1 Derivative Operators and Parallel Transport

A derivative operator Δ(sometimes called a covariant derivative) takes every differentiable tensor field of type (k,l) to a smooth tensor field of type (k, l+1) and satisfies linearity, Leib-

nitz rule, commutativity with all contraction, consistency, and torsion free(this fifth one is sometimes dropped).

Given this one can derive an expression for the commutator of two vector fields.

$$[\nu, w]^b = \nu^a \Delta_a w^b - w^a \Delta_a v^b$$

Next, the Christoffel symbol can be found as

$$\Delta_a t^b = \partial_a t^b + \Gamma^b_{ac} t^c$$

Now as mentioned earlier, parallel transport can be defined as

$$t^a \Delta_a T^{b_1 \cdots}_{c_1 \cdots} = 0$$

If one choices a coordinate system it can then be defind as:

$$t^a \partial_a v^b + t^a \Gamma^b_{ac} v^c = 0$$

Or in terms of components in the coordinate system:

$$\frac{dv^\nu}{dt} + \sum_{\mu,\nu} t^\mu \Gamma^\nu_{\mu\lambda} v^\lambda = 0$$

While many forms of derivative operators can be defined from the general structure of a manifold, but metric manifolds privilege a specific kind.

In this case the Christoffel symbol can be defined by:

$$\Gamma^c_{ab} = \frac{1}{2} g^{cd} \left(\hat{\Delta}_a g_{bd} + \hat{\Delta}_b g_{ad} - \hat{\Delta}_d g_{ad} \right)$$

Where $\hat{\Delta}$ refers to an arbitrary derivative operator.

21.0.3.2 Curvature

The Riemann Curvature Tensor is defined by:

$$\Delta_a \Delta_b \omega_c - \Delta_b \Delta_a \omega_c = R^d_{abc} \omega_d$$

From here the total change in the vector v can be found by:

$$\delta v^a = \Delta t \Delta s v^d T^c S^b R^a_{cbd}$$

Where s and t repersent infintesimal changes within the metric during the transport.

Now the key properties of the Riemann tensor are

$$R^d_{abc} = -R^d_{bac}$$

$$R^d_{[abc]} = 0$$

$$R_{abcd} = -R_{abdc}$$

And the Bianchi identity holds

Decomposing the Riemann tensor into "trace part" and a "trace tree part" can be usful. As a trace over the first two or last two can to vanish

$$R^b_{abc} = R_{ac} = R_{ca}$$

The scalar curvature is further defined as

$$R = R^a_a$$

The "trace free part" is called the Weyl Tensor C_{abcd} and is defined for manifolds $n >= 3$ by the equation

$$R_{abcd} = C_{abcd} + \frac{2}{n-2}$$

The Riemann Curvature Tensor is defined by:

$$\Delta_a \Delta_b \omega_c - \Delta_b \Delta_a \omega_c = R^d_{abc} \omega_d$$

From here the total change in the vector v can be found by:

$$\delta v^a = \Delta t \Delta s v^d T^c S^b R^a_{cbd}$$

Where s and t repersent infintesimal changes within the metric during the transport.

Now the key properties of the Riemann tensor are

$$R^d_{abc} = -R^d_{bac}$$

$$R^d_{[abc]} = 0$$

$$R_{abcd} = -R_{abdc}$$

And the Bianchi identity holds

Decomposing the Riemann tensor into "trace part" and a "trace tree part" can be usful. As a trace over the first two or last two can to vanish

$$R^b_{abc} = R_{ac} = R_{ca}$$

This defines the Ricci Tensor R_{ac}. From here the scalar Curvature, R, is defined as the trace of the Ricci tensor.

$$R = R^a_a$$

The "trace free part" is called the Weyl Tensor, C_{abcd}, and when diminsions are greater than 3.

$$C_{abcd} = R_{abcd} - \frac{2}{n-2}\left(g_{a[c}R_{d]b} - g_{b[c}R_{d]a}\right) + \frac{2}{(n-1)(n-2)} R\, g_{a[c}g_{d]b}.$$

Because it behaves easily in conformal transformations it is

sometimes referred to as the conformal tensor.

Next, Einstein tensor is found from the twice contracted Bianchi identity:

$$G_{ab} = R_{ab} - \frac{1}{2} R g_{ab}$$

This tensor is used to ensure the consistency of Einstein's equations.

21.0.3.3 Geodesics

One can define a geodesic as a curve whose tangent, T^a satisfies this equation

$$T^a \Delta_a T^b = 0$$

Actually, one does not need to force the same length so this weaker condition may be true

$$T^a \Delta_a T^b = \alpha T^b$$

Where α is an arbitrary function on the curve.

Thus, this means that a geodesic must be affine parameterized. Thus in a coordinate basis this becomes.

$$\frac{d^2 x^\mu}{dt^2} + \sum_{\sigma,\nu} \Gamma^\mu_{\sigma\nu} \frac{dx^\sigma}{dt} \frac{dx^\nu}{dt} = 0$$

Next, in order for simpler calculation, a Remmanian Normal coordinate can be used in which all geodesics are straight lines. It follows that the Christoffel symbol will vanish in this case.

In another form in which the derivative operator arises from the metric. One will use the Gaussian normal coordinate system. It is most useful when given a hypersurface. They satisfy the important property that geodesics remain orthogonal to all the hypersurfaces.

For a curve on a manifold the length can be derived to be:

$$l = \int (g_{ab} T^a T^b)^{1/2} dt$$

Where T is tangent to the curve and t is the curve parameter.

Obviously this is very similar to the traditional length formula.

For Lorenzian space, a curve is said to be timelike if the norm of its tangent is negative everywhere. Null if equal to zero. Spacelike if positive.(for signature(-+++))

When it is null, the length is obviously zero. If it is timelike, we change the sing in the square root and use the term proper times. τ

$$\tau = \int (-g_{ab} T^a T^b)^{1/2} dt$$

Note, since a geodesic tangent is parallel transported and thus the norm is constant. Meaning they will not change signatures.

Next, to find the curve whose length does not change to first order under arbitrary smooth deformations. More concisely put, the extremized length. For this we will take the example of a spacelike curve in coordinate basis.

$$l = \int_a^b \left(\sum_{\mu,\nu]} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \right)^{1/2} dt$$

Where $C(a) = p$ and $C(b) = q$ are the endpoints of the curve. The extremization problem is obviously mathematically iden-

tical to extremization problem in action or in most all cases.

$$\delta l = \int_a^b \left(\sum_{\mu,\nu} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \right)^{1/2} \sum_{\alpha,\beta} \left(g_{\alpha\beta} \frac{dx^\alpha}{dt} \frac{d(\delta x^\beta)}{dt} + \frac{1}{2} \sum_\sigma \frac{\partial g_{\alpha\beta}}{\partial x^\sigma} \delta x^\sigma \frac{dx^\alpha}{dt} \frac{dx^\beta}{dt} \right) dt$$

Because the length is paramerterized independently, we may assume the origional curve was parameterized so that

$$g_{ab} T^a T^b = 1 = \sum_{\mu,\nu} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}$$

Thus the equation can be simplified to

$$0 = \int_a^b \sum_{\alpha\beta} \left(g_{\alpha\beta} \frac{dx^\alpha}{dt} \frac{d(\delta x^\beta)}{dt} + \frac{1}{2} \sum_\sigma \frac{\partial g_{\alpha\beta}}{\partial x^\sigma} \delta x^\sigma \frac{dx^\alpha}{dt} \frac{dx^\beta}{dt} \right) dt$$

It can be further simplified, but essentially the main point is that a curve extremizes the length between its endpoints if and only if it is a geodesic. Similar proof can be used to find that the geodesic equation (with affine parameterization) can be obtained from variation of the Lagrangian

$$L = \sum_{\mu,\nu} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}$$

The final task of this section is to derive the geodesic deviation equation. Let $\gamma_s(t)$ denote a smooth one-parameter family of geodesics. Σ denote the two-dimensional submanifold spanned by the curve. Let s and t be coordinates of Σ. Vector field $T^a = (\partial/\partial t)^a$ is tangent to the family of geodesics and, thus, satisfies $T^a \nabla_a T^b = 0$. Vector field $X^a = (\partial/\partial s)^a$ repersents the deviation vector. There is "gauge freedom" in X^a. In the sense that X^a changes if one changes the affine parameterization.

One may ensure that X and T are orthogonal using the metric. Further ensuring that S is made in such a way that $X^a T_a = 0$.

Next $\nu = T^b \nabla_b X^a$ gives the rate of change along a geodesic of the displacement. Thus the relative velocity. Thus,

$$a^a = T^c \nabla_c v^a = T^c \nabla_c (T^b \nabla_b X^a)$$

Is the relative acceleration.

$$a^a = -R^a_{cbd} X^b T^c T^d$$

This, using the Riemann tensor, is known as the geodesic deviation equation.

Thus, one can find curve if $R^d_{abc} = 0$.

21.0.3.4 Methods of Computing Curvature

To begin, one must choose the coordinate system. Express the derivative operator ∇_a in terms of the ordinary derivative ∂_a of this coordinate system and the Christoffel symbol Γ^c_{ab}. For a dual vector field ω_a, we have:

$$\nabla_b \omega_c = \partial_b \omega_c - \Gamma^d_{bc} \omega_d$$

thus,

$$\nabla_a \nabla_b \omega_c = \partial_a (\partial_b \omega_c - \Gamma^d_{bc} \omega_d), \tag{21.1}$$

$$- \gamma^e_{ab} (\partial_e \omega_c - \Gamma^d_{ec} \omega_d) \tag{21.2}$$

$$- \Gamma^e_{ac} (\partial_e \omega_c - \Gamma^d_{bd} \omega_d) \tag{21.3}$$

Thus, the Riemann curvature tensor can be expressed:

$$R^d_{abc} \omega_d = [-2\partial_a \Gamma^d_{bc} + 2\Gamma^e_{ca} \Gamma^d_{be}] \omega_d$$

This can be simplified to for the coordinate system:

$$R^{\sigma}_{\mu\nu\rho} = \frac{\partial}{\partial x^{\nu}}\Gamma^{\sigma}_{\mu\rho} - \frac{\partial}{\partial x^{\mu}}\Gamma^{\sigma}_{\nu\rho} + \sum_{\alpha}(\Gamma^{\alpha}_{\nu\rho}\Gamma^{\sigma}_{\alpha\nu} - \Gamma^{\alpha}_{\nu\rho}\Gamma^{\sigma}_{\alpha\mu})$$

The Ricci tensor can then be obtained by contracting the above equation.

There are some other importaint points. One can write the coordinate basis for components, $g_{\mu\nu}$, of the metric as a matrix. Then the components $g^{\mu\nu}$ will then obviously be the inverse. We can define g to be the determinate of $g_{\mu\nu}$.

Then, the natural volume elementof the manifold is $\sqrt{|g|}dx^1....dx^n$.

A simple formula may be derived for the contracted Chrisoffel symbol.

$$\Gamma^{a}_{a\mu} = \sum_{\nu}\Gamma^{\nu}_{\nu\mu} = \frac{1}{2}\sum_{\nu,\alpha} g^{\nu\alpha}\frac{\partial g_{\nu\alpha}}{\partial x^{\mu}}$$

Thus,

$$\Gamma^{a}_{a\mu} = \frac{1}{2}\frac{1}{g}\frac{\partial g}{\partial x^{\mu}}$$

This is useful as Γ^{a}_{ab} is used in the Ricci Tensor as well as formulating the divergence of an arbitrary vector field.

21.0.3.5 Orthonormal Basis (Tetrad) Methods

In tensor calculations an orthonormal basis is in some cases advantagous, thus one develops a nonholonomic or noncoordinate, orthonormal basis of smooth vector field $(e_{\mu})^{a}$ that satisfies:

$$(e_{\mu})^{a}(e_{\nu})_{a} = \eta_{\mu\nu}$$

where $\eta_{\mu\nu} = diag(-1,..,-1,1,..,1)$. In four dimensions, $(e_{\mu})^{a}$

is called a tetrad. It is useful in the equation

$$\sum_{\mu,\nu} \eta^{\mu\nu}(e_\mu)^a(e_\nu)_b = \delta^a_b$$

Where δ is the identity map of the vectors and $\eta^{\mu\nu}$

Calculation by tetrad has sufficent differences than the coordinate basis method. In both methods three key ingrediants are needed. 1: derivative operator that is compatable with the metric, $\Delta_a g_{bc} = 0$. 2: the derivative operator is torsion free. 3: the Riemann tensor is related to the derivative operator.

21.0.3.6 Einstein's Equations

21.0.3.6.1 Special

Energy is the time component of the 4-vector momentum.

A perfact fluid's Stress-energy tensor can be defined as

$$T_{ab} = \rho u_a u_b + P(\eta_{ab} + u_a u_b)$$

where u^a is a unit timelike vector field representing the 4-velocity of the fluid. ρ and P are the mass energy density and pressure of the fluid measured in its rest frame.

a cool trick is that one can define the Euler's equations using non-relativistic limits.

The stress energy of the Lorentz force:

$$T_{ab} = \frac{1}{4\pi}\left(F_{ac}F^c_b - \frac{1}{4}\eta ab F_{de}F^{de}\right)$$

21.0.3.7 General

In general theory of Relativity, while it is possible to define background observers in such a way to define gravity for Earth. When time symmetry does not hold, such in 3-body problems, this is not the case and tidal forces are the only "force" with a meaningful definition.

From the simple exploration of physical energy conservation in gravitational fields where the dimensions is small compared to the curvature of space, it can be found that all matter obeys the equation of motion:

$$\nabla^a T_{ab} = 0$$

Through logical examinations the equation comes to be

$$G_{ab} = R_{ab} - \frac{1}{2} R g_{ab} = 8\pi T_{ab}$$

Using a trace we find that

$$R = -8\pi T$$

and thus:

$$R_{ab} = 8\pi \left(T_{ab} - \frac{1}{2} T g_{ab} \right)$$

Thus, the entirety of GR can be summed up as followed. *Spacetime is a manifold M on which there is defined a Lorentz Metric g_{ab}. The curvature of the metric is related to matter distribution via the Einstein's equation.*

21.0.3.8 Linearized Gravity

I am not sure what to write here...

21.1 Homogeneous, Isotropic Cosmology

...

21.2 The Schwarzschild Solution

21.2.1 Derivation

For an object to be static, the time-like killing vector field must satisfy the following equation:

$$\xi_a \Delta + b\xi_C = 0$$

From here, the fact that this shows that spacetime is camveriant in the time dimension leads to the metric can be defined as

$$ds^2 = -V^2(x^1, x^2, x^3)dt^2 + \sum_{\mu,\nu=1}^{3} h_{\mu\nu}(x^1, x^2, x^3)dx^\mu dx^\nu$$

Where $V^2 = -\xi_a \xi^a$, and the absence of $dt\ dx^\mu$ cross terms expresses the orthogonality of ξ^a and σ.

From here time translation and reflection are here. In most cases something that is time invarient will also be time reflected but not when rotation is involved. If one takes it to be spherically symmetric:

$$ds^2 = r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

In curved space, r does not have to be in relation to the radius, but for the purposes of this, we will take it as such.

the rest...

Now that we have the metric, we see that the solution is asymptotically flat, ie, as r approches infinity spacetime approches flat. Allowing the interpretation of an isolated body. Where C is equivalent to $-2M$.(Remember, natural units)

A striking feature is that the metric components become singular in both $r = 2m$ and $r = 0$. This is caused by either a breakdown of coordinate, or a singularity of the spacetime structure. We will see later that the singularity of 2M is simply a breakdown of coordinates while at 0 it is seen as a physical singularity.

It can also be found that for all spherically symmetric spacetimes with $R_{ab} = 0$ are static.

21.2.2 Interior Solutions

Now take a static, spherical, stress energy tensor

$$T_{ab} = \rho u_a u_b + P(g_{ab} + u_a u_b)$$

Where u^a must point in the same direction as the static killing vector.

We can now take three independent forms of T_{00}, T_{11}, and T_{22} using the metric derived earlier. To get h to be

$$h(r) = \left(1 - \frac{2m(r)}{r}\right)^{-1}$$

Here to have staticity, where $h \geq 0$, $r \geq 2m(r)$

Where m(r) isn't exactly the proper mass but rather difference between the gravitational binding mass and the proper mass.

From here they derive the Tolman-Oppenheimer-Volkoff equa-

tion or TOV. Which is the hydrostatic equilibrium with relativistic corrections. This equation comes with a limit where pressure required goes towards infinity as is the case of the singularity.

21.2.3 Geodesics

Not too much other than deriving gravitational lensing

21.2.4 Kruskal Extention

N/A

21.3 Methods for Solving Einstein's Equation

21.3.1 Stationary, Axisymmetric Solutions

Earlier we have derived that a spacetime is said to be stationary if there exists a one-parameter group of isometries whose orbits are timelike curves. Thus every stationaly spacetime possesses a timelike killing vector field.

A spacetime is called axistmmetric if there exists a one-parameter group of isometries whose orbits are closed spacelike curves. To have both

$$[\xi, \psi] = 0$$

these solutions are useful as they describe

Chapter 22

The Geometry of Kerr Black Holes by Barrett O'Neil

22.1 Beginning Kerr Spacetime*

One can define the difference between rotation types to be given by the horizon function $\Delta = r^2 - 2Mr + a^2$(this is in geometrized units)

The kerr metric does not fail at $r = 0$ like the Schwarzchild solution does. Though it adds the two horizons and the ring singularity.

Really complex math for relatively trivial solutions. I plan to go back at some point, just not today.

22.2 Maximal Extention

Traditional kerr uses ingoing principle null geodisics, but to get maximal extention, one must "glue" both the traditional ingoing formulation and outgoing.

Chapter 23

Assumptions of Physics

23.0.1 Overview

The overall goal is to explore the minimal set of physical assumptions are necessary and sufficient to derive different theories within a unified framework. This is almost identical to my Axiom project with the exeption that while not as broad their research is much deeper than mine.

It is broken into two parts

Reverse Physics: Looks at the main theories and finding the common and the different aspects of those theories.

Physical Mathematics: Connecting physical objects and theories to be mapped onto the correct mathematical objects

23.0.2 Reverse Physics

It was a great read with a lot of interesting ideas, but I don't want to write it all, esspecially since I am planning on writing something very similar in part 1.

So, I will write a summary of the summary

The first assumption is Infinitesimal Reducibility(IR). This tells us the domain of classical physics is of those that can be thought of as made of arbitrarily small parts. Mathematically this tells us that the state space of the particles is a differentiable manifold, together with a volume measure that defines the count of states for each region. We can then add the assumption IND that tells us that the system is decomposable into independent degrees of freedom. For this to work the manifold must have even dimensions. That any state variable q^i has a conjugate state p_i. This give a symplectic manifold. If we assume DR, that evolution is deterministic and reversable, we obtain Hamiltonian Mechanics. This enforces preservation of area.Thus, the displacement vector field is *H*, which corresponds with the Hamiltonian. This can be generalized to the time dependent case, which recovers relativistic features even without the noting of a metric tensor.

Assuming KE, that dynamics are recoverable from kinematics, we get Newtonian. If we assume both KE and DR we get Lagrangian.

23.0.3 Physical Mathematics

This one is a bit more abstract and newer stuff, so I will have more notes.

Chapter 24

A Concise Introduction to Geometric Numerical Integration

For this I will skip around as I already have knowledge on the subject.

24.1 Variational Integrator

Essentially, finding the discrete Lagrangian

$$L_h(q_n, q_{n+1}, h) \approx \int_{t_n}^{t_{n+1}} L(q(t), \dot{q}(t))$$

Then summing them over a discrete curve with N points

$$S_h(\{q_n\}_{n=0}^{N}) = \sum_{n=0}^{N-1} L_h(q_n, q_{n+1}, h)$$

Now assume continuous and find the variation of this sum

held fixed.

24.2 Volume-Preserving Methods

Volume preservation is used in many places; particle tracking in in-compressible flows, perturbations in Hamiltonian systems.

Take a divergence free function of velocity. Then a volume perserving flow can be described by this satisfed by the derivative $X(t) = \frac{\partial \varphi_t(x)}{\partial x}(x_0)$

$$\dot{X} = A(t)X$$

$$X(0) = I$$

Known as the variational equation. Here $A(t) = f'(x(t))$ repersenting the Jacobian matrix of f evaluated at $x(t) = \varphi_t(x_0)$.

Thus, if X is divergence free, then the determinate is equal to 1. All Hamiltonians are divergence free.

If one takes a numerical look. X becomes using the implicit midpoint rule.

$$X_{n+1} = (I - \frac{h}{2}A)^{-1}(I + \frac{h}{2}A)X_n$$

So the numerical flow is volume-perserving if.

$$det\left(\frac{\partial X_{n+1}}{\partial X_n}\right) = 1$$

24.3 Lie Group Method

Assume another differential equation of the same type as before, but A(t) and I are $d \times d$ matricies

With this equation, despite its simplisity, closed form solutions only exist in particular situations. In scalers(d=1), it is trivial. For $d > 1$, it is true when $A(t)$ and $\int_0^t A(s)ds$ commute.

In non-communtable cases expressed

$$X(t) = \exp(\Omega(t))$$

One can derive the differential equation

$$\dot{\Omega} = d\exp_\Omega^{-1}(A(t))$$

$$\Omega(0)$$

where

$$d\exp_\Omega^{-1} = \sum_{k=0}^{\infty} \frac{B_k}{k!} ad_\Omega^k(A)$$

Here B_k are Bernoulli Numbers and ad^k is shorthand for iterative commutators.

Solving, one gets a very long infinite series that I will just look up if needed(Magnus Expansion)

I think I am going to more on now.

24.4 Time-Splitting PDE

This is simply the same as traditional splitting but using time with evolution equations.

Chapter 25

Geometric Numerical Integration

[Structure-Preserving Algorithms for Ordinary Differential Equations]

25.1 Symplectic Integrators of Hamiltonian systems

25.1.1 Variational Integrators

25.1.1.1 Hamilton's Principle

The motion extremizes the action integral is Hamilton's Principle

One can then derive the equation of the derivative of the action

$$dS = p_1 dq_1 - p_0 dq_0$$

which is the basic formula for simplectically generating functions.

25.1.1.2 Discretization of Hamilton's principle

I don't think I need to go over this

25.1.1.3 Noether's theorem

Continuous symmetris in te Lagrandian lead to first integrals, and thus this connection between Lagrandian and Hamiltonian leads to more first integrals.

25.1.2 Characterization of Symplectic Methods

Also trivial

25.2 Non-Canonical Hamiltonian

25.2.1 Constrained Mechanical Systems

For the most part these are pretty simple extensions of regular symplectic integrators

25.2.2 Poisson System

25.2.2.1 Poisson Integrators

Chapter 26

Introduction to Mechanics and Symmetry

26.1 Hamiltonian Systems on Linear Symplectic Spaces

26.1.1 Introduction

For a deeper understanding of Hamiltonian's equations one can take a matrix approach where E as a real vector space and E* as its dual space. Ok, I don't know how that gives a deeper understanding

One can also view it through Symplectic and Poisson Structures, they simply define the Jacobian structure of symplectic manifolds.

Finding that:

$$\{F, G\} = B(dF, dG) = dF \cdot \mathcal{J}\boldsymbol{\nabla} \cdot G$$

The symplectic structure Ω is the bilinear form associated

to R^{-1}

Thus the hamiltonian can be written as

$$\dot{z} = X_H(z) = \Omega^{\#} dH(z)$$

Chapter 27

The Relativistic Boltzmann Equation

27.1 Special Relativity

It is convenient to define the four-velocity of a particle as

$$U^\alpha = \frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}$$

Where τ is proper time, which is scalar invariant. The contra and co variant four velocity as

$$(U^\alpha) = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{V}{\sqrt{1 - v^2/c^2}} \right)$$

$$(U_\alpha) = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{-V}{\sqrt{1 - v^2/c^2}} \right)$$

27.1.1 Minkowski Force

Four velocity is defined in the same manner

Then the four momentum contravariant component is given as

$$(p^\alpha) = (\frac{E}{c}, \mathbf{p})$$

Here we can derive the famous $E = mc^2$

Taking the acceleration we get the Minkowski force's contravariant components

$$(K^\alpha) = (\frac{\mathbf{F} \cdot \mathbf{v}}{c(1 - v^2/c^2)^{\frac{1}{2}}}, \frac{\mathbf{F}}{(1 - v^2/c^2)^{\frac{1}{2}}})$$

27.1.2 Electrodynamics in free space

27.1.2.1 Maxwell Equations

The electromagnetic field tensor is defined through:

$$F^{\alpha\beta} = (-\mathbf{E}, c\mathbf{B})$$

This tensor is antisymetric.

With its transormation laws being

$$F'^{\alpha\beta} = \lambda^\alpha_\gamma \lambda^\beta_\delta F^{\gamma\delta}$$

$$F^{\alpha\beta} = \bar{\lambda^\alpha}_\gamma \bar{\lambda^\beta_\delta} F'^{\gamma\delta}$$

$$p^\mu p_\mu = -m^2c^2 \qquad \text{(mass shell)},$$
$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu \qquad \text{(EM field)},$$
$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \qquad \partial_\mu {}^\star F^{\mu\nu} = 0 \qquad \text{(Maxwell)},$$
$$\frac{dp^\mu}{d\tau} = qF^{\mu\nu}u_\nu \qquad \text{(Lorentz force)},$$
$$dP = \frac{d^3p}{p^0} \qquad \text{(invariant measure)},$$
$$J^\mu = \sum_s q_s \int p^\mu f_s \, dP \qquad \text{(kinetic current)},$$
$$p^\mu \partial_\mu f + qF^{\mu\nu}p_\nu \frac{\partial f}{\partial p^\mu} = C[f] \qquad \text{(relativistic Boltzmann)},$$
$$T^{\mu\nu} = \int p^\mu p^\nu f \, dP \qquad \text{(kinetic stress-energy)}.$$

27.2 Relativistic Boltzmann Equation

27.2.1 Single Non-degenerate Gas

Note that momentum and number of particles in an volume element are scalar invariant.

We make the volume element

$$d\mu = d^3x d^3p$$

We can approximate the Jacobian to be

$$J = 1 + \frac{\partial F^i}{\partial p^2}\Delta t + O[(\Delta t)^2]$$

We assume the following

- Only binary collisions are considered(reasonable if the gas is dilute)
- Molecular Chaos assumption
- One particle distribution function does not vary much over the time interval

Remember the momentum four vector is

$$p^\alpha = (p^0, \mathbf{p})$$

where p^0 is

$$p^0 = \sqrt{|p| + m^2c^2}$$

where p^0 is the energy component and $\mathbf{p}$ is the 3-vector spacial momentum vector.

This is all pretty trivial and can be looked up later. I am going to skip ahead.

27.3 Vlasov

27.3.1 The Vlasov-Maxwell system

The Vlasov can be written in form

$$p^\mu \frac{\partial f}{\partial x^\mu} - \Gamma^\sigma_{\mu\nu} p^\mu p^\nu \frac{\partial f}{\partial p^\sigma} = 0$$

Here the phase space isnot a product of the two spaces but rather a sub-manifold of the tangent bundle.

Thus the derivative is taken in the direction tangent to the manifold $g_{\mu\nu}p^\mu p^\nu = m^2c^2$.

One can couple with Einstien's field quation by applying the

stress energy tensor to be:

$$T_{\mu\nu} = c \int p_\mu p_\nu f \sqrt{g} \frac{d^3 p}{p_0}$$

Creating the Vasov-Einstein system.

27.3.2 The Threshold of Black hole formation

Chapter 28

Introduction to 3+1 Numerical Relativity

28.1 3+1 formulism

28.1.1 3+1 split

The 3+1 split of the metric can be defined as

$$g_{\mu\nu} = \begin{pmatrix} -\alpha^2 + \beta_k\beta^k & \beta_i \\ \beta_j & \gamma_{ij} \end{pmatrix}$$

$$g^{\mu\nu} = \begin{pmatrix} -1/\alpha^2 & \beta_i/\alpha^2 \\ \beta_j/\alpha^2 & \gamma_{ij} - b^i b^j/\alpha^2 \end{pmatrix}$$

The volume can then be expressed as

$$\sqrt{-g} = \alpha\sqrt{\gamma}$$

with g and y being the determinate of the tensors.

The spacial metric $\gamma_{ij}$is simply defined at the metric induced

on each hypersurface Σ by the full spacetime

$$\gamma_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$$

This spacial metric is a full four-tensor but when written in adapted coordinates its time component becomes trivial. This spacial metric is simply the projection operator on the spacial hypersurface.

The lapse function is

$$\alpha = (-\boldsymbol{\nabla} t \cdot \boldsymbol{\nabla} t)^{1/2}$$

The unit normal vector space can be writen as

$$n^\mu = -\alpha \boldsymbol{\nabla}^\mu t$$

Where the minus sign enforces that the unit vector is future pointing.

28.1.2 Extrinsic Curvature

The extrinsic curvature tensor $K_{\alpha\beta}$is a measure of the change of the normal vector under such parallel transport.

Before we do that the projection operator onto the spacial hypersurface:

$$P^\alpha_\beta := \delta^\alpha_\beta + n^\alpha n_\beta$$

which is simply the induced spacial metric.

Using this, the extrinsic curvature tensor is

$$K_{\mu\nu} := -P^\alpha_\mu \boldsymbol{\nabla}_\alpha n_\nu$$

The tensor is symmetric

In the coordinate system created the Hamiltonian and Mo-

mentum constraints take the form:

$$R + K^2 - K_{ij}K^{ij} = 16\pi\rho$$

$$D_j(K^{ij} - \gamma^{ij}K) = 8\pi j^i$$

with

$$\rho := n^\mu n^\nu T_{\mu\nu}, \qquad j^i := -P^{i\mu} n^\nu T_{\mu\nu}$$

28.1.3 The ADM evolution Equation

With a lot of math I am not going to write down the Hamiltonian density can be defined as

$$\mathcal{H} = -\sqrt{\gamma}\left[\alpha(R + k^2 - K_{ij}K^{ij}) + 2\beta_i D_j(K^{ij} - \gamma^{ij}K)\right]$$

Then the hamiltonian is obviously

$$H := \int \mathcal{H} d^3x$$

Variations of this hamiltonian with respect to α and β^i yields the Hamiltonian and Momentum constraints.

The evolution equations for the gravitational field now follow to be

$$\dot{\gamma_{ij}} = \frac{\delta \mathbf{H}}{\delta \pi^{ij}} \qquad \dot{\pi^{ij}} = -\frac{\delta \mathbf{H}}{\delta \gamma_{ij}}$$

where gamma is the spacial metric and pi is the cononical momenta.

28.1.4 BSSNOK formulation

ADM are only weakly hyperbolic and those are highly unstable. BSSN is also called conformal ADM

Consider the conformal rescalling of the spacial metric of the form(where, Conformal rescaling means multiplying a metric by a positive function)

$$\tilde{\gamma_{ij}} := \psi^{-4}\gamma_{ij}$$

This can be any arbitrary chosen scalar field, but a convienent choice would be a factor that the conformal metric has a unit determinate, that is

$$\psi^4 = \gamma^{1/3} \quad \Longrightarrow \quad \psi = \gamma^{1/12}$$

With γ being the deterimate of γ_{ij}.

There are some other values to be taken into account when using differing spacetimes.

The BSSNOK also seperates the extrinsic curvature into its trace K and its tracefree part

$$A_{ij} = K_{ij} - \frac{1}{3}\gamma_{ij}K$$

Which under conformal rescalling

$$\tilde{A_{ij}} = \psi^{-4}A_{ij}$$

The crucial point is that BSSNOK intorduces three auxiliary variables known as the conformal connection functions.

$$\tilde{\Gamma^i} = \tilde{\gamma^{jk}}\tilde{\Gamma^i_{jk}} = -\partial_j\tilde{\gamma^{ij}}$$

After some more math the equation simplifies to the evolution equations

$$\frac{\mathrm{d}}{\mathrm{d}t}\gamma_{ij} = \tilde{-2\alpha A_{ij}}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi = -1/6\alpha K$$

$$\frac{d}{dt}\tilde{A}_{ij} = e^{-4\phi}\left\{-D_iD_j\alpha + \alpha R_{ij} + 4\pi\alpha\big[\gamma_{ij}(S-\rho) - 2S_{ij}\big]\right\}^{\mathrm{TF}} + \alpha\left(K\tilde{A}_{ij} - 2\tilde{A}_{ik}\tilde{A}^k{}_j\right)$$

$$\frac{d}{dt}K = -D_iD^i\alpha + \alpha\left(\tilde{A}_{ij}\tilde{A}^{ij} + \frac{1}{3}K^2\right) + 4\pi\alpha(\rho + S)$$

$$\frac{d}{dt}\tilde{\Gamma}^i = \tilde{\gamma}^{jk}\,\partial_j\partial_k\beta^i + \frac{1}{3}\tilde{\gamma}^{ij}\,\partial_j\partial_k\beta^k - 2\,\tilde{A}^{ij}\,\partial_j\alpha + 2\alpha\left(\tilde{\Gamma}^i_{jk}\tilde{A}^{jk} + 6\tilde{A}^{ij}\partial_j\phi - \frac{2}{3}\tilde{\gamma}^{ij}\partial_j K - 8\pi\tilde{j}^i\right)$$

28.2 Gravitational Waves

28.2.1 Gauge invariant Perturbations of Schwarzchild

When referring to multipoles, if the transformation is $(-1)^l$ then it is even or polar. It is odd or axial is it is instead $(-1)^{l+1}$.

The authors define the Zerilli-Mathews tensor harmonics, useful tensors for looking at quadropoles. It works spesifically for even tensors

$$Z^{l,m}_{ab} := D_aD_bT^{l,m} + \frac{1}{2}l(l+1)\Omega_{ab}Y^{l,m}$$

where the scalar sphere harmonics are eigenfunctions of the laplance operator on the sphere. The above tensor is in fact traceless, we can also find its divergence to be given by

$$D^bZ^{l,m}_{ab} = -\frac{1}{2}(l-1)(l+2)Y^{l,m}_a$$

Odd tensors can only be constructed in one way

$$X^{l,m}_{ab} = \frac{1}{2}(D_a X^{l,m}_b + D_b S^{l,m}_a)$$

With the divergence being

$$D^b X^{l,m}_{ab} = -\frac{1}{2}(l-1)(l+2)X^{l,m}_a$$

This can be expanded into multipoles that find that the vector harmonics vanish at monopoles while the tensor ones vanish at both mono and di - poles.

Next the Regge-Wheeler gauge is a choice that $G^{l,m} = H^{l,m}_A = 0$ and K and H_{AB} stay the same.

Here we have the even and odd parity master equation. I won't write it down but I am sure I can look it up quite easily.

28.2.2 Gravitational Radiation in the TT Gauge

I don't want to write all of this, it is relatively simple math(not really), but a lot of it.

28.2.3 Energy and Momentum of GRW

Take the second order pertubation to derive the Ricci Tensor as

$$R_{\mu\nu} = \epsilon R^{(1)}_{\mu\nu} + \epsilon^2 R^{(2)}_{\mu\nu}$$

Where $R^{(n)}_{\mu\nu}$ can be derived through F rather simply.

This can be used to construct a psuedo-tensor that can then be used to find the real tensors.

Chapter 29

Applications of Lie Groups to Difference Equations

29.1 Finite Differences and Transformation Groups in Space of Discrete Variables

One can take a differential operator and expand using a Taylor series.

Chapter 30

Topological Methods in Hydrodynamics

30.1 Group and Hamiltonian Structures

Chapter 31

Chaos and Structure in Nonlinear Plasma

31.1 Nonlinear Oscillations

31.1.1 Harmonic Generations in Small Amplitude Oscillations

N/A

31.2 Hamiltonian Dynamics

In plasma physics a special case for 1D would be when the Hamiltonian is equivalent to the stream function ψ.

Having a physical problem reduced to a 3-dimensional vector space is tantamount to having the problem solved do to the clear research in the area.

31.2.1 Poincare Surface of Section

31.2.2 Fixed Points and Invariant Curves

31.2.3 KAM Theory

Take the two dimensional map

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = T(x, y)$$

produced from the flows of systems such as given above are common in plasma physics. The theorems used are called KAM for conviencences based on the three founders of many of the theorems. To apply maps for twist conditions that produce monotonically increasing rotation of the point in the surface of section.

$$\frac{\partial y'}{\partial x} > 0 \quad \forall x$$

The twist conditions guarantees that around a fixed point there is a monotonic increase of the rotational angle.

31.2.4 KAM Torii

Essentially, KAM is the study of how nearly integratable Hamiltonians remain similar to their integratable counterparts. Resonance refers to ones that remain.

31.2.4.1 Ergodic Theorem

The Ergodic theorem states that the long time average of a function over phase space is equal to the ensemble average.(essentially the average of one function is the same as the average over the collection of possible states of a system)

This theorem is valid for dynamics $\theta' = \theta\alpha(r)$, provided α is irrational.

We may compute the difference between f_0 from the theorem would be

$$\delta f(\theta) = \sum_{k=1}^{\infty} \left[a_k e^{ik\theta} + a_k^* e^{-ik\theta}\right]$$

assuming that $|a_k| < \frac{a_1}{k^2}$ to eliminate discontinuous functions. Summing this we can get that the sum diverges wher $k\alpha = 2\pi m$. The sum will converge provided that

$$\left|\frac{\alpha}{2\pi} - \frac{m}{n} > \frac{\epsilon}{n^3}\right|$$

not here that m and n are integers, particular resonance between the frequencies $\alpha = (m/n)2\pi$

31.2.5 KAM theory in Labrotory Plasmas

The $1\frac{1}{2}$ D abstraction can be mapped out for electron in toriodial confinement. Also, incompressible drift motion by $E \times B$ plasma flow.

31.2.6 Visualization of the Magnetic Flux surfaces by Electron beam mapping

The winding number in the phase space becomes directly to the rotational transformation $t(r)$, this "safety factor" $q(r) \equiv 1/t(r)$ of tokamak theory.

They used this methodology to shrink instability islands in the tokomak.

31.2.7 Magnetic Islands and the Unstable-Chaotic Trajectories of Field Lines in Tokamaks

Most dangerous magnetic islands are developed by helial plasma currents from unstable resistive MHD motions.

Chapter 32

Papers

I can't believe how few papers I have written notes about, I should do better.

32.1 Operator Splitting Methods: Numerical Solutions of Ordinary Differential Equations via Separation of Variables

32.1.1 General Idea

It is deriving the mathematical underpinning of operator splitting from Koopman-Lie semigroup rather than physical systems. Generalizing it beyond Hamiltonian theory

32.1.2 Notes On Introduction

N/A

32.1.3 Strongly Continuous and bi-continuous semigroups

You know what, at the moment this paper may be beyond my graps.

32.2 Splitting methods for differential equations

32.2.1 Notes on introduction and Abstract

Some argue there are only ten big ideas in numerical analysis. They make a cool little connection to Descartes rules of logic and how splitting lays the foundation of science.

They then use a simple argument to derive strang splitting

Adjoint method refers to steping forward and then stepping backwardswith the inverse if its adjoint. Adjoint simply refers to that if $\psi_h = \varphi_1(\varphi_2)$ then $\psi_h^* = \varphi_2(\varphi_1)$

This is called time symmetric or self-adjoint method.

Some disadvantages of higher order splitting is that they go backwards in time which is troublesome in dissipative systems. extensively used in fields as distant as molecular dynamics, simulation of storage rings in particle accelerators, celestial mechanics, astronomy, quantum (statistical) mechanics, plasma physics, hydrodynamics, and Markov Chain Monte Carlo methods.

32.2.2 Higher Order Splitting and Composition

32.3 Core Collapse Supernova Modeling: The Next Ten Years

32.4 Physical, numerical, and computational challenges of modeling neutrino transport in core-collapse supernovae

32.4.1 Structure Preserving

32.5 Discontinuous Galerkin Methods

32.6 Energy conserving and well-balanced discontinuous Galerkin methods for the Euler–Poisson equations in spherical symmetry

32.6.1 Numerical Methods

Poisson equations is derived to be

$$\Phi_h(r) = \Phi_h(r_{j+1/2}) - \left(\sum_{i=1}^{k+1} \frac{g_{j,i}\tau^{i+1}}{i+1} - \frac{g_{j,-2}}{\tau} \right) \Big|_{\tau=r}^{r_{j+1/2}}$$

The euler-Poisson equation in DG

Chapter 33

Colloquia: UTK

33.1 August 24: Controlling Topological Quantum Phases through Epitaxy

33.1.1 Main notes

How can one material have topological phase? Essentially the electronic band structure. Things to modify are strain, dimensionality, crystal, structure, and composition.

He uses a molecular beam epitaxy, the growth of crystalline overlayer on a crystalline substrate.

- Phase and strain control
- Dimensionality control

33.1.1.1 Phase and strain control

First one cools to avoid bubbling.

Odd points that where later understood via computational experiments via energy ...

33.1.1.2 Strain-controlled phase stabilization

IDK

33.1.1.3 Quantum Oscillation in strained a-Sn

Heavy-hole state and surface state have negative correlation.

33.1.1.4 Dimensioality Control

IDK

33.1.1.5 Geometry and Interfaces

Changing the arranging of mask, InGaAs, InAsm and InP(001). In order to protect, enhance electron mobility, and phase coherence length.

These create kinds of pseudo-shapes like trenches and bridges that can grow continuously retaining their general shape.

You can suppress induction effects post fabrication through something(not sure)

33.1.1.6 Emerging

Decouple synthesis from integration: Find what materials best lead to growth

Reconfigure strain: Find new ways to reduce growth strain

Expand Heterogeneous integration: IDK

Main ways of doing this was through remote epitaxy and van der waals epitaxy on fledible mica

33.1.2 Connections

IDK

33.1.3 Summary

I think the general idea is an overview of creating superior method of developing superconducting topological material through epaxity. Studying the best arrangements, conditions(strain, dimensions, temp, etc). I really didn't understand beyond this vague overview.

33.1.4 Questions

What are electronic band structure?

What are buffer layers?

What are α-Sn and β-

How do these things grow?

33.2 August 31 - Electron Pairing and Fractionalization in age of AI and Quantum Computing

33.2.1 Main notes

33.2.1.1 Introduction

Takes the traditional billiard-ball electron scattering to either pairing up or quantum hall, making it so that Resistance can come to zero. Also, conduction is an exact fraction of e^2/h.

These Cooper pairs are caused by quantum pairs via photon glue. These permutate lattices that make it "easier" for other particles to do the same and move without resistance.

He introduces, the fact that the last ten years, people aren't trying to increase tempature but rather greater mathematical machinery. Looking at Chiral 2D SC. Allowing for accuracy with both counter and clockwise translations.

On the side of quantum hall and fractionalization.

Found that conductors can effectively channel $1/3$ of electrons through "anyons," a type of quasiparticle.

33.2.2 Concepts

Using a Fourier transformation can define N atoms using K space.

Solving the encoding problem with the quantum states is near impossible in classical computers so he hopes that quantum computers can do it much easier.

Current theory can only work for around ten atoms. It also becomes a struggle when the Hilbert space goes up exponentially. This is particularly troubling when real matter has 10^{23} atoms.

They work on this through using smooth Moire potential space that is more smooth than real solids.

33.2.3 Learning Hamiltonians

Learning Hamiltonian AI systems are meant to bring such material simulating to $\mathcal{O}(N)$, which is quite out there.

Encoding Hamiltonian through one section of the idea of locality as on the scale of matter, electron become local.

They have a "white box"

They have a scale of $10^6 - 10^9$.

33.2.4 Kinetic Magnetism

To lower the energy in matter, you can polarize so that all spins are facing the same direction.

Singlets entangle between spin polarization.

Connecting them all together is what develops the cooper pair.

Neural networks can also be used to compute wavefunctions. This is done by pairing-native electron antzan.

Geminal works to machine precision in very few states. Much lower then previous models.

33.2.5 Quantum Computing

Four main different methods of quantum computing.

33.2.6 Connections

Open frontier being:

Finite-q photons

One thing he mentioned is the idea that programming for quantum computers will end up something primarily done by mathematicians and physicist for a while as AI or traditional CS won't have the skills needed at the moment.

33.2.7 Summary

AI for quantum matter:

Controlled strong-coupling theory:

Quantum Proccessors:

Next comes:

Further AI-Hamiltonian

AI Wavefunction

Simulating anyon braiding and topological quantum computation

33.2.8 Questions

33.3 Curiosity and Perseverance: Martian Surface

33.3.1 Main notes

Using fluid dynamics one is able to more accurately figure out the past of mars surface. This is highly challenging due to limited data and people's tendicy to apply Earth principles to the Earth.

-

33.3.2 Connections

33.3.3 Summary

33.3.4 Questions

33.4 Weighing the Earth: The Quest to Measure the Gravitational Constant

33.4.1 Main notes

PEP acronym

- Physics
- Engineering
- Psychology

One theory is there is an additional force they are not accounting for, but this is highly unlikely.

Another being the imperfact vacuum pressure that leads to imperfact data.

Finally, the possibility of conscious or subconscious bias of researchers.

They solved it through working with false data so that you don't subconsciously match.

33.4.2 Connections

The methodology to fix bias was cool, I wonder how this can be applied in other situations.

33.4.3 Summary

33.4.4 Questions

33.5 Direct Observation of the Superallowed α-Decay of 104 Te

33.5.1 Main notes

The main project of their programme is to push outside of stability as far as possible.

General notes

- Particle collidors are several order of magnitude higher in energy useage
- Atomic collidors are about 5 orders of magnitude greater in beam intensity.

Cross section of the projectile fragmentation drops down after a certain mass number. Also, these cross section estimations have huge deviations in prediction, could be as far off as an order of magnitude.

Where the α particle comes from is still an open question not fully captured by theory.

Cluster structures in light nuclei is called the Hoyle state and is the reason stellar evolution happens past carbon 12 and is clearly established both theoretically and experimentally. On heavy nuclei, we have evidence for a alpha cluster that isn't as developed.

Observed the first superallowed alpha decay, so far this Te is the only thing that shows.

33.5.2 Connections

He talks about how his plot being logromithic doesn't show certain useful information.

33.5.3 Summary

33.5.4 Questions

Chapter 34

Coding

34.1 *Linux Terminal

34.2 *Github

34.3 *FORTRAN

34.4 *Python

34.5 *Emacs

34.6 *LaTeX

Chapter 35

Bible Reading

35.1 2026-16-02

35.1.1 Genesis 41:17-42:17

God protects Joseph through establishing gifts within him. He does the same for us.

Why does Joseph test his brothers?

35.1.2 Matthew 13:24-46

The parables of Jesus, establish stories and myth as the methodology of developing the abstract concepts of God's will.

It refers to the weeds as everything that causes sin and all who do evil, what exactly does this mean? Everyone does evil and everyone has within that which tempts them. Taking out the "weeds" prematurely will destroy the wheat.

The final stories is about how the kingdom of God gives everything for the grace of those who are lost.

35.1.3 Psalm 18:1-15

Love of God

35.1.4 Proverbs 4:1-6

Community, family, and God fosters wisdom. Wisdom that protects you.

35.2 2026-17-02

35.2.1 Old testament

Why is Joseph testing with them even after hearing what they had to say while also giving them money

35.2.2 New Testament

The kingdom of God takes in all, but not all remain.

Interesting how such a radical take on ones word is used.

35.2.3 Psalm

God rescues you and protects you from the enemies that are greater than you

35.2.4 Proverbs

Pursue wisdom over all other virtues

35.3 2026-24-03

35.3.1 Genesis 46:1-47:31

Joseph built his family and tradition up in accordance with God's will. He also did his duty with extreme diligence.

35.3.2 Matthew 15:1-28

While tradition matters, God's commandments are far above.

Your words matter and define you. Disregarding this defiles your nature. Acting in a virtuous manner in accordance with the idea matters much more than rules. Beyond that, a critical examination of such things is important as people lead one another astray.

Actions can corrupt you, they are not in isolation but rather, not just a consequence of what is beneath but create what is beneath.

Jesus was here for the Jewish people, yet his words go far beyond.

35.3.3 Psalm 19:1-14

To be truly free is to be free of sin, while this is impossible on Earth, God's commands lead to this by our creation of our own self though the actions he has put forth.

35.3.4 Proverbs 4:14-19

Don't think of evil for it will corrupt you without thought. For, while not always, that is what evil comes from, lack of thought. Impulse and action without conscious morality driving ones

will. One must then separate oneself from this evil that surrounds them.

35.4 03/25/26

35.4.1 Genesis 48:1-49:33

I continually attempt to find greater understanding of God's will for us through the traditions and things he set forth. How should I go about doing it? I mean I would make the claim being an endogamous clan based family structure does not hold deeper wisdom, but how do I make sure my cultural preferences are hidding things from me in other areas.

In this story as in others, God overturns natural human hierarchy as he choices the secondborn above the first.

In the prophecies it mentions the children's clans fate that mirror their temperament, flaws, and virtues. Meaning or fate is either our own or God has put it in fornt of us purposefully.

Juda's past is messy but he brings out the Davidic monarchy and eventually Jesus.

Joseph's blessing begins by his attack but also being saved through God, not himself.

35.4.2 Matthew 15:29-16:12

Interpret the signs of our time to understand God's will.

Those that have turned for Jesus will need signs to see, but with those with faith that is not needed.

The Pharisees only offer short 'bread' coming from gratification of tradition and lineage, to go beyond one must follow Jesus.

The disciple worry about food even after seeing the miracle Jesus had just done. In the same way we so simply forget and ignore the acts Jesus has already done for us.

35.4.3 Psalm 20:1-9

To quickly we think our victories are our own, to devote ourself to Christ is the antidote to this pride.

35.4.4 Proverbs 4:20-22

Pay attention to the wisdom of others, especially what God says. Attempt to fully understand it, beyond even what they have defined for the wisdom.

Guard your heart tightly, even listening to the world around you too closely can corrupt you. This heart, the virtue inside of you guides your entire being, your life in ways that aren't clear at first glance. Sacrificing this is far more dangerous than it seems.

Even something a trivial as degrading language can have a profound impact on our heart, for this reason one must make a plan and follow it.

35.4.5 Take-away

I have already started on my essays on self-authorship, these passages even more so highlight how important this is for my development with my relationship with Christ.

35.5 4/24/26

35.5.1 Exodus

”Am I God, that I can punish you?” Only God has the right and capacity to punish in a just manner.

One theory I had was that they misunderstood years and that is why they lived so ”long,” but according to the bible they legtitamtely lived to see multiple generations worth of people.

35.5.2 Matthew

The disiples were told that Jesus would raise himself from the grave and Jesus even talked of the cross, and yet they doubted when the time came. We can too miss what is directly in front of us.

Peter tried to control Jesus even knowing he is the son of man, just as we do without thought.

What does Jesus mean by Peter is the Rock, why does he have this authority. What does it mean for God to give the authority over heaven to a mortal man and a singular one at that?

35.5.3 Psalm

We do not have enemies for victories, we only have reflections of God’s. His victories is our honor, his enemies are ours. We are extensions of his will upon the Earth, empty without his will and might.

35.5.4 Proverbs

You will express the wisdom you have gained. While one can fake wisdom through prideful intellect, it becomes obvious over time as it falls as poison to the mind of those who hear.

35.6 4/25/26

35.6.1 Exodus

God has Moses take his shoes off to walk upon the holy ground. What does this mean?

God understands and condemns the slavery and abuse of his people. Takes upon an unworthy person and gifts him with the abstract skill of public speaking. Just as he does with all of us. Our natural talents are a product of him. It does make one question what is not a direct product and what is an indirect product of him and us.

I AM and Yahweh are God's names, his eternal names. Yahweh is sometimes directly translated to I AM, so what does this mean? The common interpretation is that he is self-referential and the beginning of all things, that he is unchanging. One thing I am thinking of is that I am is a relational function, so this blankness is complex with multiple ideas; he is everything, he is by definition relational, he is beyond categorification, he has no attributes, all attributes are defined by him, etc. The main one I am thinking of is that he is beyond relations, categories, attributes, and all things as he defines them, he even defines tautologies.

35.6.2 Matthew

It is said that John the Baptist came in the spirit and power of Elijah, what exactly does this mean; did he reincarnate, guided by Elijah, take on the power as a linage to Elijah?

"How long must I put up with you?" Jesus says this to the disciples as he criticizes their lack of faith. Our sin is insignificant compared to the fatigue of our disbelief. I, more so than most, switch this around, but the truth is in Jesus not morality dressed up.

35.6.3 Psalm

The world will tear you down completely, beyond what is thought possible, but God remains.

35.6.4 Proverbs

As people fall into lust it clouds their judgment more than most vices. Tearing themselves apart and bringing them to ruin as they stagger forword like an animal. This is what makes lust a sin, it is the very death of a man's soul, mind, and discipline if it ensnares itself. Making one a slave to others and their own flesh.

35.7 4/29/26

35.7.1 Exodus

I remember one theory that the reason he was not good with words was because he was raised speaking the Egyptian language rather than Hebrew.

When referring to the who God is by his creation of all things. He uses an infinite tense rather than on of the past. Does this suggest classical theism or occasionalism?

Why does he harden the Pharaohs heart. One interpretation is the above occasionalism as everything and every action is a result of his power. Another is simply that he permits it, and had foreknowledge of it. Another could be that hardening his heart could refer to an act of judgment. Another could be it is a narrative purpose to lead out and show his power to the Jews. Also, classical theism of God being the primary cause of all things. Lastly, a literary device to simply show God is in control.

"They are lazy," referring to slaves doing backbreaking work that kills them. We can see this in the same way throughout society and history.

35.7.2 Matthew

Jesus refers to turning from sin to be like children, but children tend to be the most hedonistic and foresake all others in their immediateness, but this is not on purpose. They are blameless not because they share no blame but because their heart and intentions have not been sorrowed by evil.

"Sorrow awaits the world." How can the world feel sorrow?

Do not let your body control you but cut it off. Do not let others tempt you but fear them.

Show forgiveness, but do not barter with those that abandon the Church.

Jesus refers to their Angles, making a case for guardian angles, at least for children.

35.7.3 Psalms

They already had inklings of God's gloabal salvation.

35.7.4 Proverbs

The Lord sees clearly everything we do, our intentions, and where it leads us. This path is the reason for all of the moral rules, we degrade ourselves which opens us up to more and corrupts our power.

He refers to your wife as yourself and another as a stranger, highlighting the extreme close bound between two spouses.

Having relations with others gives them a part of you, a power over you that you cannot completely take back. In this case, of that of a stranger, that part of you is destroyed in another.

35.8 5/24/26

35.8.1 Exodus

In this story God directly influences the people through physical force. What does this tell us about him when he doesn't use it in the same way now? Should one take this story literal or figertively. Showcasing who God is and his nature?

He also influences the Pharaoh's mind in order to show his capacity. Why? This is also related to communion. Maybe before Jesus's resurection, gentiles didn't have souls. This was a communion of his people and to bring them against an enemy to unite them. Afterwards, Jesus brings them into the community.

Later they use the term predicted rather than cause the

hardening of his heart. Maybe it means more his actions will bring about the desired goal, so hust as before the connection of a national identity that forms into the Church but without the idea that the gentiles don't have souls, that was rather silly.

35.8.2 Matthew

Forgive everyone from the heart as our grevinces towards the world are far more. Every tiny, of what we call imperfection, is a violent blast to the perfection of God. God desires ous to be social.

He also desires a holy marriage, but this life is not for everyone. Many are not meant for marriage. A great example of this would be those that are gay, but this then begs the question of whether it is the marrige, physical relations, or impulse that is wrong. Further examination is needed.

35.8.3 Psalm

...

35.8.4 Proverbs

Self-control is the heart of all things. Without it one becomes a slave of their impulses that destroy their minds and body.

35.9 5/26/26

35.9.1 Matthew

Become like children. Children are often wrong, selfish, and mildly cruel, but they do so without malice. They do actions

out of nature than concious diliberation. So does God mean be in that way, open like children, do little sin(for while they do it relitivley often their actions are small), or dependence on God and shape ourselfs in a new way unlike we have in the past. I think the last one.

35.10 6/8/26

35.10.1 Exodus

"So you can tell our children and their children"

Here he is creating Jewish identity through the pass-over. Not to waste, to share, cook fully, frugal and healthy choices, urgency in what you do, and be prepared/

35.10.2 Matthew

Serving others comes first before all. The further down the higher up.

35.10.3 Proverbs

Take initative and work for your life, rather than be absent of it all.

35.11 6/16/26

35.11.1 Exodus

All things belong to the Lord. In this he creates traditions for us to remind ourselves of this. Habits to live by and learn from.

It is filled with symbols, such as yeast symboizing moral decay.

35.11.2 Matthew

Temples should have dignity just as people do. Protected from things. The act of selling and buying is important but it is not of the Church.

Why did he make the fig tree wither? Metaphor for us doing our part in stewardship and work?

35.11.3 Psalms

Our integrity protects us as it is in the God almighty.

35.11.4 Proverbs

You will understand people as you see them. Wicked makes themselves clear without thought.

35.12 6/23/26

35.12.1 Exodus

This reminds me of an idea. That the reason God does not reveal himself is to protect us from ourselves. When we look at the Isrealites disobeing God when he is there with them in physical proof, we are outraged. It seems insane, but in many ways we do the same. For those that aren't Christian for reasons other than lack of proof, how would they live their lives with God manefest before them. Even for Christians, this would be a sudden and drastic difference.

35.12.2 Matthew

Jesus tells us for our actions and lives rather than what we boast outwardly.

Now, he will take new people rather than the Jews who had desicrated his kingdom. We should be careful in our dogma but remain in our loyalty of the Lord, this is the primary paradox of Christian teachings.

35.12.3 Psalms

Our integrity is what we have for the Lord. To protect it, one takes a further extreme than thought. To seperate oneself and mind from their enviorment and become more than a slave to their impulses and history.

35.12.4 Proverbs

Why did they write "six… no seven"? What could possibly been their purpose?

Anyways, the gifts that God has given us can just as easily be used for evil. This being the curse of our free will, our purpose and function exist but the capacity to function in its inverse is there just as well. As someone cannot be good without the capacity towards evil, nor can they truly do good.

It gives us seven things: haughty eyes, lying tongue, hands that kill inocence, heart that plots evil, feet that race to do wrong, a false witness who pours out lies, and a person who sows discord in a family.

On the other side: Eyes to observe and learn, a tongue to teach and express, hands to steward the world, a mind to meditate on what is right, feet to bridge the gap with others, witness to protect one another, and a person to build up and sharpen

a family.

35.13 Meeting

35.13.1 Order of events

- General opening
- Open the table
- Things I noticed and thought of
- questions

35.13.2 Things I have noticed

- Jesus parable of taking out the weeds will kill the crops
- Wisdom does not only come from God alone but community and family
- God has designed our nature in his image, a nature of mind and virtue, but such capacity can just as easily be in vice that defiles your nature in such the way that it is inline with it.
- Actions influence who you are and aren't just in isolation, guard your heart
- To be truly free is to be free of sin
- Be fully aware at all times as when we are happy is when our impulses sneak up on us.
- Our fate is in line with our nature, both as our actions create our lives and God creates our lives in accordence to who we are

- One must interprate the signs of our times
- Pay attention to the wisdom of others, don't just assume you know what they mean. Look into it, even if you know what they mean God may have more in it for you.
- We often try to control Jesus subcounciously, just as Peter does in Matthew trying to tell him what to do even after knowing that he is God manefest.
- We are an extiontion of God's will on Earth
- People and their nature are obvious
- God's name is I AM
- Our sin is insignificant compared to the fatigue of our disbelief, Jesus is not morality dressed up
- Why does God harden Pharoes heart
- Jesus and us acting like Children
- The passover and commnion
- the idea that God is not here to protect us from ourselves

35.13.3 Diccusion Questions

- What do you think are the signs of our time
- What does God refering to himself as I AM mean
- Why does Jesus ask us to be like children, and how does this relate to Genesis
- The relation of passover to communion
- What does it mean to be made in the image of God

- Does civilization progress towards the glory of heaven or do we delude ourselves
- What is our nature and how does that relate to the ideas presented
- Exodus and freedom, difference between freedom from and freedom for, why do the Isrelites want to go back to Eygpt
- Jesus's teachings and virtue ethics
- What does it mean for the fruit to be the knowledge of good and evil
- Do communities need rituals, and what rituals do we do or should do
- What do God's and Jesus's miricals and actions reveal about them
- what reponsibilites do we have towards society
- If one were to collapse the morality God puts out for us, what would it be and why?
- in just the same way that the stories of the bible reveal things about truth, God's nature, and morality; what does your life as a story tell?

35.14 6/24/26

35.14.1 Exodus

This story is a great symbol of why God does not make himself maniefest. People complain, challenge, question, and become

overtly used to him instead of traditional reverence. Not taking it upon themselves to become virtious.

35.14.2 Matthew

First, God calls out to all, not all come, but some who come do not there truly, and will be forced to leave.

Next, we will not be married in Heaven, what does this mean for our relationships?

35.14.3 Psalms

It is right to give our thanks and praise. Some question the fact that what he protects us from is a result of him, we should still praise for our existance and that which challenges us if for a greater purpose.

35.14.4 Proverbs

Take upon our parents wisdom and let it create who we are as a person. This is what will truly protect us, our integrity and character.

35.15 7/27/26

35.15.1 Exodus

God is coming in his form, further reading is needed for this.

35.15.2 Matthew

Call no one rabbi or teacher as their is only one teacher. I have trouble with this as I onften search for knowledge outside of religion.

35.15.2.1 Psalm

35.15.2.2 Proverbs

Lust is inherently dangerous as it makes you do objectively foolish things.

Chapter 36

Philosophy

36.1 Scope

Here I will write general notes from sources on the subject of philosophy.

36.2 Nicomachean Ethics

36.2.1 3/26/26

While I have constantly defended virtue ethics, and even Aristotle's ethics, I am a bit critical of his methods. Rather than defining a platonic ideal, he observes those around him and defines ethics based upon that. It works quite well and is something I also do but this as the primarly method tends to lead to some differences.

I am also surprised how similar his views of ethics are so similar to 21st century ethics in many ways; considering how different the views of people just a couple decades ago were.

I definitely find coherence with his idea of magnanimity with

my philotimic virtue. While I had originally conceived of the idea from listening to a summary of the idea, reading it directly from him shows that it was incredibly similar. Making me almost wonder if I should forget having my own word and just use that one. The only real difference is that I also blend philotimic with my specific views of morality and the idea of self-authorship; obviously an Aristotelian magnamonious man would value self-authorship, it isn't directly imbued with the word. Also, his spesific definition has a more prideful connotation.

Kinda funny how even in that time older folks found young comedians as vulger and inappropriate while the younger found the otherside as offensive and cruel.

Obviously his views of justice, while different are quite well.

Chapter 37

Random readings

37.1 Plan

For starting, let us first constrain ourselves by reading one paper a week. Making a list of fields would be

- Physics
- Math
- Philosophy
- Economics
- History
- Engineering
- AI
- Computer Science
- Psychology
- Anthropology

- Quantitative Finance

37.2 Doing it

37.2.1 Sept 23, 2026

Kind of boring over view look at quantitiative finance and AI. Think I will focus on physics and math

Part III

Projects

Chapter 38

Introduction

Chapter 39

Previous Research

These are brief/lazy copies/explanations of things I have written previously made projects. Copied from my last book Logos

39.1 3D Modeling of Non-Equilibrium Dynamics in Compressed Plasma Using Lattice Boltzmann Method

The construction of a theoretical model, magnetohydrodynamic lattice boltzmann method(MHD-LBM) model for 3D compressed plasma, using a finite volume scheme is constructed. The hyperbolic Maxwell equations, which satisfy the elliptic constraints of Maxwell's equations and the constraint of charge conservation, are used to simulate the electromagnetic field. The flow field and electromagnetic field are coupled to simulate a compressible plasma through the electromagnetic force and magnetic induction equations. This model can further be applied to create a quantitative simulation to model complex nonequilibrium effects of compressed plasma to provide mesoscopic physical insights into the flow mechanism of a shock

wave in a supersonic plasma.

Theoretical model, Finite volume scheme, hyperbolic maxwell equations, and nonequilibrium effects

39.1.1 3D Modeling of Non-Equilibrium Dynamics in Compressed Plasma Using Lattice Boltzmann Method Introduction

The study of nonequilibrium dynamics in compressible plasma is a critical area of research in plasma physics, with significant implications for both theoretical understanding and practical applications. Compressible plasmas are found in a variety of contexts, such as astrophysical phenomena(1), industrial processes(2,3), and fusion. Understanding the complex interactions and behaviors of plasma under nonequilibrium conditions is essential for advancing these fields.

Previous research has primarily focused on two-dimensional models(4) While these models have provided valuable insights, they often fall short in capturing the full complexity of three-dimensional plasma dynamics. The limitations of these models highlight the need for more advanced simulation techniques that can accurately represent the intricate behaviors of compressible plasma in three dimensions.

The Lattice Boltzmann Method (LBM) offers a powerful tool for modeling fluid dynamics at the microscopic level.(4,5) Unlike traditional computational fluid dynamics methods, such as the Navier-Stokes equation, the LBM is particularly well-suited for simulating nonequilibrium effects and complex boundary conditions. Its ability to handle multiphase flows and incorporate microscopic interactions makes it an ideal choice for studying compressible plasma dynamics.(5)

This paper aims to develop a comprehensive 3D model using the LBM to simulate nonequilibrium effects in compress-

ible plasma. By leveraging the strengths of the LBM, we seek to overcome the limitations of previous models and provide a more accurate and detailed representation of plasma behavior. Our research focuses on the coupling of flow and electromagnetic fields, exploring the interactions and dynamics that arise under nonequilibrium conditions.

39.1.2 Physical Model

Kinetic Equation

Li and Zhong(10) introduced the potential energy distribution function as well as a compressed DDF Lattice Boltzmann equation. A potential energy distribution function can be added so the Boltzmann BGK can obtain an adjustable specific heat ratio or Prandtl number(4)

This new Boltzmann Kinetic equation can be written as followed:

$$\frac{\partial f_k}{\partial t} + (\mathbf{e}_k \cdot \nabla) f_k + \mathbf{a} \cdot \nabla_e f_k = -\frac{1}{\tau_f}(f_k - f_k^{eq})$$

$$\frac{\partial h_k}{\partial t} + (\mathbf{e}_k \cdot \nabla) h_k + \mathbf{a} \cdot \nabla_e h_k = -\frac{1}{\tau_h}(h_k - h_k^{eq}) + \frac{z_k}{\tau_{hf}}(f_k - f_k^{eq})$$

Where k is the direction of discrete velocity. f_k is the density distribution function. h_k is the potential energy distribution function, while f_k^{eq} is the equilibrium distribution function. e_k is the discrete velocity component, and τ_f is the relaxation time of the density distribution function. τ_h is the relaxation time of the potential energy distribution function.

This can also be defined as:

$$\tau_{fh} = \frac{\tau_f}{\tau_h}$$

The force term can be approximated as:

$$\mathbf{a} \cdot \nabla_e f \approx \mathbf{a} \cdot \nabla_e f^{eq} = -\frac{a \cdot (\mathbf{e}_k - \mathbf{u})^2}{RT} f^{eq}$$

39.1.3 Density Distribution Function

$$f_k^{n+1}(x_i, y_j, z_l) = f_k^n(x_i, y_j, z_l) - \Delta t \left(\frac{F_{k,i+1/2,j,l} - F_{k,i-1/2,j,l}}{\Delta x} + \frac{F_{k,i,j+1/2,l} - F_{k,i,j-1/2}}{\Delta y} \right.$$
$$\left. -\Delta t \cdot \frac{1}{\tau}(f_k - f_k^{eq}) \right.$$

39.1.4 Potential Energy Distribution

$$h_k^{n+1}(x_i, y_j, z_l) = h_k^n(x_i, y_j, z_l) - \Delta t \left(\frac{G_{k,i+1/2,j,l} - G_{k,i-1/2,j,l}}{\Delta x} + \frac{G_{k,i,j+1/2,l} - G_{k,i,j-1/} }{\Delta y} \right.$$
$$\left. -\Delta t \cdot \frac{1}{\tau_h}(h_k - h_k^{eq}) + \frac{z_k}{\tau_{hf}}(f_k - f_k^{eq}) \right.$$

39.1.5 Hyperbolic Maxwell Equations

The previously used Maxwell equations have been enhanced with Lagrange multipliers, Ψ and Φ, to combine with the evolution equation. These are the new Maxwell equations in derivative form:

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} + \gamma \nabla \psi = 0$$
$$\frac{\partial \mathbf{E}}{\partial t} - c^2 \nabla \times \mathbf{B} + \chi c^2 \nabla \phi = -\frac{\mathbf{J}}{\epsilon_0}$$

$$\frac{\partial \Psi}{\partial t} + \gamma c^2 \nabla \cdot \mathbf{B} = 0$$
$$\frac{\partial \Phi}{\partial t} + \chi \nabla \cdot \mathbf{E} = 0$$

Where E is the electric field, B is the magnetic field, Ψ and Φ are introduced divergence variables, γ and χ are divergence error propagation speeds, c is the speed of light, and ϵ_0 is the vacuum permittivity.

39.1.6 Kinetic Non-Equilibrium Method

The nonequilibrium effects on compressed plasma are observed through differences in molecular speed. Each non-equilibrium kinetic moment can be represented as the difference between the corresponding kinetic moment and the local equilibrium kinetic moment. The equations of kinetic moments may be used to find the nonequilibrium quantities.

$$M_{f,m}^{neq} = M_{f,m} - M_{f,m}^{eq}$$
$$M_{h,m}^{neq} = M_{h,m} - M_{h,m}^{eq}$$

39.1.7 Discussion

In this paper, a 3D MHD-LBM analytical model was constructed for compressed kinetic plasma. Future research must be done in developing a computer model and validating said computer model. A third-order finite volume MUSCL could be applied to this analytical model combined with a D3Qx density equilibrium distribution function to create an easily solvable numerical solution. While electromagnetic fluxes were evaluated through Steger-Warming flux vector splitting. MHD-LBM models have shown to have greater accuracy when it comes to plasma shockwaves, therefore MHD-LBM models have good

reason for continual development.

39.2 A Hamiltonian Framework on ICF Implosions Rocket Equation Based on Rayleigh–Taylor Instabilities

39.2.1 Background

Inertial Confinement Fusion (ICF) Implosions: One of the primary methods of heating fusion environments is through lasers. In this approach, high-energy lasers heat spherical fuel capsules, causing them to implode. This implosion leads to an increase in pressure and heat on the fuel, creating the necessary environments for fusion.

Hamiltonian: A Hamiltonian is a function in classical mechanics that describes the total amount of energy in the system. From here, equations of motion are derived.

Rayleigh–Taylor Instabilities: Rayleigh–Taylor instabilities are specific types of instabilities caused by forceful interactions between higher and lower density fluids. In the context of ICF Implosions, when the higher density outer shell interacts with the lower-density inside, it creates perturbations through finger-like structures and bubbles that interfere with the efficiency of the implosion.

Mass Ablation (Rocket Effect): During the process of ICF Implosion, the interaction between the outer shell and the inner fuel causes mass to be sprayed off, giving a rocket-like effect.

39.2.2 Problem Statement

Achieving commercial fusion energy through inertial-confinement-fusion (ICF) remains one of the most significant scientific and engineering challenges of our day. One of the key obstacles is optimizing the interactions between high-energy lasers and fuel capsules, which are prone to Rayleigh-Taylor instabilities during implosions. These instabilities can lead to inefficient energy confinement and hinder the overall success of the fusion process. Traditional numerical simulations are computationally expensive and often lack the accuracy needed to address this issue. Therefore, there is a need for a robust analytical framework to model these instabilities and provide new insights for optimizing the interactions. This research aims to develop a Hamiltonian framework for ICF implosions, specifically focusing on Rayleigh–Taylor instability and mass ablation. By leveraging this theoretical approach, we seek to provide deeper insights and practical solutions for optimizing laser-capsule interactions in fusion experiments.

39.2.3 Assumptions

Thin-Shell Approximation: The thickness of the shell is assumed to be insignificant compared to the size of the imploding object. This approximation simplifies the creation of the model and the subsequent numerical analysis. It is valid given that the perturbations caused by the instabilities are much larger than the actual thickness of the shell.

Deceleration: This model is specifically for an imploding spherical shell that decelerates as it converges onto the compressed fluid within its interior.

Acceleration: This model does not consider the acceleration phase of an ICF capsule implosion at the beginning of the capsule-laser interaction.

39.2.4 Framework

In order to analyze the dynamics of the imploding shell, it must first be parameterized through two Lagrangian coordinates ϑ and ϕ. Here, ϑ corresponds to the polar angle θ, and ϕ corresponds to the azimuthal angle. The position vector is $\mathbf{X} = \mathbf{X}(t, \phi, \vartheta)$. Thus, the parameterization can be expressed as:

$$\mathbf{X} = \mathbf{X}(t, \phi, \vartheta)$$

The derivative of the surface is given by:

$$\frac{d\mathbf{X}}{dt}$$

Reference: D. E. Ruiz; Degradation of performance in ICF implosions due to Rayleigh–Taylor instabilities: A Hamiltonian perspective. Phys. Plasmas 1 December 2024; 31 (12): 122701.

39.2.5 Shell Kinematics

The force differential can be described by:

$$dF = p(t) \cdot \mathbf{dA} \times \mathbf{n}$$

where $p(t)$ is the function of pressure with respect to time, and the cross product involves the vectors following the surface of the shell. This leads to the derivation of several important quantities, such as the velocity vector field, position vector, and centrifugal forces, which are relevant to the shape and dynamics of the shell.

An example of the centrifugal force is:

$$F_{\text{centrifugal}} = m \cdot \omega^2 \cdot r$$

39.2.6 Shell Areal Density

The change in density of the shell is critical to its dynamics. Given that total mass is not conserved in this equation, deriving it becomes more complex.

$$\rho_{\text{areal}} = \frac{M(t)}{A}$$

where $M(t)$ is the mass at time t, and A is the area of the shell's surface.

39.2.7 Compressed-Fuel Pressure

The pressure that decelerates the shell can be described in several ways depending on the known information and boundary conditions. One example is:

$$P = P_0 \left(\frac{V_0}{V} \right)^{\gamma}$$

where P_0 is the initial pressure, V_0 is the initial volume, γ is the adiabatic index, and V is the current volume.

39.2.8 Variational Principles

To mathematically solve the Hamiltonian framework, given that the equation is asymptotic, variation principles are applied.

39.2.8.1 Phase-Space Lagrangian

This formulation describes the dynamics of a system by combining its configuration space (position variables) and momentum space into a single framework, called phase space.

Euler–Lagrange Equations

Through the principle of least action, a set of differential equations are derived to provide an analytical solution to complex nonlinear functions. For example, for radial velocity:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0$$

where L is the Lagrangian of the system.

39.2.8.2 Conservation Laws

Generally, conservation laws are employed to show that certain parts of the Euler-Lagrange or phase-space Lagrangian are conserved. However, due to mass ablation, parts of the fuel and capsule are lost, which complicates these conservation laws.

39.2.9 Conclusion

Summary: A Hamiltonian framework was developed to include both Rayleigh–Taylor Instabilities and mass ablation in an inertial confinement fusion implosion, leading to a set of variational principles for future work on optimizing ICF implosions in practical applications.

Significance: Analytical analysis can provide future insights into optimizing ICF implosions in fusion reactors. Additionally, nonlinear numerical models can be developed based on these principles, leading to less computationally expensive models.

Limitations:

1. This model does not account for more complex laser-plasma interaction phenomena (e.g., bremsstrahlung x-ray losses, alpha heating).

2. This model does not consider the initial acceleration phase of the interaction.
3. This model uses the thin-shell approximation, which may not be accurate when perturbations caused by instabilities are smaller than the shell thickness.

Future Works:

1. Quasilinear Models To study additional insights from this model, a quasilinear model must be developed for analytical study.
2. Acceleration Phase Incorporating the initial acceleration phase into the model will increase its accuracy.
3. Nonlinear Growth Calculations Developing nonlinear numerical simulations will be useful in generating further insights and advancing current models.

39.3 Basic MHD

39.3.1 MHD Module

The first section of it is initializing the values of things like magnetic field, pressure, and velocity. Future work will be done to have it so such values can be edited in a config file.

Next is computing the current density. The analytical equation can be written as

$$J = \frac{1}{\mu}\nabla \times B$$

Which in finite difference is:

$$J_z(i,j) = \frac{1}{\mu_0}[\frac{B_y(i+1,j) - B_y(i-1,j)}{2\Delta x} - \frac{B_x(i+1,j) - B_x(i-1,j)}{2\Delta y}$$

Next we have

Update: I have advanced the MHD simulations time steps, magnetic field calculations, current, pressure calculations, and density calculations. Beyond that I have created animations based on magnetic field and made graphical outputs for heat, velocity, temperature, and magnetic field.

I also created another one specifically made for Tokamak geometry.

39.3.2 Hamiltonain MHD

My next project was creating a Hamiltonian based MHD simulation also in FORTRAN.

I explicitly created simulation of plasma thrusters with it.

39.4 A Metriplectic Formulation of Reduced Magnetohydrodynamics with Resistivity

39.4.1 Introduction

Here I will use the Strauss formulation of RMHD. Then I will incorporate resistivity. This resistivity typically breaks Hamiltonian structure, but I will use metriplectic to preserve the geometric identities through adding a separate antisymmetric bracket.

39.4.2 Defining the Hamiltonian

For Strauss MHD, a reduced MHD model that works for strongly magnetized Plasmas, we will formulate the Hamiltonian system.

$$\mathcal{H}[\omega, \psi] = \frac{1}{2}\int (\phi\omega + \psi j) dxdy$$

Here the $\phi\omega$ represents the kinetic energy with ψj represents the magnetic energy.

Now onto the Noncannonical Poisson Structure for this ideal Hamiltonian.

$$\{F, G\} = \int \omega[\frac{\delta F}{\delta \omega}, \frac{\delta G}{\delta \omega}]dxdy + \int \psi([\frac{\delta F}{\delta \omega}, \frac{\delta G}{\delta \psi}] - [\frac{\delta G}{\delta \omega}, \frac{\delta F}{\delta \omega}])dxdy$$

39.4.3 Entropy and Dissipation

Now to add dissipation. While generally adding this will break Hamiltonian structure, we will use a couple of tools.

Though first we must define our function we are using, or more accurately funcional. This is our functional for entropy:

$$S[\psi] = \frac{1}{2}\int \psi^2 dxdy$$

39.4.4 Metric Bracket and Metriplectic Structure

To incorporate we define our symmetric bracket. We make sure to construct it so that it only affects the magnetic flux variable ψ, since resistivity only affects the magnetic field lines and not

vorticity directly.

$$(F, G) = \int \frac{\delta F}{\delta \psi} \eta \nabla^2 \frac{\delta G}{\delta \psi} dx dy$$

This has

- Symmetry $((F, G)) = ((G, F))$
- Positive semi-definiteness: $((S, S)) \leq 0$
- Energy Conservation: $((\mathcal{H}, F)) = 0$ for any F

39.4.5 Combined Dynamics

39.5 Death Star

Not to long ago(from when I wrote this section) I tried to simulate the death star using FLASH-X(which is really a testament to who I am as a person, getting my hands on government codes and the fist thing I do is play around with pop-culture based planetary destruction) anyways. It failed. The laser dynamics were based upon old code that was outdated and I couldn't figure it out. Though I did learn a lot about Flash-X as a system, which is good.

So for it... what am I doing, this doesn't make sense if I am trying to teach myself through practical action of writing but I already did the most practical action of all, doing it. This is kinda useless.

I later went back and ended up actually making the simulation.

39.6 Interstellar travel

39.6.1 Speeds and forces

Before we can get to all of the more complex stuff about design and the effects of extreme environments of specific interstellar space travel we must first look at how forces effect speeds at at relativistic speeds. Here is the math stuff in simplified form: We all know that force is the derivative of the momentum over time:

$$F = \frac{dp}{dt}$$

Where momentum can be expand for relativistic momentum:

$$p = \gamma m v$$

You can then expand force to be(this is a constant mass, we will look at that in a second)

$$F = m\frac{d(\gamma v)}{dt}$$

One thing to keep in mind for future maths is that in simplifying $\dot{\gamma}$ is equal to $\frac{\gamma v}{c^2}\frac{dv}{dt}$ With the force you can calculate acceleration with

$$a = \frac{F}{m\gamma}\frac{1}{1+\frac{\gamma^2 v^2}{c^2}}$$

Due to the fact most rockets use propulsion which lowers the mass, we must include mass variations:

$$F = \frac{\gamma m v}{dt}$$

Where you must use the product rule to expand

$$F = \dot{\gamma} m v + \gamma \dot{m} v + \gamma m \dot{v}$$

Which can be further expanded to

$$F = mv\frac{\gamma v}{c^2}\frac{dv}{dt} + \gamma v\frac{dm}{dt} + \gamma m\frac{dv}{dt}$$

With $\dot{m}$ being the infinitesimal change in mass over time.
To calculate acceleration you must find $\frac{dv}{dt}$. First subtract the middle equation without acceleration within it.

$$F - \gamma v\frac{dm}{dt} = mv\frac{\gamma v}{c^2}\frac{dv}{dt} + \gamma m\frac{dv}{dt}$$

Then factor for acceleration

$$F - \gamma v\frac{dm}{dt} = \frac{dv}{dt}(mv\frac{\gamma v}{c^2} + \gamma m)$$

Then divide the parentheses:

$$(F - \gamma v\frac{dm}{dt})\frac{1}{mv\frac{\gamma v}{c^2} + \gamma m} = a$$

Now, this is all well and good, now let us find the position. To take position from acceleration you must take a double integral, but this integral can only be solved numerically(maybe it could be approximated)

Another possible way to solve this is through Hamiltonians/Lagrangians (I will choose Hamiltonians)
Let us first start with the Hamiltonian expressed for relativistic speeds in covariant form
First the Lagrangian must be calculated: The four-momentum is

$$P^\mu = m\gamma(c, v)$$

Energy:

$$E = \gamma mc^2$$

Therefore the Lagrandian is:

$$L = -mc^2\sqrt{1 - \frac{v^2}{c^2}}$$

Now to find generalized momentum:

$$p = \frac{\partial L}{\partial \boldsymbol{v}}$$

$$p = \gamma m v$$

Now the Hamiltonian

$$H = p \cdot v - L$$

Therefore a relativistic Hamiltonain can be represented as

$$H = \sqrt{(pc)^2 + (mc^2)^2}$$

Now to include acceleration

$$H(r, p) = \sqrt{(pc)^2 + (mc^2)^2} + V(r)$$

Now to make mass variable for the rockets:

$$H(r, p, t) = \sqrt{(p(t)c)^2 + (m(t)c^2)^2} + H_{thrust}$$

Now to incorporate with equations of motion:

$$\frac{dr}{dt} = \frac{pc^2}{\sqrt{(p(t)c)^2 + (m(t)c^2)^2}}$$

Now, this works for the time for most people, now let us look at proper time

$$H(r, p, \tau) = \sqrt{(p(\tau)c)^2 + (m(\tau)c^2)^2} + H_{thrust}$$

For specific cases, these equations will be evolved to include them, but this is the general theoretical framework.

One thing I will add, is the force will likely be variable, a couple of days of 2-3 g, most of the time in g then reversing after the half-way point.

39.6.2 Magnetic Fusion Plasma Drive

39.6.2.1 Theoretical Basis

Magnetic Fusion Plasma Drive is a type of propulsion system specifically for interstellar travel.

The basis of this concept is that there is a fusion reactor undergoing Thermal Nuclear fusion:

$$D + T \rightarrow^4 He + n + 17.6MeV$$

The spent fuel then being shot out the nozzle as propulsion.

39.6.3 *Optimization

39.6.4 *Special Circumstances

39.6.4.1 *Strong Magnetic Fields

39.6.4.2 *Gravitational Fields

39.7 Kerr Black hole stuff

39.7.1 Energy (finish)

Kerr Black holes, or spinning black holes are a great source of energy. They have extreme amounts of radial kinetic energy that we are actually capable of 'bleeding' off through several methods.

First, we must ask why we are able to do this. Well, because a rotating object with a strong gravitational field 'drags' spacetime, that spacetime can affect the direction, speed, and even waves in a way that can amplify them.

First, Penrose(really like that dude) found that if you break apart an object rotating a Kerr black hole, the object that escapes gains energy. This process is challenging given that the object that escapes must have a velocity of $v > \frac{c}{2}$. There is a whole lot of math involved in proving this, but I will skip ahead.

The next that is found is that electromagnetic waves can be amplified. That if they are reflected around the black hole, they will gain energy and be amplified(this process is different from the Doppler shift and does not change the frequency.

After a lot, and I mean a lot of very complected mathematics you can find the amplification factor as

$$Z = 8r_+^2 T_{H(^{2l+1)})}(r_+ - r_-)^{2l}[\frac{\Gamma(1+l-s)\Gamma(1+l+s)}{2l+1)!!\Gamma(l+1)\Gamma(2l+1)}]^2\times,$$

$$\sinh\left(\frac{m\Omega_H}{r_+T_H}\right)\Gamma(l-\frac{im\Omega_H}{\pi r_+T_H}+1)\Gamma(l+\frac{im\Omega_H}{\pi r_+T_H}+1)$$

For more information visit here

Now the point of this is to take strange theoretical and apply it, so I will apply it for actual application. For this I will assume the average mass of 10 $M_\odot$ or ten solar masses. A spin of 0.98(if 1 refers to an extremal Kerr black hole)

First we must find out what a spin 0.98 really means for a $10M_\odot$ black hole. We can find this with a simple equation with a being the spin parameter, r being the radius, and M being the mass

$$\Omega = \frac{ac^3}{2GMr_h}$$

Though, for a kerr black hole the is slightly different to the swarzchild radius.

$$r_H = \frac{GM}{c^2}(1+\sqrt{1-\frac{ac}{GM}}$$

You know, I wanted to optimize the black hole engine, but I don't want to do that anymore, so I choice to move on.

39.7.2 *Multiversal Travel

39.7.3 *Other effects

39.8 *Extreme Theoretical Work

39.9 Astroid Game

39.9.1 Introduction

I have recently created a Python game based on the Atari game Asteroid.

39.9.2 Classical

Just like the regular game, nothing special.

39.9.3 Newtonian Gravity

This adds that the asteroids and UFO's mass, so they attract each other and add more challenges.

39.9.4 Dark Matter

Add the challenge of invisible mass attractors, though when you are close by they light up.

39.9.5 Relativistic

This adds time dilatation(time skips or slow downs), length contractions(changing shapes of objects), black holes, and even doopler effect.

Chapter 40

Omega-X

40.1 Pre-planing(part 1)

40.1.1 Ideas & Overarching Plan

Start with replicating 1D thermal fluid model, then start with more complex models, then experiment with other ways to do it, then dive deeper into the mathematics and hopefully come up with something(maybe metriplectic integrator), then create a code base, compare these systems to other traditional models, and create tokomak simulation(maybe, not sure if possible yet) that is it so far. Hopefully these ideas will evolve.

Basically, the whole idea is to create a Flash-X like project with Metriplectic formulism underlying the the dynamics rather than lagrangian or Newtonian. Obviously much simpler though.

40.1.2 Metriplectic 4-bracket algorithm for constructing thermodynamically consistent dynamical systems

40.1.2.1 Math

1: Identify dynamical variables 2: Propose energy and entropy functionals 3: Find Poisson bracket F, G for which entropy S is a Casimir invariant, $F, S = 0 \forall F$ 4: Construct metriplectic 4-bracket (F, K; G, N) via Kulkarni-Nomizu product by a new method that separates local thermodynamics from phenomenological quantities, giving the EoMs as Poisson bracket + 4-bracket

To find the Σ and M a new method is derived.

$$\partial_t \xi^\alpha = \{\xi^\alpha, H\} + \nabla \cdot J^\alpha$$

$\alpha = 1, 2...N-1$

$$\partial_t \xi^N = \{\xi^N, H\} + \nabla \cdot J^N + Z_\alpha \cdot \tilde{L}^{\alpha\beta} \cdot Z_\beta$$

where $\xi_N = \sigma$, the entropy density. Above splits Hamiltonian and conservative. Which leads to:

$$M(dF, dG) = F_{\xi N} G_{\xi N}$$

$$\Sigma(dF, dG) = \nabla F_{\xi\alpha} \frac{L^{\alpha\beta}}{H_{\xi N}} \nabla (G_{\xi\beta})$$

40.1.3 A thermodynamically consistent discretizations of 1D thermal-fluid models using their metriplectic 4-bracket structure

40.1.3.1 Math

Edit: so I don't keep forgetting, ρ is mass density, m is momentum density, and σ is entropy density. Then for the thermodynamic quantities: η is specific entropy, u is velocity, and T is temperature Let $V_h = v_h \subset H^1(\Omega)$ be the degree-p continuous Galerkin finite element space defined over a uniform grid, τ_h, on Ω: i.e.

$$V_h = v_h \in H^1(\Omega) : v_h|K \in P_p(K), \forall K \in T_h$$

The discretizations is accomplished using the method of lines by positing that all dynamical fields have spatial dependence modeled in this Galerkin subspace. However, rather than discretizing the equations of motion themselves, we discretize the weak forms implied by the metriplectic formulation.
Let $(\rho_h.m_h.\sigma_h) \in V_h \times V_h \times V_h$ So that the discretized Hamiltonian and entropy can be given as:

$$H^h[\rho_h.m_h.\sigma_h] = \int_\Omega [\frac{1m_h^2}{2\rho_h} + \rho_h U(\rho_h, \frac{\sigma_h}{\rho_h}]dx$$

$$S^h[\sigma_h] = \int_\Omega \sigma_h dx$$

Then the Metriplectic 4 Bracket can be expressed through:

$$(F^h, K^h, G^h, N^h)_h = -\frac{1}{\mathrm{Re}} \int_\Omega \frac{1}{T_h} \left[\left(K^h \partial_x F^h_{m_h} - F^h \partial_x K^h_{m_h} \right) \right],$$

$$\left(N^h \partial_x G^h_{m_h} - G^h \partial_x N^h_{m_h} \right) +,$$

$$\frac{1}{\mathrm{Pr}\gamma - 1}\frac{1}{T_h}\left(K^h\partial_x F^h - F^h\partial_x K^h\right)\left[\left(N^h\partial_x G^h - G^h\partial_x N^h\right)\right]\,dx$$

Then the poission bracket is thus:

$$\{F^h, H^h\}_h(u_h) = -(m_h\partial_x u_h, \phi_m)_{L^2} + (m_h u_h, \partial_x, \phi_m)_{L^2} - (\rho_h\partial_x\eta_h, \phi_m)_{L^2} + (\rho_h u_h, \partial_x, \phi_\rho)_{L^2} - (\sigma_h\partial_x T_h, \phi_m)_{L^2} + (\sigma_h u_h, \partial_x, \phi_\sigma)_{L^2}$$

Now, one thing I will say is that this is not technically a Poisson bracket, because it is fails to satisfy the Jacobi identity. This is because it is impossible(or at least nobody has figured out a way to discretize fluid poission brackets.

To then find equations of motion, adding them together $F^h = \{F^h, H^h\}_h + (F^h, H^h; S^h, H^h)_h$ where F^h is the observable.

In the case of momentum we must let $\phi_\rho = \phi_\sigma = 0$ so that only ϕ_ρ the momentum test function in the finite element(FE) test space V_h.

$$(\phi_m, \partial_t m_h) + (m_h\partial_x u_h, \phi_m)_{L^2} - (m_h u_h, \partial_x, \phi_m)_{L^2} + (\rho_h\partial_x\eta_h, \phi_m)_{L^2} + (\sigma_h\partial_x T_h\phi_m)_{L^2}$$

Next, I will have $\phi_m = \phi_\sigma = 0$ to get continuity equations:

$$(\phi_\rho, \partial_g\rho_h)_{L^2} - (\rho_h u_h, \partial_x\phi_\rho)_{L^2}$$

Finally the entropy equation is derived in a similar fashion:

$$(\phi_\sigma, \partial_g\sigma_h)_{L^2} - (\sigma_h u_h, \partial_x\phi_\sigma)_{L^2} + \frac{1}{Re}[(\frac{(\partial_x u_h)^2}{T_h}, \phi_\sigma)_{L^2} - \frac{1}{Pr}\frac{\gamma}{\gamma - 1}[(\frac{\partial_x T_h}{T_h}, \partial_x\phi_\sigma)_{L^2} - ($$

To prove energy conservation

$$(\frac{u_h^{n+1}-u_h^n}{\Delta t},\delta h_h)=\frac{H(u_h^{n+1})-H(U_h^n)}{\Delta t}=0$$

Positive entropy production can be found through:

$$\frac{s_h^{n+1}-s_h^n}{\Delta t}=\frac{1}{Re}[(\frac{(\partial_x u_h^n)^2}{T_h^n},1)_{L^2}+\frac{1}{Pr}\frac{\gamma}{\gamma-1}\frac{(\partial_x T_h^n)^2}{T_h^n},1)_{L^2}]\geq 0$$

Let's figure out all we need to find for each equation, starting with momentum. Start with $M_{ii}=\phi_m$, this should be equal to dx, but I will confirm. Also need the time derivative of the momentum variational, u_h variational, ρ_h variational, T_h, η_h, and $\frac{1}{Re}$, reynolds number, decide later

$$M_{ii}=dx$$

$$u_h=(\rho_h,m_h,\sigma_h)$$

ρ_h, T_h, and η_h are the respective fields, initialized simply and evolved through the equation. Obviously $\rho_h, m_h,$ and σ_h evolve through their metriplectic equations of "motion." T_h, η_h, are evolved through these equations:

$$(\eta_h+\frac{m_h^2}{2\rho_h^2}-U(\rho_h,\frac{\sigma_h}{\rho_h})-\rho_h\partial_1 U(rho_h,\frac{\sigma_h}{\rho_h})+\frac{\rho_h}{\sigma_h}\partial_2 U(\rho_h,\frac{\sigma_h}{\rho_h},\phi_\eta)_{L^2}=0$$

$$(u_h-\frac{m_h}{\rho_h},\phi_u)=0$$

$$(T_h-\partial_2 U(\rho_h,\frac{\sigma_h}{\rho_h})\phi_T)_{L^2}=0$$

Nothing new is added in continuity
Entropy adds: Pr and γ

40.1.3.2 Coding

The first thing is to use a Galerkin Method for projecting PDEs into a finite-dimensional function space. (can base off of Firedrake, FEniCS, Galerkin, or another) M_{ii} in mesh.f90

The second thing to do is to initialize the parameters and states(use mesh to calculate field) in states.f90 (Pr, Re, γ for constants; initialize ($\rho_h, \sigma_h, m_h, \eta_h, T_h, u_h$)

Then functionals: calculates pointwise functional derivatives of the Total Hamiltonian and Entropy. in functionals.f90 (Update $\rho_h, \sigma_h, m_h, u_h$)

EOS.f90 is used to update the thermodynamic points(η_h, T_h)

Time step the program, by using Gauss-Legendre implicit Runge-Kutta methods in time_integration.f90

Run the Program(using the driver and makefile)

Input the outputs in io.f90

Some notes: $(f, g)_{L^2} = \int_\Omega f(x) \cdot g(x)$

An example would be, for a point in the momentum equation $(\phi_m, \partial_t m_h)_{L^2}$, or the mass matrix times the derivative of momentum functional as a function of time. Thus it makes it so $\phi_m = M_{ii} = dx$ thus, $(\phi_m, \partial_t m_h)_{L^2} = dx \cdot \frac{m_h}{\partial t}$

40.1.3.3 Future Work

The paper presents several ways to improve, or at least look into improving through future work.

Mainly, they mention that many different structure preserving methods exist and may be more suitable.

I also need to figure out how to make this codebase work for more examples

40.1.4 Metriplectic Framework for Dissipative Magneto-Hydrodynamics

40.1.4.1 My Introduction

In my time trying to figure out how to numerically solve metriplectic equations, someone beat me to dissipative magneto-hydrodynamics. Oh well, I can build off of.

Beyond that, the abstract introduces the paper better than I could, " The metriplectic framework, which permits to formulate an algebraic structure for dissipative systems, is applied to visco-resistive Magneto-Hydrodynamics (MHD), adapting what had already been done for non-ideal Hydrodynamics (HD). The result is obtained by extending the HD sym metric bracket and free energy to include magnetic field dynamics and resistive dissipation. The correct equations of motion are obtained once one of the Casimirs of the Poisson bracket for ideal MHD is identified with the total thermodynamical entropy of the plasma. The metriplectic frame work of MHD is shown to be invariant under the Galileo Group. The metriplectic structure also permits to obtain the asymptotic equilibria toward which the dynamics of the system evolves. This scheme is finally adapted to the two-dimensional incompressible resistive MHD, that is of major use in many applications."

40.1.4.2 Notes on introduction

They have an interesting word choice, "with its noble descendants of path integral representations," or Algebrization of dynamical systems appears to be the final destination of that virtuous route.

MSTDOF means "microscopic, statistically treated, degrees of freedom"

In this introduction they derive the metriplectic existence

quite simply. Starting with the poission bracket

$$\dot{f} = \{f, g\}$$

In this case

$$0 = \{S, H\}$$

Thus it is the Casimir functionals of the Poisson bracket

$$\{C, f\} = 0 \forall f$$

From here we can define the Hamiltonian free energy.

$$F = H + \lambda C$$

Where λ is an arbitrary constant left. In MSTDOF, it is the minus of the tempeture.

To then expand the system, $\langle\langle f, g\rangle\rangle$ via F. So that $\langle\langle f, g\rangle\rangle = \{f, h\} + (f, g)$ and (f, g) is symmetric, bilinear and semi-definite.

Evolution is then given by

$$\dot{f} = \langle\langle f, F\rangle\rangle$$

(the symmetric bracket (f, g) will be defined so to cancel out the presence of the coefficient λ, removing it from the equations of motion).

(f, g) is defined so that H is conserved.

$$(H, f) = 0 \forall f$$

Thus metriplectic evolution reads:

$$\dot{f} = \{f, H\} + \lambda(f, C)$$

The Casimir to mimic entropy undergoes an evolution described

by:

$$\dot{C} = \lambda(C, C)$$

Now, I like this introduction. Clearly showing the basic understanding on Metriplectic dynamics.

40.1.5 METRIPLECTIC FORMULATION OF VISCO-RESISTIVE MHD

The system observed is fully ionized plasma. Where dissipation takes place due to the finite viscosity and resistivity of the fluid. Also, heat conductivity is finite. Here is the SO(3)-covariant form.(expressing physical equations in a way that is invariant under rotations in three-dimensional space)

$$\begin{aligned}
\partial_t \rho + \partial_i(\rho u^i) &= 0 \\
\partial_t B^i + \varepsilon^{ijk}\partial_j E_k &= 0, \quad \partial_i B^i = 0 \\
\partial_t(\rho u^i) + \partial_j(\rho u^i u^j + p\delta^{ij}) &= \varepsilon^{ijk} J_j B_k + \partial_j(\eta\, \partial_j u^i) \\
\partial_t s + \partial_i(s u^i) &= \frac{\eta}{T}\partial_i u^j \partial_i u^j + \frac{\kappa}{T^2}\partial_i T \partial_i T + \frac{\mu}{T} J^i J^i
\end{aligned}$$

There is quite a bit to this equation, but I will move on.

One thing I will add is a new notation. $\dot{H} \stackrel{\partial}{=} 0$, it simply means that means that $\dot{H}$ and 0 only differ by a boundary term.

They then describe the poission bracket for ideal MHD(when dissipation does not happen, thus traditional poission bracket).

This equation has multiple Casimir observables. The paper mentions:

$$C[\rho, s] = \int_{\mathbb{D}} \rho\varphi(s) d^3x$$

Which within is both the total mass and entropy.

$$M[\rho] = \int_{\mathbb{D}} \rho d^3x$$

$$S[\rho, s] = \int_{\mathbb{D}} \rho s d^3x$$

Thus, mass and entropy are conserved. There are some other, but obviously for this purpose, use the entropy.

Taking the dissipative equations from the to combine $D = (\vec{D}^{\rho}, \vec{D}^{v}, \vec{D}^{B}, \vec{D}^{s})$ something called an 8-uple. So we must want to define our metriplectic bracket as $\psi = (\psi, D)$ where ψ is the dynamic variables.

Uhm.. I just realized this wasn't the paper I thought it was. Good read, but I don't think I will continue these notes.

40.1.6 A least Action principle for visco-resistive Hall Magnetohydrodynamic with metriplectic reformulation

40.1.6.1 Abstract

The abstract persented says,

"We present a new variational formulation for Viscous and resistive Hall Magnetohydrodynamic. We first find a variational principle for ideal HMHD by applying the physical assumptions leading to HMHD at the lagrangian level, and then we add the viscous and resistive terms by the means of constrained variations. We also provide a metriplectic reformulation of our formulation, based on two canonical Lie-Poisson brackets for the ideal part and metric 4-brackets for the dissipative part."

They then go into all of these, I will begin with the Metriplectic for notes alone.

40.1.6.2 Metriplectic bracket, Hamiltonian and Entropy for Viscoresistive Hall MHD

They use the Inclusive curvaturelike tensor found in Morrison's recent work. Where the dissipation is included using a 4-bracket that have the same symmetries as a curvature tensor.

They take the assumption that the viscous tensor can be found to be

$$\sigma = \Lambda \nabla u_i$$

Where Λ is symmetrize and positive.

40.1.7 Variational discretizations of viscous and resistive magnetohydrodynamics using structure-preserving finite elements

40.1.7.1 Abstract

We propose a novel structure preserving discretization for viscous and resistive magnetohy drodynamics. We follow the recent line of work on discrete least action principle for fluid and plasma equation, incorporating the recent advances to model dissipative phenomena through a generalized Lagrange-d'Alembert constrained variational principle. We prove that our semi discrete scheme is equivalent to a metriplectic system and use this property to propose a Poisson spliting time integration. The resulting approximation preserves mass, energy and the diver gence constraint of the magnetic field. We then show some numerical results obtained with our approach. We first test our scheme on simple academic test to compare the results with estab lished methodologies, and then focus specifically on the simulation of plasma instabilities, with some tests on non Cartesian geometries to validate our discretization in

the scope of tokamak instabilities.

This paper

40.2 part 2

40.2.1 Coding(1D fluid)

40.2.1.0.1 Introduction My first step to creating my codebase is to create it for the simplest form. This is specifically made for the purpose of figuring things out, documenting it, and such things. So, I will be writing down each file.

40.2.1.0.2 mesh_1d Obviously, this mesh program creates the mesh for the program. The size N, the nodes x_nodes and the mass matrix M_{ii}. The rest are self explanatory but I will go into what the mass matrix is. It represents the discrete inner product of basis functions.

$$M_{ii} = (\phi_i, \phi_j)_{L^2} = \int_\Omega \phi_i(x)\phi_j(x)dx$$

So, M_{ii} becomes a metric tensor for program. It turns continuous inner products into discrete sums over coefficients.

It can be approximated through the "lumped mass matrix" where $M_{ii} = dx$ Though, I use the trapezoidal mass matrix:

$$M_{ii}(1) = 0.5d0 * dx$$

$$M_{ii}(N) = 0.5d0 * dx$$

$$M_{ii}(2 : N - 1) = dx$$

For more information visit, `https://arxiv.org/pdf/2008.03883`

Listing 40.1: mesh_1d.f90

```
module mesh_1d
  implicit none
  private
  public :: initialize_mesh, x_nodes, dx, N, Mii

  ! Parameters
  integer, parameter :: N = 100          ! Number of spatial
    grid points (nodes)
  real(8), parameter :: L = 1.0d0        ! Length of the
    domain

  ! Mesh data
  real(8), dimension(N) :: x_nodes       ! Coordinates of
    each mesh node
  real(8) :: dx                          ! Uniform cell
    spacing

  ! Mass matrix (diagonal approximation here, for full FEM
    use full M)
  real(8), dimension(N) :: Mii           ! Lumped mass
    matrix diagonal

contains

  subroutine initialize_mesh()
    integer :: i

    dx = L / (N - 1)

    ! Initialize node positions
    do i = 1, N
      x_nodes(i) = (i - 1) * dx
    end do

    ! Lumped mass matrix: each diagonal entry is dx
    Mii = dx
```

```
31
32      ! Uncomment below to use trapezoidal mass matrix
33      Mii(1) = 0.5d0 * dx
34      Mii(N) = 0.5d0 * dx
35      Mii(2:N-1) = dx
36
37    end subroutine initialize_mesh
38
39
40  end module mesh_1d
```

40.2.1.0.3 states_1d States then initializes the constants and fields. First the constants(parameters) used throughout the code. Re, Reynolds numbers. γ, compression of the fluid. Pr, Prandtl number, ratio of viscous to thermal diffusion.

Then, it initializes the discretion fields of entropy, momentum, mass, temperature, and specific entropy.

This should be modularized for the future to make this codebase iterative.

Now, what exactly to do. I think for this, have constants in a separate one. Then for the fields, I am not sure at this moment.

Listing 40.2: states_1d.f90

```
1  module states_1d
2    use mesh_1d
3    implicit none
4    private
5    public :: m_h, rho_h, sigma_h, eta_h, T_h
6    public :: initialize_states
7
8    ! Field arrays
9    real(8), allocatable :: m_h(:)       ! Momentum density
10   real(8), allocatable :: rho_h(:)     ! Mass density
```

```
  real(8), allocatable :: sigma_h(:)    ! Entropy density
  real(8), allocatable :: eta_h(:)      ! Specific entropy (diagnostic)
  real(8), allocatable :: T_h(:)        ! Temperature (diagnostic)
  real(8), parameter, public :: gamma = 1.4
  real(8), parameter, public :: Pr = 0.71
  real(8), parameter, public :: Re = 1000.0

contains

  subroutine initialize_states()
    implicit none
    integer :: i

    ! Allocate fields
    allocate(m_h(N))
    allocate(rho_h(N))
    allocate(sigma_h(N))
    allocate(eta_h(N))
    allocate(T_h(N))

    ! Set initial conditions (example: Gaussian density bump)
    do i = 1, N
      rho_h(i) = 1.0d0 + 0.2d0 * exp(-100.0d0 * (x_nodes(i) - 0.5d0)**2)
      eta_h(i) = 1.0d0
      sigma_h(i) = rho_h(i) * eta_h(i)
      m_h(i) = 0.5d0
      T_h(i) = (0.4d0) * eta_h(i)  ! Ideal gas ! Uses EOS
    end do

  end subroutine initialize_states

end module states_1d
```

40.2.1.0.4 projection_matrix_1d Here the creation of two subroutines to simplify future calculations. First, initialize_projection_ma that simply creates matrix. Next, apply_weak_derivative(f, d_proj). This does:

$$proj = \sum_{i=1}^{n}(\sum_{j=1}^{n} D_{i,j}) * f$$

Where f is the observable. For future work, I think all I need to do for this one is create multiple for different dimensions.

Listing 40.3: projection_matrix_1d.f90

```
module projection_matrix_1d
  use mesh_1d
  implicit none
  real(8), allocatable :: D(:,:)
  public :: initialize_projection_matrix,
    apply_weak_derivative

contains

  subroutine initialize_projection_matrix()
    integer :: i
    real(8) :: dphi_dx
    if (allocated(D)) then
      deallocate(D)
    end if

    allocate(D(N, N))
    D = 0.0d0

    dphi_dx = 1.0d0 / dx

    ! Loop over elements and assemble D
    do i = 2, N-1
      ! Local contributions from basis functions over each
    element
      D(i, i-1) = -dphi_dx / 2.0d0
```

```
      D(i, i  ) =  0.0d0
      D(i, i+1) =  dphi_dx / 2.0d0
    end do

    ! Optional: Neumann (zero derivative) at boundaries
    D(1,:) = 0.0d0
    D(N,:) = 0.0d0
  end subroutine initialize_projection_matrix

  ! Applies D to f to compute (phi_i, partial_x f)
  subroutine apply_weak_derivative(f, d_proj)
    implicit none
    real(8), intent(in)  :: f(N)
    real(8), intent(out) :: d_proj(N)
    integer :: i

    d_proj = 0.0d0
    do i = 1, N
      d_proj(i) = sum(D(i,:) * f(:))
    end do
  end subroutine apply_weak_derivative

end module projection_matrix_1d
```

Listing 40.4: mass_1d.f90

40.2.1.0.5 mass_1d

```
module mass_1d
  use mesh_1d
  use projection_matrix_1d
  implicit none
contains

  subroutine compute_mass_flux(N, rho_h, u_h, rho_rhs)
    integer, intent(in) :: N
    real(8), intent(in) :: rho_h(N), u_h(N)
    real(8), intent(out) :: rho_rhs(N)
```

```
    real(8) :: flux(N)
    integer :: i

    ! Compute flux
    do i = 1, N
      flux(i) = rho_h(i) * u_h(i)
    end do

    ! Apply weak derivative
    call initialize_projection_matrix()
    call apply_weak_derivative(flux, rho_rhs)

    ! Divide by dx
    do i = 1, N
      rho_rhs(i) = -rho_rhs(i) / dx
    end do

    ! Boundary conditions
    rho_rhs(1) = 0.0d0
    rho_rhs(N) = 0.0d0

  end subroutine compute_mass_flux

end module mass_1d
```

Listing 40.5: momentum_1d.f90

40.2.1.0.6 momentum_1d

```
module momentum_1d
  use states_1d, only: Re
  use mesh_1d
  use projection_matrix_1d
  implicit none
contains

  subroutine compute_momentum_rhs(N, rho_h, m_h, sigma_h,
    eta_h, T_h, rhs_m)
```

```
    integer, intent(in) :: N
    real(8), intent(in) :: rho_h(N), m_h(N), sigma_h(N),
  eta_h(N), T_h(N)
    real(8), intent(out) :: rhs_m(N)
    real(8) :: u_h(N), du_proj(N), mu(N), dmu_proj(N),
  deta_proj(N), dT_proj(N)
    integer :: i

    ! Compute velocity
    do i = 1, N
      if (rho_h(i) > 1.0d-12) then
        u_h(i) = m_h(i) / rho_h(i)
      else
        u_h(i) = 0.0d0
      end if
    end do

    ! Apply weak derivatives
    call initialize_projection_matrix()
    call apply_weak_derivative(u_h, du_proj)
    do i = 2, N-1
      mu(i) = m_h(i) * u_h(i)
    end do
    call apply_weak_derivative(mu, dmu_proj)
    call apply_weak_derivative(eta_h, deta_proj)
    call apply_weak_derivative(T_h, dT_proj)

    ! Compute RHS
    do i = 2, N-1
      rhs_m(i) = 0.0d0
      rhs_m(i) = rhs_m(i) - m_h(i) * du_proj(i) * dx
      rhs_m(i) = rhs_m(i) + dmu_proj(i) * dx
      rhs_m(i) = rhs_m(i) - rho_h(i) * deta_proj(i) * dx
      rhs_m(i) = rhs_m(i) - sigma_h(i) * dT_proj(i) * dx
      rhs_m(i) = rhs_m(i) - (1.0d0 / Re) * du_proj(i) * dx
    end do

```

```
    ! Boundary conditions
    rhs_m(1) = 0.0d0
    rhs_m(N) = 0.0d0

  end subroutine compute_momentum_rhs

end module momentum_1d
```

Listing 40.6: entropy_1d.f90

40.2.1.0.7 entropy_1d

```
module entropy_1d
  use states_1d
  use mesh_1d
  use projection_matrix_1d
  implicit none
contains

  subroutine compute_entropy_rhs(N, sigma_h, u_h, T_h,
    rhs_sigma)
    implicit none
    ! Dummy arguments
    integer, intent(in) :: N
    real(8), intent(in) :: sigma_h(N), u_h(N), T_h(N)
    real(8), intent(out) :: rhs_sigma(N)

    ! Local variables
    real(8) :: flux(N), dx_u(N), dx_T(N)
    real(8) :: visc_prod(N), heat_prod(N), heat_grad(N),
    heat_term(N)
    integer :: i

    ! Compute flux
    do i = 1, N
      flux(i) = sigma_h(i) * u_h(i)
    end do

```

```
    ! Apply weak derivatives
    call initialize_projection_matrix()
    call apply_weak_derivative(flux, rhs_sigma)
    call apply_weak_derivative(u_h, dx_u)
    call apply_weak_derivative(T_h, dx_T)

    ! Viscous and thermal terms
    do i = 1, N
      visc_prod(i) = (dx_u(i)**2) / T_h(i)
      heat_grad(i) = dx_T(i) / T_h(i)
      heat_prod(i) = (dx_T(i)**2) / (T_h(i)**2)
      heat_term(i) = (gamma / (gamma - 1.0d0)) * (heat_prod
    (i) - heat_grad(i) / dx)
      rhs_sigma(i) = rhs_sigma(i) - (1.0d0 / Re) *
    visc_prod(i) * dx
      rhs_sigma(i) = rhs_sigma(i) - (1.0d0 / (Re * Pr)) *
    heat_term(i) * dx
    end do

    ! Divide by dx
    do i = 1, N
      rhs_sigma(i) = -rhs_sigma(i) / dx
    end do

    ! Boundary conditions
    rhs_sigma(1) = 0.0d0
    rhs_sigma(N) = 0.0d0

  end subroutine compute_entropy_rhs

end module entropy_1d
```

Listing 40.7: velocity_1d.f90

40.2.1.0.8 velocity_1d

```
module velocity_1d
  use states_1d
```

```
  use projection_matrix_1d
  implicit none
contains

  subroutine compute_velocity(N, rho_h, m_h, u_h)
    implicit none

    ! Dummy arguments
    integer, intent(in) :: N
    real(8), intent(in) :: rho_h(N), m_h(N)
    real(8), intent(out) :: u_h(N)

    ! Local variables
    integer :: i
    real(8), parameter :: eps = 1.0d-12  ! Small number to
    avoid division by zero

    ! Compute velocity safely
    do i = 1, N
      if (rho_h(i) > eps) then
        u_h(i) = m_h(i) / rho_h(i)
      else
        u_h(i) = 0.0d0
      end if
    end do

  end subroutine compute_velocity

end module velocity_1d
```

Listing 40.8: EOS_1d_thermal.f90

40.2.1.0.9 EOS_1d_thermal

```
module eos_1d
  use states_1d
  implicit none
  private
```

```
  public :: compute_temperature, compute_eta



contains

  !
    ------------------------------------------------------------------

  ! Compute temperature T from specific entropy eta
  ! T = (gamma - 1) * eta
  !
    ------------------------------------------------------------------

  function compute_temperature(eta_h) result(T_h)
    real(8), intent(in) :: eta_h
    real(8) :: T_h

    T_h = (gamma - 1.0d0) * eta_h
  end function compute_temperature

  !
    ------------------------------------------------------------------

  ! Compute specific entropy eta from temperature
  ! eta = T / (gamma - 1)
  !
    ------------------------------------------------------------------

  function compute_eta(T_h) result(eta_h)
    real(8), intent(in) :: T_h
    real(8) :: eta_h

    eta_h = T_h / (gamma - 1.0d0)
  end function compute_eta

end module eos_1d
```

Listing 40.9: time_integrator_1d.f90

40.2.1.0.10 time_integrator_1d

```
module time_integrator_1d
  use mesh_1d
  use states_1d
  use velocity_1d
  use eos_1d
  use mass_1d
  use momentum_1d
  use entropy_1d
  implicit none
  private
  public :: advance_one_step

contains

  subroutine advance_one_step(dt, u_h, rhs_m, rho_rhs,
    rhs_sigma)
    real(8), intent(in) :: dt
    real(8), intent(INOUT) :: u_h(N), rhs_m(N), rho_rhs(N),
     rhs_sigma(N)
    integer :: i

    ! Step 1: Compute velocity
    call compute_velocity(N, rho_h, m_h, u_h)

    ! Step 2: calculate tempeture
    do i = 1, N-1
      T_h(i) = compute_temperature(eta_h(i))
    end do
    ! step three: calculate Eta
    do i = 1, N-1
      eta_h(i) =compute_eta(T_h(i))
    end do
    ! Step 4: Compute Galerkin RHS terms
    call compute_momentum_rhs(N, rho_h, m_h, sigma_h, eta_h
    , T_h, rhs_m)
```

```
    call compute_mass_flux(N, rho_h, u_h, rho_rhs)
    call compute_entropy_rhs(N, sigma_h, u_h, T_h,
   rhs_sigma)

    ! Step 5: Advance in time (Euler / RK1)
    do i = 1, N
      m_h(i)     = m_h(i)     + dt * rhs_m(i)
      rho_h(i)   = rho_h(i)   + dt * rho_rhs(i)
      sigma_h(i) = sigma_h(i) + dt * rhs_sigma(i)
    end do

  end subroutine advance_one_step

end module time_integrator_1d
```

Listing 40.10: io_1d.f90

40.2.1.0.11 io_1d

```
module io_1d
  use mesh_1d
  use states_1d
  use velocity_1d
  use time_integrator_1d
  implicit none
  private
  public :: write_state_to_csv

contains

  subroutine write_state_to_csv(filename)
    implicit none
    character(len=*), intent(in) :: filename
    integer :: i
    logical :: wait
    real(8) :: u_h(N)
    integer :: ierr

```

```
    ! Ensure the output directory exists
    wait = .true.  ! Command should run synchronously
    call execute_command_line("mkdir -p output", wait, ierr)
    if (ierr /= 0) then
        print *, "Error: Could not create output directory!"
        stop
    end if

    ! Compute velocity u = m / rho
    call compute_velocity(N, rho_h, m_h, u_h)

    ! Open file for writing
    open(unit=10, file=filename, status="replace", action="write", form="formatted")

    ! Header
    write(10, '(A)') 'x,rho,m,sigma,u,T,eta'

    ! Data
    do i = 1, N
      write(10, '(F12.6,1x,F12.6,1x,F12.6,1x,F12.6,1x,F12.6,1x,F12.6,1x,F12.6)') &
          x_nodes(i), rho_h(i), m_h(i), sigma_h(i), u_h(i), T_h(i), eta_h(i)
    end do

    close(10)
  end subroutine write_state_to_csv

end module io_1d
```

Listing 40.11: momentum_1d.f90

40.2.1.0.12 1D_Thermal_driver

```
program omega_x_driver_1d
```

```
  use mesh_1d
  use states_1d
  use time_integrator_1d
  use io_1d
  implicit none
  real(8) :: dt, t, t_end
  integer :: step = 1
  real(8), dimension(N) :: u_h, rhs_m, rho_rhs, rhs_sigma
  character(len=100) :: fname

  dt = 1.0d-3
  t_end = 1.0d0
  t = 0.0d0

  call initialize_mesh()
  call initialize_states()
  u_h = 0.0d0
  rhs_m = 0.0d0
  rho_rhs = 0.0d0
  rhs_sigma = 0.0d0  ! Explicitly initialize rhs_sigma here

  do while (t < t_end)
    call advance_one_step(dt, u_h, rhs_m, rho_rhs,
   rhs_sigma)
    t = t + dt
    print *, 't = ', t

    if (mod(step, 100) == 0) then
        write(fname, '(A,I4.4,A)') "output/state_", step, "
   .csv"
        call write_state_to_csv(fname)
    end if

    step = step + 1
  end do
end program omega_x_driver_1d
```

40.2.1.1 Plans

I have now create it in its most basic form, here are my next steps.

Break states up into constants and fields(multi-dimensional fields)

Make operators folder for projection, weak derivative, boundary conditions

Add checks(mainly energy conservation)

Upgrade time integration,(figure out)

Make different spacial discretization

More boundary conditions

Add more physics

Though, the main thing I need to work on is module registry + driver pattern.

40.3 part 3

40.3.1 Notes

40.3.1.1 Notes on module FORTRAN

I should have a main driver like:

Listing 40.12: Omegax.f90

```
program Omegax
        use Driver_module
        implicit none

        call Driver_init()
        call Driver_evolve()
        call Driver_finalize()

```

```
9 end program Omegaz
```

There should also be a a config file, saying which modules should be used, and a python script capable of parsing this registery file. Config would also be where parameters would be stored.

40.3.1.2 Diminsionaless programing

In the config file I can have a section to input the diminsions.

Then have all variable allocatable based upon the diminsions

then have an allocation based upon such diminsions(either preprocessor or generic shape handling)

Instead of hardcoding loops, create cases for different diminsions.

Generaliz derivatives and fluxes.

Also, while physics sections should be generalized, initializing between 3D, would be beneficial.

40.3.1.3 Generalize

Due to Metriplectics dyanmics complex mathematics, a fully generalized system, like FLASH-X, is impossible. This is because it is a dynamical theory based upon Hamiltonian Mechanics, rather than forces in Newtonian.

I am thinking the way to fix this is to create a symbolic python code to be able to generate the function I need.

Also, since these would be in py

40.3.2 Plan

I think the steps I am going to take are:

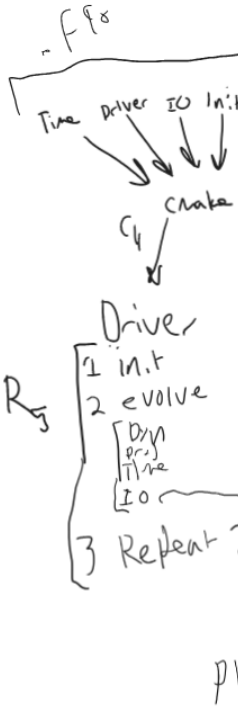
1: Develop a python code that can generate the math I need, I would also like for it to be able to read a LaTeX code for it.

2: I need to look into how to create a Config file for my problem at hand

3: Fix integration for this new system

4:

40.3.3 New Structure



Here is the basic idea of what it should look like and do:

40.4 Part 4

It has been quite some time since I had last worked on this project so I am in a way starting anew.

40.4.1 Summary On the Past

40.4.2 AI Attempt

Recently, I tried to use Claude AI to attempt to do something of a similar vein of this project. To see how capable it was, it quite surprised me. Here I will analysis its attempt; my initial view is that it misinterpretated what I wanted and relied on well developed past developments rather than the novel project I was attempting, but its ability to optimize and clearly write things out outpaces my ability, also using things like kernel development was and parallel computing was suprising.

First, to help with my analysis I used the AI tool to develop its own analysis, then I will do a separate one on it.

40.4.2.1 AI Analysis

40.4.2.2 Code

40.4.2.3 My Analysis

Chapter 41

Axiom

41.1 *Structure

41.2 LaTeX

41.2.1 Division of Structure

In this program I have used this structure to divide the book up:

```
\part{}
\chapter{}
\section{}
\subsection{}
\subsubsection{}
\subsubsubsection{} % custom made
\paragraph
\subparagraph
```

I may have to learn new ways to divide in the future, but I will try to aviod that.

41.2.2 Commands

Here are some commands that I will have here for reference:

```
\begin{verbatim} writes things exactly, used mostly
for writing LaTeX commands in this method
\label{} → Set a reference point
\ref{} → Refer to numbered items

\pageref{} → Refer to page number

\nameref{} (requires hyperref) → Refer to name/title
of chapter/section

\autoref{} (requires hyperref) → Smart referencing
(automatically adds \Chapter", \Section")
\begin{equation} ... \end{equation} → Numbered
equations

\[ ... \] → Displayed, unnumbered equations

$ ... $ → Inline math

\frac{a}{b}, \sqrt{x}, \sum, \int → Standard
math symbols

\begin{align} ... \end{align} (from amsmath)
→ Multi-line aligned equations

\begin{cases} ... \end{cases} → Piecewise functions

\bra{}, \ket{}, \braket{} →
Use physics package for quantum notation

\tensor{} → Use tensor package
```

```
for tensor indices

\dv{}, \pdv{} → Derivatives (from physics)
\begin{figure}[h!] ... \end{figure} →
Insert figures

\includegraphics[width=0.8\textwidth]{file}
→ Image inclusion (graphicx)

\caption{} / \label{} → Caption and reference

\begin{table}[h!] ... \end{table} → Tables
```

I also have begin{lstlisting} to write both FORTRAN and make. I will likely add python and Julia

Chapter 42

Linux & Customizing my Computer

Now that I am dual booting my computer I am going to make plans for customizing it and making sure I have all of the things I want installed.i

42.1 Introductory Plans

I want the libraries, package management tools, set up emacs, Terminal Multiplexing, compililers, toolchain, git, setting up Windows/Linux workflow, and automation

42.2 Package Manegment

ripgreg, fd, bat, fzf

42.3 Terminal Emulator

I am thinking of choicing alacritty

42.4 Editor

While I plan to put all of my Emacs stuff in the GitHub to transer, what I am thinking is for this time to use NeoVim and then from there decide which to use.

42.5 AI summary

Rather than write the rest of it I had ChatGPT write a summary:

42.5.1 Core System Setup

42.5.1.1 Installation

```
sudo apt update && sudo apt upgrade -y

sudo apt install -y \
build-essential cmake make ninja-build \
git curl wget \
htop tmux openssh-client openssh-server
```

42.5.1.2 Purpose

- `build-essential`: Provides compilers and standard libraries
- `cmake`: Cross-platform build system

- `tmux`: Persistent sessions for long simulations
- `ssh`: Remote access for distributed workflows

42.5.2 Compilers

42.5.2.1 Installation

```
sudo apt install -y gcc g++ gfortran clang
```

42.5.2.2 Why Multiple Compilers?

Different compilers provide:

- Different optimization strategies
- Better compatibility with legacy codes
- Access to vectorization and architecture-specific tuning

42.5.2.3 Usage Example

```
gfortran -O3 -march=native simulation.f90 -o sim
```

42.5.3 Scientific Libraries

42.5.3.1 Installation

```
sudo apt install -y \
libopenblas-dev liblapack-dev \
libfftw3-dev \
libhdf5-dev \
libopenmpi-dev openmpi-bin
```

42.5.3.2 Mathematical Role

- **BLAS/LAPACK**: Linear algebra (matrix solves, eigenvalues)
- **FFTW**: Fast Fourier transforms
- **HDF5**: Structured scientific data storage
- **MPI**: Parallel distributed computation

42.5.3.3 MPI Example

```
mpirun -np 4 ./simulation
```

42.5.4 Python Scientific Environment

42.5.4.1 Installation (Conda)

```
conda create -n physics python=3.11
conda activate physics

conda install numpy scipy matplotlib pandas numba sympy h5py
```

42.5.4.2 Role in Workflow

Python acts as:

- Data analysis layer
- Visualization interface
- Rapid prototyping environment

42.5.4.3 Example

```
import h5py
import numpy as np

f = h5py.File("output.h5", "r")
data = np.array(f["density"])
```

42.5.5 Fortran Development

42.5.5.1 Tools

```
pip install fpm
```

42.5.5.2 Why Fortran?

- Efficient array operations
- Strong numerical performance
- Widely used in legacy HPC codes

42.5.6 Visualization

42.5.6.1 Installation

```
sudo apt install gnuplot paraview
```

42.5.6.2 Purpose

- **gnuplot**: Quick plotting
- **ParaView**: Large-scale simulation visualization

42.5.7 LaTeX Environment

42.5.7.1 Installation

```
sudo apt install texlive-full
```

42.5.7.2 Purpose

LaTeX is used for:

- Research papers
- Technical documentation
- Mathematical derivations

42.5.8 Editors

42.5.8.1 Installation

```
sudo apt install emacs
```

42.5.8.2 Usage Philosophy

- Emacs: Integrated research environment

42.5.9 Containers and Reproducibility

42.5.9.1 Installation

```
sudo apt install docker.io
```

42.5.9.2 Purpose

- Reproducible environments
- Dependency isolation
- Portable experiments

42.5.10 Workflow Architecture

42.5.10.1 Full Pipeline

1. Write simulation code (Fortran/C++)
2. Compile with optimized libraries
3. Run locally or via MPI
4. Output data in HDF5 format
5. Analyze using Python
6. Visualize results
7. Document in LaTeX

42.5.11 Directory Structure

```
~/projects/
~/simulations/
~/data/
~/papers/
~/notes/
```

42.5.12 Advanced Topics

42.5.13 Performance Optimization

- Use `-O3 -march=native`
- Profile with `perf` or `gprof`
- Optimize memory access patterns

42.5.13.1 Parallel Scaling

- Domain decomposition
- MPI communication patterns
- Load balancing

42.5.13.2 Reproducibility Strategy

- Version control (git)
- Environment locking (conda)
- Containerization (Docker)

42.6 3/20/26

I had some trouble with having different scalling for different displays. I ended up figuring out to change the GNOME from X11 to Wayland.

I was not able to make sure that the cursor switching works as I wanted, as far as I can tell there is nothing I can do to fix that.

Chapter 43

Emacs

This will be a summary of some of the main packages and customizations that I have either installed or created. As well as whatever I think about

Chapter 44

manim

44.1 Notes from making WHCKL

Chapter 45

Website

45.1 Brainstorm

I had thought of a couple of ideas, but had forgot to write them down. I guess I will just explain my final one.

My plan is to have it so it can parse one of my daily org files for a schedule, then can paste that schedule into the website to view. I would like to allow for checking things off, in progress, notes, and some other things I will get into at a later point.

I would also like to access my book easily through the website

45.2 2/16/2026

I meant to document what all I did and why through my journey, but I hadn't. Thus, at the moment I am finished.

Chapter 46

Category-Theoretic Metriplectic

46.1 Plans

46.2 Papers

Here I will begin with a list of papers to use for my research in this study

- Baez, Weisbart & Yassine — Open systems in classical mechanics
- John C. Baez & Christopher L. Rogers — Categorified Symplectic Geometry and the String Lie 2-Algebra (2008/2009)
- Alan Weinstein — Symplectic categories (survey/paper, 2009)
- Brendan Fong — Decorated Cospans (2015) & related work (Baez–Fong, Baez–Fong–Pollard)

- John C. Baez — expository / historical pieces on n-categorical physics
- P. Guha — Metriplectic structure, Leibniz dynamics and dissipative systems
- Work on double categories / structured cospans for open dynamical systems (Courser thesis; Kenny Courser; "Open Systems: A Double Categorical Perspective")

46.3 Review of Metriplectic systems

46.4 Understanding of Category-Theoretic Systems

46.5 My work

Chapter 47

Gravitational Memory study

47.1 First meeting

We talked in general about a couple of things. My next "mini-project" was to create a python code to read the strain off of the gravitational wave data and use that to create some audio. I have developed the code to work for hdf5 files and proved it worked with a black hole merger, I do not have the gravitational wave data for CCSN yet.

Also, the bigger project is to work on figuring out how to use 3Gen gravitational wave detectors to find CCSN. They even estimate these new projects may be able to see throughout the Virgo Cluster. We are presumably basing our work off of the paper **Detecting Gravitational Wave Memory in the Next Galactic Core-Collapse Supernova**

47.1.1 First go: Detecting Gravitational Wave Memory in the Next Galactic Core-Collapse Supernova

General notes from my first look at the paper detecting Gravitational Wave Memory in the Next Galactic Core-Collapse Supernova. Using data from GWOSC and several CCSN simulations they use some kind of linear prediction filtering system to develop what the waves should look like and how far away we could see them.

47.1.1.1 Action Notes

- Learn Linear Prediction Filtration
- Learn librosa python module
- What is a whitened signal
- What is amplitude spectral density

I found a MITOpenCourseWare on signal theory that I will likely try to do

47.1.1.2 Code

Got some audio stuff working

47.1.1.3 General Readings/Videos

Convolution is defined via

$$\sum_{k=-\infty}^{\infty} x[k]h[x-k] = (x \star h)[n]$$

There is also a lot of other information listed but I am not sure what is important, the equations are generally pretty simple so I think I will just look them up.

47.2 September 10

add whiten to gw, zero out wavelets

Looking into trying without interpolation.

Look into generating gravitational waves from a 2d checkpoint file, calculate quadropole moments. Density variable multiply by coordinates. Stencile for time derivative or look into Pedro.

47.2.1 Code

I got the code for the audio of the CCSN. I did successfully whiten the gravitational wave data but was unable to get the black hole merger right. I verified what I did using the graph though

47.2.2 Quadrupoles

Calculating the quadrupole mass moment is fairly simple

$$I_{2m} = \frac{16\sqrt{3}\pi}{15} \int \rho Y^{\star}_{2m} r^2 dV$$

The star superscript means we are using the complex conjugate.

Rather than write everything out, calculate mass moment very simply, find its second order time derivative, then plug that into the gravitational wave equation.

47.3 September 24

47.3.1 Meeting overview

Next step: would be noise generation in time series forms. Weighted Gaussian noise with the amplitude estimated.

PRL template, short formate so it could look like a short PRD

Inputting generated noise, bandpass filter, correlations.

I will be working on Austrian detector, deseco, and cosmic explorer(can't do lisa or Einstein Telescope)

47.4 October 1

Look into Burner Shuuts

Gorgoulon General Relativity

47.5 October 15

Chapter 48

Website

48.1 Brainstorm

I had thought of a couple of ideas, but had forgot to write them down. I guess I will just explain my final one.

My plan is to have it so it can parse one of my daily org files for a schedule, then can paste that schedule into the website to view. I would like to allow for checking things off, in progress, notes, and some other things I will get into at a later point.

I would also like to access my book easily through the website

48.2 2/16/2026

I meant to document what all I did and why through my journey, but I hadn't. Thus, at the moment I am finished.

Chapter 49

YouTube

49.1 Scope and Goals

As a kid, I had always wanted a YouTube Channel. Now I will be working towards that, on a bit more pratical level. For the purposes of learning more personally, leveling up my skill as a

49.2 Getting Started

49.2.1 Ideas

At some point I want a dive into Kerr Black holes but I am not ready for that.

- Numerical Hamiltonian(done)
- A Physical Look Into Hamiltonian Mechanics
- Hidden Assumptions in Modern Physics
- Journey through General Relativity

- Positive Mass Warp Drive
- Axioms of Hamiltonian Mechanics
- Gravitational Waves
- MagnetoHydrodynamics Propulsion
- MagnetoHydrodynamics in a nebulua
- Deeper Numerical Methods
- Scientific Programme
- Destroying a Universe with a punch(done)
- Best way to simulate a solar system(done)
- Creating hydrodynamics from Boltzmann
- New Universe
  - Creating new force from guage symmetry
  - Creating modifed General Relativity(symplectic manifold instead of Remmanian)
  - Creating New Fluid Dynamics
- Metriplectic series
  - Why Hamiltonians don't like dissipation
  - Metriplectic 4-bracket examination
  - Numerical Recipes for Metriplectic
  - Putting it all together/making our own equations
  - Other methods
- In-depth operator splitting(done*)
- Deriving Plasma kinetic equations

- Simple GR solver
  - Kerr
  - TOV
  - Accretion
  - Quasi-star
  - Additional(maybe stuff)
  - Warp drive
  - Cosmic censorship
  - CCSN
- ADM
- Symplectic geometry: A physicist's perspective
- DG CCSN(done)
- Deriving Gravitational Wave Memory
- Geometric Particle in cell
- Non-canonical MHD
- Multiscale kinetic
- Chaos theory for MHD vs RMHD or other forms
- Geometric integration of Vlasov-Maxwell
- Structure preserving Discretization in kinetic PDEs
- Collision operators and nonlinear kinetic entropy
- Phase-space of kinetic turbulence
- MHD singularities

- Magnetic topology
- Plasma waves as Hamiltonians
- The Abomination that is generally covariant Newtonian Mechanics
- Spectral Algorithms
- Rational Krylov and contour-integral algorithms
- Infinite-Dimensional Geometric integrators
- multisymplectic

49.3 Introduction to Symplectic Integrators

49.3.1 General overview of direction

- Introduction to me
- Motivation
- Hamiltonian Mechanics introduction
- Geometric perspective
- Symplectic Euler compared to traditional Euler
- Leapfrog/Verlet
- Practicule guidence

49.3.2 Opening

Before I begin this video; I am not an expert. Must of my understanding comes from Problems In Hamiltonian Dynamics by (.....). This is more for my own education than others, for a better understanding of the concepts and development of my teaching abilities. I highly recommened continue to learn far beyond this simple introduction I will lay out. (picture of the book)

49.3.3 Motivation

Symplectic Numerical methods are focus inheritily on the structure and methods of Hamiltonian mechanics. Thus, to understand them, one must understand why. Why it is used at all. At first glance, it seems quite contrived. Hamiltonian Mechanics is quite mathematically abstract and complicated. Beyond that, expanding it in differing ways such as dissipative or entropic spaces requires creative solutions. Thus, begs the question, why is it such a developed field?

(put really complex hamiltonian)

The short answer is that Hamiltonian Mechanics geometry enforces energy conservation in a way others formulisms do not. The long answer... is well this video.

(put graph showing energy conservation)

49.3.4 Hamiltonian Mechanics Review

First, a quick review on Hamiltonian Mechanics. If you are comfortable feel free to skip ahead.

A hamiltonian system is defined by the function H(q,p,t) - the hamiltonian - which repersents something analogus to total energy. Not exactly though, hopefully in a future video we

will get to that, but for not thinking of it in energy works great for this purpose.

$$H(q,p,t) = T + V$$

Here q repersents generalized position, which is the minimum amount of set parameters that fully spesify where a system is in configuration space. Where configuration space is a space whose points repersent all possible configurations. Meaning that generalized position is not confined to distance, but also things like angles, diminsionless quantaties, or what ever other convenient quantities that can be used.

The q, repersents generalized momenta. Which holds the same rules as position does, but repersenting momenta instead of position.

The evolution can thus be derived to be

$$\frac{dq}{dt} = \frac{\partial H}{\partial p}$$

$$\frac{dp}{dt} = -\frac{\partial H}{\partial p}$$

Let us look at a concrete example: a simple pendulum

(picture of pendulumn)

$$H(q,p) = \frac{p^2}{2m} + \frac{1}{2}kq^2$$

Such, one can take the partial derivatives to find:

$$\frac{dq}{dt} = \frac{p}{m}$$

$$\frac{dp}{dt} = -kq$$

In phase-space, the space of all generalized pairs, the system traces out a tragectory. For harmonic ossilators these are

ellipses. Each ellipse corresponds to a different energy level. (show phase-space)

Here are some key points.

- energy remains constant along the tragectories. As tragectories are defined by conserved energy

- the phase-space volume is conserved. This is essentially (shade in one region)

-finally, the most importaint for this, the manifold retains a symplectic structure. Which is defined at the 2-form $\omega = dq \wedge dp$. This will be disgussed further shortly.

49.3.5 Why standard models fail

To understand the symplectic methods, one must find their motivation. Why traditional methods fail in certain senerios, no matter the type. To understand this, we will look at the simplest integrator. Forward Euler

$$q_{n+1} = q_n + h * \frac{\partial H}{\partial q}_n$$

$$p_{n+1} = 1_n + h * \frac{\partial H}{\partial q}_n$$

Where h is the time step.

Let us apply this to our harmonic ossilater and see what happens... Tragectory spirals outwards! This is simply because traditional methods aren't made to conserve energy, they are meant to model physics. The main reason is that tiny errors add up without the ability to contain them. Also, traditional methods add unphysical correction terms that over time lead to unphysical effecs. Being dissipate in certain areas and adding energy in others. Adding artificial energy to the system in order to have temporary Now, obviously higher order correc-

tions help with these problems and making the models more physical helps, but these problems are fundemental. Adding higher order corrections simply increases the computational workload without ever fully resolving the problem.

Even sofisticated methods like RK4 have this problem. It is accurate to the 4th order, but that means that the error per step is $O(h^5)$. This is tiny, but adds up. For something like a celestial simulation over millions or even billions of years, this creates distorted simulations.

(Log-log plot showing error growth)

49.3.6 Symplectic Integrators

(illistration of symplectic maps)

Symplectic integrators fix this problem by peserving the symplectic structure I mentioned earlier.

$$\omega = dq \wedge dp$$

The symplectic form is a mathematical object that mesures oriented areas in phase-space. For a simple system with one degree of freedom in the position and momentum each one gains this equation.

This is a 2-form, when integrated over any region of phase space, you get an oriented area. (physical deformation)

As the hamiltonian flow evolves, the region over phase-space gets stretched, rotated, and deformed. But, the symplectic form remains perserved.

From this basic axiom, one can derive something called the Shadowing Theorem, that says:

A tragectory computed by symplectic integrator, while not exactly true, stays close to the tragecory of a slightly modified Hamiltonain.

(visual: compared integration to true)

Meaning, standard integrators essentially create a new equation that is close to the origional.

Symplectic integrators create another hamiltonian that is exactly equal to the origional hamiltonain plus the order of the error.

(energy conservation plot)

Such, the energy does not increase monotonically, it ocilates within a bounded region. Over exponentially long time, the error grows logorithmicly rather than linearly.

49.3.7 Symplectic Euler's method

To compare, we will once again take the simpliest possible case.

Take the traditional Forward Euler

$$q_{n+1} = q_n + h * \frac{\partial H}{\partial q}_n$$

$$p_{n+1} = 1_n + h * \frac{\partial H}{\partial q}_n$$

Now turn it into what is called a backward Euler

$$q_{n+1} = q_n + h * \frac{\partial H}{\partial q}_n$$

$$p_{n+1} = 1_n + h * \frac{\partial H}{\partial q}_n$$

Notice what has changed, we updated p using the old position but the momentum by using the new momentum. One thing to note, backwards or implicit integration is commonly used outside of symplectic integration, it is simply a general requirments for symplectic integration. There are some special

cases where explcit numerical methods satisfy the requirments of a symplectic integrator. Another requirment being that one using a Hamiltonian or Hamiltonian like formulism such as Numbu mechanics.

Now let us see why it is symplectic:

$$H(q,p) = T(p) + V(q)$$

where T is kinetic energy and V potential.

The symplecti Euler step can be decomposed into pure position-step, which is the flow of the potential. The a pure momentum step, which is exactly the flow of kinetic energy. The composition of these is the exact flow of the Hamiltonian, which is by definition symplectic.

The exact requirments are quite involved and mathematically difficult to derive, so an independent video on the topic is reqiured. For the purposes of this video the general rules described and the types of integrators are suffient.

Let us implement this for the harmonic oscillator.

In this, the orbit remains stable. Energy oscilates but doesn't drift.

One can also make the equation implicit by updating q before p.

49.3.8 The LeapFrog/Verlet Method

The most commonly used sympletic integrator is the leapfrog method. It is second order accurate.

The core idea: Evaluate positions and momenta at staggered time points.

(equation)

(visual)

Think of it as half-step momentum backward. Full step position forward, using the half step. Then half-step momentum backward again.

This is symmetric: if you run it backward, you get the exact same algorithm. This time-reversibillity is what gives second order accuracy.

There is an equivlent position formulism that is often more convenient. (equation)

Where v is velocity and a is acceleration. Note, that while it uses variables commonly used in Newtonian dynamics, it is still fundementally hamiltonian. (visual) One can compare performence after 1000 orbits. For a timestep of 0.01 Forward euler has energy error of 15.3%, symplectic euler 0.8%, and leapfrog 0.002%.

As you can see much smaller, as the error scales by the square of the time step. Allowing for larger time steps for the similar accurecy.

A derivation of this equation is outside of the scope of this video but is an interesting idea that may be develed into in the future.

49.3.9 Splitting Method

To build higher order symplectic integrators, one must learn operator splitting.

The key idea: if you can write the Hamiltonian as a sum

$$H = H_A + H_B$$

and you know how to integrate each piece exactly, you can build approximate integrators by composing these exact functions of these two.

where $\phi_A(h)$ is the exact flow generated by H_A over time h.

and ϕ_A is the exact flow over H_B.

Then the true hamiltonian flow is:

where $\{\cdot,\cdot\}$ denotes the Poisson Bracket.

The simpliest approximation is obtained by composing the two exact flows, this is known as the Lie-Totter splitting, this method is first otrder accurate. Higher order methods can be built using splitting, such as the 4th-order Forest-Ruth:

$$\psi_4(h) = \psi_2(\theta h) \circ \psi_2((1-2\theta)h) \circ \psi_2(\theta h)$$

A derivation of this is outside the scope of this video but is something we will get into.

49.3.10 Beyond the Basics

Symplectic integration goes far beyond the introduction given in this video.

- Dynamics on a manifold
- veriational integration
- Backward error analysis
- Adaptive symplectic methods
- Methods for Hamiltonian adjacent frameworks(ie. Nambu, metriplectic, etc)

Some of these I would like to explore in future videos but many of them are currently beyond my expertice.

49.4 Physical Interpretations of Hamiltonian Dynamics and Byond

49.4.1 Planning

1. Opening
2. What is the Hamiltonian
3. What physically is phase space
4. Dissipation and the Non-Hamiltonian World
5. Beyond Symplectic: Metriplectic and Generic
6. Beyond Symplectic: Contact Geometry and Geometrization of Irreversibility
7. Beyond Symplectic: Poisson Manifolds, Casimirs, and Nambu
8. Structural Realism
9. Open Problems

49.4.2 Opening

Often in higher level mathematical physics, you will here many phrases thrown around with generally little to no explanation. "the hamiltonian is analougus to total energy but isn't literally", "Hamiltonian mechanics forbids dissipation", and many more.

These complex ideas are pushed aside for latter, well if you are like me, you think that later is now, so that is the purpose of this video.

This video will require a good intuitive and understanding of Hamiltonian mechanics and the basic structural underpinnings. Also, be warned, this is a fun project, mistakes are entirely possible.

49.4.3 What is the Hamiltonian?

In most courses, you will be introduced to the Hamiltonian functions through this equation

$$H = V + T$$

where the Hamiltonian is the addition of kinetic and potential energy, and is thus the total energy.

A few years later you may here terms like generator of the flow, with only the explanation that H is a function that generates the correct observables, what ever those are.

So, how do these definitions fit together, and what do they even mean.

Essentially, the idea is that H uses Noether's theorem's time translation to make it so that the Hamiltonian is always equal to:

$$\frac{\partial A}{\partial t} = \{A, H\}$$

where H is the observable. An observable is a pretty self explanatory concept. Something that can be physically observed; momentum, energy, velocity, entropy, or even non-physics measurements like personality. Though classical Hamiltonians have trouble with entropy as we will get into.

As you can figure out with some careful playing; the Hamiltonian is only equal in a special case; in classical systems where the observables are position and momenta. In other

cases, it no longer is that. It could be something else, or simply a number with no traditional physical analogy.

49.4.4 What is Phase Space?

Now, to actually use the Hamiltonian, knows we need phase space. So, what physically is that.

Well, a good analogy is imagine the universe is frozen in time, what information do you need to predict what happens next?

Now that is just an analogy, if you are watching this video I suspect you want something more than that.

Well, we must first start with a state, which is really just a property, the properties needed for physics. Newton defined these properties through forces, mass, position, time, and their respective derivatives to arbitrary orders.

Hamilton, defined it through configuration momenta and position. It really can be anything but as we will motivate later in most physical systems these are the generally the most useful. Then, he found you are able to for every possible physical state, there could be one point in this phase space of position and momenta that corresponds to it. Such as a 2D body can be ploted in 4D space through

$$(x, y, p_x, p_y)$$

There, objects no longer viewed as moving through space but rather tragectories in phase space. Generally this view is considered to be a mathematically significant but metaphysically trivial change, but some people disagree, which is something that will be covered later in the video.

49.4.5 Is Momentum more Fundamental than Velocity

The question of which is more fundemental; momentum or velocity seems quite arbitrary, and in many ways it is, but we will look into it regardless.

In traditional Newtonian Mechanics; momentum is traditionally derived by velocity, but the inverse is just as true.

In hamiltonian mechanics, this isn't the case. Momentum is more than just mass and velocity, it also includes think like electric fields or even other fundemental forces like gravity in some cases. In quantum mechanics this connection becomes even more muddled for obvious reasons.

Now to really answer the question. I am sad to say that it depends on what you are looking for. Velocity is much more intuitive, but momentum is a generator of translations and naturally connected with symplectic structures. Likely not a very satisfying answer, but the only one I have.

49.4.6 Hamiltonian Mechanics and Teleology

49.4.7 Noether's Theorem

49.4.8 Canonical Transformations

49.4.9 The Question of Dualism

49.4.10 Hamiltonian Branch offs

49.4.11 Hamiltonian and Dissipation

49.4.11.1 Metriplectic and Others

49.4.12 Structural Realism

49.5 Variational Integrator for RMHD

49.5.1 Planning

1. Opening
2. Introduction
3. RMHD
4. Formal Lagrangian
5. Conservation Laws
6. Electron Inertia
7. Symmmetrisation
8. Discrete Action Principle
9. Discrete Conservation Laws

10. Numerical Methods

11. Validation

12. Random Testing

I am thinking for random testing either a video of one of the earlier simulations or do something random.

49.5.2 Opening

Before I begin, as always, I am not an expert. Most of my information comes from the paper "Variational Integrators for Reduced Magnetohydrodynamics", by Michael Kraus, Emanuele Tassi, and Danueka Grasso. For more information their paper will be in the description. Anyways, on to the video.

49.5.3 Introduction

For all purposes, the subject of this video is the creation of a numerical method that takes a simplified model of magnetized plasma and preserves the conservation laws with greater accuracy than most other methods. The tool used to do so is called a variational integrator. That descretizes the action, as in the action in Lagrandian Mechanics, rather than the equations of motion.

49.5.4 Variational Integrators

Variational integrators work in the inverse of traditional methods. In traditional methods you start with the model

$$F = kx$$

then define the equations of motion

$$\ddot{x} + \frac{k}{m}x = 0$$

From here one descretizes the equation of motion.

For variational derivatives one takes the action:

$$S[q] = \int_{t_0}^{t_1} L(q, \dot{q})dt$$

The nature selects where

$$\delta S = 0$$

From here derivation of things such as

- Euler-Lagrange
- Hamiltonian
- Conservation laws

From here instead of discretizing the differential equation, one discretizes the lagrangian

$$S_d = \sum_{k=1}^{N-1} L_d(q_k, \dot{q_{k+1}})$$

Where

$$L_d \approx \int_{t_k}^{t_{k+1}} Ldt$$

From here, varing across the sequence rather than a smooth curve gives you the Discrete Euler-Lagrange Equation. Here the equation is already an algorithm, no additional methods needed(though we will use some in this video)

49.5.5 RMHD

To do this, we must define what RMHD is, to do that we must first define MHD. MHD treates plasma as a single conductive fluid coupled with a magnetic field. While it is a massive simplification of "Billions of charged particles" it is still quite accurate and obviously much less computationally expensive.

RMHD, further simplifies, for a spesific, useful physical situation. In plasma with a strong backgroud magnetic field, like in a tokamak, where the field mostly points one way (call it "toroidal") and everything interesting happens in the plane perpendicular to it. Under those conditions, the fast physics along the strong field direction basically stops mattering, and you're left with an effectively two-dimensional problem.

With only four fields remaining as useful:

- ϕ, the stream function, describes how the fluid flows
- ω, the vorticity, how the flow swirls
- ψ, the magnetic potential
- **j**, the current density, how the magnetic field swirls

The equation then simplifies to:

$$\omega_t\{\phi,\omega\} + \{j,\psi\} = 0, \quad -\Delta\phi = \omega$$

$$\psi_t + \{\phi,\psi\} = 0, \quad -\Delta\psi = j$$

With the poisson bracket being simply two dimensional so that

$$\{\phi,\omega\} = \phi_x\omega_y - \phi_y\omega_x$$

Here, is can be seen that RMHD has Casimir Invariants, quantaties that are conserved, not from traditional physical

symmetries but from the structure of the Lie algebras. There are families with infinite many subsets, for the pusposes of thisl we will focus on Magnetic helicity(how tangled the magnetic fields are) and Cross helicity(how aligned the flow is with the magnetic field)

Now, for this paper to work, rather than use Hamiltonian structure, it uses Lie-Poisson, Hamilton's weird cousin. It still conserves energy but standard symplectic methods simply do not work.

This is because fluid mechanics observables do not naturally describe motion along a phase space, but live on a Lie Algabra with symmetries within it. So the Poisson operator changes with the state rather than is a constant symplectic structure.

49.5.6 Formal Lagrandian

From here comes a snag, variational integrators need a Lagrandian, an action to discreize. But, for RMHD, there is no obvious Eulerian-Lagrandian. So what does one do?

They make one up. There is a trick called a formal lagrandian. Essentialy, take every equation equation in your system and multiply is by a made up variable called an auxiliary variable that has no physical meaning. Then add them up

$$L = \xi\left(\omega_t + \{\phi, \omega\} + \{j, \psi\}\right) + \mu\left(\omega + \Delta\phi\right) + \chi\left(\psi_t + \{\phi, \psi\}\right) + \zeta\left(j + \Delta\psi\right)$$

Here four variables ξ, μ, χ, ζ, are all ghost variables that only exist to make the arithmatic work.

This helps by giving stationary variables to allow for one to vary along the physical variables using Hamilton's Principle leading to a brand new "adjoint equation."

Leading to the Euler-Lagrandian:

Where φ denotes all of the variables. Many of the auxillary variables can be found to be to be functions of the origional variables. So that φ can be defined as

$$\varphi = (\omega, \psi, \phi, j, \psi\omega, 0, 0)$$

49.5.7 Conservation Laws

Now, the whole puspose of the creation of the Lagrandian was to apply Noether's theorem. We restrict our attention to conservation laws which are generated by vertical transformations, that is transformations which only affect the field variables but leave the base space invariant. From here with some complex math I won't get into, it is found that the symmetries are

- Magnetic Helicity
- L^2 norm of ψ
- Cross Helicity
- Energy

49.5.8 Electron Inertia

This current RMHD, has a defect. It does not include magnetic reconnection, field line snapping and reconnecting. Which tend to be one of the most importaint studies in plasma physics. So, we will add a small correction, electron inertia. Real electrons have mass, just a small one, ones you treat them as such a new term develops in Ohm's law, one that allows reconnection. Thus the new model changes ψ to $\overline{\psi} = \psi + d_e^2 j$ to give the electron canonical momentum where d_e denotes the electron skin depth.

That's it, that is the whole mortification. From here, small bit of derivation shows that the Hamiltonian stays the same, so does the Casimir invariants and the conservation laws, only expressed in for of $\overline{\psi}$ rather than ψ.

49.5.9 Symmetrisation

Before we can put anything into our computer, there is some tedious house keeping needed. Our formal Lagrandian has a second derivative, which is highly challenging to discretize directly and even more costly to compute. To do this we use symmetrisation, essentially integration by parts but for functional rather than function's. We do this on the terms that produce the Laplancians, and get an equivalent Euler-Lagrange equation that only depends on the first order derivative of the fields.

Further care can be taken when discretising the Poisson Bracket. It can be seen through integration by parts when assuming appropriate boundary conditions that the cyclic permutations of the functions in the integrandare all identical

Insted of selecting on of those forms, a convex combination can be considered.

$$\int \xi\{\phi,\omega\}dxdy = \int [\alpha\xi\{\phi,\omega\} + \beta\phi\{\omega,\xi\} + \gamma\omega\{\xi,\phi\}]$$

Where $\alpha = \beta = \gamma = \frac{1}{3}$ to conserve symmetry. From here we find the motified Lagrangian

$$L'''(\varphi, \varphi_t, \varphi_x, \varphi_y) = \frac{1}{2}\,\varphi_t^T \Lambda\, \varphi \tag{49.1}$$

$$+ \frac{1}{3}\left(\xi\{\phi, \omega\} + \phi\{\omega, \xi\} + \omega\{\xi, \phi\}\right) \tag{49.2}$$

$$+ \frac{1}{3}\left(\xi\{j, \psi\} + j\{\psi, \xi\} + \psi\{\xi, j\}\right) \tag{49.3}$$

$$+ \frac{1}{3}\left(\chi\{\phi, \psi\} + \phi\{\psi, \chi\} + \psi\{\chi, \phi\}\right) \tag{49.4}$$

$$+ \mu\,\omega - \nabla\mu \cdot \nabla\phi + \zeta\, j - \nabla\zeta \cdot \nabla\psi. \tag{49.5}$$

49.5.10 Discrete Action Principle

As mentioned earlier we will discretize the action rather than the equation of motion, for simplicity we will use Veselove-type finite difference to split into two space and one time. But the extention to more elaborate approaches is fairly straight forwards from here.

Here we can quite easily find that

A large deal of algebra allows us to get to the point of employing the Hamilton-Pontryagin Principle. Which states when transfering from Lagrandian to Hamiltonian we should not assume velcity and momentum's relation but derive it after the fact of treating them as independent.

The using what is called the Arakawa's discretization for the Poisson bracket, which is given by

$$A_{i,j}(\phi, \omega) = \frac{1}{3}\Big(A_{i,j}^{++}(\phi, \omega) + A_{i,j}^{+\times}(\phi, \omega) + A_{i,j}^{\times+}(\phi, \omega)\Big). \tag{49.6}$$

Where the bracket is the averaging of the A^{++}, the centered difference of both X and y. $A^{+\times}$, the diangonal interaction. And $A^{\times+}$, another diangonal rotated relative to the other.

Using these principles and a lot of mathematics we gain these massive equations.

```
0 &=
\frac{\omega^{n+1}_{i,j} - \omega^{n}_{i,j}}{h_t}
+ \frac{1}{4}\left[
    A_{i,j}(\phi^{n+1}, \omega^{n+1})
  + A_{i,j}(\phi^{n},   \omega^{n+1})
  + A_{i,j}(\phi^{n+1}, \omega^{n})
  + A_{i,j}(\phi^{n},   \omega^{n})
\right] \nonumber \

\[1em]
&\quad + \frac{1}{4}\left[
    A_{i,j}(j^{n+1}, \psi^{n+1})
  + A_{i,j}(j^{n},   \psi^{n+1})
  + A_{i,j}(j^{n+1}, \psi^{n})
  + A_{i,j}(j^{n},   \psi^{n})
\right], \tag{86a}
\

\[1em]
0 &=
\frac{\psi^{n+1}_{i,j} - \psi^{n}_{i,j}}{h_t}
+ \frac{1}{4}\left[
    A_{i,j}(\phi^{n+1}, \psi^{n+1})
  + A_{i,j}(\phi^{n},   \psi^{n+1})
  + A_{i,j}(\phi^{n+1}, \psi^{n})
  + A_{i,j}(\phi^{n},   \psi^{n})
\right], \tag{86b}
\

\[1em]
\omega^{n+1}_{i,j} &=
- \Delta_x \phi^{n+1}_{i,j}
```

```
- \Delta_y \phi^{n+1}_{i,j}, \tag{86c}
\

\[1em]
j^{n+1}_{i,j} &=
- \Delta_x \psi^{n+1}_{i,j}
- \Delta_y \psi^{n+1}_{i,j}. \tag{86d}
```

If you wish to see the derivation out, I would reccomend reading the paper.

49.5.11 Discrete Conservation Laws

Using the same methodology we can find our earlier conservation laws

Total energy

$$E = \frac{hx\,hy}{2} \sum_{i,j} (\phi_{i,j}\,\omega_{i,j} + \psi_{i,j}\,j_{i,j}) = \text{const.}$$

Magnetic Helicity

$$C_{\text{MH}} = hx\,hy \sum_{i,j} \psi_{i,j} = \text{const.}$$

L^2 norm of ψ

$$C_{L^2} = hx\,hy \sum_{i,j} \psi_{i,j}^2 = \text{const.}$$

And cross Helicity

$$C_{\text{CH}} = hx\,hy \sum_{i,j} \omega_{i,j}\,\psi_{i,j} = \text{const.}$$

49.5.12 Numerical Methods

So now we have out implicit forms tangled in non-linearity.

To solve this we use Newton's Method linearize the residual, solve the resulting linear system, update your guess, repeat until the residual is essentially zero. That linear system at each Newton step gets handed off to GMRES, an iterative Krylov-subspace solver, and to make GMRES converge fast, they build a custom, physics-informed preconditioner — following earlier work by Chacón and collaborators — that exploits the specific structure of the vorticity and Ohm's law equations.

The elliptic bits — inverting the Laplacian to get the varibles — get handled by whatever general-purpose elliptic solver you have on hand (they use PETSc); if your domain happens to be doubly periodic, there's an even cleaner trick available, since periodic boundary conditions mean the discrete Laplacian is exactly diagonalized by a Fast Fourier Transform, so you can invert it directly and exactly in Fourier space, no iteration required.

I mimic this in python by setting up a file to create the initial condition, the operators of the spectral solver and Arakawa's method, and integrator file to run it through the actual equations derived earlier.

49.5.13 Validation

Here I make sure my model matches the paper through Orszag-Tang Vortex

(image)

Current sheet

(image)

Current sheet with Electron Inertia

(image)

Here the models match that of the paper, and matches within machine persicion the analytical solution.

49.5.14 Random Test

Now that we have a working model, let's use it for something cool. Here is Kinetic-Alfven Trublence

(movie here)

49.6 Why: Developing Physics

49.6.1 Planing

Essentially, one person teaching another parabolic tragectories but the student keeps asking why in a generally eurdite way. Analogous to Socratic Dialog.

1. Introduction to Parabolic tragectories

49.6.2 Introduction

Here we set the stage, a stage. An introductory physics teacher is having a one-on-one lesson with a young student. A highly intelligent student that wants to get to the base of everything. The most fundemental of the fundemental.

49.7 Can you Destroy a Universe with a Force?

49.7.1 Introduction

I am sure everyone knows of the extreme forces created by fictional characters. Blowing up planets, stars, galaxies, and even Universes. Now these obviously aren't physically realistic, but it begs the question, how much? What would such extreme forces actually look like?

49.7.2 Planck Force

To start with this, we will start with the theoretical highest force that still works with modern physics. The Planck Force. To explain this, the Planck Units are essentially normalizing and using units and physics starting with only the speed of light, gravitational constant, and Planck's Constant. Beyond the simple exersice of mathematical reasoning and the possibility of simpler mathematics. As is the case with other Natural Units, these units have been found to also coincide with when both quantum effects and gravity become significant. Making traditional theories fall apart, as no one knows how these two things will interact.

Using this method, the force unit becomes

$$\frac{c^4}{G}$$

Obviously an extremely large number, in newtons it is roughly two times ten to the power of 44.

Many have conjectured that this is the largest possible force that can occur. Others have disputed. Others have found that

according to traditional General Relativity it is simply a fourth of the Planck Force.

Until a theory of Quantum Gravity has been resolved, no one will know.

49.7.3 Striking the Earth with a Planck Force

While obviously, without a theory of quantum gravity, no one really know what will happen with a Planck Force, but using General Relativity to guess should be fun.

I used Einstein's toolkit, a very helpful open source numerical general relativistic code base, and added a thorn to strike an object at a point with a force. Then I edited their Neutron Star simulation to be Earth adjacent, obviously not exact but with such extreme cases I doubt it would make much of a difference.

Doing this, was surprisingly difficult. I thought it would be the easiest part of this video, but I was wrong.

after running it I go this

(Insert movie)

While you likely can't read the numbers, and they are very strange units at that. So I will explain, very quickly the of contact very quickly explodes into a vacum at relativistic speeds. Then, as the relativistic mass adds up, gravity slowly overcomes the pressure wave. Collapsing it into a small point. A point roughly the size of a single mesh in the simulation. Thus, including things such as the likely fusing of particles, neutron degenerate matter, or if it would collapse into a black hole would require a more advanced simulation.

There is another problem. The force being so great would lead to particles reaching super-luminal speeds before the next time step. Leading to the simulation coming to errors as the

mass become imaginary. The creators of the toolkit obviously came to a solution, but I am not exactly sure what their solution is. Thus, how accurate this simulation is, is knowledge beyond mine.

I tried to decrease the computational complexity and decrease the time-step substantially. I made the simulation symmetric so that I only had to simulate one eighth of the Earth, 2 dimensional rather than 3d, and increase the size of the mesh.

I eventually got to a single time step being one ten trillionth of a second. Still getting the same warning. So, I eventually had to accept the fact that I don't know. I would think that a collapse would happen but not into a black hole. Likely a mini-star that would eventually explode out, but that is something that I can't confirm.

To do so would require much more than my computer and messing with Einstien toolkit.

49.7.4 Assumptions about a Universe

While that was rather disappointing, there is still the question of the universe. Which is surprisingly much simpler.

Anyways, the we need to make assumptions about said universe.

One, our current observable universe, 93 billion light years in diameter, is the size we will use. Assuming that matter outside of such volume would have a negligible effect on the simulation.

Two, we assume that spacetime is mostly flat. Most observations put that the universe has an extremely small overall curvature, but the small change likely wouldn't change the results of the simulation much.

Three, take dark energy to be constant, such as a vacum

energy model. While recent studies despute this, they have trouble coming up with exact values so for now I will leave it as is to simplify.

Four, with dark matter, we take the estimated values of both Hubble's estimations of dark matter and Shoe's as well.

Five, we take a relatively isotropic density in the universe. Meaning a relatively constant one, rather than variable.

Six, when we later talk about the energy of the force, assume that it is some kind of magic or from another universe. As it is "creating" or adding energy to the universe that was not there previously.

Now that we have these.

49.7.5 Modeling a Universe

Now, how to actually destroy a universe. Being filled with so much empty space it is pretty obvious you can't just punch it and blow it up. The small quarks in the plasma might hit a couple of things, but would be pretty useless and definitely wouldn't make it to other galaxies. So, that leads us to our solution. In the previous simulation, the Earth collapsed from gravity, so it is conceivable that a more intense force would collapse into a black hole.

So, first we work backwards. To get force we can take energy, and to get energy we can find the analogous mass needed to overcome dark energy.

Starting with the FRW Friedmann equation:

$$H(a)^2 = H_0^2 \left(\Omega_{r0}\, a^{-4} + \Omega_{m0}\, a^{-3} + \Omega_{\Lambda 0}\right)$$

Here we will compress the parentheses into $E(a)$ for simplification.

Next, we take the Schwarzschild-de Sitter metric function

for a relatively flat expanding space time.

$$f(r) = 1 - \frac{2GM}{c^2 r} - \frac{\Lambda r^2}{3}$$

Here, one can take the derivative and find when that derivative is equal to 0 to find the point in which the universe becomes static to be

$$r_c = \left(frac3GM\Lambda c^2\right)^{\frac{1}{3}}$$

Then solving for mass when the radius is the current size of the universe we get

$$M_{crit} = \frac{\Lambda c^2}{3G} R_{obs}^3 = \frac{\Omega_{\Lambda 0} H_0^2}{G} R_{obs}^3$$

Now for the exact solution we take

$$ds^2 = -c^2 dt^2 + \frac{R'(r,t)^2}{1 + 2E(r)/c^2} dr^2 + R(r,t)^2 d\Omega^2$$

Here we take the Lemaitre-Tolman-Bondia metric, an exact solution for Einstein's equation for a lump of matter with no assumed symmetry.

(show algebraic manipulation)

Here we see that the structure of the shell makes no difference, only the total mass of it.

From here simple algebra shows allows us to to find the energy needed to lead to an eventual collapse and the function that can be solved to find when it crosses.

To actually solve, we need values. For the hubble constant and the mass and energy of the universe. Using the two values from Hubble telescope and Shoe, we use Bayes uncertainty to create a Monte Carlo simulation to find the list of values.

49.7.6 Results

From here, we run the simulation. Getting the below results

(results)

with the total energy needed to over come

49.7.7 Interprating the Results

Now, the purpose of this, wasn't to find the energy, but rather the force. I know most people watching this know the general work formula, but in the case of relativity, this doesn't always work. Though, with a couple of simplifications, we will end up with a just as simple formula.

We start with a practically flat spacetime to make the calculations simple.

From here, the fact of the conservation of energy and momentum, makes it so that the stress energy tensor, which has both energy and momentum is clear. Derived to be

$$\nabla_\mu T^{\mu\nu} = f^\nu$$

Where we take the contravairiant derivative of the stress energy tensor to get the force.

where we can expand to

$$f^\nu = \frac{\partial T^{0\nu}}{\partial x^0} + \frac{\partial T^{1\nu}}{\partial x^1} + \frac{\partial T^{2\nu}}{\partial x^2} + \frac{\partial T^{3\nu}}{\partial x^3}$$

where we can take the first term to find that this is simply the energy added by the force by volume. Then the other three terms simplify to be

$$f^i = \frac{1}{c}\frac{\partial T^{0i}}{\partial t} + \partial_j T^{ij}$$

which s essentially the change in momentum density plus the momentum flowing through boundary.

To simplify the calculations, we will focus just on the energy density portion. The result would be the same just differing math. Say the change in energy is 3.6 times ten to the power of 71. This force is spread through a volume of 1 meter, energy density and thus force would be 3.6 times 10 to the power of 71 Newtons.

As I have mentioned, this is an extreme simplification, but when working with energies such as these, there is really no physical object that would lead to the initial energy or the pressure gradient to have any non-negligible effect.

So now we have the force. 36 followed by 70 zeros Newtons.

About 67 orders of magnitude beyond the average punch of 1,000 Newtons. About 66 orders of magnitude above the highest ever recorded punch of 69,000 Newtons.

For comparasion, this is 25 orders of magnitude above the estimated energy release of the big bang.

On the order of comics, the best feat that I could come up with a real number too, not estimations based on the wacky comic physics would be superman pulling a chain of planets as very quick speeds. With other people's estimates coming at 10^{38} to 10^{40} newtons. Still on the upper end being 31 orders of magnitude away. Humans are closer to the superman's feat than superman is to the calculated feat.

49.7.8 General Questions

There are a couple of questions that I can anticipate. If gravity moves at the speed of light, how would that changes things. The answer is not an insane amount. Red shifting and losing energy is only for waves, the constant gravity would not lose energy. Our model make sure the gravity overcomes dark en-

ergy by definition. The main change would be the change in the radius of the universe and the possibility of parts of the currently observable universe going past the cosmic horizen where gravity would never be able to reach it. The two aspects conteract one another, and since we are working with such large numbers I assume they wouldn't change much. Though, as always, I could always be wrong.

Thank you for watching

49.8 Over Engineering Celestial Mechanics: WHCKL model

49.8.1 Planning

1. Introduction
2. Scope
3. Deriving WH
4. WH Maping
5. Deriving Symplectic correctors
6. using variations
7. Splitting method
8. Lazy Kernel
9. General Improvements
10. Comparing to other methods
11. video of solar system

49.8.2 Introduction

(general animate introduction)

Hi everyone

49.8.3 Scope

Essentially the scope and goals of this project are to develop a celestial mechanics numerical model that is mathematically complex enough to be a challenge. To slightly constrain the scope, the goal will be to allow for highly efficient long term keplarian celestial mechanics. On the order of millions of even billions of years with relative accuracy.

After some research, the clear, for very long term celestial mechanics the clear modern winner is Wisdom-Holman

Then, when researching this program I found that REBOUND, an open source code for celestial mechanics found ways to improve its accuracy that introduced new fun challenges. Such as symplectic correctors, variational methods, splitting methods, and even something they call a lazy kernel. So, let us get into it.

49.8.4 Deriving WH

To derive Wisdom-Holman, we start with a traditional N-body hamiltonian in inertial cartesian coordinates

$$H(q,p) = \sum_{i=0}^{N-1} \frac{P_i^2}{2m_i} - G\sum_{i<j} \frac{m_i m_j}{\|q_i - q_j\|}$$

With i = 0 to be the central star, and the rest being the planets.

From here, the hamiltonian is not sperable, so we make

a canonical transformation into heilocentric canonical coordinates.

Where we define $Q_i = q_i - q_0$ where essentially Q is the position of the planets in refrence to the sun. The corresponding canonical momenta doesn't change. $P_i = p_i$.

With the exeption of the sun which changes by the transformation. $P_0 = p_0 + \sum_{i=0}^{N-1} P_i$.

Then we define capital M to be the cumulative mass

From here it is simply derived that the Hamiltonian becomes

$$H = \frac{P_0^2}{2M} + \sum_{i=1}^{N-1}\left[\frac{P_i^2}{2m_i} - \frac{Gm_0M_i}{Q_i}\right] + \sum_{i=1}^{N-1}\frac{P_i^2}{2m_0} - G\sum_{i<j,i,j>0}\frac{m_im_j}{\|Q_i - Q_j\|}$$

Here we can divide the hamiltonian into $H = H_{sun} + H_{kepler} + H_{int}$.

This is the traditional method; though, for later reasons we will get into, rather than Heilocentric coordinates we will use Jacobi, meaning in refrence to the center of mass rather than the central body.

Where the center of mass of all particles interior to the i-th particle is R_{i-1}, thus their coordinates become $r_i' = r_i - R_{i-1}$.

Because jacobi coordinates are linear functions of cartesian, the rest of the quantities, such as velocity and acceleration change in the same way. Momenta is different however.

$$p_i' = m_i'\dot{r'}$$

where the reduced mass is equavlent to.

$$m_i' = m_i\frac{M_i - 1}{M_i}$$

Using these coordinates the N-body hamiltonian instead becomes

$$H = \sum_{i=0}^{N-1} \frac{p_i'^2}{2m_i'} - \sum_{i=0}^{N-1} \sum_{j=i+1}^{N-1} \frac{G m_i m_j}{\|r_i - r_j\|}$$

Keeping the potential term in cartesian coordinates and only writing the kinetic in jacobi.

Next, for the sake of the algebra we add and subtract the term

$$H_{\pm} = \sum_{i=1}^{N-1} \frac{G m_i' M_i}{\|r_i'\|}$$

This grouping leads to the Hamiltonian

$$H = \underbrace{\frac{(p_0')^2}{2m_0'}}_{H_0} + \underbrace{\sum_{i=1}^{N-1} \left(\frac{(p_i')^2}{2m_i'} - \frac{G\, m_i' M_i}{|r_i'|} \right)}_{H_{\text{Kepler}}} + \underbrace{\sum_{i=1}^{N-1} \frac{G\, m_i' M_i}{|r_i'|} - \sum_{i=0}^{N-1} \sum_{j=i+1}^{N-1} \frac{G\, m_i m_j}{|r_i - r_j|}}_{H_{\text{Interaction}}}.$$

Here one can further split the kepler term into

$$(H_{ekpler})_i = \frac{p_i'^2}{2m_i'} - \frac{G m_i' M_i}{\|r_i'\|}$$

each Hamiltonian describes the keplarian motion of the i-th particle around the center of mass of all interior particles.

After some more algebra we can split the jacobian into two parts, one that can be easily computed in cartesian and the other in Jacobi coordinates

$$H_{int} == \sum_{i=2}^{N-1} \frac{G m_i' M_i}{\|r_i'\|} - \sum_{i=0}^{N-1} \sum_{j=i+1}^{N-1} \frac{G m_i m_j}{\|r_i - r_j\|}$$

We now have three speratable equations that a all much easier to integrate than the highly non-linear one we stated with. Where H_0 is motion along a straight line, H_{Kepler} is the hamiltonian for kepler orbits, and H_{int} is the interaction of particles outside of the i-th particle. To note, this methodology only works for near keplarian orbits, in say a three body binary this method would be quite unphysical quite quickly.

49.8.5 WH Maping

We do this splitting as the origional equation's equation of motion have no analytical solutions.

$$H(q,p) = \sum_{i=0}^{N-1} \frac{P_i^2}{2m_i} - G\sum_{i<j} \frac{m_i m_j}{\|q_i - q_j\|}$$

Thus, approximate solutions are needed. Many methods exist but the method we have been developing is obviously splitting the Hamiltonian into smaller parts that can be integrated easily.

Analytical solutions can be found for the evolution of the system under each individual Hamiltonian.

Thus, to create the symplectic integrator using operator split method.

To do this we start with the familiar Liouville operator

$$L_H f = \{f, H\}$$

If you are not familiar, this is simply the same as saying that

$$\dot{f} = \{f, H\}$$

This obviously looks like an ordinary differential equation that has the formal solution of

$$f(t+h) = e^{hL_H} f(t)$$

Because this equation is for a differential operator not a variable, this exponential opperator is instead defined through:

$$e^{hL_h} = I + hL_H + \frac{H^2}{2!} L_H^2 + ...$$

where the key difference is that differential operators do not commute as traditional variables do. This here is the backbone of the symplectic integrator, to approximate this non-commutable equation.

Using this we can take our origional Hamiltonian in Poisson form to be:

$$L_H = L_0 + L_{Kep} + L_{int}$$

let us turn those into A, B, and C:

$$L_H = A + B + C$$

Then add A and B to make a and turn C into B

$$L_H = A + B$$

then the exact propegator is obviously

$$e^{h(A+B)}$$

Now that we have that we can use the generic strang method.

$$\boxed{e^{\Delta t(A+B)} \approx e^{\frac{\Delta t}{2}A} e^{\Delta t B} e^{\frac{\Delta t}{2}A}}$$

Or the drift-kick-drift. Drifting along the kepler motion and motion of the center of mass and kicking using the inteaction

between planets.

49.8.6 Symplectic Correctors

Another method we will add are symplectic correctors. A symplectic corrector is simply a small adjustment to the coordinates that one does before and after running a symplecitc integrator so that the numerical trajectory more closely matches the real physical trajectory.

Because they are only applied at the very beginning and end, one can add arbitrarily higher order correctors with negligible effects on the speed.

We choose an eleventh order corrector that effectively brings the effective order of the integrator from second order to fifth, while barely changing the time it takes to run the program.

The exact benefits of the correctors change. As in Earth mass planets, beyond a fifth order corrector there is little change, but in Jupiter mass ones an eleventh order method is quite helpful.

49.8.7 Building the Code

When building this code, taking optimization to heart I used FORTRAN. A computational science coding language that is very fast. Optimized it is comparable and arguably faster than optimized C++, it also has the added benefit of being much easier to read.

I found a paper that already derived the best ways to optimize the subcalculations, specifically coordinate transformations between Jacobi and Cartesian, Newtons Method, c-functions, and a couple of other things.

I won't bore you with the details, but I eventually added the fiften of the most massive bodies that orbit the sun. The

eight planets, Ceres, Pluto, Haumea, Makemake, Gonggong, and Erirs. Yes, those are real names of dwarf planets.

I didn't add moons as this simulation, by using the kepler hamiltonian, assumes that the body orbits the center of mass of interior object, but moons obviously don't do that. Such, the simulation is not as physical for say the Earth whose moon is a significant compared to its mass. But it still works as a fun project with beyond that being very accurate.

After that I set the time step to roughly 18 days, the max speed while still being accurate. I did underestimate how many years the simulation was able to complete so Mercury did become unbounded near the end

49.8.8 Results

Here we now come to the fruits of our labor. I really should have worked on another program to have something to compare to, as these numbers are pretty meaningless on their own. Just that they are about as efficient as it gets...

Well, the purpose was the process not the result

49.8.9 Watching it

Though, to get a little extra out of it, let us watch the animation of a couple of the planets.

49.9 Lie Operator Splitting: One of the ten Pillars

49.9.1 Introduction

Hi, welcome to a semi-advanced Introduction into Lie Operator splitting, a mathematical Approach.

49.9.2 Scope

Essentially, the scope of this video being an introduction into Lie Operator splitting, a method that I have used several times in past videos.

A good introduction to its importance is a quote by Dr. Strang, "If, as has sometimes been argued, there are only ten big ideas in numerical analysis and all the rest are merely variations on those themes, splitting is undoubtedly one of them." as it is at the foundation of so much of numerical analysis. I primarily have used it for symplecic integrators, but its uses are far beyond that. Other structure preserving methods, statistical mechanics, nearly integratable systems, optimization, quantum mechanics, and really anything that can be described by differential equations without easily found analytical solutions.

n fact, one could say that, at least since Descartes stated his four rules of logic in the Discours de la méthode, the notion of sub-dividing a complicated problem into its simpler constituent parts, solving each one of them separately and combining those separated solutions in a controlled way to get a solution to the original overall problem constitutes a guiding principle in all areas of science and philosophy.

49.9.3 Motivating Operator Splitting

As in most cases in applied mathematics, the easiest start to defining new mathematics is to motivate their existence. To confine this agumentation, let us look at the simpliest case of a non-integratable Hamiltonian. An Anharmonic oscillator defined through this equation:

$$H(q,p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 + \lambda q^4$$

Here, this equation's derivatives have no elementary closed-form solution

But we can split it into its potential and kinetic energy

$$H_T = \frac{p^2}{2m}$$

$$H_V = \frac{1}{2}m\omega^2 q^2 + \lambda q^4$$

Here these problems have trivial solutions.

Thus, if one wishes to solve them together one can trivially see that the Hamiltonian vector field is

$$X_H = X_T + X_V$$

(show vector fields adding together)

The corresponding Liouville operators are

$$L_H = L_T + L_V$$

Then if we introduce a phase-space vector

$$z = \begin{pmatrix} q \\ p \end{pmatrix}$$

Solving the differential equation we get

$$z = e^{h(L_T+L_V)}$$

Solving this may seem easy. Your first instinct would probably be solve the inside the parenthasies, but this is obviously the same thing as integrating the whole, making the earlier splitting useless.

Another methodology would be through commuting

$$e^{hL_T}e^{hL_V}$$

But, as this doesn't work as they do not commute.

$$[L_T, L_V] \neq 0$$

We can see this directly for solving for q

$$[L_T, L_V]q = -\frac{V'(q)}{m} - 0$$

Thus this take naive one-step composition to be problematic.

I will note, this traditional splitting can work, it is only first order accurate and has a local error of $O(h^2)$

49.9.4 Deriving Strang Splitting

Taking our differential equation earlier, replacing T and V with A and B:

$$\dot{z} = (A + B)z$$

Using taylor expansion we can expand to:

$$z(t+h) = \left[I + h(A+B) + \frac{h^2}{2}(A^2 + AB + BA + B^2) \right] z + O(h^3)$$

Thus, any second order method must reproduce the middle section.

To do that, let us assume a three state composition that has factors a and b

$$S(h) = e^{ahA} e^{bhB} e^{ahA}$$

We can thus expand the exponentials to

$$e^{ahA} = I + ahA + \frac{a^2 h^2}{2} A^2 + O(h^3)$$

$$e^{bhB} = IbhB + \frac{b^2 h^2}{2} B^2 + O(h^3)$$

Multipliying together we get

$$S(h) = I + h(2aA + bB) + O(h^3)$$

Where as we previously noted that the exact solution should be

$$I + h(A+B) + O(h^3)$$

Therefore, a should equal one half while b should equal 1.

Creating a second order method with a local error of $O(h^3)$.

An importaint thing to note, while if you assume that the lower case a's could have differing values, we gain infinitely many solutions, but these solutions lose symmetry that would then lead to another first order equation.

49.9.5 Adjoint formulism

self adjoint formulisms are very simple, they are when a numerical method satisfies

$$\psi_h = \psi_h^\star = (\psi_{-h})^{-1}$$

Where psi star is the inverse composition of psi. ie if $\psi = x^1 \circ x^2$ then $\psi^\star = x^2 \circ x^1$.

These self-adjoin is sometimes called time-symmetric as ...

49.9.6 Higher Order Splitting and Composition

"it is a meta-theorem of numerical analysis that second order methods often achieve the right balance between accuracy and complexity." - Strang

While most often 2nd order is used, it isn't the only one. There are methods for developing higher order methods, particularly used many places. The most obvious being simulations of the solar system.

So, if we start with the strang splitting

$$S_h^{[2]} = \varphi_{h/2} \circ \varphi_h \circ \varphi_{h/2}$$

The composition is then

$$\psi_h = S_{\gamma_s h} \circ S_{\gamma_{s-1} h} \circ ... \circ S_{\gamma_1 h}$$

The order is atleast three if

$$\sum_{j=1}^{s} \gamma_j = 1$$

$$\sum_{j=1}^{s} \gamma_j^3 = 0$$

The smallest number of j that satisfies this equation is three. By imposing a symmetry where $\gamma_3 = \gamma_1$, one can relatively easily derive a solution for γ_1 and γ_2.

Futher, in general one can take the equation

To find arbitrarily high orders of order 2k where $k \geq 2$.

There are many other ways to find higher order splitting, a more detailed observation of these will be in the description.

49.10 Discontinuous Galerkin Method: Self Gravitating Euler Equations

49.10.1 Planning

- Introduction
- Scope
- Motivating DG
- General Introduction to DG
- Physical Model Model
- Deriving Projection
- Structure preserving LDG
- Time Discretization
- Ossilation Elimination
- Structure Preserving Properties

- The Code itself(Flowchart)
- Results

49.10.2 Introduction

Hi, welcome to Discontinuous Galerkin Method: a focus on Self-Gravitating Euler Equation

49.10.3 Scope

Here we address the scope of this video. It is meant as an introduction into DG, using an example to help facilitate understanding. For both the purposes of challenging myself, I have chosen a particularly challenging one I have recently read about.

Taking a self-gravitating fluid and using the structure preserving capacity of DG to preserve positivity, ie that pressure, mass, energy, etc are all positive at all points, and well balanced, ie compressible fluid that admits hydrostatic equilibrium. For this video I won't go to far into the math of these more advanced subjects.

Also, for the rest of the video I will refer to the DG are "DG"

49.10.4 Introduction to DG

Before we get into the math, we will take a more intuitive look before we get into the math. A good way to start is to look at the whole family tree. I assume anyone who is watching this has heard of and understands finite difference models; approximate spacial derivatives with stencils on point values. Further, we get to finit volume which trackes cell averages

and evolve them via fluxes computed at cell faces from a Riemann solver. Then we come to continuous finite element or Galerkin who represents the solution as one global piecewise-polynomial function, continuous across element boundaries. Higher order and structural preserving methods are baked into the equations. The separation between DG and CG is that these polynomials are discontinuous, exactly as you would expect. The reason for this is that in many cases, the physical situations are partially discontinuous. Think of an explosion in a fluid, it takes time for the explosion to effect outer areas of the fluid, more time than would be found through a continuous polynomial.

essentially, we divide the space into elements

$$\Omega = \cup_e \Omega_e$$

We approximate the solution to the differential equation using polynomials

$$u_h(x,t) \sum_{j=0}^{N} u_j(t)\phi_j(x)$$

Next we write the PDE in its weak form

$$\frac{\partial u}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{F} = S(u)$$

Multiply the test function ϕ then integrate by parts

$$\int_\Omega \phi \frac{\partial u}{\partial t} dV - \int_\Omega \boldsymbol{\nabla}\phi \cdot \mathbf{F} dV + \int_{\partial\Omega} \phi \mathbf{F} \cdot \mathbf{n} ds = S(u)$$

They communicate between elements via fluxes, that can be defined via your physical system

Then we can generalize the fluxes into the residuals from the weak form equation used previously used to rearange into

$$\mathbf{R}(\mathbf{U}) = \mathbf{R}_{\text{vol}} + \mathbf{R}_{\text{surf}} + \mathbf{R}_{\text{src}} \tag{49.7}$$

$$\mathbf{R}_{\text{vol}} = \int_K \nabla \boldsymbol{\phi} \cdot \mathbf{F}(\mathbf{u}_h)\, dV \tag{49.8}$$

$$\mathbf{R}_{\text{surf}} = -\int_{\partial K} \boldsymbol{\phi}\, \mathbf{F}^*\, dS \tag{49.9}$$

$$\mathbf{R}_{\text{src}} = \int_K \boldsymbol{\phi}\, \mathbf{S}(\mathbf{u}_h)\, dV \tag{49.10}$$

$$\mathbf{R}(\mathbf{U}) = \int_K \nabla \boldsymbol{\phi} \cdot \mathbf{F}(\mathbf{u}_h)\, dV - \int_{\partial K} \boldsymbol{\phi}\, \mathbf{F}^*\, dS + \int_K \boldsymbol{\phi}\, \mathbf{S}(\mathbf{u}_h)\, dV \tag{49.11}$$

$$M \frac{d\mathbf{U}}{dt} = \mathbf{R}(\mathbf{U}) \tag{49.12}$$

from here, using the basis functions from the mesh we get the mass matrix which is the matrix representation of the inner product on your finite-dimensional approximation space.

From here, the derivative of the functions as a function of time can be derived via

$$\frac{\partial u}{\partial t} = M^{-1} rhs(u_h)$$

Now, listening to all of this may seem fairly confusing, this is doubled by my admitted brevity, but I allow this brevity as I am going to take a concrete example as mentioned previously.

49.10.5 Physical Model

As mentioned earlier, we take the Euler-Poisson for a self-gravitating fluid, such as a toy model of a star. While keeping the simplicity of the model, I add a personal challenge by adding constraints of energy conservation and positivity. Using the general Euler-Poisson equation

insert long equation here

Then adding Polytropic steady-states and Lane-Emden equations in order to enforce EOS and positivity.

equation

I know I once again was very brief, but that is simply because the actual equations don't matter too much, just the general shape and view of them so that we can create the code.

49.10.6 Mesh and Projection

We start by assuming a uniformly space square/cube meshing

(image of square meshing)

We can then define n^+, n^- as the normal of the flux.

Next to define the projection, but to do that we define our test problem. Simply a weighting function within the polynomial space that allows us to integrate more easily.

(show examples in traditional methods)

Then we can define the projection matrix P via the L^2 projection:

$$\int_K uvdx = \int_K \mathbf{P}(u)vdx$$

Now we can start moving backwards by rearanging the equa-

tion to get

$$P(u)(x) = \sum c_j v_j(x)$$

By defining the mass matrix as

$$M_{ij} = \int_k \phi_i v_j$$

where phi of i is the basis function, we use the nodal basis function, coefeciencts fairly easily looked up.

we can also define the right side vector as

$$b_j = \int_K u v_j dx$$

We can rearagne our origional projection function

$$P(u)(x) = \sum (M^{-1}b)_j v_j(x)$$

49.10.7 Flux

We start with a basic peicewise function for flux of a fluid.

$$F^{hllc}(U_L, U_R; \mathbf{n}) = \begin{cases} \mathbf{F}(U_L), \text{if } 0 \leq S_L, \\ \mathbf{F}_{\star L}, \text{if } S_L \leq 0 \leq S_\star, \\ \mathbf{F}_{\star R}, \text{if } S_\star \leq 0 \leq S_R, \\ \mathbf{F}(U_R), \text{if } 0 \geq S_R, \end{cases}$$

This is the standard HLLC or Harten–Lax–van Leer–Contact flux solver. As this is more an introduction into DG than fluid dynamics I won't go through and expand all of these out, but essentially, it approximates the Remanian problem to approximate the flux of conserved quantities like energy, momentum, mass, etc.

I also slightly modify it in order to enforce Well-balencedness by adding the earlier Lane-Emden equation to the flux equation.

49.10.8 Scheme

Now building the actual numerical method. Before we get to the well-balanced, energy-conserving version, I want to build the plain scheme first, so that when I show you exactly how it fails near equilibrium later, you can point to the specific term responsible, rather than just taking my word for it.

Let's write out our conserved variables, flux, and source term the way we did generically before, but now specifically for the self-gravitating Euler system in d spatial dimensions:

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho\mathbf{u} \\ E \end{bmatrix}, \qquad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho\mathbf{u} \\ \rho\mathbf{u}\otimes\mathbf{u} + p\mathbf{I}_d \\ (E+p)\mathbf{u} \end{bmatrix}, \qquad \mathbf{S}(\mathbf{U}, \boldsymbol{\nabla}\phi) = \begin{bmatrix} 0 \\ -\rho\boldsymbol{\nabla}\phi \\ -\rho\mathbf{u}\cdot\boldsymbol{\nabla}\phi \end{bmatrix}.$$

Here $E = \rho e + \frac{1}{2}\rho|mathbfu|^2$ is the usual total (non-gravitational) energy, e is the specific internal energy fixed by an equation of state, and for an ideal gas $p = (\gamma-1)\rho e$. With this notation, our full model compresses down to

$$\mathbf{U}_t + \boldsymbol{\nabla}\cdot\mathbf{F}(\mathbf{U}) = \mathbf{S}(\mathbf{U}, \boldsymbol{\nabla}\phi), \qquad \Delta\phi = 4\pi G\rho.$$

ϕ is the gravitational potential and G the gravitational constant. That second equation, Poisson's equation, is the entire reason this problem is harder than an ordinary Euler solve: it isn't fixed but rahter generated by the fluid, it changes shape every time the density field changes, and every single timestep we need to re-derive a self-consistent gravitational field from whatever the density happens to look like at that instant.

49.10.8.1 Why local DG, specifically

There are two broad strategies exist for handling the gravity term numerically. One is to treat the Poisson equation as a pseudo-hyperbolic problem, giving it an artificial wave speed and a relaxation time so the same Riemann-solver machinery built for the Euler equations can be reused directly on the gravity sector too. The other, which is the one this scheme takes, is to solve the elliptic Poisson equation directly. Direct solves used to be avoided in this setting because they historically meant either an FFT-based Poisson solver (fast, but restricted to very particular, usually periodic, domain shapes) or a multipole expansion (flexible, but expensive and awkward to make high order). Local DG sidesteps both complaints: it was originally invented specifically to let a DG method handle equations with second derivatives without ever forming a second derivative explicitly, and it inherits all of the DG machinery's flexibility with boundary conditions and geometry for free. That's the actual technical justification for reaching for LDG here rather than, say, a finite-difference Poisson solver bolted onto a DG hydro code.

49.10.9 Standard LDG

So concretely: how do we get ϕ out of a DG method when Poisson's equation has a second derivative in it, and DG's whole weak-form machinery is built assuming one derivative per equation? This is exactly the "Local" in Local DG. Following Cockburn and Shu's original construction, we introduce an auxiliary variable standing in for the gravitational force itself,

$$\mathbf{g} = \boldsymbol{\nabla}\phi,$$

and rewrite the whole system as a genuinely first-order system, no second derivatives anywhere:

$$\begin{cases} \mathbf{U}_t + \boldsymbol{\nabla} \cdot \mathbf{F}(\mathbf{U}) = \mathbf{S}(\mathbf{U}, \mathbf{g}), \\ \mathbf{g} = \boldsymbol{\nabla}\phi, \\ \boldsymbol{\nabla} \cdot \mathbf{g} = 4\pi G\rho. \end{cases}$$

Now every equation is first order, so we throw the exact same weak-form recipe from earlier at each piece independently. We work on a mesh $\mathcal{T}_h$ of elements K, with $\mathcal{E}_I$ the interior faces and $\mathcal{E}_B$ the domain boundary. On a shared face $\mathcal{E}$ between two elements K^+ and K^-, with outward normals $\mathbf{n}^+, \mathbf{n}^-$, define the usual average and jump of a scalar function u and a vector function $\mathbf{q}$:

$$\{\{u\}\} = \tfrac{1}{2}(u^+ + u^-), \qquad [[u]] = u^+\mathbf{n}^+ + u^-\mathbf{n}^-,$$

$$\{\{\mathbf{q}\}\} = \tfrac{1}{2}(\mathbf{q}^+ + \mathbf{q}^-), \qquad [[\mathbf{q}]] = \mathbf{q}^+ \cdot \mathbf{n}^+ + \mathbf{q}^- \cdot \mathbf{n}^-.$$

On a single element K, testing the three equations against $v \in V_h^k$, $\mathbf{w} \in \Sigma_h^k$, and $\psi \in V_h^k$ respectively (piecewise polynomial spaces of degree up to k, scalar and vector valued), and integrating by parts exactly the way we did for the single scalar conservation law earlier, we get:

$$\int_K (\mathbf{U}_h)_t \, v \, d\mathbf{x} - \int_K \mathbf{F}(\mathbf{U}_h) \cdot \boldsymbol{\nabla} v \, d\mathbf{x} + \sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} \widehat{\mathbf{F}} \, v^{\mathrm{int}(K)} \, ds = \int_K \mathbf{S}(\mathbf{U}_h, \mathbf{g}_h) \, v \, d\mathbf{x},$$

$$\int_K \mathbf{g}_h \cdot \mathbf{w} \, d\mathbf{x} = \sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} \widehat{\phi}_h \, \mathbf{w}^{\mathrm{int}(K)} \cdot \mathbf{n}_{\mathcal{E},K} \, ds - \int_K \phi_h \boldsymbol{\nabla} \cdot \mathbf{w} \, d\mathbf{x},$$

$$\int_K \mathbf{g}_h \cdot \boldsymbol{\nabla}\psi \, d\mathbf{x} = \sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} \psi^{\mathrm{int}(K)} \, \widehat{\mathbf{g}}_h \cdot \mathbf{n}_{\mathcal{E},K} \, ds - \int_K 4\pi G\rho_h \, \psi \, d\mathbf{x}.$$

The superscript "$\mathrm{int}(K)$" just means: take the value from

inside the element K we're currently integrating over, as opposed to the value carried over from the neighbor on the other side of the face ("ext(K)"). This distinction matters constantly in DG because our solution is discontinuous across faces by construction, so "the value at the face" is genuinely ambiguous until we pick a numerical flux to resolve it.

The first equation is exactly the fluid solve we already built, with $\widehat{\mathbf{F}}$ the HLLC flux from the previous section, evaluated using $\mathbf{U}_h^{\mathrm{int}(K)}$ and $\mathbf{U}_h^{\mathrm{ext}(K)}$ as the left and right Riemann states. The other two equations are the "elliptic half" of the scheme: a mixed formulation of Poisson's equation.

49.10.9.1 Fluxes for the elliptic part

Here's a subtlety that trips people up the first time they see LDG for an elliptic problem: gravity, unlike a fluid wave, doesn't propagate in a preferred direction. There's no upwind side to a Poisson equation the way there's an upwind side to a shock. So HLLC, or any Riemann-solver-based flux, simply doesn't apply here. Instead, following the standard LDG-for-elliptic-problems recipe, we use plain averages of ϕ_h and $\mathbf{g}_h$ across the face, corrected by a jump-penalty term that keeps the resulting linear system well posed:

$$\widehat{\mathbf{g}}_h = \{\{\mathbf{g}_h\}\} - C_{11}[[\phi_h]] - \mathbf{C}_{12}[[\mathbf{g}_h]], \qquad \widehat{\phi}_h = \{\{\phi_h\}\} + \mathbf{C}_{12} \cdot [[\phi_h]],$$

with C_{11} a scalar penalty and $\mathbf{C}_{12}$ a vector penalty, both free parameters we get to choose. On a tensor-product (rectangular) mesh, a standard, well-behaved choice is $C_{11} = 1$ with $\mathbf{C}_{12} = (0.5, 0.5)^T$ in 2D or $(0.5, 0.5, 0.5)^T$ in 3D.

If you assemble the full linear system for $(\mathbf{g}_h, \phi_h)$ implied by

these two weak forms, it has a block, saddle-point structure,

$$\begin{bmatrix} K & B \\ -B^T & S \end{bmatrix} \begin{bmatrix} \mathbf{g}_h \\ \phi_h \end{bmatrix} = \begin{bmatrix} g \\ f \end{bmatrix},$$

which is annoying to solve directly because it's not symmetric. The standard move is a Schur complement: eliminate $\mathbf{g}_h$ algebraically to get a smaller system purely in ϕ_h,

$$A\phi_h = \left(S + B^T K^{-1} B\right) \phi_h = f + B^T K^{-1} \mathbf{g}_h,$$

and now A is symmetric positive definite, which means a Cholesky factorization or a (preconditioned) conjugate-gradient solve both work reliably. Since our mesh never moves or changes over the course of the simulation, that Cholesky factorization can be computed exactly once, at the very start of the run, and reused at every single timestep afterward — the sparsity pattern and the matrix entries here depend only on geometry and the penalty parameters, never on the current fluid state. (For a periodic domain the block structure changes slightly, the reduced system loses its symmetry, and an LU factorization is used instead — same idea, slightly different linear algebra.) This is a genuinely nice practical property of building gravity this way: the expensive linear-algebra setup work happens once, not once per timestep.

Initial conditions for the fluid variables are set by the L^2 projection $\mathbf{P}$ we derived earlier: $\rho_h^0 = \mathbf{P}\rho^0$, $(\rho\mathbf{u})_h^0 = \mathbf{P}(\rho\mathbf{u})^0$, $E_h^0 = \mathbf{P}E^0$. Wrapping the whole thing in a forward-Euler step in time gives a complete, working numerical scheme:

$$\int_K \left(\frac{\mathbf{U}_h^{n+1} - \mathbf{U}_h^n}{\Delta t} \right) v \, d\mathbf{x} - \int_K \mathbf{F}(\mathbf{U}_h^n) \cdot \boldsymbol{\nabla} v \, d\mathbf{x} + \sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} \widehat{\mathbf{F}}^n \, v^{\mathrm{int}(K)} \, ds = \int_K \mathbf{S}(\mathbf{U}_h^n, \mathbf{g}_h^n) \, v \, d\mathbf{x}.$$

Density, momentum, and energy evolve; gravity is re-derived

from the freshly updated density at the top of the next step, and the loop repeats. This is what I'll call the *standard scheme* for the rest of the video. Let me be fair to it: this is a real, working, high order accurate method. Run it on a smooth manufactured solution and you'll see the convergence rates you'd expect from the polynomial degree you chose. Run it on a genuine shock and, with an appropriate limiter, it captures the shock. It is not a broken scheme.

It does, however, have one specific, well-defined blind spot, and that blind spot is the entire reason the rest of this video exists.

49.10.10 Structure preserving DG

A huge fraction of astrophysically interesting configurations spend almost all of their time sitting extremely close to hydrostatic equilibrium: gravity pulling material inward, pressure pushing back outward, velocity essentially zero, the two forces balanced to enormous precision. That exact balance is

$$\mathbf{u} = \mathbf{0}, \qquad \boldsymbol{\nabla} p = -\rho \boldsymbol{\nabla} \phi, \qquad \Delta \phi = 4\pi G \rho.$$

A common and analytically tractable family of such equilibria is the *polytropic* equilibrium, where pressure and density are locked together by $p = \kappa \rho^{\gamma}$ for some positive constant κ. Substitute this and the hydrostatic balance into Poisson's equation and, after the classic Lane-Emden change of variables $\rho = \lambda \theta^n$ with $\gamma = (n+1)/n$, you land on the Lane-Emden equation. In spherical symmetry, with r the (rescaled) radius, it reads

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = -\theta^n.$$

This has closed-form solutions only for a few special integer values of n: $\theta_0(r) = 1 - r^2/6$ for $n = 0$ (an incompressible, $\gamma = \infty$

fluid); $\theta_1(r) = \sin(r)/r$ for $n = 1$ (a $\gamma = 2$ fluid, and the one this video's numerical tests will actually use, since it's smooth, non-trivial, and exact); and $\theta_5(r) = 1/\sqrt{1 + r^2/3}$ for $n = 5$ ($\gamma = 6/5$). For any of these, the full equilibrium profile is recovered from

$$\rho^e = \lambda\,(\theta(\mathbf{x}))^n\,, \qquad \mathbf{u}^e = \mathbf{0}, \qquad p^e = \kappa(\rho^e)^\gamma, \qquad \phi^e = -\frac{\kappa\gamma}{\gamma - 1}(\rho^e)^{\gamma-1}.$$

This is the analytic "star in a box" that our scheme will be asked to hold perfectly still.

Here's the actual failure mode, stated precisely. The standard scheme discretizes the pressure-gradient contribution to the momentum flux and the gravity source term $-\rho\boldsymbol{\nabla}\phi$ using two completely independent numerical procedures: one is a DG surface/volume flux integral, the other is a DG volume source integral. There is no algebraic reason these two, separately truncated approximations should cancel exactly, even when the exact continuous quantities they're approximating do cancel exactly. What's left behind is $O(h^{k+1})$ discretization error sitting right on top of the equilibrium, at every single timestep. If the physical phenomenon you actually care about — a stellar oscillation mode, a slowly growing instability — is smaller in amplitude than that leftover truncation error, it's simply drowned out, and the only fix available to the standard scheme is to keep refining the mesh until the noise floor drops below your signal, which gets expensive fast, especially in 3D. This is the well-balanced (WB) problem, and a scheme that avoids it is called well-balanced.

There is a second, related failure worth separating out clearly, because it has a genuinely different cause: total energy conservation (TEC). Because gravity here is self-generated rather than externally imposed, the gravitational potential energy is honestly part of the system's total energy budget, and near equilibrium it's typically a large negative number nearly can-

celling a large positive kinetic-plus-internal energy. Any small, systematic energy leak in the numerical scheme is no longer a rounding-error-level nuisance; it can be comparable to, or larger than, the actual dynamical energy of the perturbation you're trying to resolve, especially over long integrations or during genuine gravitational collapse, where nonlinear growth will happily amplify a small numerical energy leak into a qualitatively wrong answer.

We want both properties simultaneously, and it turns out they require two largely independent fixes.

49.10.10.1 Fix one: splitting the potential for well-balancedness

The core idea is to split both density and gravitational potential into a known, static equilibrium piece and an evolving perturbation piece:

$$\rho = \rho^e + \rho^\delta, \qquad \phi = \phi^e + \phi^\delta, \qquad \mathbf{g}^e = \boldsymbol{\nabla}\phi^e, \quad \mathbf{g}^\delta = \boldsymbol{\nabla}\phi^\delta,$$

with ρ^e, ϕ^e computed once at initialization from the analytic equilibrium above (via the L^2 projection: $\rho_h^e = \mathbf{P}\rho^e$, and ϕ_h^e from actually solving the discrete Poisson problem for $4\pi G\rho_h^e$) and then frozen, unchanged, for the entire simulation. Only the perturbation fields $\rho_h^\delta, \phi_h^\delta$ are recomputed every timestep. On its own this split buys us nothing; the trick is entirely in how the numerical flux and source term are built around it.

49.10.10.1.1 The modified HLLC flux.

Recall the contact property of HLLC: feed it two states at rest with equal pressure, $\mathbf{U}_L = (\rho_L, \mathbf{0}, p/(\gamma-1))^T$ and $\mathbf{U}_R = (\rho_R, \mathbf{0}, p/(\gamma-1))^T$, and it collapses exactly to

$$\mathbf{F}^{hllc}(\mathbf{U}_L, \mathbf{U}_R; \mathbf{n}) = (0,\ p\,\mathbf{n}^T,\ 0)^T.$$

You can check this directly from the definitions: with zero velocity on both sides, the estimated wave speeds satisfy $S_L = -c_L \leq 0 \leq c_R = S_R$ and the middle speed $S_\star$ works out to exactly 0, which collapses the star-region states to something that leaves only the pressure term surviving in the momentum flux, with mass and energy fluxes vanishing identically. This "nothing is happening, only pressure pushes on the face" behavior is precisely the discrete analogue of hydrostatic balance, and it's the property the well-balanced modification is built to exploit.

The fix: instead of feeding HLLC the raw conserved states, we rescale each side by the ratio of a face-averaged equilibrium pressure to that side's own cell-interior equilibrium pressure:

$$\widehat{\mathbf{F}} = \mathbf{F}^{hllc}\left(\frac{p_h^{e,\star}}{p_h^{e,\mathrm{int}(K)}}\,\mathbf{U}_h^{\mathrm{int}(K)},\quad \frac{p_h^{e,\star}}{p_h^{e,\mathrm{ext}(K)}}\,\mathbf{U}_h^{\mathrm{ext}(K)}\;;\;\mathbf{n}_{\mathcal{E},K}\right), \qquad p_h^{e,\star} = \tfrac{1}{2}\left(p_h^{e,\mathrm{int}(K)} + p_h^{e,\mathrm{ext}(K)}\right)$$

Why this particular rescaling? Because at exact equilibrium, $\mathbf{U}_h^{\mathrm{int}(K)}$ and $\mathbf{U}_h^{\mathrm{ext}(K)}$ are both zero-velocity states with pressures $p_h^{e,\mathrm{int}(K)}$ and $p_h^{e,\mathrm{ext}(K)}$ respectively, and generally $p_h^{e,\mathrm{int}(K)} \neq p_h^{e,\mathrm{ext}(K)}$ — the equilibrium pressure genuinely varies across a face, that's just what a stratified star looks like. The contact property only fires cleanly when both sides share the *same* pressure. So we rescale each side up (or down) to the common reference value $p_h^{e,\star}$ before calling HLLC, apply the contact property there (where it gives exactly $(0,\, p_h^{e,\star}\mathbf{n}^T,\, 0)^T$), and this becomes the discrete flux we actually use. The rescaling is undone implicitly, because it's baked directly into the momentum source term derivation next, so the two exactly cancel at equilibrium.

49.10.10.1.2 The well-balanced source term. On the momentum equation, the full source is $\mathbf{S}^{[2]} = -\rho\boldsymbol{\nabla}\phi = -\rho\boldsymbol{\nabla}\phi^e - \rho\boldsymbol{\nabla}\phi^\delta$. The perturbation piece is discretized in the ordinary, direct way. The equilibrium piece is where the actual balanc-

ing act happens, and it's worth walking through the algebra in full, because every line of it exists for a specific reason:

$$\int_K -\rho \boldsymbol{\nabla}\phi^e \, v \, d\mathbf{x} = \int_K \rho \, \frac{\boldsymbol{\nabla} p^e}{\rho^e} \, v \, d\mathbf{x}$$

(using the exact equilibrium relation $\boldsymbol{\nabla} p^e = -\rho^e \boldsymbol{\nabla}\phi^e$ to trade $\boldsymbol{\nabla}\phi^e$ for $\boldsymbol{\nabla} p^e/\rho^e$, a quantity we can evaluate directly from the known equilibrium profile). Now add and subtract the cell-average ratio $\overline{\rho}_K / \overline{\rho^e}_K$ inside the integral:

$$= \int_K \boldsymbol{\nabla} p^e \left(\frac{\rho}{\rho^e} - \frac{\overline{\rho}_K}{\overline{\rho^e}_K} \right) v \, d\mathbf{x} \; + \; \frac{\overline{\rho}_K}{\overline{\rho^e}_K} \int_K \boldsymbol{\nabla} p^e \, v \, d\mathbf{x}.$$

The first term is the genuinely new part of the perturbation source, and notice it vanishes identically the instant ρ/ρ^e is constant across the cell — exactly what happens at equilibrium, where $\rho_h = \rho_h^e$ everywhere. The second term is integrated by parts, using precisely the same weak-form move that built DG in the first place, turning a volume integral of a gradient into a boundary flux minus a volume integral against $\boldsymbol{\nabla} v$:

$$\frac{\overline{\rho}_K}{\overline{\rho^e}_K} \int_K \boldsymbol{\nabla} p^e \, v \, d\mathbf{x} = \frac{\overline{\rho}_K}{\overline{\rho^e}_K} \left(\sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} p^e \, v \, \mathbf{n}_{\mathcal{E},K} \, ds - \int_K p^e \, \boldsymbol{\nabla} v \, d\mathbf{x} \right).$$

Putting it together, and writing $\langle \mathbf{S}^{[2]}, v \rangle_K$ for the fully quadrature-discretized version (quadrature points $\mathbf{x}_K^{(q)}$ with weights $w_K^{(q)}$ inside K, and $\mathbf{x}_{\mathcal{E}}^{(\mu)}$ with weights $w_{\mathcal{E}}^{(\mu)}$ on each face):

$$\langle \mathbf{S}^{[2]}, v \rangle_K = \sum_q w_K^{(q)} \left(\frac{\rho_h(\mathbf{x}_K^{(q)})}{\rho_h^e(\mathbf{x}_K^{(q)})} - \frac{\overline{\rho}_K}{\overline{\rho^e}_K} \right) \boldsymbol{\nabla} p_h^e(\mathbf{x}_K^{(q)}) \, v(\mathbf{x}_K^{(q)})$$

$$+ \frac{\overline{\rho}_K}{\overline{\rho^e}_K} \left(\sum_{\mathcal{E} \in \partial K} \sum_{\mu} w_{\mathcal{E}}^{(\mu)} p_h^e(\mathbf{x}_{\mathcal{E}}^{(\mu)}) \, v^{\mathrm{int}(K)} \, \mathbf{n}_{\mathcal{E},K} - \sum_q w_K^{(q)} p_h^e(\mathbf{x}_K^{(q)}) \, \boldsymbol{\nabla} v(\mathbf{x}_K^{(q)}) \right) - \sum_q w_K^{(q)} \, \rho_h$$

That last line is just the ordinary perturbation-gravity contribution, $-\rho\boldsymbol{\nabla}\phi^{\delta}$, tacked back on with no special treatment. The whole point of everything above it is that, at exact equilibrium, this discrete source term is built term-for-term to exactly cancel the modified HLLC flux from the previous paragraph. That's not a coincidence or a lucky numerical accident; it's a designed algebraic identity, and it's the reason we can actually *prove* the well-balanced property later instead of just observing it empirically on a few test cases.

49.10.10.2 Fix two: a conservative form for total energy

For energy conservation, the move is to stop evolving the ordinary non-gravitational energy E directly, and instead evolve a total energy that folds in gravitational potential energy from the start:

$$E_{tot} = E + \tfrac{1}{2}\rho\phi.$$

Why does this help? Because with some care you can rewrite the evolution equation for E_{tot} in genuinely conservative form — divergence of a flux, with no leftover volumetric source term at all:

$$\frac{\partial E_{tot}}{\partial t} + \boldsymbol{\nabla}\cdot\Big((E+p)\mathbf{u} + \mathbf{F}_g\Big) = 0,$$

where the new piece, the *gravity energy flux*, is

$$\mathbf{F}_g = \frac{1}{8\pi G}\Big(\phi\,\boldsymbol{\nabla}\dot{\phi} - \dot{\phi}\,\boldsymbol{\nabla}\phi\Big) + \rho\mathbf{u}\phi.$$

It's worth actually seeing why this identity is true, rather than taking it on faith, because the derivation is short and it explains where every piece of $\mathbf{F}_g$ comes from. Start from the time derivative of the gravitational energy piece, $\partial_t(\frac{1}{2}\rho\phi)$, and add and subtract the divergence of the proposed flux and the ordi-

nary source term $-\rho\mathbf{u}\cdot\boldsymbol{\nabla}\phi$ from the plain energy equation:

$$\frac{\partial}{\partial t}\left(\tfrac{1}{2}\rho\phi\right)+\boldsymbol{\nabla}\cdot\mathbf{F}_g-\rho\mathbf{u}\cdot\boldsymbol{\nabla}\phi=\tfrac{1}{2}\dot{\rho}\,\phi+\tfrac{1}{2}\dot{\phi}\,\rho+\frac{1}{8\pi G}\Big(\phi\,\Delta\dot{\phi}-\dot{\phi}\,\Delta\phi\Big)+\boldsymbol{\nabla}\cdot(\rho\mathbf{u})\,\phi.$$

Now use mass conservation, $\dot{\rho}=-\boldsymbol{\nabla}\cdot(\rho\mathbf{u})$, to cancel the $\phi\,\dot{\rho}$ term against the last term above, and use $\Delta\phi=4\pi G\rho$ together with its time-derivative $\Delta\dot{\phi}=4\pi G\dot{\rho}=-4\pi G\,\boldsymbol{\nabla}\cdot(\rho\mathbf{u})$ to rewrite the remaining two terms. Every surviving term cancels in pairs, and the whole expression collapses to exactly zero. Add this identity to the ordinary energy equation and you land on the conservative form above. It's a genuinely satisfying piece of algebra: the messy, seemingly ad hoc combination of terms in $\mathbf{F}_g$ isn't arbitrary at all, it's precisely what's needed to make two independent physical statements (mass conservation and Poisson's equation) conspire to produce a clean conservation law for energy.

There's an obvious wrinkle: $\mathbf{F}_g$ contains $\dot{\phi}=\partial\phi/\partial t$, a time derivative of a field, which isn't something we normally track explicitly in a DG code. The way around it is to note that $\dot{\phi}$ itself satisfies another Poisson equation, obtained by differentiating $\Delta\phi=4\pi G\rho$ in time and using mass conservation again:

$$\Delta\dot{\phi}=4\pi G\,\dot{\rho}=-4\pi G\,\boldsymbol{\nabla}\cdot(\rho\mathbf{u}).$$

So we solve one additional elliptic problem, every stage, using the exact same LDG mixed formulation as before, now with $-4\pi G\,\boldsymbol{\nabla}\cdot(\rho\mathbf{u})_h$ as the source instead of $4\pi G\rho_h$, to obtain both $\dot{\phi}_h$ and its gradient $\dot{\mathbf{g}}_h$.

There's one more, purely numerical, subtlety worth being upfront about, because it's the kind of thing that will cost you a full order of accuracy if you don't know to watch for it. If you discretize the right-hand side $-4\pi G\,\boldsymbol{\nabla}\cdot(\rho\mathbf{u})_h$ naively — literally differentiating the DG polynomial $(\rho\mathbf{u})_h$ pointwise inside each cell — you lose an order of accuracy, because differen-

tiating a degree-k polynomial approximation generically only gives you a degree-$(k-1)$-accurate approximation of the true derivative. The fix, again, is to reach for the DG weak form itself rather than a pointwise derivative: apply summation-by-parts to move that derivative off of $(\rho\mathbf{u})_h$ and onto the test function ψ instead, at the cost of picking up a boundary term built from the already-computed mass flux $\widehat{f}^{[1]}$:

$$\int_K \dot{\mathbf{g}}_h \cdot \boldsymbol{\nabla}\psi \, d\mathbf{x} = \int_{\mathcal{E}} \psi^{\mathrm{int}(K)} \, \widehat{\dot{\mathbf{g}}}_h \cdot \mathbf{n}_{\mathcal{E},K} \, ds + \int_K 4\pi G \, (\rho\mathbf{u})_h \cdot \boldsymbol{\nabla}\psi \, d\mathbf{x} - \sum_{\mathcal{E}\in\partial K} \int_{\mathcal{E}} 4\pi G \, \widehat{f}^{[1]}_{\mathbf{n}_{\mathcal{E},K}} \, \psi^{\mathrm{int}(K)} \, ds$$

This one small technical patch restores the full design order of accuracy for $\dot{\phi}_h$, and by extension for the whole energy flux; skip it and you'll still get a scheme that conserves energy (that property doesn't depend on this fix), but one whose overall convergence rate quietly degrades, especially on coarser meshes.

With E_{tot} as the evolved variable, we can repackage the whole system uniformly:

$$\mathbf{W} = \begin{bmatrix} \rho \\ \rho\mathbf{u} \\ E_{tot} \end{bmatrix}, \quad \mathbf{F}(\mathbf{W}) = \begin{bmatrix} \rho\mathbf{u} \\ \rho\mathbf{u}\otimes\mathbf{u} + p\mathbf{I} \\ (E+p)\mathbf{u} \end{bmatrix}, \quad \mathbf{G}(\mathbf{W}) = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{F}_g \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ -\rho\boldsymbol{\nabla}\phi \\ 0 \end{bmatrix},$$

$$\frac{\partial \mathbf{W}}{\partial t} + \boldsymbol{\nabla}\cdot\mathbf{F}(\mathbf{W}) + \boldsymbol{\nabla}\cdot\mathbf{G}(\mathbf{W}) = \mathbf{S}.$$

One bookkeeping detail that turns out to matter enormously in practice: at every stage we still need the ordinary energy E (to get pressure out of the equation of state), but we're now evolving E_{tot}. The conversion is $E_h = (E_{tot})_h - \mathbf{P}\big(\tfrac{1}{2}\rho_h\phi_h\big)$ — and that $\mathbf{P}$, the L^2 projection, is not optional decoration. If you instead compute $\tfrac{1}{2}\rho_h\phi_h$ pointwise at quadrature points and skip projecting the product back into the DG polynomial space V_h^k, the formula *looks* identical but the well-balanced cancellation from the previous section silently breaks, because the pres-

sure recovered this way is no longer guaranteed to equal p_h^e exactly at equilibrium. This is exactly the kind of easy-to-miss implementation detail that separates "it looks right and mostly works" from "it's provably well-balanced," and it's the sort of thing I'd genuinely only have caught by trying to prove the property myself and watching the proof break.

49.10.11 Time Discretization

Everything above was written semi-discretely: space is discretized into the DG polynomial spaces, but time is still continuous, giving us an ODE system of the schematic form

$$\mathbf{W}_h^{n+1} = \mathbf{W}_h^n + \Delta t\, \mathcal{H}(\mathbf{W}_h^n, t^n),$$

where $\mathcal{H}$ is shorthand for the entire right-hand side we just spent two sections deriving: the modified HLLC flux, the well-balanced source term, and the gravity energy flux. (The standard, non-structure-preserving scheme has an analogous operator, which I'll call $\mathcal{L}(\mathbf{U}_h^n, t^n)$.) Forward Euler, as written, is only first order accurate in time, which would badly undercut all the care we just took getting high spatial order — there's no point running a P^3 DG method in space if a first-order time integrator is going to wash out the accuracy anyway. The standard fix, and the one used throughout this scheme, is strong-stability-preserving Runge-Kutta (SSP-RK), specifically chosen because it's built as a convex combination of forward-Euler steps, which means any stability or positivity property you can prove for forward Euler carries over automatically to the full multistage method, with no extra work.

Second order SSP-RK:

$$\mathbf{W}_h^{(1)} = \mathbf{W}_h^n + \Delta t\, \mathcal{H}(\mathbf{W}_h^n, t^n), \qquad \mathbf{W}_h^{n+1} = \tfrac{1}{2}\mathbf{W}_h^n + \tfrac{1}{2}\Big(\mathbf{W}_h^{(1)} + \Delta t\, \mathcal{H}(\mathbf{W}_h^{(1)}, t^n + \Delta t)\Big).$$

Third order SSP-RK:

$$\mathbf{W}_h^{(1)} = \mathbf{W}_h^n + \Delta t\, \mathcal{H}(\mathbf{W}_h^n, t^n),$$

$$\mathbf{W}_h^{(2)} = \tfrac{3}{4}\mathbf{W}_h^n + \tfrac{1}{4}\Big(\mathbf{W}_h^{(1)} + \Delta t\, \mathcal{H}(\mathbf{W}_h^{(1)}, t^n + \Delta t)\Big),$$

$$\mathbf{W}_h^{n+1} = \tfrac{1}{3}\mathbf{W}_h^n + \tfrac{2}{3}\Big(\mathbf{W}_h^{(2)} + \Delta t\, \mathcal{H}(\mathbf{W}_h^{(2)}, t^n + \tfrac{1}{2}\Delta t)\Big).$$

Each stage is literally just another forward-Euler-style evaluation of $\mathcal{H}$, meaning the entire elliptic solve, modified HLLC flux, and gravity-energy-flux pipeline from the previous section gets called again, unchanged, at every stage. This reusability is one of the more pleasant features of how the scheme was built: none of the well-balanced or energy-conserving modifications care what order of Runge-Kutta method is wrapped around them, so raising the temporal order later is purely a matter of adding stages, with zero changes to the spatial operator itself.

Practically, the timestep is chosen from a standard CFL condition based on the fastest wave speed present in any cell,

$$\Delta t = \mathrm{CFL} \cdot \frac{h}{\max_K \left(|\overline{\mathbf{u}}_K|_\infty + \overline{c}_K\right)}, \qquad \overline{c}_K = \sqrt{\gamma\, \overline{p}_K / \overline{\rho}_K},$$

with h the smallest element diameter in the mesh and $\overline{(\cdot)}_K$ denoting the cell average. Notice gravity doesn't directly enter this CFL restriction — the elliptic solve is implicit (we're solving a linear system exactly, not marching it forward explicitly), so the only explicit wave-speed restriction left is the ordinary hydrodynamic one.

49.10.12 Ossilation Elimination

High order polynomials are wonderful at representing smooth flow and genuinely terrible at representing a shock. Near a real discontinuity — the blast wave from our earlier explosion example, say — a high-degree DG polynomial will ring: classic Gibbs-phenomenon over- and under-shoots that, in the worst case, can drive density or pressure negative at a quadrature point and crash the whole simulation outright.

The traditional fix here is a slope limiter, but limiters can be blunt instruments: applied indiscriminately, they can easily destroy the very fine cancellations we spent the last section engineering into the well-balanced source term and modified flux. Instead, this scheme reaches for a more recent idea, the oscillation-eliminating (OE) technique, which works by letting the high-degree moments of the polynomial solution decay through an artificial damping ODE, applied as a separate fractional step tacked on after each Runge-Kutta stage, rather than by clipping or reconstructing the polynomial directly

$$\frac{\mathrm{d}\mathbf{U}}{\mathrm{d}t}+\mathbf{L}_f(\mathbf{U})+\boldsymbol{\Sigma}(\mathbf{U})\mathbf{U}=0 \quad\Longrightarrow\quad \frac{\mathrm{d}\mathbf{U}}{\mathrm{d}t}=-\mathbf{L}_f(\mathbf{U}), \qquad \frac{\mathrm{d}\mathbf{U}_\sigma}{\mathrm{d}t}=-\boldsymbol{\Sigma}(\mathbf{U})\mathbf{U}_\sigma.$$

The first piece is just the ordinary DG spatial operator we already have. The second, $\boldsymbol{\Sigma}(\mathbf{U})\mathbf{U}_\sigma$, is a purely local, per-cell damping term, and its coefficient ϑ_K^m is built from a smoothness indicator, summed over each cell's faces:

$$\vartheta_K^m(\mathbf{U}_h)=\sum_{e\in\partial K}\beta_e\,\frac{\sigma_{e,K}^m(\mathbf{U}_h)}{h_{e,K}},$$

where β_e is a local estimate of the maximum wave speed across face e (the spectral radius of the flux Jacobian in the normal direction), $h_{e,K}$ is roughly the distance from face e to the far side of the cell, and the indicator itself, measures how big the jump

in the m-th derivative across that face is, relative to how much the solution varies overall. In a smooth region, that jump is tiny compared to the solution's global variation, so ϑ_K^m stays small and the damping barely touches anything. Right next to a shock, the jump is large, ϑ_K^m spikes, and the damping aggressively suppresses the offending high-order moments — but, and this is the important structural feature, it leaves the lowest moment, the cell average, completely untouched by construction (ϑ_K^0 effectively doesn't act on the mean). The damping equation actually has a closed form: writing the solution in a hierarchical orthogonal basis $\phi_K^{(\alpha)}$, each higher moment simply decays exponentially at a rate set by the local indicator,

$$\mathcal{F}_{\Delta t}\mathbf{U}_h = \mathbf{U}_K^{(0)}\phi_K^{(0)}(\mathbf{x}) + \sum_{j=1}^{k} e^{-\Delta t \sum_{m=0}^{j} \vartheta_K^m(\mathbf{U}_h)} \sum_{|\alpha|=j} \mathbf{U}_K^{(\alpha)}\phi_K^{(\alpha)}(\mathbf{x}).$$

Because the cell average $\mathbf{U}_K^{(0)}$ is left alone, mass, momentum, and total energy on each cell are completely unaffected by the OE step; only the shape of the polynomial *within* the cell gets smoothed. That's exactly the property we need to keep conservation intact.

The remaining piece of care, specific to this well-balanced scheme, is where the damping is allowed to act. It's applied only to the perturbation part $\mathbf{W}^\delta$ of the solution, never to the frozen equilibrium background $\mathbf{W}^e$:

$$\mathbf{W}_h = \mathbf{W}_h^e + \mathcal{F}_{\Delta t}(\mathbf{W}_h^\delta).$$

If the damping were instead applied to the full solution, including the equilibrium piece, it would gradually smooth away the very steady-state structure the well-balanced property exists to protect — a hydrostatic star has real, physically meaningful higher moments in its density profile, and those aren't the "oscillations" we're trying to eliminate. Restricting the damping to

the perturbation alone means: on a problem that's exactly at equilibrium, $\mathbf{W}^{\delta} \equiv 0$, the damping has literally nothing to act on, and the well-balanced property survives the OE step trivially. On a genuinely perturbed or shocked problem, one can show the damping only introduces an error of the same order as the underlying DG truncation error itself, so smooth problems don't quietly lose accuracy — you only pay a price near an actual discontinuity, which is exactly where you want to be paying it.

For the very strongest shocks — the blast wave test later, where the central pressure jumps by two orders of magnitude — OE alone isn't quite enough to guarantee density and pressure stay positive everywhere, so it's paired with a separate, more surgical positivity-preserving (PP) limiter, which only activates at a quadrature point if density or pressure there is actually about to go negative, and otherwise leaves the solution untouched.

49.10.13 Structure Preserving Properties

It's worth stating plainly, and proving carefully, exactly what all of this extra machinery buys us, because the algebra above is dense enough that the punchline can get lost.

49.10.13.1 Theorem: well-balancedness

Claim. If the scheme is initialized exactly at a known hydrostatic equilibrium — $\mathbf{u}^0 = \mathbf{0}$, $p^0 = -\rho^0 \boldsymbol{\nabla}\phi^0$, $\Delta\phi^0 = 4\pi G\rho^0$ — then for every subsequent timestep n,

$$\rho_h^n = \rho_h^e, \qquad (\rho\mathbf{u})_h^n = \mathbf{0}, \qquad p_h^n = p_h^e, \qquad \phi_h^n = \phi_h^e,$$

exactly, up to floating-point round-off, regardless of mesh resolution.

Proof sketch, by induction on n. Assume the claim holds at step n. Walk through the flowchart from the next section term by term. Since $\rho_h^n = \rho_h^e$, the perturbation density is exactly zero, so the perturbation Poisson solve returns $\phi_h^\delta = 0$ and hence $\phi_h^n = \phi_h^e$ exactly. From this, the recovered non-gravitational energy is $E_h^n = p_h^e/(\gamma - 1)$ and the recovered pressure is exactly p_h^e — this is precisely where the consistent L^2-projection bookkeeping flagged earlier becomes essential; without it this line fails. Feeding $\mathbf{U}_h^{n,\mathrm{int}(K)} = (\rho_h^{e,\mathrm{int}(K)}, \mathbf{0}, p_h^{e,\mathrm{int}(K)}/(\gamma-1))^T$ (and the analogous exterior state) into the modified HLLC flux, the contact property fires and gives $\widehat{\mathbf{F}}^n = (0,\, p_h^{e,\star}\mathbf{n}^T,\, 0)^T$ — in particular the mass-flux component $\widehat{f}^{n,[1]} = 0$. Since the mass flux is zero everywhere, the second Poisson solve for $\dot{\phi}_h, \dot{\mathbf{g}}_h$ has a zero right-hand side, so $\dot{\phi}_h^n = 0$, $\dot{\mathbf{g}}_h^n = \mathbf{0}$, and the entire gravity energy flux vanishes identically, $(\widehat{\mathbf{F}_g})^n = 0$. Finally, substitute all of this into the momentum update: the four resulting pieces — the pressure-gradient volume term, the "add-and-subtract" perturbation term from the source, the rescaled-pressure boundary term, and the perturbation-gravity term — cancel exactly in pairs, term for term, leaving $(\rho\mathbf{u})_h^{n+1} = (\rho\mathbf{u})_h^n = \mathbf{0}$. Since the mass and energy fluxes were already shown to vanish, $\rho_h^{n+1} = \rho_h^e$ and $(E_{tot})_h^{n+1} = (E_{tot})_h^n$ follow immediately, closing the induction. ■

The proof is mechanical once you have it laid out, but the punchline is genuinely strong: this isn't an asymptotic statement about the error shrinking as $h \to 0$. The equilibrium is preserved exactly, on any mesh, at any polynomial order, because every single term that could disturb it has been shown to vanish identically rather than merely become small.

49.10.13.2 Theorem: total energy conservation

Claim. Under periodic boundary conditions, or compactly-supported boundary conditions (velocity vanishing on the bound-

ary), the scheme conserves discrete total energy exactly at every timestep:

$$\int_\Omega (E_{tot})_h^{n+1}\, d\mathbf{x} = \int_\Omega (E_{tot})_h^{n}\, d\mathbf{x}.$$

Proof sketch. Take the discrete E_{tot} update equation, set the test function $v = 1$, and sum over every cell K in the mesh. Every interior face appears in this sum exactly twice, once from each of the two cells sharing it, with opposite orientations of the outward normal — so every interior-face contribution from the energy flux and the gravity energy flux cancels exactly against its neighbor: whatever leaves one cell enters the other, unchanged. What's left after that cancellation is purely a sum over boundary faces $\mathcal{E}_B$. Under periodic boundary conditions, boundary faces are identified in matching pairs on opposite sides of the domain and cancel each other the same way interior faces do. Under compactly-supported boundary conditions, velocity vanishes identically on $\mathcal{E}_B$, which makes both the ordinary energy flux and the gravity energy flux vanish there directly, since every term in each flux carries at least one explicit factor of $\mathbf{u}$. Either way, the total boundary contribution is exactly zero, and we're left with $\int_\Omega (E_{tot})_h^{n+1} = \int_\Omega (E_{tot})_h^{n}$. ■

Notice this proof didn't need the well-balanced machinery at all — TEC is a genuinely independent property, coming entirely from writing the energy equation in conservative form. In practice this shows up as a total-energy trace that's flat to machine precision on a well-resolved, stable configuration, and — arguably more usefully — as an energy diagnostic you can actually trust while something dramatic is unfolding, like a gravitational collapse, precisely the regime where the standard scheme's energy error tends to grow fastest and become least trustworthy.

49.10.14 The Code itself (Flowchart)

Putting all of the above in the order you'd actually implement it, here is one full Runge-Kutta stage, advancing from a known state $\left(\rho_h^n, (\rho\mathbf{u})_h^n, (E_{tot})_h^n\right)$ to the corresponding stage value:

- **Step 0 — initialization (once, at $t = 0$ only).** Project the physical initial conditions into the DG space: $\rho_h^0 = \mathbf{P}\rho^0$, $(\rho\mathbf{u})_h^0 = \mathbf{P}(\rho\mathbf{u})^0$, $E_h^0 = \mathbf{P}E^0$. Solve the equilibrium Poisson problem once, $(\mathbf{g}_h^e, \phi_h^e) = \mathcal{D}_1(4\pi G\rho_h^e)$, and freeze it for the rest of the run. Solve the initial perturbation Poisson problem to get ϕ_h^0, and set $(E_{tot})_h^0 = E_h^0 + \mathbf{P}\left(\frac{1}{2}\rho_h^0\phi_h^0\right)$ — using the projection here, not a pointwise product, for exactly the reason flagged above.

- **Step 1 — gravity.** Given ρ_h^n, solve the perturbation Poisson problem $(\mathbf{g}_h^\delta, \phi_h^\delta)^n = \mathcal{D}_1\left(4\pi G(\rho_h^n - \rho_h^e)\right)$ and recombine with the frozen equilibrium piece to get the total field, $(\mathbf{g}_h^n, \phi_h^n) = (\mathbf{g}_h^e, \phi_h^e) + (\mathbf{g}_h^\delta, \phi_h^\delta)^n$. Separately, solve the summation-by-parts-corrected Poisson problem $\mathcal{D}_2$ for the time-derivative fields, $(\dot{\mathbf{g}}_h^n, \dot{\phi}_h^n) = \mathcal{D}_2\left(4\pi G\,\boldsymbol{\nabla}\cdot(\rho\mathbf{u})_h^n\right)$. Recover the ordinary energy, $E_h^n = (E_{tot})_h^n - \mathbf{P}\left(\frac{1}{2}\rho_h^n\phi_h^n\right)$, and from it the pressure via the equation of state.

- **Step 2 — fluid flux.** At every interior face, compute the equilibrium-pressure-rescaled states on each side and evaluate the modified HLLC flux $\widehat{\mathbf{F}}^n$ from those rescaled states.

- **Step 3 — gravity energy flux.** Using $\phi_h^n, \mathbf{g}_h^n$ from Step 1, $\dot{\phi}_h^n, \dot{\mathbf{g}}_h^n$ also from Step 1, and the mass-flux component $\widehat{f}^{n,[1]}$ from Step 2, assemble both the volume contribution to $\mathbf{F}_g^n$ inside each element and its face contribution $(\widehat{\mathbf{F}_g})^n$.

- **Step 4 — assemble and update.** Combine the fluid flux (Step 2), the well-balanced momentum source term (built the same way as in Step 1's rescaling), and the gravity

energy flux (Step 3) into the full right-hand side $\mathcal{H}$, and take one Runge-Kutta stage update for $\big(\rho_h, (\rho\mathbf{u})_h, (E_{tot})_h\big)$.

- **Step 5 — oscillation control.** Apply the OE damping (and, for strongly discontinuous problems, the positivity-preserving limiter) to the perturbation part of the newly updated solution only, leaving the frozen equilibrium piece untouched.

Repeat Steps 1 through 5 once per Runge-Kutta stage (twice for the second-order method, three times for third order), then move on to the next timestep with a freshly computed CFL-limited Δt. The only genuinely expensive part of this whole loop is the pair of elliptic solves in Step 1 — but because the mesh is fixed for the entire simulation, the factorization of the underlying stiffness matrix (Cholesky for the compactly-supported case, LU for the periodic case) is computed exactly once, before the time loop even starts, and reused at every stage of every timestep, which keeps the per-step cost dominated by the same sparse triangular solves you'd need for any implicit elliptic problem, not by repeated factorization.

49.10.15 Results

Here I have the results in the form of fun cool animations

49.11 What is Velocity(joke video)

- Basic
- Average speed
- Derivative form
- Vector form

- Newtonian
- Newtonian + inertial refrence frame
- Newtonian Covariance/Galilean Relativity
- Newtonian-Cartan/ general covariance
- Fluid
- Special Relativity
  - Coordinate Velocity
  - Four-Velocity
  - Proper Velocity
  - Rapidity
  - Relative Velocity
- General Relativity
  - Coordinate velocity
  - GR four-velocity
  - Tangent vector
  - Covariant velocity
  - Geodesic
  - Warp Drive
- Dynamics
- Lagrandian
- Hamiltonian
- Phase-Space Flow
- Poisson Structure
- Symplectic flow

49.11.1 Introduction

Let's define velocity, I mean its a pretty simple concept, you could ask a fifth grader what it is and they could define it with ease. I mean it is the simple equation...

$$
\begin{aligned}
V \equiv (Q_\xi, X) &= \mathfrak{X}_{Q_\xi}[X] \\
\delta[Q_\xi] &= K(\delta, \mathfrak{X}_{Q_\xi}) \\
&= \Omega_{\text{ext}}(\delta, \mathfrak{X}_{Q_\xi}) - G(\delta, \mathfrak{X}_{Q_\xi}) \\
&= \Omega(\delta, \mathfrak{L}_\xi) + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner\, \vartheta(\delta) \\
&\quad + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner\, \vartheta(\delta) - G(\delta, \mathfrak{L}_\xi) \\
&= \Omega(\delta, \mathfrak{L}_\xi) + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner\, \vartheta(\delta) \\
&\quad + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner\, \vartheta(\delta) \\
&= \Omega_{\text{ext}}(\delta, \mathfrak{L}_\xi) - G(\delta, \mathfrak{L}_\xi) \\
&= \Omega(\delta, \mathfrak{L}_\xi) + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\delta) \lrcorner\, \vartheta(\mathfrak{L}_\xi) \\
&\quad - \Omega(\mathfrak{L}_\xi, \delta) - Q_{[\mathbb{X}(\mathfrak{L}_\xi), \mathbb{X}(\delta)]} - \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner\, \vartheta(\delta) \\
&\quad - \left[- \oint_{\partial\Sigma} \mathbb{X}(\delta) \lrcorner\, \vartheta(\mathfrak{L}_\xi) - \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner\, \vartheta(\delta) \right]
\end{aligned}
$$

$$
\begin{aligned}
&= \delta\left[\int_\Sigma (\vartheta(\mathfrak{L}_\xi) - \xi \lrcorner L)\right] - Q_{\delta[\xi]} - \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta) \\
&\quad + Q_{[\mathbb{X}(\delta),\mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta) \\
&= \delta\left[\int_\Sigma \big(\vartheta(\mathcal{L}_\xi g_{ab}, \mathcal{L}_\xi \psi^I) - \xi \lrcorner L\big)\right] - Q_{\delta[\xi]} \\
&\quad + Q_{[\mathbb{X}(\delta),\mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta) \\
&= \delta\left[\int_\Sigma \big(\vartheta(2\nabla_{(a}\xi_{b)}, \mathcal{L}_\xi \psi^I) - \xi \lrcorner L\big)\right] - Q_{\delta[\xi]} \\
&\quad + Q_{[\mathbb{X}(\delta),\xi]} + \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta)
\end{aligned}
$$

$$= \delta \left[\int_\Sigma \big(\vartheta(\nabla_a \xi_b + \nabla_b \xi_a, \mathcal{L}_\xi \psi^I) - \xi \lrcorner L \big) \right] - Q_{\delta[\xi]}$$
$$+ Q_{[\mathbb{X}(\delta),\xi]} + \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta)$$

$$= \delta \left[\int_\Sigma \big(\vartheta(\partial_a \xi_b + \partial_b \xi_a - 2\Gamma^c{}_{ab} \xi_c, \mathcal{L}_\xi \psi^I) - \xi \lrcorner L \big) \right]$$
$$- Q_{\delta[\xi]} + Q_{[\mathbb{X}(\delta),\xi]} + \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta)$$

$$= \delta \left[\int_\Sigma \big(\vartheta(\partial_a \xi_b + \partial_b \xi_a - g^{cd}(\partial_a g_{db} + \partial_b g_{da} - \partial_d g_{ab}) \xi_c, \mathcal{L}_\xi \psi^I) - \xi \lrcorner L \big) \right]$$
$$- Q_{\delta[\xi]} + Q_{[\mathbb{X}(\delta),\xi]} + \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta)$$

$$\dot{F} = (Q_\xi, F)$$
$$(Q_\xi, g_{ab}) = \mathcal{L}_\xi g_{ab} = 2\nabla_{(a} \xi_{b)}$$
$$(Q_\xi, \psi^I) = \mathcal{L}_\xi \psi^I$$
$$(Q_\xi, x^\mu) = \xi^\mu$$

$$\Gamma^c{}_{ab} = \frac{1}{2} g^{cd}(\partial_a g_{db} + \partial_b g_{da} - \partial_d g_{ab})$$
$$g^{ac} g_{cb} = \delta^a{}_b, \qquad \nabla_c g_{ab} = 0$$

$$(Q_\xi, Q_\xi) = -G(\mathfrak{L}_\xi, \mathfrak{L}_\xi)$$
$$= \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\mathfrak{L}_\xi)$$

... uhm, well... maybe I am wrong about that.

49.11.2 Basic

Ok, well then, let's start at the beginning. Velocity can generally be defined by the change in the distance traveled divided by the time it takes to travel.

$$v = \frac{\Delta x}{\Delta t}$$

If we take the infintesimal transformation of such we get

$$v = \frac{\mathrm{d}x}{\mathrm{d}t}$$

But, as your middle school science teacher likely drilled in your head, velocity isn't just speed, it also has a direction, making a vector the easiest way to represent it.

$$\dot{\mathbf{r}} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \langle \frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}, \frac{\mathrm{d}z}{\mathrm{d}t} \rangle$$

49.11.3 Newtonian

Now that we have a basic understanding of the math behind velocity we can start putting into context. Likey eveyone here knows newtons second law

$$\mathbf{F} = m\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$$

solving for velocity gives

$$\dot{r} = \int \frac{\mathbf{F}}{m} dt$$

Now this is all nice and good, but it forgets relativity, I don't mean special or general, I mean Galilean relativity. Keeping

that in track is very simple

$$\dot{r}' = \int \frac{\mathbf{F}}{m} dt - \dot{r}_{\text{refrence frame}}$$

This brings us to our next problem, as having Newtonian mechanics rely on frames of refrences comes up with several problems, I am sure many people have heard high school schience teachers telling you that there is no centrifugal forces, as it is a fictitious force created by frames of refrence, to solve this problem we make Newtonian mechanics general covariant by using the Newton-Cartan method. Essentially taking a space-time approach similar to general relativity but making time absolute rather than relative.

Defining velocity as

$$V^{\mu} = \frac{\mathrm{d}x^{\mu}}{\mathrm{d}dt} = (1, \frac{\mathrm{d}x^{i}}{\mathrm{d}dt})$$

While this equation may look pretty simple, it is an extreme simplification. I really don't know the purpose of the Newton-Cartan other than to simply prove Einstien wrong.

49.11.4 Special Relativity

This talk or relativity branches cleanly into special relativity, as velocity truly begins to change. As time becomes another relative dimension to move through, which is a really funny idea to think about. Running through time.

Anyways, the velocity vector obviously becomes 4-d adding time to the mix

$$U^{\mu} = \frac{\mathrm{d}x^{\mu}}{\mathrm{d}\tau}$$

Where τ is proper time, or the subjective time of the travelers rather than those observing them. Take the tiny people on a relativistic baseball as compared to the batter waiting for them.

We can relatively easily then expanded this to our traditional relaitive speed.

Taking galelian relativity

$$v_{\text{rel}} = v_1 - v_0$$

and adding relativistic corrections

$$v_{\text{rel}} = \frac{v_1 - v_0}{1 - v_1 v_2 / c^2}$$

Because why not

49.11.5 GR

But, this method isn't enough, there is now the problem of absolute acceleration, creating paradoxes and and preferential frames of refrence, something physically nonsensical in a relative world without absolute space or time. Though, using our earlier covariant newtonian framework and combining that with Einstienian proper time we get an equation that looks the same

$$U^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$$

but now

$$g_{\mu\nu} U^\mu U^\nu = -c^2$$

Thus adding a curved geometry of space-time that adds further separation between observers. The rest also stays the same as SR but with further corrections, and can be expressed through the language of remanian geometry rather than Lorentzian

or geodesics, I won't get into all of that though.

Though, before we move on, I want to talk about the velocity of warp drives. If you haven't heard, a solution of GR allows for apperent faster than speed of light movement, really messing with our definition of velocity which in all cases must be less than the speed of light.

It doesn't really break anything as most solutions have it where the matter doesn't move at all but spacetime shrinks and grows behind it. Making its actual movement 0 m/s yet to an outside observer moving several times the speed of light as it moves a great "distance" in a very short time. Making definitions a bit confusing.

49.11.6 Dynamics

Ok, all of this spacetime stuff is really confusing and make a great deal of a mess. Lets move onto a completely different approach. Moving away from kinematics and into dynamics. Involving phase-space, where there aren't some defined set of dimensions of space and time, but an infinite amount of dimenisons of anything you could observe; momentum, position, time, heat, entropy, well... not really traditionally entropy. Anyways, rather than taking an infinite amount of dimensions, let's shrink it down to two, the two most generally used, momenta and position. Then we will add mass but as it is constant in our situation it is infinitely then and thus can be expressed in 2d

(show graph)

Here, momenta and position are not like their Newtonian cousians, but rather generalized. They are related pairs and aren't constrained in the way position and momentum generally are. Position can mean angle, position plus direction rolled in one, or all three dimensions rolled into one vector field. The

same with momenta, but to make this example concret, we will take the traditional picture of position and momenta to give that if one represents the flow of the vector field as H, or commonly refered to as the Hamiltonain, we can express the derivative of position as a function of time as

$$v = \dot{q} = \frac{\partial H}{\partial p_i}$$

Why, that is a really big question, just trust me.

Thus saying that velocity is a part of the vector field on phasespace

(show on graph)

but what is phase space, well... that is an even bigger question but I will take a quick jab at it.

First we introduce something called a Poisson Structure

$$\dot{f} = \{f, H\} = \frac{\partial f}{\partial q^i}\frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial H}{\partial q^i}$$

Where the entire vector field can thus be written as

$$X_H = \{\cdot, H\}$$

Where the dot represents all possible observables, like position, momenta, or really anything else. The geometry behind the vector field is the phase-space. This geometry is something called symplectic, but I won't get into that.

49.11.7 Other

Now, let's speed run some others. Fluid velocity is a vector function that attaches a newtonian velocity to every point in the velocity.

Quantum mechanics adds fuzzy probability to velocity

$$\mathbf{v} = \frac{\mathbf{j}}{|\psi|^2}$$

Kinetic theory combines probability of QM with the continuum of fluids

Geometric mechanics defines it as a "natural element of a tangent bundle," which seems hard to explain

49.12 Developing Numerical General Relativity: Part 1, Vacuum Kerr Black Whole

49.12.1 Planning

- Introduction
- Scope
- Kerr SpaceTime
- 3+1 split
- ADM Formulism
- BSSN Formulism
- Guage Conditions
- Initial data
- Cartoon Method
- Excision

- Method of lines + RK4
- Boundary conditions
- Code
- Validation
- Problems I had
- Next steps
- Animations

49.12.2 Introduction

Welcome to my first video on developing a numerical general relativistic code, starting with a vacuum Kerr Black Hole.

49.12.3 Scope

Before getting to the scope of this video, I will mention my overall scope of my broader project. I plan to create a FORTRAN code that uses General Relativity, Relativistic Fluid dynamics, and I am thinking about adding additional tools like magnetic or Nuclear EOS.

The main goal is for the final output to be a toy model of a quasi-star, a theoretical extremely large star with a black hole within its center but the accretions from the black hole powers the star just like fusion in a traditional star, this obviously combines a lot of extreme relativistic and fluid tools.

Rather than jumping straight into this extremely large project, I will do it in steps. I will start with a fairly basic one that uses just GR, a kerr black hole in a vacuum. That is what I am doing.

Also, for berivity I will be using geometrized units.

$$G = c = 1$$

49.12.4 Kerr SpaceTime

Kerr black holes are essentially spinning black hole, now that may seem pretty simple in the face of itself, but I ask you this, in a black hole with an infinitely small singularity, what is spinning?

Well spacetime is spinning. To give it more intuitive newtonian vocabulary, objects have a gravitational pull in a swirl around the black hole, this proccess is called frame dragging. Efficetively, this means that it will change the shape of the event horizen, as certain bits of light previously unable to escape the black hole now will. From here we get the following metric space In Boyer–Lindquist coordinates:

$$ds^2 = -\left(1-\tfrac{2Mr}{\Sigma}\right)dt^2 - \tfrac{4Mar\sin^2\theta}{\Sigma}\,dt\,d\phi + \tfrac{\Sigma}{\Delta}dr^2 + \Sigma\,d\theta^2 + \left(r^2+a^2+\tfrac{2Ma^2r\sin^2\theta}{\Sigma}\right)\sin^2\theta\,d\phi^2$$

with $\Sigma = r^2 + a^2\cos^2\theta$ and $\Delta = r^2 - 2Mr + a^2$. The horizons sit where $\Delta = 0$, $r_\pm = M \pm \sqrt{M^2 - a^2}$, and the ergosphere is at $r_E = M + \sqrt{M^2 - a^2\cos^2\theta}$. Notice $g_{rr} \to \infty$ at the horizon. That is a coordinate singularity of these

coordinates, not of the spacetime, and a simulation cannot cross it. So instead we use the Kerr–Schild form, which is regular across the horizon:

$$g_{\mu\nu} = \eta_{\mu\nu} + 2H\,l_\mu l_\nu, \qquad \eta = \mathrm{diag}(-1,1,1,1),$$

$$H = \frac{Mr^3}{r^4 + a^2z^2}, \qquad l_\mu = \left(1,\ \frac{rx+ay}{r^2+a^2},\ \frac{ry-ax}{r^2+a^2},\ \frac{z}{r}\right),$$

where r is defined implicitly by

$\frac{x^2+y^2}{r^2+a^2} + \frac{z^2}{r^2} = 1$. Explicitly, with $R^2 = x^2 + y^2 + z^2$,

$$r^2 = \frac{1}{2}\left[(R^2 - a^2) + \sqrt{(R^2 - a^2)^2 + 4a^2z^2}\right].$$

The vector l_μ is null, and the spatial part satisfies $l_i l^i = 1$.

Here we define our unit for angular momentum to be

$$a = \frac{J}{M}$$

Where J is the traditional angular momentum, we are able to do this because we are using geometic units where all units are in terms of a kind of length. Turning angular momentum that is l^2 and mass at l, turning a into l. This is generally much easier to use in general relativity, but sometimes people will use the traditional angular momentum or even a dimensionless quantity.

We can also define the SpaceTime through the metric tensor.

$$g_{\mu\nu} = \eta_{\mu\nu} + 2Hl_\mu l_\nu$$

H is the scalar function

$$H = \frac{Mr^3}{r^4 + a^2z^2}$$

where r isn't traditional euclidian space as space is obviously curved

$$\frac{x^2 + y^2}{r^2 + a^2} + \frac{z^2}{r^2} = 1$$

l being the null direction

$$l_\mu = \left(1, \frac{rx + ay}{r^2 + a^2}, \frac{rx - ay}{r^2 + a^2}, \frac{z}{r}\right)$$

49.12.5 3+1 Split

To solve most physics simulation, we derive them as initial value problems as a function of time and then numerically integrate. Those that have an understanding of GR can see how this is troubling. Space and Time are just two of the same types of dimension, also time and space can switch nature depending on the curciumstance, such as Black holes have space behave timelike and time behave space like.

Thus we work to foliate the spacetime

(image)

The method we use is the 3+1 formulism, foliating the space into sets of spacelike surfaces that can be combined to make the spacetime structure.

(image)

Here we define our variables to fully represent this new foliated space structure.

Obviously, γ or spacial metric defines the metric structure of the space at any given time.

The α or lapse defines the change in proper time between slices.

The β or shift controls how the spacial coordinates change.

Finally K or extrinsic curvature defines how the spacial hyper surface fits in spacetime.

This turns our metric into

$$ds^2 = \alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt)$$

This defines how we evolve the metric, but defining how to evolve these variables

$$\alpha, \gamma_{ij}, K_{ij}, \beta^i$$

is the hard part.

I won't go over it in depth but for ease of access I will place the

49.12.6 ADM Formulism

The Arnowitt–Deser–Misner (ADM) formalism is the Hamiltonian (canonical) formulation of general relativity obtained from the 3+1 split. It turns Einstein's equations into a constrained initial-value problem that is ideally suited to numerical evolution.

Each spacelike surface Σ_t

we have the induced metric γ_{ij}, the unit normal n^μ which directs towards the future, and the extrinsic curvature K.

The Einstein tensor projected onto and orthogonal to the slices yeilds two sets of equations (image)

The Hamiltonian Constraint which enforces conservation of energy

$$R + K^2 - K_{ij}K^{ij} = 16\pi\rho$$

Where ρ represents the energy density measured by the Eulerian observer.

We also gain a further momentum constraint

$$D_j(K^{ij} - \gamma^{ij}K) = 8\pi S^i$$

Where S^i is the momentum densisty in the same form the energy density was represented earlier.

Finally the actual evolution equation that advance the geometry from one slice to another can be defined as

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij}$$

where the funny looking l is the lie derivative.

Next

$$\partial_t K_{ij} = -D_i D_j \alpha + \alpha(R_{ij} + K K_{ij} - 2K_{il}K^l_j) - \mathcal{L}_\beta K_{ij} + 4\pi\alpha((S-\rho)\gamma_{ij} - 2S_{ij})$$

An evolution with 6 components.

Looking at this, it may seem to have too many components, especially when the current constraints seem redundant as you can only determin γ_{ij} and K_{ij}.

But this isn't the case, rather the constraints are guaranteed to be satisfied if the solution is solved precisily, the evolution equations will always satisfy the constraints if the initial conditions satisfy the constraints.

Now before I continue, I will note that when I say always, I am refering to analytical solutions, numerically you still need other methods to enforce the constraints.

Now we have 12 compoenents, but four are constrained via the constraint equations, another four correspond to the gauge freedom. Leaving us with four, metrics and extrinsic curvature.

Because of the additional terms even in linear form, GR is especially hard to solve even for hyperbolic equations.

If we take linearized Einstein equation for a slightly pretured spacetime

$$\gamma_{ij} = \eta_{ij} + h_{ij}; \quad |h_{ij}| \ll 1$$

Anyone who has looked at gravitational waves would know that

$$\ddot{h_{ij}} = \Delta h_{ij} + (h_{ik,kj} - h_{jk,ki} + h_{kk,ij})$$

(parentheses now to show special part change in video)

From this, we find that in ADM formalism, if constraints are violated even slightly, the error will increase in time with no limit, even in nearly flat spacetime.

Thus we need something different.

49.12.7 BSSN formalism

Because of the problem we just mentioned, we must reformulate the formulsism.

At least on a linear level, the wave equations must be derived, where we define two new variables

$$F_i = h_{ij,j}\Phi = h_{ii}$$

Thus rewriting our earlier equation as

$$\ddot{h_{ij}} = \delta h_{ij} - F_{i,j} - F_{j,i} + \Phi_{,ij}$$

Because of our constraint equations, we can use the fact that the momentum equations states that

$$\dot{F_i} - \dot{\Phi}_{,i} = 0$$

And the hamiltonian constraint states

$$\Delta\Phi - F_{i,j} = 0$$

Thus

$$\ddot{h_{ij}} = \delta h_{ij}$$

Similar definitions of new variables is also possible in nonlinear cases.

I won't go over the full proof but I will link it in the video.

49.12.8 Guage Conditions

Due to General Relativity's general covariance, there are degrees of freedom in choosing coordinates, the more mathematical and broad term for this is gauge freedom.

The slicing condition where we may prescribe the lapse function and the shift condition where we choose the shift vector

(earlier image)

Einstien's equations say nothing about how the gauge condition should be imposed, thus choosing them is something we can do, we should do it because gauge conditions can make the numerical simulations easier or harder.

Some examples of this is that improper slicing can lead to the appearence of singularities in simulations without black holes, another aspect is that carefully choosen shift vectors can allow one to do a compact binary merger without the extra computation from frame dragging.

Two failures show why the gauge matters. With geodesic slicing ($\alpha = 1$), observers fall toward the hole and the slices hit the singularity in finite time. In shift conditions that do not adapt, coordinate points get dragged around by the shift and the grid becomes badly distorted.

We use the standard "moving puncture" pair.

Slicing: 1+log. This slicing avoids the singularity by slowing the lapse where curvature is strong:

$$\partial_t \alpha = -2\alpha K + \beta^k \partial_k \alpha.$$

Shift: Gamma-driver. This drives the shift so that $\tilde{\Gamma}^i$ stays roughly constant, which keeps the coordinates from stretching:

$$\partial_t \beta^i = \tfrac{3}{4}\tilde{\Gamma}^i - \eta\,\beta^i, \qquad \eta \sim 1/M.$$

The damping η stops the shift from growing without bound and

keeps the system from oscillating.

For a first test one can also freeze the gauge at the analytic Kerr–Schild α, β^i. That isolates the numerical method from the gauge dynamics, and the spacetime should then stay static to within truncation error. A caution for dynamic gauge with excision: the gauge waves travel at speeds that can exceed the speed of light, so the excision surface must sit deep enough that every characteristic, including gauge modes, points into the hole. Check this numerically instead of assuming it.

49.12.9 Initial Data

As mentioned earlier, BSSN formulism automattically enforces its constraints as long as the initial data fits the constraints and satisfies Einstien's field equation. Traditionally this is a challenging thing to do, but because the kerr solution in a vacuum is one of the few analytical solutions to GR, we can just use that. The equations we used earlier very simply and easily work.

I have read some technically more efficeint and better methods of formulating the initial conditions, but for my purposes they are a bit too wieldy.

Kerr–Schild coordinates are already in 3+1 form, so we can read off every ADM variable analytically (with $l^i = \delta^{ij} l_j$):

$$\alpha = \frac{1}{\sqrt{1+2H}}, \quad \beta^i = \frac{2H}{1+2H}\, l^i, \quad \gamma_{ij} = \delta_{ij} + 2H\, l_i l_j, \quad \gamma^{ij} = \delta^{ij} - \frac{2H}{1+2H}\, l^i l^j, \quad \det\gamma =$$

The solution is stationary, so $\partial_t \gamma_{ij} = 0$, and the evolution equation for γ_{ij} gives the extrinsic curvature directly:

$$K_{ij} = \frac{1}{2\alpha}\big(D_i\beta_j + D_j\beta_i\big).$$

To feed the BSSN variables we do the conformal decomposition:

$$\phi = \tfrac{1}{12}\ln\det\gamma, \quad \tilde{\gamma}_{ij} = e^{-4\phi}\gamma_{ij}, \quad K = \gamma^{ij}K_{ij}, \quad \tilde{A}_{ij} = e^{-4\phi}\big(K_{ij} - \tfrac{1}{3}\gamma_{ij}K\big), \quad \tilde{\Gamma}^i = -$$

Because this is an exact vacuum solution, the constraints hold analytically. On the grid they hold only to truncation error, and that residual is our baseline for the validation section.

49.12.10 Cartoon Method

Essentially, the cartoon method is a way to run an axisymmetric grid in a 2d Cartesian grid. This is because traditionally the way to take out a dimentsion through axisymmetric systems is to use cylindrical or spherical coordinates, but one this adds more work in trying to get it into this form, and two these systems have the tendincy to create coordinate singularities.

(image animation)

The mathematical trick is to rotate all of the vectors and tensors around the z axis so that instead of an x, y, z axis we have a $\rho, 0, z$ axis where $\rho = \sqrt{x^2 + y^2}$. This then slightly changes the numerical derivatives of the y-axis. Take the traditional form

$$\partial_y \beta^x = \frac{\beta^x(x, +\Delta y, z) - \beta^x(x, -\Delta y, z)}{2\Delta y}$$

Take your space

$$(x, +\Delta y, z)$$

Take the corresponding point in the computational plane

$$(\rho, 0, z)$$

where

$$\rho = \sqrt{x^2 + \Delta y^2}$$

thus backwards one can get the our change in y for the derivative.

49.12.11 Excision

While everyone knows about the physical singularity inside of the black hole, there is also a coordinate singularity at the event horizen.

While analytically, one can get around these problems, doing so numerically is a bit more challenging.

There are multiple possible solutions.

- Puncture
- Horizon penetration
- Hyperbolic-penetrating
- Characteristic evolution
- Black-hole membrane
- Excision

For the purposes of this project we will use Excision, which is what it sounds like. Cutting the black hole out of the spacetime.

To cut the apperent horizen out, note I said apperent not event. The apperent horizen is the observed event horizen by an observer, a true event horizen is the horizen for all possible observers.

To do this, we need to know where it is. In this simplified model, the solution is analytical and thus no numerical machinery is needed, in other problems this is not the case.

Simply finding where the null ray line is, I won't go into detail about that but I will be using it in the future.

Excision means we stop evolving points inside a surface $r < r_{\mathrm{exc}}$ in the interior of the hole. Why is this allowed? Inside the event horizon, information can only flow inward, so what we do to the interior cannot affect the exterior, provided every characteristic at the surface points into the hole. For Kerr in Kerr–Schild coordinates the horizon is the spheroid $r = r_+$ given above, so no horizon finder is needed. In general one solves for the apparent horizon, where the outgoing null expansion $\Theta = D_i s^i - K + K_{ij} s^i s^j$ vanishes. (The event horizon needs the whole future evolution and cannot be located on a slice.) I choose r_{exc} safely inside r_+ (for example $1.5M$ for $a = 0$, where $r_+ = 2M$), using the r formula from the initial-data section for the mask, and the mask also encloses the ring singularity at $r = 0$.

The subtlety is the boundary of the excised region. Finite-difference stencils near it reach into the excised points, and the stencil "domain of dependence" is wider than the light cone, so errors near the singularity can leak outward even though the physics is causal. Two things I found (edit to match your runs):

- Simply freezing the interior values was not enough. It slowed the growth of errors but did not stop it, and a slow exponential divergence (e-folding time about $0.9M$) appeared in longer runs.

- Refilling the excised points every step by extrapolation from the outside, using a Jacobi-style iteration, fixed it. The refilled values do not need to be accurate, only smooth and non-growing.

This is a low-order treatment and it does not converge at the excision surface. That is acceptable because the region is causally

disconnected from the exterior, but it means convergence tests should exclude a thin shell around the surface.

49.12.12 Method of lines + RK4

We discretize space and leave time continuous, so the evolution equations become one big system of ODEs, $\partial_t u = F(u)$, where u is every evolved variable at every grid point. Spatial derivatives use centered finite differences (second order to start, fourth order is the natural upgrade), except the advection terms $\beta^k \partial_k$, which use one-sided stencils biased against the direction of the shift. Then classical RK4:

$$k_1 = F(u^n), \qquad k_2 = F(u^n + \tfrac{\Delta t}{2} k_1),$$
$$k_3 = F(u^n + \tfrac{\Delta t}{2} k_2), \qquad k_4 = F(u^n + \Delta t\, k_3),$$

$$u^{n+1} = u^n + \tfrac{\Delta t}{6}\,(k_1 + 2k_2 + 2k_3 + k_4).$$

Stability requires the CFL condition $\Delta t \le \lambda\, \Delta x$; I use $\lambda \approx 0.25$–0.5, and a safe choice accounts for the fastest gauge speed too. Every RK stage, not just the step, must first apply the boundary conditions, fill the cartoon ghost planes, and refill the excised region.

Finite-difference noise at the grid scale can grow, especially near the excision surface and outer boundary. We add a small high-frequency filter to every right-hand side, for a fourth-order scheme

$$\partial_t u \;\mathrel{+}=\; \frac{\epsilon}{64\,\Delta x}\big(u_{i+3} - 6u_{i+2} + 15u_{i+1} - 20u_i + 15u_{i-1} - 6u_{i-2} + u_{i-3}\big)$$

(this is the 6th-derivative form; a 4th-derivative stencil goes with 2nd-order schemes), with $0 < \epsilon < 1$, typically around 0.1. It removes only the noisiest modes without changing the convergence order.

49.12.13 Boundary Conditions

The outer boundary is artificial, so we want waves and gauge pulses to leave without reflecting. The options:

- Dirichlet (freeze the fields) and Neumann/copy both reflect waves and, in the copy case, break the $1/r$ falloff of the Kerr field.
- Periodic is unphysical here.
- Constraint-preserving and reflection-free conditions (e.g. Bayliss–Turkel, Buchman–Sarbach) are best, but heavy for a first code.
- Sommerfeld (radiative) conditions are simple and, crucially, exactly preserve a static $1/r$ tail.

The Sommerfeld condition assumes each field approaches its flat-space value f_0 and that deviations are outgoing spherical waves $\propto 1/r$, i.e. $(\partial_t + v\,\partial_r)\big(r(f - f_0)\big) = 0$:

$$\partial_t f = -v\Big[\frac{x^k}{r}\partial_k f + \frac{f - f_0}{r}\Big],$$

with $v = 1$ for most fields and $v = \sqrt{2}$ for the 1+log lapse. The flat values are $\alpha = 1$, $\tilde{\gamma}_{ij} = \delta_{ij}$, and zero for $\phi, K, \tilde{A}_{ij}, \tilde{\Gamma}^i, \beta^i$. Derivatives at the boundary use one-sided stencils. Check that a static Kerr tail $f - f_0 = c/r$ gives $\partial_t f = 0$: the two terms in the bracket cancel exactly. It is an approximation, not a true characteristic condition, so place the boundary far away (tens of M or more) and confirm the results do not depend on where it is.

49.12.14 Code

Now, putting it all together, while simple, is hard to find information on. For the most part books on the topic just assume you know enough to put it together but I am going to spell out the entire algorithm.

(image of flow chart)

49.12.15 Validation

49.12.16 Next Steps

Now we have the basic SpaceTime code working, the next steps is trying to figure out how to incorporate hydrodynamics into the code, a pretty big problem for next time.

49.12.17 Results

Here we can enjoy the results

49.13 Metriplectic Part 1: Why Can't Ordinary Hamiltonians be Dissipative

- Introduction
- Example of dissipative Hamiltonian that fails(air resistance)
- What is a Hamiltonian
- Why this disallows dissipation

- Introducing Reversibility
- Entropy
- Metriplectic
- Conclusion and a look towards the future

49.13.1 Introduction

Hi, and welcome to this video on why Hamiltonian dynamics does not permit dissipation and how modern advances get around this fact, spesifically looking towards Metriplectic dynamics.

49.13.2 Example

This might seem strange, I mean air resistance dissipation is one of the first problems introduced in physics. Taking linear drag to even further simplfy and no gravity to even futher simplify.

Taking say a baseball released at a given speed v, with a drag represented via

$$m\ddot{x} = -d\dot{x}$$

The rest is trivial to get what ever you want; velocity at a given point, kinetic energy at a given time, total energy.

So, to create the Hamiltonain, we define the momentum as

$$p = m\dot{x}$$

thus

$$\dot{p} = -\frac{b}{m}p$$

Such that the solution is

$$p(t) = p_0 e^{-bt/m}$$

Because of this, and the fact that there is no position dependent function or potential energy, the Hamiltonian is

$$H(q, p, t) = \frac{p^2}{2m} e^{-\frac{b}{m}t}$$

This allows us to very easily say that it calculates the position trivially, yet it claims that momentum never changes. Further, when taking the function of the Hamiltonian we find that it doesn't match the energy function in the Newtonian case.

(graph images)

As you can clearly see these are miss matches. Now one can say that the Hamiltonian isn't energy and the conjugate momenta is not momentum, this is true, but then we got the right answer only because we went backwards, with the solution to get the Hamiltonian, making the Hamiltonian mostly useless. As we will further get into this can have much more devastative consequences beyond simply unphysical phase-space.

49.13.3 Why?

So why did this happen? A Hamiltonian is not just a function. It is a function together with a geometric structure on phase space, and that structure cannot accommodate dissipation. Conservation of energy and conservation of phase-space volume look like two separate facts, but they are two shadows of the same one.

49.13.4 What a Hamiltonian System Actually Is

The state space is a smooth manifold M of dimension $2n$ with a symplectic form ω, a closed and nondegenerate two-form. By Darboux's theorem there are always local coordinates in which

$$\omega = \sum_{i=1}^{n} \mathrm{d}q^i \wedge \mathrm{d}p_i.$$

A function H then generates a vector field X_H through

$$\iota_{X_H}\,\omega = \mathrm{d}H \quad \Longleftrightarrow \quad \dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

Dually, the inverse of ω is an antisymmetric bivector J^{ij}, which defines the Poisson bracket

$$\{f, g\} = \partial_i f \, J^{ij} \, \partial_j g, \qquad \dot{f} = \{f, H\}.$$

The bracket has three properties. It is antisymmetric, $\{f, g\} = -\{g, f\}$. It obeys the Leibniz rule, $\{f, gh\} = \{f, g\}h + g\{f, h\}$. And it obeys the Jacobi identity,

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0.$$

When J is invertible, Jacobi is exactly the statement $\mathrm{d}\omega = 0$.

Now relax one assumption. Allow $J^{ij}(z)$ to be degenerate, as long as it stays antisymmetric and satisfies Jacobi. This is a *Poisson manifold*. A degenerate J has *Casimir functions*, meaning functions C with $\{C, f\} = 0$ for every f, or equivalently $J\nabla C = 0$. They are constant along all Hamiltonian motion, and their level sets slice M into *symplectic leaves*. Each leaf is a genuine symplectic manifold, and a Hamiltonian flow never leaves the leaf it starts on. Hold on to this picture, because the metriplectic bracket is built from it.

49.13.5 Why This Disallows Dissipation

There are three independent obstructions, and all three follow from the structure above.

Energy. Antisymmetry gives

$$\dot{H} = \{H, H\} = \nabla H \cdot J \nabla H = 0.$$

As most cases the Hamiltonian is energy, this means energy is conserved and thus dissipation cannot happen.

Volume. By Cartan's formula,

$$\mathcal{L}_{X_H}\,\omega = \mathrm{d}\,\iota_{X_H}\omega + \iota_{X_H}\mathrm{d}\omega = \mathrm{d}^2 H + 0 = 0.$$

So the flow ϕ_t satisfies $\phi_t^*\omega = \omega$, which means it preserves ω^n. This is Liouville's theorem, and it also preserves every Poincaré integral invariant $\int_\Sigma \omega^k$. Now compare the physical drag flow $X = \frac{p}{m}\partial_x - \frac{b}{m} p\,\partial_p$ on $\omega = \mathrm{d}x \wedge \mathrm{d}p$:

$$\iota_X\omega = \tfrac{p}{m}\,\mathrm{d}p + \tfrac{b}{m} p\,\mathrm{d}x \quad\Longrightarrow\quad \mathcal{L}_X\omega = -\frac{b}{m}\,\omega \quad\Longrightarrow\quad \phi_t^*\omega = e^{-bt/m}\omega.$$

The drag flow is only *conformally* symplectic.

Spectrum. Linearize about a fixed point, so that $\dot{\xi} = A\xi$ with $A = JS$ and $S = \operatorname{Hess} H$ symmetric. Then $J^{-1}AJ = SJ = -A^T$, so A is similar to $-A^T$ and its spectrum is symmetric under $\lambda \to -\lambda$. In particular $\operatorname{tr} A = 0$. Every decaying mode has a growing partner, so a Hamiltonian system has no asymptotically stable fixed points. For the drag problem, $\operatorname{tr} A = -b/m \neq 0$.

Combined with Poincaré recurrence on bounded phase space, the conclusion is that Hamiltonian flows have no attractors.

(visual: phase-space blob keeps its area, versus the blob contracting under drag)

49.13.5.1 Loophole 1: Hide the Dissipation in Time

The Caldirola–Kanai Hamiltonian $H = \frac{p_c^2}{2m} e^{-bt/m}$ is a perfectly good symplectic system in the variables (x, p_c) with $p_c = m e^{bt/m} \dot{x}$. The physical momentum is $p = e^{-bt/m} p_c$, and the map $\psi_t(x, p_c) = (x, e^{-bt/m} p_c)$ satisfies

$$\psi_t^*(\mathrm{d}x \wedge \mathrm{d}p) = e^{-bt/m}\, \mathrm{d}x \wedge \mathrm{d}p_c.$$

The canonical flow conserves area, the physical area shrinks, and the discrepancy lives entirely in a time-dependent, non-symplectic change of variables. The cost shows up when you quantize. The commutator is $[x, p_c] = i\hbar$, so the physical variables satisfy

$$[x, p] = i\hbar\, e^{-bt/m} \to 0,$$

and the uncertainty bound for the physical variables decays away (see Dekker's review). The Hamiltonian is not describing the physical phase space.

49.13.5.2 Loophole 2: Give Up the Canonical Structure

Time dependence is not even necessary. The drag flow has a first integral $c = x + p/b$, since $\dot{c} = p/m - p/m = 0$. In the coordinates $(c, u = \ln|p|)$ the flow is $\dot{c} = 0$, $\dot{u} = -b/m$, a constant vector field, so it preserves the area form $\mathrm{d}c \wedge \mathrm{d}u$. In the original variables this is

$$\omega' = \frac{\mathrm{d}x \wedge \mathrm{d}p}{p}, \qquad \{x, p\}' = p, \qquad \tilde{H} = \frac{bx + p}{m},$$

and one checks directly that $\dot{x} = \{x, \tilde{H}\}' = p/m$ and $\dot{p} = \{p, \tilde{H}\}' = -bp/m$. So drag *is* autonomously Hamiltonian, on the Lie–Poisson structure of the affine algebra. But look at the price:

- The bivector $J = p$ vanishes exactly on $p = 0$, which is

the attractor. The rank drops to zero there, and the symplectic leaves are the two half-planes $p \gtrless 0$ plus a line of points.

- Liouville is not violated. The ω'-area of the strip $\varepsilon < p < 1$ is $\ln(1/\varepsilon) \to \infty$, so the attractor sits infinitely far away in symplectic terms.
- $\tilde{H}$ is just the conserved quantity c in disguise. It knows nothing about kinetic energy, heat, or why the ball stops.

This is the real lesson. Locally, any planar flow without fixed points is Hamiltonian, since flow-box coordinates give $\dot{c} = 0$, $\dot{s} = 1$. So the existence of *a* Hamiltonian is cheap. What is expensive is a Hamiltonian that is the physical energy, on the canonical phase space, with a nondegenerate bracket. That combination is what dissipation forbids.

49.13.6 Reversibility

Time reversal is an *anti*-symplectic involution Θ, with $\Theta^2 = 1$ and $\Theta^*\omega = -\omega$. In canonical coordinates $\Theta(q, p) = (q, -p)$. If $H \circ \Theta = H$, then $\Theta_* X_H = -X_H$, which gives $\Theta \circ \phi_t = \phi_{-t} \circ \Theta$. Hamiltonian flows are conjugate to their own inverse.

Split the drag field into $X_{\text{rev}} = \frac{p}{m}\partial_x$ and $X_{\text{diss}} = -\frac{b}{m} p\, \partial_p$. Then

$$\Theta_* X_{\text{rev}} = -X_{\text{rev}}, \qquad \Theta_* X_{\text{diss}} = +X_{\text{diss}}.$$

The reversible terms are odd under time reversal and the dissipative term is even, so a reversed trajectory solves $\ddot{x} = +\frac{b}{m}\dot{x}$, which is anti-damping. Remember this parity split, because it comes back in the metriplectic structure.

There is also an algebraic version of this. A Hamiltonian flow $\{\phi_t\}_{t\in\mathbb{R}}$ is a *group* of diffeomorphisms, so it is injective and distinct states can never merge. The drag flow $p \mapsto e^{-bt/m}p$ is

only a *semigroup* on $t \geq 0$. As $t \to \infty$, all initial momenta collapse to the same point, and information about the initial condition is lost.

49.13.7 Entropy

Microscopic laws are reversible, yet friction exists. The resolution is that the ball is not a closed system. A standard model is Caldeira–Leggett, in which the ball couples linearly to N harmonic oscillators:

$$H_{\mathrm{tot}} = \frac{p^2}{2m} + V(x) + \sum_{k=1}^{N} \left[\frac{p_k^2}{2m_k} + \frac{m_k \omega_k^2}{2} \left(q_k - \frac{c_k}{m_k \omega_k^2}\, x \right)^2 \right].$$

This is exactly Hamiltonian, symplectic and energy-conserving. Projecting out the bath, which is Mori–Zwanzig with a projector $\mathcal{P}$ onto the ball's variables, gives a generalized Langevin equation:

$$m\ddot{x} = -V'(x) - \int_0^t \gamma(t-s)\, \dot{x}(s)\, \mathrm{d}s + \xi(t), \qquad \langle \xi(t)\xi(s) \rangle = k_B T\, \gamma(|t-s|).$$

The second relation is the fluctuation–dissipation theorem. For a bath with an Ohmic spectral density $J(\omega) \propto \omega$ in the limit $N \to \infty$, the kernel collapses to $\gamma(t) \to 2b\,\delta(t)$ and we recover $-b\dot{x}$. The order of limits matters here: for any finite N the full system has Poincaré recurrences, and dissipation is emergent only as $N \to \infty$ followed by $t \to \infty$.

What does entropy do in this picture? For a density ρ on M, Liouville's equation $\partial_t \rho = \{H, \rho\}$ says ρ is transported by a volume-preserving flow. So the Gibbs entropy

$$S_G[\rho] = -k_B \int \rho \ln \rho \quad \frac{\omega^n}{n!}$$

is exactly constant. Entropy growth requires coarse-graining. If $\mathcal{P}\rho$ is the projection of ρ onto the reduced description, a Pythagorean identity holds:

$$S_G[\mathcal{P}\rho] = S_G[\rho] + k_B\, D_{\mathrm{KL}}(\rho, —, \mathcal{P}\rho) \quad \geq \quad S_G[\rho].$$

The entropy gain is the relative entropy of what we discard, which is information about the bath. The reduced state z then carries an entropy function $S(z)$, in Boltzmann's sense of the log-volume of microstates consistent with z, and the second law reads $\dot{S} \geq 0$ along the reduced dynamics while the total energy E stays exactly conserved.

That sets our design target on the reduced space. We need a dynamics with $\dot{E} = 0$ and $\dot{S} \geq 0$ that holds for *any* E and S satisfying the right compatibility conditions.

49.13.8 Metriplectic Dynamics

The idea, due to Kaufman and Morrison in 1984, with the full paradigm in Morrison 1986 and independently as GENERIC by Grmela and Öttinger, is to carry two brackets on the same space: the Poisson bracket $\{\cdot,\cdot\}$ for reversible physics, and a symmetric, positive-semidefinite bracket $(\cdot,\cdot)$ for irreversible physics. In matrix form,

$$\dot{z} = L(z)\,\nabla E + M(z)\,\nabla S,$$

or in bracket form,

$$\dot{f} = \{f, E\} + (f, S).$$

The requirements are as follows.

- $L^T = -L$ and L satisfies the Jacobi identity.
- $M^T = M \succeq 0$, which is Onsager's reciprocity, from mi-

croscopic reversibility, together with positivity, from the second law. Geometrically, M^+ is a Riemannian metric on the range of M.

- The *degeneracy conditions*

$$L\nabla S = 0, \qquad M\nabla E = 0.$$

The degeneracy conditions have a clean interpretation. The first says S is a *Casimir* of the Poisson bracket, so reversible motion stays on a leaf of constant entropy. The second says E is a Casimir of the metric bracket, so irreversible motion stays on a surface of constant energy. Equivalently, $\dot f = \{f, F\} + (f, F)$ with generator $F = E + S$, since the cross terms vanish. The two laws then follow in one line each:

$$\dot E = \underbrace{\{E, E\}}_{0} + \underbrace{(E, S)}_{=\nabla E\cdot M\nabla S\,=\,0} = 0, \qquad \dot S = \underbrace{\{S, E\}}_{0} + (S, S) = \nabla S\cdot M\nabla S \geq 0.$$

The geometric picture is that reversible motion slides *along* symplectic leaves, irreversible motion slides *across* leaves within an energy shell, and S increases monotonically in the crossing direction. Under time reversal the L-term is odd and the M-term is even, exactly the parity split we found for drag. The name *metriplectic* just means metric plus symplectic.

(visual: energy shell in (x, p, s) space, foliated by constant-s symplectic leaves, with a horizontal reversible arrow and an upward-tilting irreversible arrow)

Now the baseball. Add one variable s, the entropy of the ball and surrounding air, with an internal energy $U(s)$ and temperature $T = U'(s) > 0$. On $z = (x, p, s)$, take

$$E = \frac{p^2}{2m} + U(s), \qquad S = s, \qquad \nabla E = \begin{pmatrix} 0 \\ p/m \\ T \end{pmatrix}, \quad \nabla S = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

with the canonical bracket on the (x,p) block and the dissipative matrix

$$L=\begin{pmatrix}0&1&0\\-1&0&0\\0&0&0\end{pmatrix},\qquad M=\frac{b}{T}\,vv^{T},\qquad v=\begin{pmatrix}0\\T\\-p/m\end{pmatrix}.$$

The degeneracy conditions hold. $L\nabla S=0$ because s is a Casimir of L, and $M\nabla E=\frac{b}{T}\,v\,(v\cdot\nabla E)=0$ because $v\cdot\nabla E=Tp/m-Tp/m=0$. The equations of motion are

$$\dot{x}=\frac{p}{m},\qquad \dot{p}=-\frac{b}{m}\,p,\qquad \dot{s}=\frac{b\,p^{2}}{m^{2}T}=\frac{b\,\dot{x}^{2}}{T}.$$

The first two are the original drag equations, now with the physical momentum, no explicit time dependence, and a nondegenerate bracket on the mechanical block. The third is $\mathrm{d}S=\delta Q/T$, with dissipated power $b\dot{x}^{2}$. Energy conservation checks out explicitly:

$$\dot{E}=\frac{p\dot{p}}{m}+T\dot{s}=-\frac{bp^{2}}{m^{2}}+\frac{bp^{2}}{m^{2}}=0.$$

Finally, $\dot{S}=\frac{b}{T}(p/m)^{2}$ vanishes only at $p=0$, so by LaSalle's invariance principle every trajectory converges to the equilibrium set $\{p=0\}$. The final entropy is fixed by energy conservation through $U(s_{\infty})=E_{0}$. The attractor that Hamiltonian mechanics forbids now exists, and the phase-space volume lost in (x,p) has been moved into s rather than destroyed.

49.13.9 Conclusion and a Look Ahead

A Hamiltonian system is a Poisson structure plus a generator. Antisymmetry of the bracket gives $\dot{H}=0$, Cartan's formula gives $\mathcal{L}_{X_H}\omega=0$, and the spectrum is symmetric under

$\lambda \to -\lambda$. Every attempt to write drag as a Hamiltonian system pays somewhere: with explicit time dependence and non-symplectic coordinates, or with a Poisson structure that goes singular on the attractor. Real dissipation is projection, which is information flowing into degrees of freedom we do not track, and entropy is the ledger. Metriplectic dynamics encodes this by joining a Poisson bracket to a metric bracket under the two degeneracy conditions, and the first and second laws follow by construction.

Several directions open from here. Morrison's 4-bracket replaces the symmetric bracket with a curvature-like object $(f, g; k, n)$ with Riemann-tensor symmetries, which makes the structure more systematic. For continuum systems, L becomes a Lie–Poisson bracket and M becomes the Landau collision operator in plasmas, or viscosity and heat conduction in fluids. On the numerical side, structure-preserving discretizations such as discrete-gradient methods conserve E and make S monotone at the discrete level. In the next video we will construct these brackets for a concrete PDE, and see what goes wrong, and how badly, when the degeneracy conditions are broken even slightly. Thanks for watching.

49.14 Numerically Simulating Navier-Stokes Singularity

- Introduction
- Why
-

49.15 Developing Numerical Relativity: Simulating a Star with TOV

- Introduction
- Scope
- Einstien + Matter
- Matter Model
- Relativistic Hydrodynamics
- TOV equations
- Initial data for Star
- Matter term to BSSN
- Coupling systems
- Convergence
- Slightly perturbed

Chapter 50

GHMDH: Kerr black hole inside of Star

50.1 Scope

Here I outline my plan to create a simulation of a kerr black hole inside of a star, in order to develop my understanding of BSSN while adding th extra challenge of MHD.

50.2 Stage 1: Vacuum Kerr Black Hole

50.2.1 Initializing Values

We start with taking in the data, then transferring it into ADM and then BSSN.

By follutiong the 4D hypersurface by

$$M = \cup_t \Sigma_t$$

Taking

$$g_{\mu\nu} \rightarrow (\gamma_{ij}, \alpha, \beta^i)$$

The spatial metric, the lapse, and the shift.

Also the introducing the extrinsic curvature.

The decomposing it.

Next we introduce the BSSN formalism, which creates the unit-determinant condition, or the contradiction of the conformal Christoffel symbol.

50.2.2 Hamiltonian Constraints

From here we end up at the Hamiltonian constraints, that force conservations of momentum and energy just like traditional hamiltonians, they just aren't the only enforcement in BSSN that keep things only partially hyperbolic.

50.2.3 Integration

50.2.3.1 RK4

We start with a traditional RK4 model for integration. Inside each section we have the traditional residual, Sommerfeld radiative boundary conditions, and k0 dissipation

50.2.3.2 RHS

Pretty trivial

50.3 Theorizing about Hydrodynamics

For the set up I need to add a Valience TOV to initialize the star in the first place

Add more primitive and conservative values for the purposes of conseriving the matter fields.

Edit the constraints for the TOV

The CFL should be expanded for the speed of sound in the star rather than simply the grid and the speed of light.

Add the hydrodynamics seperately, using the same boundary condition and such but for the Valience fluid dynamics.

Add Con2prim

I think I am going to start with building the initialization

50.3.1 Initialization

Add to the parameter building that EOS settings like gamma or such.

Solve the TOV ODE for the swarzchild metric.

Map to isotropic

Add matter data points to the initial data.

Take the primitives to write out conservative variables

Edit the constraints for matter.

Figure out how to skip Excision with no black hole, I might need to go about adding a black hole finder but that seems exsesive for now.

50.4 Orientation

50.4.1 What this code is

The code evolves the vacuum Kerr spacetime in the **BSSN** formulation of general relativity, under the assumption of **axisymmetry about the z-axis**, using the **Cartoon method** to represent the y-direction derivatives, and **excising** a region inside the inner horizon so that no boundary condition is ever needed at the black hole.

The test problem is exact: Kerr–Schild initial data, $M = 1$, $a = 0.8$. Because this is an exact vacuum solution, the continuum constraints vanish identically everywhere. Any constraint violation the code reports is therefore, by construction, *pure discretisation error*. This makes the code an unusually clean testbed: there is no physics to blame.

The domain is the half-plane $x \in [0, x_{\max}]$, $z \in [-z_{\max}, z_{\max}]$, with the y-direction reconstructed on demand.

50.4.2 Why the design is shaped this way

Three decisions dominate the architecture, and each exists for a good reason:

1. **BSSN rather than ADM.** ADM's hyperbolic, constraint-preserving properties degrade badly in practice; BSSN adds 21 extra variables (a conformal factor, a traceless metric, a connection, a driver) and buys a genuinely constraint-satisfying evolution. The cost is stiffness and a much larger set of equations to discretise consistently.

2. **A Cartoon grid rather than full 3D.** A full N^3 grid is wasteful when the solution is axisymmetric. The Cartoon method stores only $y = 0$, $x \geq 0$ and synthesises the rest

by interpolating in radius and rotating. The price is that y-derivatives are only as good as the radial interpolation, which in practice limits them to second order.

3. **Excision rather than a puncture condition.** Rather than imposing regularity at a coordinate singularity, the code simply never evolves points inside a surface $r_{\rm KS} < r_{\rm excise}$ and fills them by extrapolation. The price is that stencils near the mask degrade, and that degradation is where this code's problems live.

50.4.3 Quick start

```
# Build the production binary and run the full test suite (the
    gate).
make check

# Run the production configuration.
make run

# Run with runtime overrides, keeping output out of the tree.
make run RUN_ENV='GR_NSTEPS=700 GR_HORIZON_EVERY=0 GR_HZ_ABORT=0'
     RUN_DIR=/tmp

# Run a different parameter file.
make run PARFILE=parameter_files/vacuum_kerr_192.par
```

A typical short run looks like:

```
$ GR_NSTEPS=700 GR_HORIZON_EVERY=0 ./bin/vacuum_kerr
    parameter_files/vacuum_kerr.par
 Grid: Nx=48  Nz=96
 dx= 0.255  dy= 0.255  dz= 0.253
 Excision radius r_excise= 1.000  (horizon at r_+= 1.600M;
    interior frozen)
 step   700  t=  0.6252  max|H|(ext)=  1.2656E-02  rms|H|(ext)=
    2.0211E-03
   max|M|(ext)=  2.8728E-02  alpha(probe)=  0.910348
    near-mask r_KS <   3.0000E+00  n=   192  max|H|=  1.9493E+00
```

```
8     max|M|=  1.4920E+00  worst r=  1.0227E+00
```

Notice the near-mask line. The `max|H|(ext)` and `near-mask max|H|` differ by a factor of 154, and the two numbers are about completely different things. Chapter 50.9 explains why both exist, and why the first one is structurally incapable of seeing the region that matters.

50.4.4 Repository map

50.4.4.1 Core evolution path

These files, and only these, constitute the production solver:

File	Role
`helper/kinds\_mod.F90`	The single kind parameter. Eight lines.
`init/param.F90`	All runtime parameters and their environment overrides.
`init/grid.F90`	Grid coordinate arrays.
`init/vars.F90`	The state vector layout: 24 indices and their meanings.
`init/init\_data.F90`	Kerr–Schild initial data and the conformal connection.
`helper/cartoon.F90`	The Cartoon method: interpolation and rotation.
`helper/stencil.F90`	All finite differences, with excision-aware fallbacks.
`helper/geometry.F90`	Conformal geometry: Ricci tensors, Laplacian, connections.
`integrators/bssn\_rhs.F90`	The BSSN evolution equations.

File	Role
integrators/ko_diss.F90	Kreiss–Oliger dissipation.
integrators/sommerfeld.F90	Outer boundary condition.
helper/excision.F90	The excision mask, ghost fill, and characteristic check.
helper/excision_sat.F90	Level-set characteristic SAT (available, off).
helper/constraints.F90	Hamiltonian and momentum constraint evaluation.
integrators/time_step.F90	RK4, the algebraic projection, the time loop.
helper/horizon.F90	Horizon location finder.
helper/output.F90	HDF5 output.
drivers/vacuum_kerr.F90	The program: setup, time loop, diagnostics.

50.4.4.2 Not on the production path

Several substantial modules exist that are *not* used by the production run. Knowing which is which saves a great deal of confusion:

- helper/fd_excision.F90, helper/fo_*.F90, integrators/fo_*.F90, init/fo_*.F90 — a first-order-in-time ("FO") reduced formulation with a boundary SAT built on fo_sbp.F90. These are a parallel development track. The SBP module does not currently satisfy the SBP property (see Section 50.11.1), so this track is not usable.

- `helper/equilibrium\_1pluslog.F90`, `helper/eq\_schwarzschild\_data.F90` — special equilibrium data used only by tests.

- `helper/primitive\_recovery.F90` — hydrodynamics recovery. Not exercised by the vacuum problem, though the BSSN RHS does accept an optional matter source `msrc`.

50.4.4.3 Tests and diagnostics

`make check` runs the full gate. Separately there are diagnostic programs, each a standalone `program` linked against the same modules:

Target	Purpose
`test-cartoon`	Verifies the Cartoon reconstruction against known fields.
`test-constraint-conv`	Hamiltonian-constraint convergence on exact data.
`test-momentum-conv`	Momentum-constraint convergence; a real gate.
`test-horizon-drift`	Validates the horizon finder against exact Kerr.
`test-gamma-driver-rhs`	Checks the gauge RHS is stationary on KS data.
`diagnose-ks-stationarity`	RHS residual on frozen Kerr–Schild.
`diagnose-momentum-point`	Stored vs. analytic state, term by term.
`diagnose-inner-boundary`	Fill perturbation-depth scan.
`diagnose-ghost-accuracy`	Ghost fill vs. exact; derivative errors.

50.5 The Physics

50.5.1 From ADM to BSSN

50.5.1.1 ADM

The starting point is the Arnowitt–Deser–Misner decomposition of spacetime into a spatial metric g_{ab}, an extrinsic curvature K_{ab}, and a lapse α and shift β^a, with

$$\mathrm{d}_\perp \alpha = -\alpha^2 K, \tag{50.1}$$

$\mathrm{d}_\perp = \partial_t + \beta^a \nabla_a$ denoting the derivative along the normal direction to the slices.

50.5.1.2 The conformal split

BSSN writes the physical metric as

$$g_{ab} = e^{2\tilde{\phi}}\, \tilde{\gamma}_{ab} = e^{4\tilde{\phi}} \hat{\gamma}_{ab}, \tag{50.2}$$

with $\tilde{\phi}$ the conformal factor and $\tilde{\gamma}_{ab}$ the conformal metric. The lapse is split as $\alpha = e^{\tilde{\phi}} A$ (here $\hat{A}$, stored as `IALPHA` with `conformal_factor_mode =` so $\tilde{\phi} = \frac{1}{12}\ln\det g$ and $\hat{A} = \alpha e^{-\tilde{\phi}}$), and the shift into $\hat{A}^a = \hat{g}^{ab}\beta_b$ plus a vector $\hat{\Gamma}^a$.

$\tilde{\phi}$ evolution

$$\partial_t \tilde{\phi} = -\tfrac{1}{6}\alpha K + \beta^a \partial_a \tilde{\phi} + \tfrac{1}{6}\beta^a \partial_a \beta^a, \tag{50.3}$$

which is `rhs_out(IPHI)` in `integrators/bssn\_rhs.F90`.

$\tilde{\gamma}$ evolution

$$\partial_t \tilde{\gamma}_{ab} = -2\alpha \tilde{A}_{ab} + \beta^c \partial_c \tilde{\gamma}_{ab} + \tilde{\gamma}_{ac}\partial_b \beta^c + \tilde{\gamma}_{bc}\partial_a \beta^c - \tfrac{2}{3}\tilde{\gamma}_{ab}\,\partial_c \beta^c, \tag{50.4}$$

with $\tilde{A}_{ab}$ the traceless part of the extrinsic curvature in conformal form, $\tilde{A}_{ab} = e^{-4\tilde{\phi}}\left(K_{ab} - \frac{1}{3}K g_{ab}\right)$. Writing $X_{ab} = -\tilde{\nabla}_a\tilde{\nabla}_b\hat{A} + \alpha\tilde{R}_{ab}$ and stripping its trace, $\bar{X}_{ab} = X_{ab} - \frac{1}{3}\tilde{\gamma}_{ab}\,\mathrm{tr}\,X$, the conformal extrinsic curvature obeys

$$\partial_t\tilde{A}_{ab} = e^{-4\tilde{\phi}}\bar{X}_{ab} + \alpha\left(K\tilde{A}_{ab} - 2\tilde{A}_{ac}\tilde{A}^c{}_b\right) + \beta^c\partial_c\tilde{A}_{ab} + \tilde{A}_{ac}\partial_b\beta^c + \tilde{A}_{bc}\partial_a\beta^c - \tfrac{2}{3}\tilde{A}_{ab}\,\partial_c\beta^c. \tag{50.5}$$

$S^a_b = K^a_b - \frac{1}{3}K\delta^a_b$ is the trace-reversed extrinsic curvature. The code needs the derivative $\nabla_a\nabla_b\beta^c$, obtained from `S%d2xy`, `S%d2xz`, `S%d2yz` assembled by `fill_d2`.

Algorithm: `compute_rhs_interior`

1. For each evolved point (i,k), set $\det\tilde{\gamma} = 1$ and $\tilde{\gamma}^{-1ab}$ from S.
2. Compute the conformal geometry: $\tilde{\gamma}^{-1ab}$, $\tilde{R}^a{}_{bc}$, $\tilde{R}_{ab}$ (metric part), $\tilde{R}^{\phi}_{ab}$, $\tilde{\Gamma}^a$, $\hat{A}^a$, $\tilde{\nabla}_a\tilde{\nabla}_b\hat{A}$.
3. $\partial_t\tilde{\phi}$ from (50.3).
4. $\partial_t\tilde{\gamma}_{ab}$ from (50.4).
5. $\partial_t K = -\tilde{\gamma}^{-1ab}\tilde{\nabla}_a\tilde{\nabla}_b\hat{A} + \alpha\left(\tilde{A}_{ab}\tilde{A}^{ab} + \frac{1}{3}K^2\right) + \beta^a\partial_a K$.
6. $X_{ab} = -\tilde{\nabla}_a\tilde{\nabla}_b\hat{A} + \alpha\tilde{R}_{ab}$, strip the trace: $X_{ab} \to X_{ab} - \frac{1}{3}\tilde{\gamma}_{ab}\,\mathrm{tr}\,X$, then (50.5).
7. $\partial_t\tilde{\Gamma}^a$ from $\tilde{\Gamma}^a{}_{bc}\tilde{A}^{bc} + \frac{2}{3}\tilde{\gamma}^{ab}\hat{A}^c\partial_b\partial_c\hat{A}^a$.
8. Gauge: $\partial_t\hat{A}$, $\partial_t\beta^a$ per (50.8) and (50.11).

50.5.2 Constraints

The code evaluates both, for diagnostics; *neither is projected*.

Hamiltonian, `constraints.F90`

$$\mathcal{H} = \tilde{\gamma}^{ab}\tilde{R}_{ab} + \tfrac{2}{3}K^2 - \tilde{A}_{ab}\tilde{A}^{ab}, \qquad \tilde{R}_{ab} = \tilde{R}^{\gamma}_{ab} + \tilde{R}^{\phi}_{ab}. \tag{50.6}$$

Momentum, `constraints.F90`, ADM form

$$\mathcal{M}^a = \nabla_b S^{ab}, \qquad S^{ab} = e^{-4\tilde{\phi}} \left[\tilde{A}^{ab} - \tfrac{2}{3}\tilde{\gamma}^{ab} K \right]. \tag{50.7}$$

Gotcha 1. *The original code assembled* (50.7) *as* $\tilde{\gamma}^{ab}\partial_b \tilde{A}_{ab} - \frac{2}{3}K\partial_b\tilde{\gamma}^{ab} - \ldots$ *using the* conformal *connection. That form is cited in the literature but is not equal to the ADM expression above unless* $\det\tilde{\gamma} = 1$ *is imposed. Since the state drifts from* $\det\tilde{\gamma} = 1$*, the two differ. Rewriting it in ADM form changed the measured momentum residual from a non-converging* $\approx 1.1 \times 10^{-1}$ *to a cleanly second-order* $4.1 \times 10^{-2} \to 3.2 \times 10^{-2} \to 2.1 \times 10^{-2} \to 1.7 \times 10^{-2}$*.*

The correction was made only *in the diagnostic routine. An A/B test that corrupts the evolved state deliberately confirmed the diagnostic was not on the evolution path, so no stability risk was incurred.*

Gotcha 2. *`test_momentum_conv` had no assertion in it. It printed numbers and passed unconditionally. A real gate requiring monotonically decreasing RMS was added, and verified to fail when a sign is flipped.*

50.5.3 Gauge

$$\textit{1+log:} \quad \partial_t \hat{A} = -2\alpha K + \beta^a \partial_a \hat{A}, \tag{50.8}$$

$$\textit{maximal:} \quad \partial_t \hat{A} = -\alpha K + \beta^a \partial_a \hat{A}, \tag{50.9}$$

$$\textit{frozen:} \quad \partial_t \hat{A} = 0. \tag{50.10}$$

$$\textit{legacy:}\quad \partial_t \beta^a = \tfrac{3}{4} B^a, \tag{50.11}$$

$$\partial_t B^a = \partial_t \tilde{\Gamma}^a - \eta B^a, \tag{50.12}$$

$$\textit{puncture:}\quad \partial_t \beta^a = \tfrac{3}{4} B^a, \qquad \partial_t B^a = -\eta\,(B^a - \Gamma^a), \quad \Gamma^a = s\,\frac{2Mx^a}{r^3}, \tag{50.13}$$

$$\textit{frozen:}\quad \partial_t \beta^a = 0. \tag{50.14}$$

Gotcha 3. *In* (50.13)*,* $\partial_t B^a$ *has* no $\partial_t \tilde{\Gamma}^a$ *term: in a* Γ*-driver the driver is driven purely toward the analytic profile. Feeding it the* $\tilde{\Gamma}$ *RHS as well — which this code once did — couples* B^a *to a different variable and lets the shift drift instead of being anchored. The code comments record that fixing this alone made the run* worse *in the legacy mode (inner-boundary failure at step 14055 instead of 18285), which is why the legacy mode is still available but the puncture mode is the default.*

Gotcha 4. *Setting the gauge to frozen (modes 2/2) is mandatory before measuring any discretisation residual. If the lapse or shift is evolving, their right-hand sides contaminate the measurement. Several early conclusions in this project were confounded this way before the control was added.*

50.6 The Discretisation

50.6.1 The grid

```
real(dp), allocatable :: xg(:)   ! x(1) = 0, x(Nx) = xmax
real(dp), allocatable :: zg(:)   ! z(1) = -zmax, z(Nz) = +zmax
```

Fortran indices run $1..N$*, so* $\mathrm{i} = 1$ *is the symmetry axis* $x = 0$ *and* $\mathrm{k} = 1$ *is* $z = -z_{\max}$*. In the production configuration* $N_x = 48$*,*

$N_z = 96$, $x_{\max} = z_{\max} = 12\,M$, *giving* $dx \approx 0.2553$, $dz \approx 0.2526$, *and* $dy = dx$.

Gotcha 5. *The symmetry axis is* $x = 0$ *and the* z*-axis is the spin axis. The code mirrors in* x *via a rotation by* π *in the* xy*-plane, which is the correct axisymmetry-about-*z *operation. It is easy to misread the axis conventions from the variable names;* `valid_x` *treats* $ii < 1$ *as "past the symmetry axis" and synthesises it.*

50.6.2 The Cartoon method in full

50.6.2.1 What is stored

Only the half-plane $y = 0$, $x \geq 0$. *Given a request for the value at* (X, Y, z_k), *the code:*

1. *Reduces to the cylindrical radius* $\varpi = \sqrt{X^2 + Y^2}$ *and the azimuth* $\theta = \arctan 2(Y, X)$.
2. *Interpolates the stored row* $u(\cdot, k, \cdot)$ *radially onto* ϖ.
3. *Rotates scalars (no-op), vectors (one rotation), and tensors (full two-index rotation).*

50.6.2.2 The radial interpolation

Quadratic, three-point Lagrange:

```
function interp_radial(u, k, varpi) result(fvals)
  j = floor(varpi / dx) + 1
  if (j < 1) j = 1
  if (j > Nx - 2) j = Nx - 2
  x0 = xg(j); x1 = xg(j+1); x2 = xg(j+2)
  L0 = (varpi - x1) * (varpi - x2) / ((x0 - x1) * (x0 - x2))
  L1 = (varpi - x0) * (varpi - x2) / ((x1 - x0) * (x1 - x2))
  L2 = (varpi - x0) * (varpi - x1) / ((x2 - x0) * (x2 - x1))
  fvals = L0 * u(j, k, :) + L1 * u(j+1, k, :) + L2 * u(j+2, k, :)
```

Gotcha 6. *This is **second order**, and it is the accuracy ceiling for the entire y-direction. The x- and z-derivatives use fourth-order stencils; the y-derivatives cannot exceed second order. In a curvature calculation where y-second-derivatives enter on equal footing with x- and z-, this mixed order is the reason convergence studies in this code show the y error falling more slowly than the others.*

50.6.2.3 Rotation

For a vector V with components (V_x, V_y),

$$V'_x = \cos\theta\, V_x - \sin\theta\, V_y, \qquad V'_y = \sin\theta\, V_x + \cos\theta\, V_y. \tag{50.15}$$

For a symmetric tensor,

$$T'_{xx} = c^2 T_{xx} - 2cs\, T_{xy} + s^2 T_{yy}, \tag{50.16}$$
$$T'_{xy} = cs(T_{xx} - T_{yy}) + (c^2 - s^2) T_{xy}, \tag{50.17}$$
$$T'_{yy} = s^2 T_{xx} + 2cs\, T_{xy} + c^2 T_{yy}, \tag{50.18}$$
$$T'_{xz} = cT_{xz} - sT_{yz}, \qquad T'_{yz} = sT_{xz} + cT_{yz}, \qquad T'_{zz} = T_{zz}. \tag{50.19}$$

The rotation is applied to $\tilde{\gamma}$, $\tilde{A}$, $\tilde{\Gamma}$, β, and B. It is not *applied to $\tilde{\phi}$, K, or $\hat{A}$, which are scalars.*

Gotcha 7. *`get_neighbor` takes $y_{\mathrm{val}} = 0$ as a sentinel meaning "plain index shift". This is why x- and z-neighbours — which do not need the y-direction at all — still go through `get_neighbor`: it is the single choke point through which every neighbour access passes, and having one choke point is what makes the excision mask easy to enforce consistently.*

50.6.3 Finite differences and the fallback cascade

50.6.3.1 Every stencil in the code

Derivative	*Stencil*	*Order*
$\partial_x f$	$[f_{i-2} - 8f_{i-1} + 8f_{i+1} - f_{i+2}]/12dx$	*4th*
	$[f_{i+1} - f_{i-1}]/2dx$	*2nd, centred*
	$[-3f_i + 4f_{i+1} - f_{i+2}]/2dx$	*2nd, forward*
	$[3f_i - 4f_{i-1} + f_{i-2}]/2dx$	*2nd, backward*
$\partial_x^2 f$	$[-f_{i+2} + 16f_{i+1} - 30f_i + 16f_{i-1} - f_{i-2}]/12dx^2$	*4th*
	$[f_{i+1} - 2f_i + f_{i-1}]/dx^2$	*2nd, centred*
	$[f_i - 2f_{i+1} + f_{i+2}]/dx^2$	***1st**, forward*
	$[f_i - 2f_{i-1} + f_{i-2}]/dx^2$	***1st**, backward*
$\partial_{xz}^2 f$	$[f_{++} - f_{+-} - f_{-+} + f_{--}]/4\,dx\,dz$	*2nd*
$\partial_y f$	$[f(\varpi, +dy) - f(\varpi, -dy)]/2dy$	*2nd*
$\partial_y^2 f$	$[f(\varpi, +dy) - 2f_i + f(\varpi, -dy)]/dy^2$	*2nd*
$\partial_{xy}^2 f$	*differencing ∂_y in x*	*2nd*
Kreiss–Oliger	$[f_{-2} - 4f_{-1} + 6f_0 - 4f_{+1} + f_{+2}]$	*2nd*

50.6.3.2 The fallback logic

For each derivative direction the code asks whether the required neighbours are usable, where usable *means "in bounds and not `EXCISED`":*

```
xp_ok  = valid_x(i,k,+1,mask)      ! mask == EXCISED -> unusable
xm_ok  = valid_x(i,k,-1,mask)
```

```
xp2_ok = valid_x2(i,k,+1,mask)      ! needs BOTH i+1 and i+2 usable
zm2_ok = valid_z2(i,k,-1,mask)
```

It then takes the highest-order stencil all of whose neighbours are available, cascading down as necessary. The mixed xz, xy, yz derivatives each test their own availability and have their own one-sided fallbacks.

Gotcha 8. *The one-sided* second *derivative $(f_i - 2f_{i+1} + f_{i+2})/dx^2$ is **first-order accurate**, not second-order. It is easy to assume symmetry with the one-sided first derivative $\frac{-3f+4f_1-f_2}{2dx}$, which genuinely is second-order. So near a boundary the code degrades to first order in second derivatives while remaining second order in first derivatives — an inconsistency that limits local convergence rate.*

Gotcha 9. *`valid_x` returns* true *for $ii < 1$, on the grounds that `get_neighbor` can synthesise the mirror point via the Cartoon rotation. An earlier version returned false, which silently forced a one-sided first-order stencil right on the axis. The current behaviour is correct.*

50.6.3.3 The variable mask contract

`compute_stencil(u, i, k, mask, S)` receives a mask with the contract

```
mask(i,k) == EXCISED   ->  point is inside the excision region, unusable
mask(i,k) == anything else -> usable (physical point or filled ghost)
```

This is why `excision_status` is re-derived from `excision_geom` at the top of every `extrapolate_excised` call: carrying last step's classification forward would let a point that was filled last step be treated as reliable when it is not.

50.7 Boundaries

50.7.1 Inner: excision

50.7.1.1 The mask

*A point is excised if its **Kerr radius** is below the cutoff:*

```
logical function is_excised(i,k)
  is_excised = (kerr_radius_squared(xg(i), zg(k)) < r_excise**2)
end function
```

with

$$r_{\mathrm{KS}}^2 = \frac{1}{2}\left(\rho^2 - a^2 + \sqrt{(\rho^2 - a^2)^2 + 4a^2 z^2}\right), \qquad \rho^2 = x^2 + z^2. \tag{50.20}$$

In the production configuration $a = 0.8$, $M = 1$, so the outer horizon is at $r_+ = M + \sqrt{M^2 - a^2} = 1.6$ and the inner horizon at $r_- = 0.4$. The default $r_{\mathrm{excise}} = 1.0$ therefore sits between the two horizons — inside the black hole but outside the Cauchy horizon, which is the standard and safe choice.

Gotcha 10. *$r_{\mathrm{KS}} \neq \sqrt{x^2 + z^2}$ whenever $a \neq 0$ and $z \neq 0$. The code base contained ten diagnostic filter sites that used the cylindrical radius where the mask uses the Kerr radius. Because the two disagree, the driver's reported "exterior" worst point could lie* inside *the mask — and at every resolution tested, it did ($r_{\mathrm{KS}} \approx 0.7$–$0.8$). This invalidated a substantial body of earlier conclusions; see Chapter 50.12.*

50.7.1.2 The ghost fill

*Points inside the mask are never evolved. They are given values so that exterior stencils have something to read. The fill extrapolates the **perturbation from a frozen background**, not the field itself:*

```
! d1..d4 are (u - background) at 1,2,3,4 cells along a ray
if (excision_order >= 4 .and. all four points usable) then
  u(i,k,:) = b0 + 4.0_dp*d1 - 6.0_dp*d2 + 4.0_dp*d3 - d4   ! order 4
if (excision_order >= 3 .and. first three usable)  then
  u(i,k,:) = b0 + 3.0_dp*d1 - 3.0_dp*d2 + d3              ! order 3
if (excision_order >= 2 .and. first two usable)    then
  u(i,k,:) = b0 + 2.0_dp*d1 - d2                          ! order 2
if (usable_point(status,i+si,k+sk))                 then
  u(i,k,:) = b0 + d1                                      ! order 1
```

The background is set once from the initial data by `set_excision_backgro`

Gotcha 11. *Because the background is the exact Kerr–Schild solution and the true Kerr–Schild solution is* stationary*, the perturbation $\delta = u - u_{\mathrm{KS}}$ is identically zero at $t = 0$. Therefore* every *order of extrapolation returns the exact answer, to machine precision. This was measured: the worst ghost-fill error anywhere is 4.4×10^{-16}.*

This has two consequences that took a while to appreciate. First, it explains why the fill order makes almost no difference at early times. Second, it means a static $t = 0$ test of the fill is structurally blind *— any perturbation-depth scan at $t = 0$ that perturbs the background will measure the extrapolation of a non-zero field, which is a different question.*

50.7.1.3 Onion-peeling and the frozen snapshot

The fill runs in sweeps, and each sweep gates every usability check on a frozen snapshot *of the mask taken before the sweep begins:*

```
status_snapshot = status          ! freeze BEFORE filling anything
do k = 1,Nz
  do i = 1,Nx
```

```
    if (status_snapshot(i,k) /= EXCISED) cycle
    call fill_one_point(u,status_snapshot,i,k,filled,order)
    if (.not. filled) cycle
    status(i,k) = GHOST
```

Gotcha 12. *Without the snapshot, a point whose best-aligned outward neighbour is not yet ready falls back,* within the same sweep, *to a less well-aligned direction that chains through points filled moments earlier in that same raster pass. Each link in such a chain reapplies the* $[4, -6, 4, -1]$ *extrapolation, whose* L_1 *norm is 15, so a chain a few links deep amplifies an* $O(1)$ *boundary value into* $O(10^4)$ *garbage. The snapshot forces genuine onion-peeling: a point only ever uses data resolved as of the end of the previous sweep.*

50.7.1.4 Positivity enforcement

After filling, any ghost whose metric is not positive definite is blended back toward the background:

```
do n = 1, 40
  theta = 0.5_dp * theta
  trial = (1.0_dp-theta)*gb + theta*g
  if (metric_is_spd(trial)) exit
end do
```

This is a safety net, not a physical model. It did not trigger in any run examined.

50.7.1.5 The characteristic SAT

`helper/excision\_sat.F90` *implements a level-set boundary SAT. It is compiled and unit-tested, and enabled by* `excision_sat = true` *in the parameter file. It is* off *in production. Tests show it slightly damps the inner boundary (lower* `max|H|`*) but does not*

change survival, and the puncture gauge is not actually driving anything — see the `KNOWN LIMITATION` note at the top of the Makefile.

50.7.2 Outer: Sommerfeld

On the outer rows and column the RHS is replaced by a first-order outgoing-wave condition applied to every one of the 24 variables:

$$\partial_t f = -c\,\partial_r f - n\,\frac{f - f_0}{r}, \tag{50.21}$$

with $c = 1$ (in units where the wave speed is unity), f_0 the asymptotic flat-space value (1 for the diagonal metric components and $\hat{A}$, 0 otherwise), and n the asymptotic falloff power.

The legacy condition uses $n = 1$ for every field. Measurement showed that for exact Kerr–Schild data the true powers are far steeper: $\tilde{\gamma} \to 1$ (no $1/r$ term at all), $\tilde{\phi}$ and $\hat{A} \sim r^{-3}$, and K, $\tilde{A}$, $\tilde{\Gamma}$, $B \sim r^{-4}$. Setting `GR_SOMMERFELD_DECAY=1` switches to per-field powers.

Gotcha 13. *The per-variable decay powers were tested and are **worse**: RMS constraint violation degrades by 8.5× when the correct powers are used. The $1/r$ term is not an asymptotic-correctness knob — it acts as a damper, and a weaker damper lets more error in. This was a clean refutation of a plausible hypothesis.*

Gotcha 14. *A single-layer, all-field, first-order Sommerfeld condition is* not *a characteristic boundary condition. It overwrites the RHS of all 24 variables in one layer regardless of which characteristics actually leave the domain. A proper treatment would impose conditions on the incoming characteristic fields only. The apparent second-order convergence in domain size L that this condition exhibits is reflection, not accuracy.*

50.7.3 Time integration

Classical RK4 with an algebraic projection after each stage:

```
utmp = u + 0.5_dp*dt*k1
call extrapolate_excised(utmp)
call project_bssn_constraints(utmp, stepnum)
... k2, k3, k4 likewise ...
u = u + (dt/6.0_dp)*(k1 + 2.0_dp*k2 + 2.0_dp*k3 + k4)
call project_bssn_constraints(u, stepnum)
```

Each stage applies, in order: the interior RHS, the excision SAT, Kreiss–Oliger dissipation, and the Sommerfeld boundary. Note that there is no *projection after stage 1 — the state entering k_1 was projected at the end of the previous step.*

50.7.3.1 The timestep

```
dt = cfl * min(dx, dz) / sqrt(2.0_dp)
```

with `cfl = 0.005`*, so $dt/dx \approx 0.0035$ — about 1.4% of the classical RK4 advection stability limit. This is extraordinarily conservative and was verified* not *to be the problem: the failure is not a time-stepping instability.*

50.7.3.2 The algebraic projection

```
scale = det**(-1.0_dp/3.0_dp)
u(i,k,IGXX) = scale*gxx;  ... (all six components)
! then remove the trace of Atilde in the same loop
```

Gotcha 15. *The projector was the leading suspect for manufacturing its own constraint violation, since it rescales $\tilde{\gamma}$ to $\det = 1$ four times per step. It was instrumented (`GR_PROJ_DIAG`) and measured. The correction is $\max|\det(\tilde{\gamma}) - 1| \approx 10^{-5}$, growing only linearly, five orders of magnitude below the violation it would have to explain.* ***Falsified.***

50.8 Initial Data

Kerr–Schild, from `kerr_schild_point`*:*

```
rho2 = x*x + y*y + z*z
r2   = 0.5_dp*( (rho2 - a2) + sqrt((rho2 - a2)**2 + 4.0_dp*a2*z*z) )
H    = M * r2 * sqrt(r2) / (r2*r2 + a2*z*z)
l(1) = (r*x + bh_spin*y) / (r2 + a2)
l(2) = (r*y - bh_spin*x) / (r2 + a2)
l(3) = z / r
gij  = I + 2.0_dp * H * outer(l, l)
alpha = 1.0_dp / sqrt(1.0_dp + 2.0_dp*H)
beta_lower = 2.0_dp * H * l
```

The conversion to BSSN variables:

```
twoH = max(detg - 1.0_dp, 0.0_dp)
phi  = log(max(detg,1.0e-300_dp)) / 12.0_dp     ! conformal_factor_mode 0
gtij = exp(-4.0_dp*phi) * gij
Ktr  = sum(gijinv*Kij)
Atij = (Kij - (gij*Ktr)/3.0_dp) / exp(4.0_dp*phi)
```

The extrinsic curvature is obtained by central differences of the analytic metric with an internal step $h = 10^{-4}$, plus a Christoffel-symbol correction to $\partial_a \beta_b$.

50.8.0.1 The conformal connection

`compute_conformal_connections` *fills* `IGTX..IGTZ` *from*

```
dginv(1,:,:) = (ginvxp - ginvxm) / (2.0_dp*dx)
dginv(2,:,:) = (ginvyp - ginvym) / (2.0_dp*dy)
dginv(3,:,:) = (ginvzp - ginvzm) / (2.0_dp*dz)
Gt(1) = -(dginv(1,1,1) + dginv(2,1,2) + dginv(3,1,3))
```

that is, $\tilde{\Gamma}^a = -\partial_b \tilde{g}^{ab}$.

Gotcha 16. *This is a* ***second-order*** *central-difference path, entirely separate from the fourth-order* `compute_stencil` *that the*

curvature operator uses. So the state field feeding $\tilde{R}$'s $\tilde{\Gamma} \cdot \partial\tilde{\Gamma}$ term is two orders less accurate than its neighbours, by construction, and its loop bounds $i = 2..N_x - 1$, $k = 2..N_z - 1$ carry no excision-mask guard on the read range.

*This was the leading remaining suspect and has now been measured: the error is $9.07 \times 10^{-2} \to 3.05 \times 10^{-2} \to 7.68 \times 10^{-3}$ for $N_x = 48, 96, 192$, i.e. it **converges** at roughly second order and sits within the range spanned by its neighbours. It is lower-order than its surroundings, but it is not an outlier and does not explain a non-converging error.*

50.9 Constraints and Diagnostics

50.9.1 The margin trap

This is the single most instructive defect found in the project, and it is worth dwelling on because it invalidated a large body of earlier work.

The driver computes `max|H|(ext)` over points satisfying

```
if (sqrt(kerr_radius_squared(xg(i), zg(k))) < r_excise +
    diag_margin) cycle
```

with `diag_margin` defaulting to 1.0. That correctly excludes $r_{\rm KS} < 2.0$, the region in which the code is known to be inaccurate. But it means the reported number says nothing about $r_{\rm excise} < r_{\rm KS} < r_{\rm excise} + 1.0$.

And that band is exactly where the constraint violation is largest. Sweeping the margin at $N_x = 48$, $t \approx 0.36$:

diag_margin	*monitors from*	*max|H|*	*rms|H|*
2.0	$r_{\rm KS} = 3.0$	4.06×10^{-3}	3.51×10^{-4}
1.0 (default)	2.0	1.17×10^{-2}	7.10×10^{-4}
0.5	1.5	1.53×10^{-1}	4.09×10^{-3}
0.2	1.2	3.19×10^{-1}	1.09×10^{-2}
0.05	1.05	5.25×10^{-1}	1.60×10^{-2}

A factor of 129 between the far field and one cell outside the mask. The default setting reports the small number and calls it the constraint.

Gotcha 17. *For a long time this was interpreted as "the initial data violate the constraint by $O(1)$ at $t = 0$, and this slowly propagates outward." The 1.18 figure that motivated that reading came from the driver's unfiltered $x = 0$ profile, which samples $r_{\rm KS} \approx 0.38$ — deep inside the mask, on filled ghost points. It was an artifact of the radius bug. The qualitative conclusion (a steep near-mask gradient exists) survived; the specific numbers did not.*

50.9.2 The near-mask band

The driver therefore now reports the band explicitly:

```
! GR_NEAR_MASK_W, default 2.0, 0 disables
if (nm_rr < r_excise .or. nm_rr >= r_excise + near_mask_w) cycle
nm_npts = nm_npts + 1
nm_Hmax = max(nm_Hmax, abs(Hgrid(i,k)))
nm_Mmax = max(nm_Mmax, sqrt(Mgrid(i,k,1)**2 + Mgrid(i,k,2)**2 +
    Mgrid(i,k,3)**2))
```

This costs one extra pass over arrays that have already been computed — no additional stencil work, no effect on the evolution. Typical output:

```
step 700  ...  max|H|(ext)= 1.2656E-02  rms|H|(ext)= 2.0211E-03
    alpha(probe)= 0.910348
   near-mask r_KS < 3.0000E+00  n= 192  max|H|= 1.9493E+00  max|
   M|= 1.4920E+00  worst r= 1.0227
```

The momentum residual deserves attention: `max|M|` *is* 1.49 *inside the band against* 2.87×10^{-2} *outside — a factor of 52. Since the band is where the fill injects its error into the stencil, a large* $\max|M|$ *there is the signature of a fill/stencil inconsistency rather than a gauge problem. That is a useful discriminator between two failure families that are otherwise hard to tell apart.*

50.9.3 The diagnostic programs

50.9.3.1 diagnose-ghost-accuracy

The most capable instrument in the project. It reports, for any resolution:

1. ***Ghost fill error*** *vs. exact Kerr–Schild, binned by depth inside the mask. Result:* 4.4×10^{-16} *everywhere.*

2. ***max—H— per radial bin*** *outward from the mask, so the spatial structure of the error is visible.*

3. ***Per-derivative errors*** *against high-order central differences of the exact analytic field, for* $\tilde{\gamma}_{xx}$ *and* $\tilde{\phi}$*, in* x*,* y *and* z*.*

4. ***Decomposition*** $H = \tilde{R} + \frac{2}{3}K^2 - \tilde{A}^2$*, split further into* $\tilde{R}^\gamma$ *and* $\tilde{R}^\phi$*. Since the* K *and* $\tilde{A}$ *terms are pointwise algebraic and carry no discretisation error, this localises the problem to* $\tilde{R}$ *immediately.*

5. Γ***-tilde accuracy*** *against the same formula evaluated on the exact analytic metric.*

6. ***A replica of the driver's own*** $t = 0$ ***scan****, so the two can be cross-checked digit for digit.*

50.9.3.2 diagnose-momentum-point

Prints the stored BSSN state beside the analytic state built independently at the same coordinate, component by component, plus the seven additive pieces of the momentum constraint at a chosen point and at a same-radius control away from the axis. Agreement to 10^{-16} *in the data stage is the precondition for blaming the derivatives instead.*

50.9.3.3 diagnose-inner-boundary

Perturbs the background and measures how the perturbation propagates into the exterior as a function of fill depth. Shows it bounded and saturating.

50.9.3.4 A methodological warning

Gotcha 18. *Every number in this project that was measured with a diagnostic operating on the wrong set of points had to be retracted. So did two numbers that were produced by bugs in the diagnostic itself. Specifically:*

- *A second-derivative reference built as* $(f(+h) - f(0))/h^2$ *— a forward difference scaling as* f'/h *— reported a spurious* $|\partial_y^2|$ *error of* $\sim 10^3$ *that looked exactly like the smoking gun. The correct central difference gives* $\sim 10^{-2}$ *and converges normally.*

- *A diagnostic that called* `set_kerr_schild_id` *but not* `compute_conformal` *left* $\tilde{\Gamma}$ *identically zero, so every Hamiltonian it reported was computed with a term of the curvature silently dropped.*

Adding the call moved the innermost-cell $\max|H|$ *at* $N_x = 96$ *from* 4.93×10^{-1} *to* 1.14×10^{-1}.

The general lesson, which cost real time: when a measurement contradicts an expectation, suspect the instrument before believing the result, and prefer two independent instruments that agree over one instrument that agrees with your prior.

50.10 Build and Test

50.10.1 The parity gate

```
check: check-parity $(TARGET) test

check-parity:
        @missing=0; \
        srcs=' $(SOURCES) '; \
        for f in $$(grep -ho '[a-z_]*/[a-z_]*\.F90' Makefile |
    sort -u); do \
          case "$$f" in drivers/*|helper/test_*|helper/diagnose_
    *|init/test_*|helper/probe_*) continue;; esac; \
          case "$$srcs" in \
            *" $$f "*) ;; \
            *) echo "check-parity: $$f is built by a test but
    missing from SOURCES"; missing=1;; \
          esac; \
        done; \
        ...
```

This is a genuinely good idea and worth copying. It scans the Makefile for every source file any test target compiles, and asserts that each also appears in `SOURCES` *— the list the production binary is built from. Without it, a test can pass against a copy of the physics that production does not actually use, and you would never know.*

50.10.2 Sources must be listed in dependency order

```
SOURCES := \
        helper/kinds_mod.F90 \
        helper/fo_sbp.F90 \
        ...
        init/param.F90 \
        init/grid.F90 \
        init/vars.F90 \
        ...
        drivers/vacuum_kerr.F90
```

gfortran will not resolve module dependencies automatically across a single invocation, so this list must be topologically ordered. This is a maintenance burden; add new modules before the files that `use` them.

Gotcha 19. *The Makefile notes that GNU make predefines `FC = f77`, so a plain `FC ?= gfortran` never takes effect and every recipe silently builds with `f77`. The fix is*

```
ifeq ($(origin FC),default)
FC      := gfortran
endif
```

so that an explicit `make FC=...` still wins. This is a classic and a good reminder to check what your build system already defines.

50.10.3 Environment overrides

Parameters can be overridden at runtime without editing files, which is what makes A/B experiments cheap and reproducible:

Variable	*Effect*
GR_NSTEPS	*Number of steps.*
GR_DIAG_MARGIN	*Width of the excluded near-mask band.*
GR_NEAR_MASK_W	*Width of the reported near-mask band; 0 disables.*
GR_SLICING_MODE	*0 = 1+log, 1 = maximal, 2 = frozen.*
GR_GAMMA_DRIVER_MODE	*0 = legacy, 1 = puncture, 2 = frozen, 3 = advective.*
GR_EXCISION_ORDER	*1–4; the ghost extrapolation order.*
GR_EXCISION_SAT	*Enable the level-set characteristic SAT.*
GR_KO_EPS	*Kreiss–Oliger strength.*
GR_GAUGE_ETA	*Γ-driver damping.*
GR_REFRESH_BG	*Re-anchor the excision background every step.*
GR_SOMMERFELD_DECAY	*Per-field asymptotic decay powers.*
GR_PROJ_DIAG	*Report the algebraic projection correction.*
GR_HZ_EVERY	*Horizon-find interval; 0 disables.*
GR_HZ_ABORT	*Abort when the horizon drifts.*

50.11 A Tour of Every File

File	*What it does and why*
helper/kinds_mod.F90	*`dp = selected_real_kind(15,307)`. One place to change precision.*
init/param.F90	*Every runtime parameter with a default, a parser, and an environment override.*
init/grid.F90	*Builds `xg` and `zg`. Fortran 1-based; x starts at 0, z is centred.*

File	*What it does and why*
init/vars.F90	*The 24 state indices, plus ICONFMET/IATILDE component tables for looping over symmetric tensors, and is_tensor_var/is_vector_var for the Cartoon rotation.*
init/init_data.F90	*Kerr–Schild point data, conversion to BSSN, and the conformal connection.*
init/stencils.F90	*Precomputed finite-difference weights used by the SAT and equilibrium modules.*
init/fd_weights.F90	*FD_RADIUS and weight tables.*
helper/cartoon.F90	*get_neighbor, radial interpolation, vector and tensor rotation.*
helper/stencil.F90	*The stencil engine: every derivative, every fallback, the excision-aware validity tests.*
helper/geometry.F90	*compute_conformal_geometry: $\tilde{\gamma}$, $\tilde{\gamma}^{-1}$, Christoffels, $\tilde{R}^{\gamma}_{ab}$, $\tilde{R}^{\phi}_{ab}$, $\tilde{\nabla}_a\tilde{\nabla}_b\hat{A}$, $\hat{A}^a$.*
integrators/bssn_rhs.F90	*The evolution equations. Pure functions of the stencil and geometry; no state mutation.*
integrators/ko_diss.F90	*Kreiss–Oliger, applied in all three directions, skipping points inside $\sqrt{x^2+z^2} < r_{\rm excise}$.*
integrators/sommerfeld.F90	*Outer boundary condition, with the per-variable decay probe.*
helper/excision.F90	*Mask geometry, frozen background, the onion-peeling fill, positivity enforcement, the incoming-characteristic check, and kerr_radius_squared for external use.*
helper/excision_sat.F90	*Level-set characteristic SAT. Compiled and tested; off in production.*

File	What it does and why
helper/constraints.F90	Hamiltonian and momentum, both with optional term decompositions for diagnosis.
integrators/time_step.F90	rk4_step and project_bssn_constraints.
helper/horizon.F90	Locates the horizon from the state and reports its drift.
helper/output.F90	HDF5 snapshots and diagnostics.
drivers/vacuum_kerr.F90	Setup, time loop, all reporting, horizon tracking, abort logic.

50.11.0.1 The first-order track

File	Role
helper/fo_sbp.F90	SBP operator for the first-order formulation.
helper/fo_sat.F90	Simultaneous approximation term.
helper/fo_constraints.F90	First-order constraint operators.
helper/fo_constraint_damping.F90	Constraint damping.
helper/fo_curved_sat.F90	SAT on a curved (non-constant-coefficient) boundary.

File	*Role*
helper/fo_normal_system.F90	*Reduction of the SAT to a normal system.*
helper/fd_excision.F90	*Finite-difference-level excision helpers.*
init/fo_vars.F90, *init/fo_complete_vars.F90*	*Reduced and full state layouts.*
integrators/fo_rhs.F90, *fo_complete_rhs.F90*	*Reduced RHS operators.*

50.11.1 SBP-SAT: why it is not finished

For a well-posed hyperbolic boundary treatment one wants a simultaneous approximation term built on a discretisation satisfying the summation-by-parts property. The existing `fo_sbp` module does not: it was machine-checked against the SBP diagonal-dominance test and the maximum eigenvalue of $hD + D^{\top}h$ came out at $+1.685$, where the property requires it to be non-positive. That is a genuine failure of the property, not a tolerance issue.

Making this production-ready is a from-scratch build, not a hardening job: a valid SBP operator, the principal symbol of the production system, and the SAT on a curved Kerr-radius boundary, for 24 fields. This is a multi-week project and has deliberately not been started.

50.12 Findings

This chapter records the investigation of the long-time failure, including the mistakes. It is long because the mistakes are as instructive as the results.

50.12.1 The failure

In the production configuration the run aborts at $t \approx 5.63$ (step 6331, live gauge) when the conformal metric loses positive definiteness:

```
Conformal metric failure det= -1.067321E+11 components= ...
Invalid state at step 7536 i=7 k=48 x= 1.53191 z= -0.12632
```

This is not a blow-up in the dynamics; it is the projection's positivity check catching a state that has drifted far from a Lorentzian metric.

50.12.2 What was established

50.12.2.1 Negative results, all measured

Hypothesis	***Result***
Frozen gauge	*Falsified. Exact stationary Kerr–Schild gauge still drifts and fails (rate $4.1\times10^{-2}/t$ vs $6.8\times10^{-2}/t$).*
Outer boundary dominant	*Resolution-dependent. Real and dominant at $dx \approx 0.10$; vanishes at the production $dx \approx 0.25$. Enlarging the domain $12M \to 18M$ at fixed dx extends survival 39% but relocates the failure to the inner boundary.*

Hypothesis	*Result*
$1/r$ *Sommerfeld falloff*	*Refuted. Correct per-field powers make the constraint* ***8.5*** $\times$ ***worse****.*
Cartoon y*-derivative*	*Measured; scalar and vector exact, tensor* 1.3×10^{-2} *at* $x = dx$*. Only 1–2% of the ring error.*
Algebraic projection	*Falsified.* $\max\lvert\det\tilde{\gamma} - 1\rvert \approx 10^{-5}$*.*
Ghost fill	*Exact to* 4.4×10^{-16}*.*
$\tilde{\Gamma}$ *on the second-order path*	*Converges at* ~ 2*nd order; not an outlier.*
Fill order	*Mitigates, does not fix (see below).*
Stale background	*Null, bit-identical.*
Excision order	*Mitigates only.*
Momentum constraint form	*Was genuinely wrong; fixed, and proven not to be on the evolution path.*

50.12.2.2 The live picture

The constraint violation is real, large, and confined to the first one or two cells outside the mask. Over physical points only, $\max|H|$ *is* 7.77×10^{-1}*,* 1.42×10^{-1}*,* 6.21×10^{-2} *at* $N_x = 48, 96, 192$ *— decreasing, but at observed orders* ~ 2.4 *then* ~ 1.2*, and the worst point sits at a different physical location each time, so the apparent non-convergence may itself be an artifact of comparing maxima at different points.*

Every individual derivative quantity examined converges at roughly second order. The assembled H *exceeds its inputs by a factor of 5–8, consistent with the curvature contraction's* $1/\gamma^2$ *and* $\tilde{\Gamma} \cdot \partial\tilde{\Gamma}$ *terms. The mechanism is therefore* not *localised to a single defective field.*

50.12.3 Excision order: a trade-off, not a win

Raising `excision_order` *reduces the near-mask violation, but not monotonically in run lifetime:*

`excision_order`	*live fail step*	*live fail* t	*near-mask* $\max\lvert H\rvert$
2	*6331*	*5.63*	***1.289***
3	***6848***	***6.07***	*1.949*
4	*4364*	*3.84*	*2.068*

Order 3 is the production default: it gains 8% in lifetime and improves all three exterior metrics. Order 4 improves the constraint further but costs 31% of survival — a more accurate fill that destabilises. Note the near-mask column runs opposite *to the margin-based column used earlier to justify order 4; the ordering of the three settings depends on which radial band is measured, and that is recorded rather than smoothed over.*

The honest summary is that `excision_order` *is a mitigation worth about a factor of two, and that even the best setting leaves the near-mask error near* 1.1 *against a far-field level of* $\sim 10^{-3}$*. No setting of this knob fixes the problem.*

50.12.4 How this investigation went wrong, repeatedly

Worth recording in full, because the failure mode is common and expensive.

1. ***Generalising from one regime.*** *The outer-boundary conclusion was drawn at* $dx \approx 0.10$ *and asserted for the code; at the production* $dx \approx 0.25$ *the effect is absent. Every earlier conclusion had been drawn at* $t \approx 0.5$*, while the run dies at* $t \approx 5.6$*.*

2. ***Trusting a diagnostic's default threshold.*** *`max|H|(ext)` with the default margin could not see the region that mattered. Every "null result" measured that way had to be re-examined.*

3. ***Bugs in the instrument producing confident nonsense.*** *Twice, as detailed in Section 50.9.*

4. ***A wrong radius in a filter.*** *Ten sites. This made a garbage maximum look like a physical result and made the "non-convergence" of* $\max|H|$ *an artifact.*

5. ***Asserting strict dominance from a single metric.*** *Order 3 was recommended as strictly better than order 2 on the strength of one radial band; a second band reversed the near-mask ordering.*

   The through-line is that every conclusion rested on a measurement whose fitness for purpose had not been verified. The single most useful discipline would have been, before interpreting any diagnostic, to confirm it is measuring the quantity you think it is — by cross-checking against a second, independently written instrument.

50.13 Writing Your Own

50.13.1 Where to start

If you are writing a new code in this style, do it in this order. Each step is independently testable, and each has a natural test that will catch you if you get it wrong.

1. ***Kerr–Schild initial data and the BSSN conversion.*** *Verify the stored state against an independently computed*

analytic state, component by component. Expect agreement to round-off. If it does not, your conversion is wrong and everything downstream will be too.

2. ***The stencil engine on a uniform grid with no mask.*** *Write a test that differentiates a manufactured smooth function and confirms the observed order matches the nominal order. Do this for every derivative, including the mixed ones. You will find your mixed-derivative stencils are wrong more often than you expect.*

3. ***Cartoon.*** *Differentiate a manufactured axisymmetric function, including a tensor field, and measure the order. Expect second order in y, and expect it to be worse than x and z because of the radial interpolation.*

4. ***Conformal geometry.*** *Compute $\tilde{R}_{ab}$ two ways: once via the connection-function form used here, and once from Christoffel symbols directly, on a random metric. They must agree to round-off. Getting the index order of* `Chr_low` *wrong is easy and produces a result that is wrong by $\sim 10\%$ without looking obviously wrong.*

5. ***Constraints on exact data.*** *Compute $\mathcal{H}$ and $\mathcal{M}^a$ on exact Kerr–Schild and confirm convergence under refinement. The continuum values are zero, so any non-convergence is entirely your error. This is the single most valuable test in the project and it should come before any evolution work.*

6. ***Excision.*** *Only then add the mask. Confirm that a frozen-gauge configuration stays stationary. If it does not, the problem is somewhere in your fill or stencil fallback, and you can localise it by the bin-by-radius diagnostic described in Section 50.9.*

7. ***Evolution.*** *Turn the gauge on. Change one thing at a time, and after every change re-run the frozen-gauge stationarity*

check.

50.13.2 Design lessons worth stealing

Keep one copy of the physics. *Every reduced or alternate operator should call the same point-level routine rather than reimplementing it. The codebase does this deliberately:* `compute_rhs_point` *exists precisely so that no second, subtly different BSSN system can grow.*

Gate production against tests. *The* `check-parity` *target is a good pattern: it verifies that what the tests exercise is what production builds.*

Make diagnostics opt-in and inert by default. *Every probe in this codebase is behind an environment variable, and the default path is verified bit-identical with the probe enabled. A diagnostic that changes the answer is worse than no diagnostic.*

Write the regression gate, then verify the gate fails. *Several gates here had to be deliberately broken to confirm they would catch a regression.*

Keep a findings document and date it. `baseline/` *in this repository records each investigation, including the retractions.*

Restate the baseline whenever a diagnostic changes. *Numbers of record are only meaningful relative to the exact code that produced them.*

50.13.3 Where the physics is genuinely unfinished

In rough order of value:

1. ***The near-mask error.*** *This is the actual bug. It is an $O(1)$ Hamiltonian violation in the first two cells outside the mask, present in the initial discretisation, not caused by the projection, the fill, the connection, or the gauge. It is what kills the run.*

2. ***A real characteristic boundary treatment.*** *The outer Sommerfeld condition is not characteristic and not accurate; the second-order convergence it appears to exhibit is reflection.*

3. ***SBP–SAT.*** *Blocked on a genuinely invalid SBP operator.*

4. ***An active puncture gauge.*** *The* `KNOWN LIMITATION` *note is accurate: the puncture gauge does not currently drive anything, and the horizon drifts.*

50.14 Deriving BSSN, and Why It Exists

This chapter fills in the reasoning that is usually skipped. If you want to write your own code you will eventually want a variable you have not seen before, and the only way to know whether it is necessary is to understand what problem each addition solves.

50.14.1 The failure of ADM in practice

The ADM system, written on the spatial metric and extrinsic curvature, is a first-order-in-time hyperbolic system whose evolution is constraint preserving in the continuum. *That is, if you start on a surface satisfying the Hamiltonian and momentum constraints, the exact continuum evolution keeps you on that surface.*

The qualification — in the continuum *— is the entire problem. In discrete form the constraints are violated at the trunca-*

tion error of whatever difference operator you used, the errors grow, they feed back through the equations, and the run eventually fails. This is the constraint-violation / constraint-error feedback mechanism, and it is why naive ADM codes have a reputation for blowing up.

50.14.2 What the constraint propagation mechanism needs

For a formulation to damp constraint violations rather than amplify them you need, in rough order of importance:

1. ***A semigroup estimate that propagates the constraint violation away.*** *A violation of size ϵ should lead to a violation of size $\epsilon + O(\Delta t^3)$ — growing at the local truncation rate, not exponentially.*

2. ***Constraint terms of the right size in the evolution equations.*** *The constraint itself should appear in the equations, so that the same mechanism that produces a violation also produces a restoring force.*

3. ***Coupling between adjacent constraint-violation components.*** *A violation at one point should not be able to grow without limit; the diffusion part of the system must dominate at long wavelengths.*

4. ***A parameter family that can be dialled.*** *This is the single most practical of the four, and the reason the theory is parameterised: different schemes are stable over different ranges of the parameter, and the existence of* some *stable regime is what matters.*

ADM fails the second condition badly: the constraints do not appear in the evolution equations at all, so nothing generates a

restoring force. Violations grow at the full truncation rate and eventually destroy the metric.

50.14.3 Where each BSSN piece comes from

Addition	***The problem it solves***
Conformal factor $\tilde{\phi}$ and conformal metric $\tilde{\gamma}_{ab}$	*Divides the metric into a well-scaled part and a scalar. Th bulk of the field content goes into $\tilde{\gamma}$ near unity, so the finite difference stencils see a well-conditioned, slowly varyin function rather than a metric that varies strongly. The confo mal factor absorbs the rest. This is the single most importan structural change, and it is why BSSN tolerates black hole at all.*
Traceless extrinsic curvature $\tilde{A}_{ab}$	*Makes the dynamical degree of freedom trace-free, which i what allows the constraint to appear in the evolution equatio for $\tilde{A}$. The trace is carried separately as K, which satisfies simple, stable equation.*
Connection $\tilde{\Gamma}^a$	*Encodes the conformal connection as an evolved variabl rather than a quantity recomputed from first derivatives. Di ferentiating a connection introduces third derivatives; evolv ing it introduces none. This is a discretisation-driven desig decision, not a mathematical one, and it is the clearest exam ple in the codebase of a variable that exists because of finit differences.*
Driver B^a	*Provides the memory needed to integrate the second-orde shift equation $\ddot{\beta}^a = \frac{3}{4}\partial_a B^a$ without effectively making the sys tem third order in the differentiation. The price is that B^a mus be evolved too, and the whole family of Γ-driver and B-drive gauges is a choice of how to close that system.*

Addition	*The problem it solves*
Shift split $\hat{A}^a$	*The difference between the contravariant and covarian is a vector whose magnitude is comparable to the radius rying $\hat{A}^a$ instead of β^a means the evolved quantity is O erywhere rather than growing with radius, which matt a code that must resolve both the near-hole and far-fi gions on one grid.*

Gotcha 20. *Each of these is a* derived *quantity in the continuum, so it is tempting to think of them as convenient reformulations. They are not: they are extra evolution equations with their own truncation errors, their own stability constraints, and their own consistency requirements. Any code that adds a variable must add a test that the variable's analytic value is actually preserved. In this codebase,* `compute_conformal_connections` *is that test for $\tilde{\Gamma}$, and it lives in the production initial-data path because the variable is genuinely evolved.*

50.14.4 Why the code writes $\tilde{R}$ in connection form

The Ricci tensor of the conformal metric appears in the K and $\tilde{A}$ equations. It can be computed from Christoffel symbols directly, or in the "connection-function" form the code uses, in which the second derivatives appear only through the Laplacian;

$$\begin{aligned}\tilde{R}^{\gamma}_{ab} = &-\tfrac{1}{2}\tilde{\gamma}^{cd}\nabla_c\nabla_d\tilde{\gamma}_{ab} + \tfrac{1}{2}\left(\tilde{\gamma}_{ac}\partial_b\tilde{\gamma}^{cd} + \tilde{\gamma}_{bc}\partial_a\tilde{\gamma}^{cd}\right)\\ &+\tfrac{1}{2}\left(\tilde{\Gamma}^{c}{}_{ab}\partial_c\tilde{\gamma}_{cd} + \tilde{\Gamma}^{c}{}_{bd}\partial_c\tilde{\gamma}_{ac}\right) - \tilde{\gamma}^{ce}\tilde{\gamma}^{df}\left(\tilde{\Gamma}^{c}{}_{ae}\tilde{\Gamma}^{f}{}_{bd} + \cdots\right).\end{aligned} \tag{50.22}$$

Gotcha 21. *There is an index-order trap here that the code comments specifically warn about, and it is worth spelling out be-*

cause it is invisible in the output. The lowered-index Christoffel-like object is stored symmetric in its last two *slots:*

```
Chr_low(k,i,j) = Gam^k_{ij}       ! symmetric in i,j -- NOT in k,i
```

A naive contraction `Chr_low(:,i,j)` *silently computes a different tensor. It agrees with the correct answer only when* $\Gamma^{\text{low}}_{kij}$ *happens to be symmetric under a first-slot swap, which is not true in general. The code comments record that the correct version was verified against a direct from-Christoffels computation on a generic metric to machine precision, while the naive version is wrong at the* $\sim 10\%$ *level. A* 10% *error in an interior quantity is the kind of thing that survives review for months.*

50.15 A Worked Example: the Cartoon Method

The Cartoon method is the piece most newcomers find hardest, so here it is derived from scratch, implemented, and validated — the sequence you should follow when adding it to your own code.

50.15.1 The problem

You have solved a 2D problem on the half-plane $y = 0$, $x \geq 0$, *storing* $u(x, y = 0, z)$ *for each of 24 fields. Your equations, however, are genuinely three-dimensional: they contain* ∂_y *and* ∂_y^2. *You need the field off the plane.*

50.15.2 Step 1: establish the symmetry correctly

For axisymmetry about the z-axis, the isometry is $(x, y, z) \mapsto (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta,\ z)$. A solution satisfies

$$u(x, y, z) = u(x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta, z). \tag{50.23}$$

For $\theta = \pi$ this maps $(x, y) \mapsto (-x, -y)$, so the stored half-plane $x \geq 0$ determines the whole plane. This is why the code mirrors in x using a rotation by π *and not a reflection: a reflection in x alone is not an axisymmetry of this problem.*

Gotcha 22. *It is easy to write the mirror as a reflection and get a plausible-looking answer, because for* scalar *fields the two agree on the $y = 0$ slice. They diverge as soon as you rotate a vector or tensor. Test the tensor case.*

50.15.3 Step 2: reduce to radius and interpolate

Write the target point as

$$X = \varpi\cos\theta, \qquad Y = \varpi\sin\theta, \qquad \varpi = \sqrt{X^2 + Y^2}, \quad \theta = \arctan 2(Y, X). \tag{50.24}$$

Then

$$u(X, Y, z) = u(\varpi, 0, z) \tag{50.25}$$

and $u(\varpi, 0, z)$ is, by construction, the function you stored. So the problem reduces to one-dimensional interpolation along x at fixed z.

Quadratic Lagrange on the three points bracketing ϖ:

$$u(\varpi, 0, z) = \sum_{m=0}^{2} L_m(\varpi)\, u(x_m, z), \qquad L_m(\varpi) = \prod_{n\neq m} \frac{\varpi - x_n}{x_m - x_n}. \tag{50.26}$$

Gotcha 23. *The bracketing index needs clamping. The code does* `j = floor(varpi/dx) + 1`*, then clamps to* $[1, N_x - 2]$*. Without the clamp, a stencil reaching past the outer boundary indexes out of bounds, and depending on how your compiler treats that you get either a crash or a plausible wrong answer. Clamp every bracketing index in every interpolator.*

50.15.4 Step 3: rotate by rank

Scalars are unchanged. Vectors transform as $V' = RV$*. Rank-2 symmetric tensors transform as* $T' = RTR^\top$*, which expands to the six formulas in Section 6 — or, more safely, you can implement the matrix product and let the compiler expand it.*

Gotcha 24. *Implement the rotation as a matrix product for* tensors *rather than writing out the six terms by hand. Hand-expanded index algebra is exactly where the* $\sim 10\%$ *errors of Section 50.14 come from. If you do write them out, then test them against the matrix form on random inputs.*

50.15.5 Step 4: validate, in this order

1. ***Scalar, first derivative.*** *Take* $f(x) = x^3$ *at* $z = 0$*. Compare* $f(\varpi\cos\theta, \varpi\sin\theta)$ *against* $f(\varpi) = \varpi^3$*. Should agree to round-off — this checks only the interpolation and the rotation of a scalar.*

2. ***Scalar, second derivative.*** *Take* $f = e^{-r^2}$ *with* $r^2 = x^2 + y^2$*. Then* $\partial_y^2 f = (4y^2 - 2)e^{-r^2}$*, computable analytically. Refine* dx *and confirm* ***second*** *order.*

3. ***Vector.*** *Take a vector field with a known covariant/rotating behaviour and check the rotation in isolation by evaluating at* $\varpi = x_i$ *so the interpolation is exact and only the rotation acts.*

4. ***Tensor.*** *Same, with a symmetric tensor, against the matrix form.*

5. ***Mixed.*** *Finally check $\partial^2/\partial x \partial y$, which is where the code uses a fallback and where errors hide.*

50.16 Debugging Numerical Relativity Code

This chapter is about method rather than mathematics, and it is included because the investigation recorded in Chapter 50.12 went wrong more often because of how it was measured than because of what it measured.

50.16.1 Rule 1: verify the instrument before interpreting it

Before you believe any diagnostic number, confirm that the diagnostic is computing the quantity you think. The cheapest way is to make it computable two different ways and check they agree.

In this project, the decisive technique was building a byte-for-byte replica of the production diagnostic inside a separate program and comparing the numbers. They agreed to every digit — which is what allowed the real bug (a filter using the wrong radius) to be found and localised in one step.

Gotcha 25. *If a standalone instrument and the production code disagree, the disagreement is a bug report about one of them. Do not average, do not pick the one whose answer you prefer, and do not assume the production path is the correct one — in this case it was not.*

50.16.2 Rule 2: cross-check against an independent truth

Exact Kerr–Schild data gives you an independent truth for the data stage: you can build the BSSN state analytically at any coordinate and compare. Agreement to round-off localises the problem to the derivatives. Disagreement localises it to the conversion. This single test divides a large search space in two.

Likewise, exact vacuum data gives $\mathcal{H} = 0$ and $\mathcal{M}^a = 0$ identically, so any constraint residual is entirely discretisation error and must converge under refinement. A constraint that does not converge is telling you about a defect, not about physics.

50.16.3 Rule 3: check the order of accuracy you claim

Measure convergence before believing accuracy. The expected order for each stencil is in the table in Section 50.6.3; if the measured order is lower, you have found something.

Concretely, in this code: the one-sided second derivative is first order while the one-sided first derivative is second, so a boundary-adjacent point sees mixed orders; the Cartoon y-derivatives are second order while x and z are fourth; and $\tilde{\Gamma}$ is computed by a second-order path while its neighbours use fourth order. All three are real, all three are invisible without an explicit convergence test, and only the third was ever measured.

Gotcha 26. *When you refine, ask whether you are comparing the same physical quantity. In this project the "same" metric — $\max|H|$ over a radial band — selected different physical points at each resolution, because the band is defined in units of dx. An apparent non-convergence can be entirely an artifact of that.*

50.16.4 Rule 4: know which knobs are inert

Before an A/B test, confirm the knob is connected. A null result is only evidence if the switch demonstrably changes something.

Two examples from this project. A background-refresh switch produced bit-identical *output, which was the correct and useful result — but only because bit-identity was checked, rather than "the numbers look the same". And a projection diagnostic initially reported all zeros, because the field it was differentiating had never been initialised in the diagnostic's own copy of the state; that was an instrument bug, not a physics result.*

Gotcha 27. *Establish the positive control first: make the switch do something obviously different, confirm the diagnostic shows it, then turn it off and confirm the diagnostic goes quiet. Only then is a null meaningful.*

50.16.5 Rule 5: gates must be able to fail

A test that prints numbers and has no assertion always passes, and will pass forever while the code is broken. This project had one such test (`test_momentum_conv`). The fix is not just to add an assertion but to demonstrate that the assertion fails, *by deliberately breaking the physics it checks.*

The same applies to the parity gate: it exists to catch tests that validate code production does not build. Break the `SOURCES` list deliberately and confirm it complains.

50.16.6 Rule 6: keep a dated findings log, including retractions

Every investigation in this project produced a markdown file under `baseline/`, dated, and stating what was measured — in-

cluding the measurements that turned out to be wrong. Several findings had to be retracted.

The retractions are the valuable part. A project that deletes its mistakes leaves behind a set of confident claims with no indication of which were shaky, and the next person repeats the work. Writing "this was wrong, and here is why the measurement was invalid" is worth more than the original conclusion was.

50.16.7 Rule 7: restate the baseline whenever anything moves

Numbers of record are meaningful only relative to the exact code that produced them. When a diagnostic changes what it measures, or a parameter default changes, the baseline must be remeasured and the old value annotated — not quietly replaced.

In this project the baseline was restated twice: once when the diagnostic radius was corrected, and once when `excision_order` *changed. Both times the previous value is recorded in the commit message and the findings document.*

50.17 Where the Numbers Come From

Every quantitative claim in this document, and where it was measured.

50.17.1 The baseline of record

Production configuration, $N_x = 48$, $N_z = 96$, *live gauge,* `excision_order` *= 3, 700 steps:*

max|H|(ext)	1.2656×10^{-2}
rms|H|(ext)	2.0211×10^{-3}
max|M|(ext)	2.8728×10^{-2}
alpha(probe)	0.910348
near-mask max|H|	$1.9493 \times 10^{+0}$
near-mask max|M|	$1.4920 \times 10^{+0}$

50.17.2 Lifetimes

Configuration	*fail step*	*fail* t	*fail* r_{KS}
12M, N_x=48, *frozen*	*7536*	*6.73*	*1.315*
18M, N_x=72, *frozen*	*10495*	*9.34*	*1.015*
order 2, live	*6331*	*5.63*	—
order 3, live	*6848*	*6.07*	—
order 4, live	*4364*	*3.84*	—
order 2, frozen	*7536*	*6.70*	—

The frozen order-2 entry reproduces a previously recorded value exactly, which is what validates the measurement harness.

50.17.3 Near-mask measurements

Quantity	*value*	
Ghost fill error (worst anywhere)	4.44×10^{-16}	*machine prec*
Projection $\max\lvert\det\tilde{\gamma} - 1\rvert$	$9.8 \times 10^{-6} \to 2.3 \times 10^{-5}$	*400*
$\tilde{\Gamma}$ *error at* $dr/dx < 1$	$9.07 \times 10^{-2} \to 7.68 \times 10^{-3}$	N_x *48, 96*

50.17.4 Driver worst-point locations, corrected filter

N_x	*driver* `max|H|`	*replica*	*worst* r_{KS}	*status*
48	8.2028×10^{-1}	*same*	*0.8255*	GHOST
96	$1.8057 \times 10^{+0}$	*same*	*0.8125*	GHOST
192	5.0928×10^{-1}	*same*	*0.7092*	GHOST

Every reported worst point was inside the excision region. After the fix, the same scan reports 1.86×10^{-2} *at* $r_{\mathrm{KS}} = 2.158$.

.1 Traps, Collected

Every one of these cost real time in this project.

#	*Trap*
1	*Kerr radius* $\neq$ *cylindrical radius. Use* `kerr_radius_squared` *for anything that must agree with the mask.*
2	*A diagnostic that excludes a band is blind in that band. State what it covers, and report the excluded region separately.*
3	*One-sided second derivatives* $(f_i - 2f_{i+1} + f_{i+2})/h^2$ *are* first-*order accurate.*
4	*Extrapolate the perturbation from a background, not the field. Extrapolating the field amplifies its own truncation error.*
5	*Freeze the mask snapshot at the start of each fill sweep, or chains within a sweep amplify by* 15^n.
6	`get_neighbor` *treats* $y_{\mathrm{val}} = 0$ *as a sentinel for a plain index shift.*

#	*Trap*
7	*`Chr_low` is symmetric in its* last *two slots. Contracting against the first silently computes the wrong tensor — wrong by* $\sim 10\%$ *and not otherwise suspicious.*
8	*Never freeze only one of lapse/shift while measuring discretisation residuals.*
9	*A forward difference is not a second derivative. It scales as* f'/h *and will produce spectacular, entirely fictional results.*
10	*A test with no assertion always passes.*
11	*`FC ?= gfortran` never fires in make; `FC` is predefined.*
12	*Fortran is case-insensitive: `Q`/`q` and `Gam`/`gam` collisions cause repeated compile failures.*
13	*Fortran I/O is line-buffered when connected to a terminal but block-buffered when redirected. Use `stdbuf -oL` for long runs.*
14	*A test that honours its argument versus one that is hard-wired to a resolution will silently waste a run.*
15	*`make check` failing for a pre-existing reason trains you to ignore it.*
16	*Distances in cells versus in* M *are not interchangeable near a mask; a bin labelled "*$\mathrm{d}r/\mathrm{d}x$*" is a different set of points at every resolution.*

.2 Environment Overrides and Reproduction

Command	*What it reproduces*
`make check`	*The gate: parity, production build, full test suite.*

Command	*What it reproduces*
`make run`	*The production configuration.*
`./bin/diagnose\ _ghost\ _accuracyparameter\ _files/vacuum\_kerr.par`	*Ghost-fill accuracy, derivative errors, H decomposition, $\tilde{\Gamma}$ accuracy.*
`GR\_DIAG\_MARGIN=0.05./bin/vacuum\ _kerr\ldots`	*The margin sweep.*
`GR\_NEAR\_MASK\_W=0./bin/vacuum\_kerr\ ldots`	*Silences the near-mask line, to confirm it is inert.*
`GR_SLICING_MODE=2 GR_GAMMA_DRIVER_MODE=2 ...`	*Frozen Kerr–Schild gauge; the stationary control.*
`GR\_PROJ\_DIAG=1./bin/vacuum\_kerr\ ldots`	*Projection correction report.*
`GR_EXCISION_ORDER=`n `...`	*The order sweep.*
`GR_SOMMERFELD_DECAY=1 ...`	*Per-field asymptotic powers.*

.2.0.0.1 Environment. *gfortran with `-fcheck=all -fbacktrace` for the test builds; `h5fc` for HDF5; a TeX distribution for this document. The verification commands used throughout are `make check` for the suite and the 700-step run for the baseline of record:*

```
GR_HORIZON_EVERY=0 GR_HZ_ABORT=0 GR_NSTEPS=700 \
  ./bin/vacuum_kerr parameter_files/vacuum_kerr.par
```

Appendix A

Different Numerical Methods for Metriplectic Simulations

Appendix B

DG CCSN

B.1 Scope and goal

In general, I want to have a psuedo-toy simulation of a CCSN that uses DG. To constrain myself further, for simplification I will make it fully Newtonian(no GR or SR) but to add challenge I enforce positivity and well-balanced.

B.2 Mathematical Model

Rather than writing down everything, in general the math starts with the Compressible Euler, adds Polytropic steady-states and Lane-Emden equation to the hydrostatic equilibrium equation. Finally, add boundary conditions in order to enforce energy conservation.

B.3 General DG

First one creates a mesh of the given space to calculate the projection martrix.

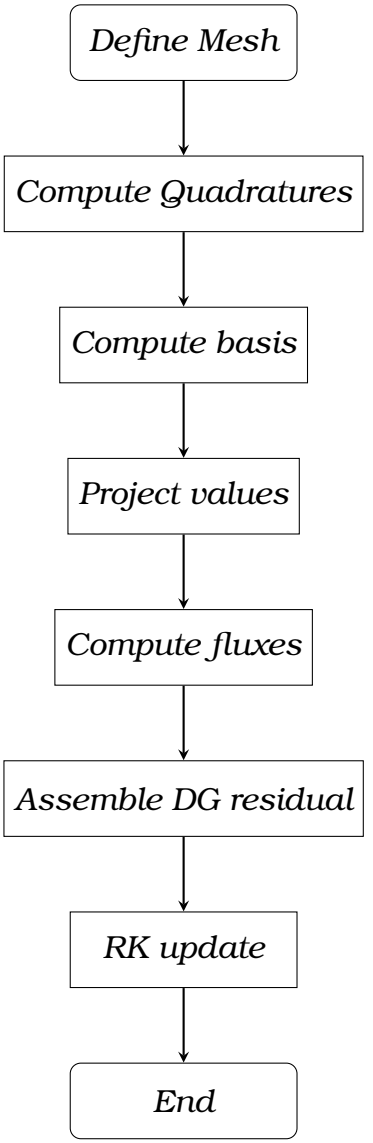
DG projects a PDE onto a functional space of polynomials using the projection matrix. Compute the fluxes of the fluid, then fluxes of energy. Then combine this all together for the residual.

Finally, time step with choosen model.

B.4 My DG

Essentially what I have talked about with the equations I use. For the sake of brevity the link to the paper that has this all writen out is here.

One thing I will add is a flow chart for my code



B.4.1 AI overview

Admittedly, much of this code is written by AI, I had it create documentation which I have here.

B.5 General Notes

Running the algorithm is very slow, far slower than anticipated, taking in total n hours to run the solution for first order polynomial, 16 x 16 x 16, 3rd order Runge-Kutta, and Newtonian iteration.

B.6 Results

Appendix C

WarpDrive Collision

C.1 Scope

Using Einstienstoolkit, I plan to develop a simulation of a warp drive, then use that to collide with an Earth like object.

Appendix D

Projects for YouTube

D.1 Planck Force on Earth

D.2 Collapsing Universe

D.3 Post-Newtonian Simulations

D.3.1 General idea

To test the effects of differing modified GR theories. I would like to create N-body post-Newtonian codes. From here I would like to simulate a simple version of the solar system.

I would like to test:

- *f(R)*
- *Higher dimensional Brans-Dicke gravity*
- *Kaluza-Klein Compact dimensions with that are large enough to effect the solar system*

D.3.2 Notes on Brans-Dicke

D.4 CCSN via DG

- *Newtonain gravity*
- *hydrodynamics*
- *SFHo EOS for Nuclear effects*
- *I need to transfer energy to tempeture for the SFHo*
- *I am going to use Nodal-LGL for my basis*

Appendix E

Hamiltonian, Boltzmann, to Navier stokes

Sources:

- *https://www.unm.edu/ aierides/505/bbgky3.pdf?*
- *https://journals.aps.org/rmp/pdf/10.1103/RevModPhys.52.569*
- *Introduction*
- *Scope*
- *Brief overview of the three*
- *Liouville to BBGKY*
- *The BBGKY Hierarchy*
- *The Boltzmann equation*
- *The H-Theorem*
- *Boltzmann to Navier-Stokes*
- *Hydro limit*

E.1 Introduction

Hello, and welcome to another video. On the connection between Hamiltonian dynamics, Boltzmann equation, and navier-stokes

E.2 Scope

Here I plan to derive the Navier-stokes using the Boltzmann, and the Boltzmann using the N-body Hamiltonian. Then using how we did that, examine their connection and when it is appropriate to use one and not the other.

E.3 Brief Overview

....

E.4 From Liouville to BBGKY

Taking an N-body Hamiltonian

$$H = \frac{1}{2m}\sum_{i=1}^{N} \vec{\mathbf{p_i}}^2 + \sum_{i=1}^{N} V\vec{\mathbf{r_i}} + \sum_{i<j}^{N} U(\vec{\mathbf{r_i}} - \vec{\mathbf{r_j}})$$

Where the Hamiltonian constains an external force $\vec{\mathbf{F}} = -\nabla V$, that acts equally on every particle. There is also two-body interaction between particles captured by the potential energy term. Later on we will make the assumption that this potential is short ranged. I will also be notating position as r rather than q for convenience.

For our interests, we wish to examine it as a probobility distribution, like all probobilitity distributions they are normalized as follows

$$\int dV\, f(\vec{\mathbf{r}}, \vec{\mathbf{p}}, t) = 1 \quad dV = \prod_{i=1}^{N} d^3 r_i d^3 p_i$$

Further, since the probability is locally conserved, the equation most follow continuity. Any probability change in one point in phase-space must by its nature be compensated by neighboring points.

This fairly trivially corresponds to the equation

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial \vec{\mathbf{r}}_\mathbf{i}} \cdot (\dot{\vec{\mathbf{r}}}_\mathbf{i} f) + \frac{\partial}{\partial \vec{\mathbf{p}}_\mathbf{i}} \cdot (\dot{\vec{\mathbf{p}}}_\mathbf{i} f) = 0$$

Using the nature of what a Hamiltonian and its derivatives we can transform this equation

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial \vec{\mathbf{r}}_\mathbf{i}} \cdot (\frac{\partial H}{\partial \vec{\mathbf{p}}} f) + \frac{\partial}{\partial \vec{\mathbf{p}}_\mathbf{i}} \cdot (\frac{\partial H}{\partial \vec{\mathbf{r}}_\mathbf{i}} f) = 0 \longrightarrow$$

$$\frac{\partial f}{\partial t} + \sum_{i=1}^{N} \frac{\partial f}{\partial \vec{\mathbf{r}}_\mathbf{i}} \cdot \frac{\partial H}{\partial \vec{\mathbf{p}}_\mathbf{i}} + \frac{\partial f}{\partial \vec{\mathbf{p}}_\mathbf{i}} \cdot \frac{\partial H}{\partial \vec{\mathbf{r}}_\mathbf{i}} = 0$$

This final equation is the Liouville's equation. It is the statement that probability doesn't change as you follow it along any trajectory in phase space, as is seen by writing the Liouville equation as a total derivative

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial t} + \sum_{i=1}^{N} \frac{\partial f}{\partial \vec{\mathbf{r}}_\mathbf{i}} \cdot \frac{\partial H}{\partial \vec{\mathbf{p}}_\mathbf{i}} + \frac{\partial f}{\partial \vec{\mathbf{p}}_\mathbf{i}} \cdot \frac{\partial H}{\partial \vec{\mathbf{r}}_\mathbf{i}} = 0$$

we can better understand this equation using the familiar Liouville theorem that we have used in the past to express in pois-

son brackets defined at

$$\{A, B\} \equiv \frac{\partial A}{\partial \vec{\mathbf{r_i}}} \cdot \frac{\partial B}{\partial \vec{\mathbf{p_i}}} + \frac{\partial A}{\partial \vec{\mathbf{p_i}}} \cdot \frac{\partial B}{\partial \vec{\mathbf{r_i}}}$$

With this definition one can clearly see the above equation finds that

$$\frac{\partial f}{\partial t} = \{H, f\}$$

Meaning that the equilibrium distribution is explicitly time independent.

To make this statement true, we must make f a function of H, the most famous example being Boltzmann's distribution $f \sim e^{-\beta H}$, there is nothing that requires this as there are several such solutions given this limited definition.

Now, let us add a new function $A(\vec{\mathbf{r}}_i, \vec{\mathbf{p}}_i)$ on the phase space.With the expectation value

$$\langle A \rangle = \int dV\, A(\vec{\mathbf{r}}_i, \vec{\mathbf{p}}_i) f(\vec{\mathbf{r}}_i, \vec{\mathbf{p}}_i, t)$$

We can then find the time dependence of the expectation value of A by

$$\begin{aligned}
\frac{\mathrm{d}\langle A \rangle}{\mathrm{d}t} &= \int dV\, A \frac{\partial f}{\partial t} && \text{(E.1)} \\
&= \int dV\, A \left(\frac{\partial f}{\partial \vec{\mathbf{r_i}}} \cdot \frac{\partial H}{\partial \vec{\mathbf{p_i}}} + \frac{\partial f}{\partial \vec{\mathbf{p_i}}} \cdot \frac{\partial H}{\partial \vec{\mathbf{r_i}}} \right) && \text{(E.2)} \\
&= \int dV \left(\frac{\partial A}{\partial \vec{\mathbf{r_i}}} \cdot \frac{\partial H}{\partial \vec{\mathbf{p_i}}} + \frac{\partial A}{\partial \vec{\mathbf{p_i}}} \cdot \frac{\partial H}{\partial \vec{\mathbf{r_i}}} \right) f && \text{(E.3)}
\end{aligned}$$

where we hav0e integrated by parts to get to the last line,

throwing away boundary terms which is justified in this context because f is normalized which ensures that we must have $f \to 0$ in asymptotic parts of phase space. Finally, we learn that

$$\frac{\mathrm{d}\langle A\rangle}{\mathrm{d}t} = \int dV\, \{A, H\} f = \langle \{A, H\} \rangle$$

This looks very closely to the quantum expectation values.

E.4.1 The BBGKY Hierarchy

Now, that we have it in a distribution formulism, we still have a problem, we still have a function with an obscene amount of variables, assuming we have $N \approx 10^{23}$, so we will focus on one particles, but since all of the particles are identical, we aren't actually privileging one particle over the others. From we find that a series of equations; energy flux, average density, average velocity, etc all require a function f_1, we define f_1 time derivative as:

$$\frac{\partial f_1}{\partial t} = N \int \prod_{i=1}^{n} d^3 r_i d^3 p_i \{H, f\}$$

Using the Hamiltonian from earlier this becomes

$$\frac{\partial f_1}{\partial t} = N \int \prod_{i=2}^{N} d^3 r_i\, d^3 p_i \left(-\frac{\mathbf{p}}{m} \cdot \frac{\partial f}{\partial \mathbf{r}} + \frac{\partial V(\mathbf{r})}{\partial \mathbf{r}} \cdot \frac{\partial f}{\partial \mathbf{p}} + \sum_{k=2}^{N} \frac{\partial U(\mathbf{r} - \mathbf{r}_k)}{\partial \mathbf{r}} \cdot \frac{\partial f}{\partial \mathbf{p}} \right)$$

Now, whenever $j = 2, \ldots N$, we can always integrate by parts to move the derivatives away from f and onto the other terms. And, in each case, the result is simply zero because when the derivative is with respect to $\vec{\mathbf{r}}_j$, the other terms depend only on $\vec{\mathbf{p}}_i$ and vice-versa. We're left only with the terms that involve derivatives with respect to $\vec{\mathbf{r}}_1$ and $\vec{\mathbf{p}}_1$ because we can't integrate

these by parts. Let's revert to our previous notation and call $\vec{\mathbf{r}}_1 \equiv \vec{\mathbf{r}}$ *and* $\vec{\mathbf{p}}_1 \equiv \vec{\mathbf{p}}$*. We have*

$$\frac{\partial f_1}{\partial t} = N \int \prod_{i=2}^{N} d^3 r_i \, d^3 p_i \left(-\frac{\mathbf{p}}{m} \cdot \frac{\partial f}{\partial \mathbf{r}} + \frac{\partial V(\mathbf{r})}{\partial \mathbf{r}} \cdot \frac{\partial f}{\partial \mathbf{p}} + \sum_{k=2}^{N} \frac{\partial U(\mathbf{r} - \mathbf{r}_k)}{\partial \mathbf{r}} \cdot \frac{\partial f}{\partial \mathbf{p}} \right) \tag{E.4}$$

$$= \{H_1, f_1\} + N \int \prod_{i=2}^{N} d^3 r_i \, d^3 p_i \sum_{k=2}^{N} \frac{\partial U(\mathbf{r} - \mathbf{r}_k)}{\partial \mathbf{r}} \cdot \frac{\partial f}{\partial \mathbf{p}}. \tag{2.10}$$

We find this expression can be defined by a Liouville-like equation with an added collision term.

$$\frac{\partial f_1}{\partial t} = \{H_1, f_1\} + \left(\frac{\partial f_1}{\partial H} \right)_{coll}$$

The first term is sometimes referred to as the streaming term. It tells you how the particles move in the absence of collisions. The second term, known as the collision integral, is given by the second term in In fact, because all particles are the same, each of the $(N-1)$ *terms in* $\sum_{k=2}^{N}$ *from earlier are identical and we can write*

.... continue

E.5 Motivating the Boltzmann Equation

Now that we have the structure of the Boltzmann equation

$$\frac{\partial f_1}{\partial t} = \{H_1, f_1\} + (\mathcal{P} f_1 t)$$

but now we need to create an expression for the collision integral in terms of f_1*.*

Appendix F

The Eudaimonia Society

F.1 Initial Exploration

F.1.1 General Thoughts & Scope

Originally, I had begun this exercise by thinking of, if I were to create an institution like the Freemasonry, what would it look like. What doctrine would I lay at the foundation, traditions I would create, values to instill, how to even go about it, and much more. This is an attempt to write it out. One more pragmatic train of thought, I would likely use the general idea in creating a club at school, very similar to my previous SVA, but more evolved. This club would likely be more constrained, focusing on intellectual virtues and their relation to research rather than the broad idea of a society based upon my fundamental values and then using them to grow to be something more.

I will break this up into Moral development, intellectual development, social cohesion, and social impact.

I am thinking of using this idea for the YouTube Channel to an extent. Maybe...

F.1.1.1 Moral Development

The core of the entire project is to take the fundamental approach towards perfection to be the absolute goal. To take all things seriously, never do things half-heartedly, and don't mistakenly take irony to that which is sacred.

That we are all within our own personal design and can take control of our lives and personalities through the continuous cultivation of habits.

On a more direct statement; take accountability for all things(even that which many would consider beyond you), never do anything anti-social(in the psychological sense) even in the most slight way, value truth, be accurate and develop yourself epistemically and intellectually, pursue the development of your community in your own image and values, develop clearly defensible and defined values that are not vague bromides, hold yourself to your standards rigorously, think in the long term for your community, only acception to the anti-social rule would be when it conflicts with written values, and be fully conscious at all times, fully aware of your actions and their relation to your moral identity.

I personally find virtue ethics to be my way of thinking, so this may be a core tenet or may be too constricting.

Beyond that general view of altruism, self-relience, and logic being virtues to pursue.

F.1.1.2 Intellectual Development

To understand things at their most fundamental level. To understand epistemology, statistical thinking, critical thinking, interdisciplinary thinking, metaphysics, and have a fully simulated world-view rather than implicitly taking on assumptions. This is the core tenent, not to take on others assumptions passed through pop culture.

F.1.1.3 Social Cohesion

Generally thinking a somewhat brotherhood of intellectuals or in close enough fields. To discuss and understand each other and build one another up.

Beyond that many questions arise of what structure would best work for this.

F.1.1.4 Social Impact

Any general social impact will do

F.1.2 First layer of Refinement

The first questions I want to ask consist of:

- *Mission Statement*
- *Structure*
- *Symbols(if any)*
- *Meetings*

F.1.3 Mission Statement

IDK...

F.1.4 Structure

A both heirercical and collaborative. With strict order on tasks; programmers of the mission, chapter directors, etc. These people have clearly defined stewardships and authority with complete authority over their specific stewardship; though, a culture of

collaboration. Meetings will be scheduled to have democratic-like discussion on actions.

F.1.5 Meetings

In the action of the goal, meetings consists of discussions, debates, and lectures.

Appendix G

NEP

G.1 Preliminary plans

G.1.1 Second week

G.1.1.1 Goals for next week

We are currently working to establish what thruster type we will use. Likely our next steps will be propellant and creating design targets. I will assign

- *Find NASA resources to create target*
- *Xeon(Hall, ion, RF)*
- *Krypton(hall or ion)*
- *Iodine(Hall, Ion)*
- *Argon(MHD or RF)*
- *Hydrogent(MHD, RF)*
- *Research Rover*

- *Research Atomic in Orbit*
- *Research Atomic at Lagrangian*

G.1.1.2 Goal

Fly by, Rover, Atomic in Orbit, or Lagrandian

This document compares four candidate mission architectures for an undergraduate nuclear-electric-propulsion (NEP) project centered on the design and computational evaluation of a Hall-effect thruster: (1) an atomic-clock spacecraft in Mars orbit, (2) an atomic-clock spacecraft near the Mars–Sun first Lagrange point, (3) a spacecraft performing a Mars flyby, and (4) a Mars rover mission.

The comparison separates mission utility from propulsion-system suitability. These objectives overlap, but they are not equivalent. A mission may have substantial scientific value while imposing requirements that are poorly matched to the team's intended propulsion study. Conversely, a comparatively modest mission may provide a better engineering case if its trajectory, payload, power demand, and operational lifetime can be translated into well-defined thruster requirements.

The central conclusion is conditional. A Mars-orbit atomic-clock mission has a strong connection to established concepts in deep-space radiometric navigation and Mars communications infrastructure. A clock near Mars–Sun L1 could provide different measurement geometry, but its advantage must be demonstrated using a specified navigation architecture rather than inferred from the Lagrange-point location alone. A flyby is attractive for a focused transfer and encounter study, whereas a rover mission offers an obvious surface exploration objective but introduces atmospheric entry, descent, landing, and surface-operation requirements that can obscure the propulsion question.

No architecture can be declared the best Hall-thruster application without a quantitative mission model. The most defensible selection is the one for which the team can establish a traceable chain from mission objectives to spacecraft mass, trajectory, electrical power, propellant consumption, thrust, and thruster operating conditions.

G.1.1.2.1 Scope, Assumptions, and Decision Criteria

G.1.1.2.1.1 The actual decision *The decision is not simply which mission is "most useful." Four distinct questions must be answered:*

1. ***Mission utility:*** *What scientific, operational, or infrastructure capability does the mission provide?*

2. ***Trajectory compatibility:*** *Does the mission benefit from continuous low thrust, or can it be accomplished more simply using conventional chemical propulsion?*

3. ***Thruster-design value:*** *Does the mission impose sufficiently concrete requirements to guide channel geometry, magnetic-field configuration, discharge power, propellant flow, thermal management, and cathode integration?*

4. ***Undergraduate feasibility:*** *Can the group evaluate the mission with accessible trajectory tools, published reference data, and a defensible simulation hierarchy without needing to solve every spacecraft subsystem?*

These questions should be scored separately before a final mission is selected. Otherwise, a mission may be chosen because it sounds scientifically compelling even though its requirements do not meaningfully constrain the thruster design.

G.1.1.2.1.2 Assumed propulsion architecture *The baseline assumption is a spacecraft using a Hall-effect thruster powered by an electric power system ultimately associated with a nuclear-electric-propulsion concept. The project focuses on the thruster and its interaction with the mission, rather than on designing a nuclear reactor.*

The relevant spacecraft-level variables include

- *initial wet mass,* m_0*;*
- *dry spacecraft mass,* m_{dry}*;*
- *propellant mass,* m_p*;*
- *thruster electrical input power,* P_{in}*;*
- *thrust,* T*;*
- *specific impulse,* I_{sp}*;*
- *total thruster operating time,* t_{on}*;*
- *required mission* Δv*;*
- *mission duration and operational lifetime;*
- *payload power and spacecraft bus power;*
- *propellant type and storage requirements.*

The analysis assumes that the team is evaluating a plausible mission concept, not proposing a fully qualified flight system. A first-order trajectory and propulsion analysis is appropriate, provided its assumptions and limitations are explicit.

G.1.1.2.1.3 Performance relations *For an idealized electric thruster, the effective exhaust velocity is*

$$v_e = g_0 I_{\mathrm{sp}},$$

where $g_0 \simeq 9.80665\ \mathrm{m\,s^{-2}}$.

The thrust is related to propellant mass flow by

$$T = \dot{m} v_e = \dot{m} g_0 I_{\mathrm{sp}},$$

where $\dot{m}$ *is the propellant mass-flow rate. For a steady operating point, an idealized jet-power relation is*

$$P_{\mathrm{jet}} = \frac{1}{2}\dot{m} v_e^2 = \frac{T v_e}{2}.$$

If the total thruster efficiency is η_t*, then*

$$P_{\mathrm{in}} \simeq \frac{P_{\mathrm{jet}}}{\eta_t},$$

and therefore

$$T \simeq \frac{2\eta_t P_{\mathrm{in}}}{v_e} = \frac{2\eta_t P_{\mathrm{in}}}{g_0 I_{\mathrm{sp}}}.$$

These equations are useful for preliminary sizing, but they do not replace a real Hall-thruster performance map. Efficiency, discharge voltage, mass utilization, divergence, cathode losses, magnetic configuration, and thermal constraints all affect the actual operating point.

The rocket equation gives a useful first-order estimate of propellant requirements for a specified ideal velocity increment:

$$\Delta v = v_e \ln\left(\frac{m_0}{m_f}\right),$$

where m_f is the final mass after propellant expenditure. This equation is not, by itself, a trajectory solver. In particular, a low-thrust trajectory must account for the direction and duration of acceleration, changing spacecraft mass, planetary gravity, solar gravity, and mission boundary conditions.

G.1.1.2.1.4 What constitutes a good mission for this project? *The most useful mission is one that constrains the thruster rather than merely carrying it.*

For example, a mission duration of several years may make high specific impulse attractive because propellant mass is expensive. However, increasing specific impulse at fixed electrical power usually decreases thrust. A longer transfer may reduce the required propellant or improve payload fraction while simultaneously increasing the required spacecraft lifetime and total time spent operating the thruster.

The mission therefore creates a trade between

propellant mass, *electrical power*, *thrust*, *transfer time*, *spacecraft*

A strong mission concept makes these competing requirements quantifiable.

G.1.1.2.2 Mission A: Atomic Clock in Mars Orbit

G.1.1.2.2.1 Concept and potential purpose *The spacecraft enters a bound orbit around Mars and carries a high-stability atomic clock. Depending on the communications and navigation architecture, the spacecraft could support one or more of the following:*

- *one-way radiometric tracking and spacecraft orbit determi-*

nation;

- *autonomous or partially autonomous navigation;*
- *precision timing for a distributed Mars communications or navigation system;*
- *radio-science measurements and improved orbit knowledge;*
- *timing and navigation experiments supporting future Mars surface and orbital missions.*

The clock is not, by itself, a navigation system. Its value depends on the signals measured, the stability of the clock, the spacecraft's orbit, the availability of external references, and the algorithms used to infer position and velocity.

NASA's Deep Space Atomic Clock work explicitly examined one-way radiometric tracking, autonomous navigation concepts, Mars-orbiter orbit determination, and radio-science applications [1]. This provides a direct technical precedent for an atomic-clock spacecraft in the Mars environment.

G.1.1.2.2.2 Arguments in favor *A. The mission has a concrete operational context*

A clock-equipped Mars orbiter can be connected to a recognizable operational need: accurately determining spacecraft position and time while operating far from Earth.

Conventional deep-space tracking often relies on measurements involving signals transmitted from Earth and received or retransmitted by the spacecraft. Such tracking depends on ground assets, scheduling, signal geometry, communications links, and the ability to interpret the returned measurements.

A sufficiently stable onboard clock can support one-way radiometric measurements. These can reduce dependence on a

round-trip measurement architecture and may permit more flexible tracking and navigation operations. The potential benefit is not that the clock removes the need for tracking, but that it changes which measurements can be used and how often they can be collected.

The technical advantage is therefore conditional on the complete measurement system. The clock must be sufficiently stable over the relevant measurement interval, and the radio system must provide measurements with suitable accuracy and geometry.

For a student project, this is valuable because the scientific objective can be expressed in testable terms. For example:

> *Determine whether a specified onboard clock stability and one-way radio-tracking architecture can achieve a target orbit determination accuracy for a spacecraft in a specified Mars orbit.*

That question can be studied using published clock stability specifications, an orbital dynamics model, simulated radio measurements, and an orbit-estimation algorithm.

B. Mars already has an operational relay infrastructure

Mars exploration is not limited to isolated landers. Orbiters have served as communications relays between the Martian surface and Earth. NASA describes the Mars Relay Network as a collection of orbiters that transmit data from surface missions back to Earth [3].

This provides a plausible operational environment for studying how precise timing could complement existing Mars infrastructure. A clock-equipped spacecraft could potentially be designed as a new node in a broader system rather than as an entirely isolated scientific experiment.

There is an important distinction, however, between a communications relay and a precision-navigation node. A spacecraft that forwards data does not automatically provide the timing stability, ranging capability, signal coverage, or orbit knowledge needed for navigation. The mission architecture must specify which role the clock-equipped spacecraft is expected to perform.

A defensible concept might therefore define the spacecraft as a precision timing and navigation demonstrator that also supports communications, rather than assuming that the clock alone creates a complete Mars navigation service.

C. The orbit provides repeated access to the Mars environment

Unlike a flyby, a Mars orbiter can make repeated observations of the same planetary environment. It can remain available for multiple navigation experiments, repeated radio links, and comparisons between predicted and observed spacecraft states.

This repeated access is useful for evaluating whether clock performance improves orbit estimates over time. It also provides a natural reason to consider long operational lifetimes, which may be relevant to an electric-propulsion spacecraft.

The orbital design remains a major decision. A low Mars orbit, high Mars orbit, eccentric orbit, and polar orbit differ in coverage, orbital period, radiation environment, communications geometry, and station-keeping requirements. The best orbit depends on the mission's intended service, not simply on the fact that the spacecraft is near Mars.

D. The mission has a relatively clear connection to a Hall-thruster design

An orbital mission allows the team to connect propulsion decisions to a well-defined arrival condition. The spacecraft must depart Earth, travel to Mars, and enter or achieve the desired

bound orbit.

A low-thrust trajectory model can estimate the required thrust history and propellant consumption. Those results constrain the spacecraft's electrical power system and thruster operating point.

The mission also creates a meaningful distinction between thruster operation during transfer and operation during the science phase. The thruster might be used primarily during interplanetary cruise, during orbit insertion, for orbit changes, or for some combination of these functions. The team should not assume that the thruster must remain continuously active after arrival.

A useful engineering study would compare at least two cases:

1. *chemical or other conventional injection into the desired Mars orbit, followed by electric-propulsion station-keeping or orbit adjustment;*

2. *electric propulsion used for a substantial fraction of the Earth-to-Mars transfer and, if required, arrival orbit acquisition.*

This comparison determines whether the NEP system creates a meaningful mass or mission-duration advantage rather than simply adding complexity.

E. The mission can be scoped around a measurable navigation objective

A strong undergraduate project should define a baseline architecture and then evaluate one change at a time.

For example, the team could compare a conventional tracking case against a case with a specified onboard clock and one-way radiometric tracking. Performance metrics could include

- *position and velocity estimation error;*

- *sensitivity to clock frequency drift;*
- *measurement availability;*
- *dependence on ground-station tracking;*
- *navigation performance under communication outages;*
- *sensitivity to uncertainty in the spacecraft's initial state.*

The group would not need to build or validate a real atomic clock. It would need to use defensible published clock specifications and model their effect on navigation measurements.

G.1.1.2.2.3 Arguments against *A. One clock-equipped orbiter is not a Mars GPS system*

The most important limitation is that a clock does not provide position by itself. Navigation requires measurements that constrain the spacecraft's state.

An onboard clock can support timing and one-way radiometric tracking, but the achievable navigation accuracy depends on the measurement architecture, signal geometry, orbit model, clock noise, radio-system errors, and reference information.

A single spacecraft also does not automatically provide a global positioning service to rovers or other orbiters. A Mars-local navigation network would require a defined architecture with appropriate signal sources, timing references, communications links, coverage, and orbit knowledge. Depending on the design, multiple spacecraft or additional ground and inter-spacecraft measurements may be necessary.

Consequently, the claim that "an atomic clock in Mars orbit will provide Mars GPS" is too strong unless a specific system is designed and analyzed.

A better claim is:

A clock-equipped Mars orbiter could demonstrate a component of a future Mars navigation architecture, provided that the measurement system and its navigation performance are specified.

B. The payload's benefit may not justify a dedicated spacecraft

The clock could potentially be hosted on an existing orbiter or included as a secondary payload. If so, a dedicated spacecraft may not be necessary to demonstrate the clock's scientific utility.

This creates a challenge for the mission justification. The team must explain why a new NEP spacecraft is useful instead of placing the clock on an existing or otherwise planned platform.

There are several possible answers: the mission could test a new orbit, demonstrate autonomous navigation, support a future network, or provide an independent measurement geometry. But each answer should be tied to an identifiable technical requirement.

If the only objective is to demonstrate that an atomic clock can operate near Mars, the proposed mission may be unnecessarily ambitious. If the objective is to demonstrate an end-to-end navigation architecture with specific performance targets, the dedicated spacecraft has a stronger justification.

C. The trajectory may not require nuclear electric propulsion

An atomic-clock payload does not intrinsically require an electric thruster. A clock-equipped spacecraft could reach Mars using a chemical launch and transfer architecture, potentially with chemical orbit insertion.

Whether NEP is advantageous depends on spacecraft mass, power, launch constraints, transfer duration, the desired Mars orbit, and the propellant mass saved by electric propulsion.

The project must avoid circular reasoning: it should not as-

sume that an NEP spacecraft is needed merely because the group wants to design an NEP thruster.

A proper comparison would evaluate the same payload and mission requirements under at least two propulsion architectures. If NEP significantly improves payload fraction or enables a trajectory that the baseline cannot support, the propulsion case is stronger. If the benefit is small, the mission may still be useful for navigation research but weak as a justification for the thruster.

D. Orbit insertion and mission duration may dominate the design

A bound Mars orbit requires the spacecraft to arrive with a suitable velocity and energy. A low-thrust arrival architecture can be more complicated than simply adding a cruise Δv to a transfer estimate.

The trajectory must account for the desired arrival orbit, the available thrust, planetary gravity, and the time at which thrust is applied. The feasibility of capture or orbit acquisition depends on the specific trajectory and propulsion design.

For an undergraduate project, this creates a risk of spending most of the effort on trajectory optimization rather than on the Hall thruster itself. The group can manage this risk by selecting a reference orbit, using an established trajectory tool, and treating detailed orbit design as a bounded subsystem study.

E. The clock payload does not necessarily demand large propulsion capability

A small clock instrument may not create a particularly large spacecraft mass or power requirement. The required NEP system could therefore be driven more by transfer duration and spacecraft architecture than by the payload.

This is not automatically a disadvantage. It does mean, however, that the team must not use the clock payload as a substitute for a mass and power budget for the project.

The clock's actual electrical power, thermal control needs, radio equipment, antenna, data handling, attitude control, and spacecraft bus must be estimated separately. The thruster should then be sized around the complete spacecraft, not around the payload label.

G.1.1.2.2.4 Assessment *The Mars-orbit atomic-clock mission is strongest when defined as a precision navigation or timing demonstrator with a clear measurement objective. It is less convincing when defined only as "put an atomic clock around Mars."*

Its strongest advantages are the direct connection to Mars operations, an established technical precedent for clock-enabled deep-space navigation, repeated access to the Martian environment, and a reasonably clear path from trajectory requirements to thruster sizing.

Its principal weaknesses are the need for a real navigation architecture, the possibility that an existing spacecraft could host the payload, and the fact that NEP must be justified independently of the clock's scientific value.

G.1.1.2.3 Mission B: Atomic Clock near Mars–Sun L1

G.1.1.2.3.1 Clarifying the location *In this section, "L1" means the first collinear Lagrange point of the Mars–Sun system, not the Earth–Sun L1 point and not a point in the Mars–spacecraft system.*

The idealized Mars–Sun L1 point lies approximately between Mars and the Sun. In a rotating three-body model, it is an equilibrium location in the frame rotating with the two massive bodies. It is not a location where all forces vanish in an inertial frame.

A spacecraft would generally not be placed at a perfectly fixed point. A practical mission might use a bounded orbit near L1, with its actual trajectory and station-keeping needs determined by the dynamical model and mission requirements.

The precise orbit matters. A mission described only as "at L1" is not yet a complete trajectory specification.

G.1.1.2.3.2 Arguments in favor *A. It provides a different geometric location for navigation measurements*

The most defensible argument for the L1 architecture is that it can occupy a location different from a Mars orbiter. If the spacecraft is part of a distributed timing and navigation system, that separation may create measurement geometry that is unavailable from a Mars-centered orbit alone.

For example, suppose a future network includes multiple Mars orbiters, surface assets, and spacecraft with known or estimated states. A clock-equipped spacecraft near Mars–Sun L1 could potentially provide an additional measurement node. Depending on the geometry and measurement types, this node might improve the conditioning of the navigation problem or help distinguish some state errors from others.

However, spatial separation is not automatically useful. A measurement contributes most when it adds information that is not already supplied by other measurements. A second spacecraft with nearly redundant geometry may add less value than one positioned to resolve an existing ambiguity.

The claim should therefore be tested through an estimation model. A suitable comparison would use the same navigation system with and without the L1 spacecraft, then compare the resulting state-estimation errors and robustness.

B. It creates a more distinctive research question

The L1 architecture can be intellectually attractive because it

requires the group to consider the interaction between interplanetary dynamics and a distributed timing system.

The project could examine how spacecraft position, clock stability, radio-link geometry, and orbit uncertainty influence navigation performance. It could also compare different bounded orbits near L1, subject to a clearly defined mission requirement.

This is a stronger research question than merely asking whether an atomic clock can operate near Mars. It asks whether a specific spacecraft placement creates a measurable advantage in a larger system.

For a technically ambitious group, this can lead to a useful systems-engineering study involving

- *three-body orbital dynamics;*
- *trajectory design and station-keeping;*
- *clock error and frequency stability;*
- *radio-ranging or Doppler measurement models;*
- *navigation observability and covariance analysis;*
- *sensitivity to spacecraft orbit uncertainty.*

The main value is the opportunity to formulate and test a nontrivial systems-level hypothesis. It is not evidence that the L1 architecture is inherently superior.

C. It may support a future distributed infrastructure architecture

A dedicated node near L1 could be considered as part of a future Mars communications, timing, or navigation infrastructure. Its usefulness would depend on what the rest of the system contains.

For instance, the architecture might include multiple spacecraft with stable clocks, cross-links, radio ranging, known ephemerides, and a method for distributing time or navigation solutions to Mars orbiters and surface assets. Under such an architecture, the L1 node could contribute an additional reference or measurement path.

This is a plausible concept, but it is a system-level proposition. It should not be confused with an immediate operational requirement for Mars exploration.

The group would need to explain why the node belongs near L1 rather than in Mars orbit, in a different solar orbit, or on another spacecraft already serving the network. The answer should follow from the geometry and performance analysis.

G.1.1.2.3.3 Arguments against *A. L1 does not provide an intrinsic atomic-clock advantage*

There is no general principle that an atomic clock becomes more accurate or useful merely because its spacecraft is near L1.

The clock's frequency stability is determined by the clock technology, environmental conditions, and its implementation. The mission location affects the system through dynamics, communications, radiation, thermal conditions, and measurement geometry, but the location alone does not guarantee improved timing performance.

The L1 architecture therefore needs an external justification: a navigation or timing task that benefits from the location. Without that task, the mission risks being a technically interesting placement exercise without a sufficiently clear operational purpose.

B. A single clock at L1 does not create a navigation network

The same limitation applies here as in Mars orbit, but it is particularly important for the L1 concept. A clock-equipped spacecraft near L1 does not automatically provide position fixes to Mars orbiters or rovers.

A navigation service requires defined signals and measurements. Depending on the design, these might include one-way or two-way radio links, Doppler measurements, ranging, crosslinks, known ephemerides, and clock synchronization or calibration.

If the spacecraft does not communicate with other nodes in a way that improves their state estimates, its location near L1 has no operational navigation benefit.

A convincing proposal must answer at least four questions:

1. *Which spacecraft or surface assets receive a benefit?*
2. *Which measurements does the L1 spacecraft provide?*
3. *What information does it add beyond existing nodes?*
4. *How much does that information improve a stated navigation or timing metric?*

If these questions cannot be answered, the L1 mission is not yet justified as a navigation mission.

C. The dynamical environment introduces additional trajectory requirements

A practical L1 mission needs an actual reference trajectory, not just a nominal equilibrium point. The spacecraft may follow a bounded orbit around the point, and the trajectory can be sensitive to injection errors and perturbations.

The exact station-keeping requirements depend on the chosen orbit, the dynamical model, the mission duration, and acceptable position error. They cannot be established merely by noting that L1 is a Lagrange point.

The team must also define how the spacecraft reaches the operational orbit. A transfer from Earth to the vicinity of Mars–Sun L1 is not the same problem as a transfer to Mars orbit. The spacecraft may never need to enter a Mars-centered bound orbit, but it still needs to reach the target region with appropriate position and velocity.

For the Hall-thruster project, the trajectory can become a substantial study in its own right. This may be desirable if the team wants to explore three-body dynamics and low-thrust control, but it creates scope risk if the primary deliverable is a thruster design.

D. It may be difficult to demonstrate a direct operational need

Mars orbiters already have clear roles in imaging, communications relay, atmospheric science, and other Mars operations. An L1 clock spacecraft needs a similarly explicit function.

If the mission's only proposed benefit is improved navigation, then the group should compare the L1 node with alternatives: another Mars orbiter, a surface timing node, a different interplanetary orbit, or an additional clock on an existing spacecraft.

The L1 architecture becomes compelling only if it produces a meaningful advantage that these alternatives do not provide as effectively. Otherwise, it may be more complicated without being more useful.

E. The propulsion benefit is not guaranteed

The L1 location does not, by itself, imply that Hall propulsion will be better than conventional propulsion. The answer depends on the transfer trajectory, spacecraft mass, available power, flight time, and target-orbit requirements.

A high-specific-impulse system can reduce propellant consumption, but its lower thrust at fixed power may increase transfer time. A station-keeping requirement may favor continuous or intermittent electric propulsion, but that must be demonstrated

for the chosen trajectory.

The project should not assume that an L1 mission is an especially good NEP application simply because it is unusual. It must show a quantitative benefit relative to a reasonable baseline.

G.1.1.2.3.4 Assessment *The L1 architecture has its strongest case as a research-oriented distributed-navigation concept. Its potential advantage is different measurement geometry and a possible role in a broader infrastructure system.*

Its weakness is that these advantages are conditional. Without a specified network, measurement model, bounded orbit, and quantitative performance comparison, "L1" is a location rather than a demonstrated mission requirement.

For a project whose main goal is Hall-thruster design, the L1 option should be selected only if the group is prepared to make the navigation and trajectory analysis part of the central research question.

G.1.1.2.4 Mars Orbit versus Mars–Sun L1

G.1.1.2.4.1 The fundamental difference *The Mars-orbit concept begins with a relatively direct operational context: a spacecraft operates in the Martian environment and may support navigation, timing, or communications services associated with Mars.*

The L1 concept begins with a different proposition: a spacecraft is placed at a distinct location in the Mars–Sun dynamical environment in order to create a potentially useful measurement or infrastructure node.

The first has a clearer immediate application. The second has a more conditional systems-level argument.

Neither location inherently produces better navigation. The

relevant quantity is the information available to the navigation system after accounting for all measurements, errors, and uncertainties.

G.1.1.2.4.2 A quantitative way to compare them *Let the spacecraft state be represented by*

$$\mathbf{x} = \begin{bmatrix} \mathbf{r} \\ \mathbf{v} \\ b \\ d \end{bmatrix},$$

where $\mathbf{r}$ *is position,* $\mathbf{v}$ *is velocity,* b *is clock bias, and* d *is clock drift. A measurement model can be written abstractly as*

$$\mathbf{y} = h(\mathbf{x}) + \boldsymbol{\epsilon},$$

where h *maps the state to predicted measurements and* $\boldsymbol{\epsilon}$ *represents measurement error.*

The navigation value of an additional spacecraft depends on how its measurements constrain the unknown state. In a linearized estimation framework, the measurement sensitivity matrix is

$$H = \frac{\partial h}{\partial \mathbf{x}}.$$

If the measurement-noise covariance is R*, a simplified information contribution is proportional to*

$$\mathcal{I} = H^{\mathsf{T}} R^{-1} H.$$

This expression is not a complete navigation solution. It illustrates the key point: a spacecraft is useful when its mea-

surements add sufficiently independent information relative to the errors and information already present in the system.

The comparison should therefore hold the clock assumptions, measurement types, radio-system assumptions, and estimator constant, then compare the navigation performance obtained from the two candidate locations.

G.1.1.2.4.3 Recommended comparison experiment *The team could formulate the following study:*

1. *Define a baseline navigation network consisting of a ground reference and a specified set of Mars orbiters.*
2. *Define a clock stability model and a radio measurement model using published or explicitly assumed parameters.*
3. *Simulate the baseline navigation performance without the proposed clock spacecraft.*
4. *Add the clock-equipped Mars orbiter and calculate the resulting position and velocity estimation errors.*
5. *Repeat the experiment with a spacecraft on a specified bounded orbit near Mars–Sun L1.*
6. *Compare accuracy, measurement availability, sensitivity to clock drift, sensitivity to ephemeris error, and robustness to measurement outages.*
7. *Perform a separate trajectory and propulsion analysis for each architecture.*

The experiment must not assume that the two spacecraft provide identical measurements from different locations unless that is physically plausible. The measurement links, visibility, signal strength, and time dependence should follow from the geometry.

G.1.1.2.4.4 Decision *If the team wants the clearest mission justification and the most direct connection to established Mars operations, Mars orbit is the stronger baseline.*

If the team wants a more novel research problem and is prepared to investigate navigation geometry, three-body dynamics, and station-keeping, L1 is a defensible alternative.

The L1 concept should not be selected solely because it sounds more advanced. It should be selected if a quantitative analysis shows that the location provides a useful capability that a Mars-orbit spacecraft does not provide as effectively.

G.1.1.2.5 Mission C: Mars Flyby

G.1.1.2.5.1 Concept and mission objective *In a Mars-flyby mission, the spacecraft approaches Mars, performs a close or distant encounter, and then continues along an outbound trajectory. It does not enter a long-term bound Mars orbit and does not land.*

A flyby could carry a scientific payload, perform a technology demonstration, collect measurements during the encounter, or use Mars as a gravity-assist body to alter its heliocentric trajectory. These are different mission concepts and should not be treated as interchangeable.

A scientific flyby needs a defined encounter objective. A gravity-assist mission needs a defined post-encounter trajectory. A mission that combines both should identify the primary objective and show how the other supports it.

NASA's descriptions of gravity-assist trajectories explain how a planetary encounter can change a spacecraft's heliocentric velocity by exchanging orbital energy and angular momentum with the planet [4]. A flyby does not provide unlimited propulsion: the useful change depends on the encounter geometry and the planet's motion.

G.1.1.2.5.2 Arguments in favor *A. The mission avoids the requirement to enter a long-term Mars orbit*

A flyby does not require the spacecraft to remain bound to Mars after the encounter. This can simplify the mission relative to an architecture requiring Mars orbit insertion, long-term orbit maintenance, or a surface landing.

The spacecraft must still reach the required encounter state. The arrival position, velocity, closest-approach altitude, and outbound state are constrained by the desired mission. However, the spacecraft does not have to satisfy the additional requirement of remaining in a selected Mars orbit.

For a preliminary study, this provides a relatively clean mission boundary condition: deliver the spacecraft to a specified Mars encounter, then determine the resulting trajectory.

This is attractive for an undergraduate group because the team can focus on the relationship between transfer trajectory, thruster performance, and total propellant use without designing a full Mars-orbit operations system.

B. A flyby provides a clear trajectory-analysis problem

A flyby naturally lends itself to a comparison between propulsion architectures. The team can estimate the transfer requirements, calculate a feasible encounter geometry, and compare a chemical baseline against a low-thrust trajectory.

The required analysis may include

- *launch and departure conditions;*
- *heliocentric transfer trajectory;*
- *thrust magnitude and pointing history;*
- *Mars-relative hyperbolic excess velocity;*
- *closest-approach distance and encounter time;*

- *outbound trajectory and final mission conditions.*

The flyby can therefore provide a focused way to investigate whether the selected Hall thruster can deliver the required integrated velocity change within the available mission time.

C. A gravity assist may reduce the propulsion burden of a larger mission

If the spacecraft uses Mars as a gravity-assist body, the planet can redirect the trajectory and alter the spacecraft's heliocentric energy. Under suitable geometry, this may reduce the propulsion needed to reach a later destination or achieve a desired outbound state.

This can make a flyby useful as part of a larger interplanetary mission, particularly when the propulsion system has limited available power or when propellant mass is a major constraint.

The benefit must be assessed against the trajectory that would be used without the gravity assist. It is not enough to claim that Mars supplies free Δv. The flyby changes the spacecraft's velocity relative to the Sun through a gravitational encounter with a moving planet; the result depends on the incoming asymptote, encounter geometry, and outgoing asymptote.

A flyby can also be combined with electric propulsion. For example, the thruster may shape the approach and departure trajectory, while Mars provides an additional gravity-driven change in heliocentric velocity. Whether this is advantageous depends on the mission's actual boundary conditions.

D. The mission can be used to test a propulsion system over a finite operating campaign

A flyby mission may be designed around a finite cruise and encounter phase rather than a long period of Mars-local operations. That can be useful if the project aims to study thruster operating life, accumulated discharge time, or propellant con-

sumption during a single transfer.

The thruster requirements remain mission-dependent. A short mission with a tight arrival-time constraint may favor higher thrust, while a longer mission with relaxed timing may allow lower thrust and higher specific impulse. The team should calculate this trade rather than assuming that a flyby is automatically easier for electric propulsion.

E. The payload can be kept simple

A flyby can carry a relatively compact instrument or technology demonstrator. If the main objective is propulsion validation, the payload may be represented as a specified mass and power budget.

This allows the team to isolate the propulsion design from a large surface system, relay network, or long-term orbital operations architecture. A simplified payload also reduces the number of spacecraft subsystems that must be estimated.

G.1.1.2.5.3 Arguments against *A. A flyby may not create a compelling need for NEP*

The absence of a Mars-orbit insertion requirement does not automatically make a flyby a good electric-propulsion mission.

A conventional chemical trajectory may already accomplish the flyby with acceptable mass and duration. If the mission has a relatively modest velocity requirement, a high-power NEP system may add complexity without a proportionate benefit.

The group must compare the required launch mass, propellant, power system, trajectory duration, and arrival conditions against a credible baseline.

A flyby is a useful project when the analysis shows a meaningful advantage for electric propulsion or when the primary objective is to demonstrate low-thrust trajectory design. It is weaker when the mission could be completed more simply by a

conventional architecture and no further benefit is established.

B. The encounter is a time- and geometry-constrained event

Unlike an orbiter, a flyby spacecraft has a limited interval during which it is near Mars. If the science objective depends on a close encounter, the trajectory must meet the required closest-approach conditions.

Errors in arrival state can change the encounter geometry, measurement opportunity, and outbound trajectory. The group must define acceptable error margins and account for navigation and maneuver requirements.

A flyby also creates a one-time operational window. There is no opportunity to repeat the encounter simply by waiting for the next orbit. A second encounter would require a new trajectory or a different mission design.

This can make the mission operationally simple after the encounter but demanding in the approach phase.

C. The spacecraft does not provide sustained Mars-local infrastructure

A flyby spacecraft cannot serve as a persistent Mars navigation node or routine communications relay after it departs. It may collect useful data during the encounter, but its continued availability near Mars is limited.

That is a major disadvantage if the intended purpose of the atomic clock is to support Mars operations over years. The clock might demonstrate one-way tracking or contribute to a technology experiment, but it would not be available as a long-term Mars-local timing reference.

The flyby architecture is therefore poorly matched to a mission whose main value depends on repeated coverage, sustained communications, or long-term orbit determination at Mars.

D. A gravity assist is not a guaranteed propulsion sub-

stitute

The value of a gravity assist depends on the planetary encounter. A flyby can increase or decrease heliocentric energy, depending on the geometry. It cannot provide an arbitrary change in velocity while preserving any desired outgoing trajectory.

The team must solve for the required incoming and outgoing states. A gravity-assist trajectory that is excellent for one destination may be unsuitable for another.

Furthermore, a low-thrust spacecraft may spend considerable time and propellant shaping the encounter trajectory. The gravity assist must be evaluated as part of the complete trajectory, not as a standalone source of free performance.

E. A flyby can be less informative for long-duration thruster operations

A flyby may be suitable for studying integrated propellant consumption and finite-duration thruster performance. It is less directly suited to a project whose objective is to assess years of operation, repeated starts, long-term erosion, or the integration of the propulsion system with a persistent Mars-orbit service.

These effects can still be studied using a flyby mission, but the connection is less direct. If the group wants to evaluate long operational lifetime, a long-duration orbital or infrastructure mission may provide a clearer requirements basis.

G.1.1.2.5.4 Assessment *A Mars flyby is the cleanest of these options for a focused trajectory and encounter study. It avoids long-term Mars-orbit operations and surface landing, and it gives the team a clear set of encounter conditions to target.*

Its primary weakness is that it offers little sustained Mars-local utility. It is also not automatically a strong NEP application: the mission must demonstrate that electric propulsion provides a meaningful benefit over a conventional baseline.

The flyby is most appropriate if the group prioritizes a bounded propulsion and trajectory problem rather than a persistent Mars navigation or communications capability.

G.1.1.2.6 Mission D: Mars Rover

G.1.1.2.6.1 Concept and mission objective *A Mars rover mission delivers a mobile surface platform to Mars to conduct geological, environmental, astrobiological, or other surface investigations. The rover may carry scientific instruments, communications equipment, autonomous navigation systems, and power systems needed for surface operation.*

The rover is an obvious mission concept because its scientific purpose is easy to explain. A rover can travel to a selected region, investigate local terrain, and return data through Mars orbiters or a direct-to-Earth communications link.

However, a rover mission is a coupled transportation and surface-system problem. It requires not only interplanetary transport but also a viable Mars arrival and landing architecture.

G.1.1.2.6.2 Arguments in favor ***A. The scientific and operational objective is concrete***

A rover has a direct, understandable purpose: reach a specified location and perform surface science or exploration.

That gives the project a useful basis for a payload mass budget, mission duration, surface power estimate, and communications requirement. The team can specify a particular science region or surface task rather than just a general spacecraft destination.

A concrete payload also makes it easier to define the spacecraft configuration. The rover, landing system, cruise stage, and communications equipment each contribute mass and operational

requirements.

For mission-level propulsion sizing, this can produce a meaningful payload-driven problem: the propulsion system must transport the complete vehicle, including the hardware needed to land and operate on Mars.

B. Electric propulsion may reduce the mass of the interplanetary transportation stage

For a sufficiently long transfer, high-specific-impulse electric propulsion may reduce the propellant required for interplanetary transportation relative to a chemical system with comparable mission constraints.

If propellant mass is reduced, some of the saved mass may be available for the rover, scientific instruments, landing hardware, or power system. The actual result depends on the spacecraft architecture, the available electrical power, the transfer duration, and the required arrival conditions.

This is an important distinction: electric propulsion does not automatically increase payload mass. The power system has its own mass, and lower thrust can lengthen the transfer. The correct comparison must include the entire propulsion and power subsystem rather than just the propellant tank.

A useful design question is whether a nuclear-electric transport stage can deliver a specified rover and landing system to Mars with a better payload fraction, lower total mass, or improved mission flexibility than the selected baseline architecture.

C. The rover creates a real mass and power budget

Unlike a purely abstract payload, a rover has identifiable subsystems. The mission analysis can assign masses to the rover chassis, wheels, instruments, onboard computing, communications equipment, thermal control, and surface power.

The landing system adds a separate mass contribution. De-

pending on the architecture, it may include an aeroshell, thermal protection, parachute, propulsion for terminal descent, landing legs, or other hardware.

These contributions make the total spacecraft mass more meaningful and can constrain the propulsion design.

The group should distinguish between

$$m_{\text{spacecraft}} = m_{\text{cruise}} + m_{\text{rover}} + m_{\text{landing}} + m_{\text{payload}} + m_{\text{propellant}},$$

where the exact categories must be defined to avoid double-counting. The equation is a bookkeeping framework, not a substitute for a detailed mass budget.

D. The mission offers a clear case for relay communications

Mars surface assets often rely on orbiters to relay substantial amounts of data to Earth. NASA describes the Mars Relay Network as an important part of the communications architecture for surface missions [3].

A rover mission can therefore be paired with a broader communications or navigation concept. For example, the rover might benefit from an orbital network that provides command and telemetry relay, and a future navigation system could potentially improve knowledge of spacecraft or rover states.

An atomic clock could be studied as a separate component of that system, but it should not be assumed that the rover itself requires a dedicated clock-equipped spacecraft. The group would need to identify the specific measurement or operational benefit.

E. It has substantial engineering depth

A rover mission provides opportunities to study interactions between propulsion, spacecraft power, payload mass, thermal

control, and landing architecture.

For an undergraduate team, this can be an advantage if the goal is a multidisciplinary spacecraft design project. The project could include a top-level mass budget, an interplanetary trajectory, a propulsion trade, and a preliminary system architecture.

It also creates a clear division of responsibilities. One subgroup can develop the Hall thruster and its operating map, another can estimate the transfer trajectory, and another can establish the rover and landing-system mass budget.

This division is only useful if the interfaces are defined early. Otherwise, each subsystem may evolve around incompatible assumptions.

G.1.1.2.6.3 Arguments against *A. Atmospheric entry, descent, and landing are separate major engineering problems*

The most important drawback is that a rover must survive arrival at Mars and reach the surface safely. This introduces a mission phase that is not solved by the interplanetary propulsion system.

The landing system must manage entry heating, aerodynamic loads, descent dynamics, and terminal landing conditions. The detailed architecture depends on rover mass, landing site elevation, atmospheric assumptions, terrain, and acceptable risk.

An electric thruster used during interplanetary cruise does not automatically solve these problems. The team should not conflate a low-thrust transfer trajectory with the atmospheric entry and landing design.

If the main project deliverable is a Hall thruster, designing a rover and its full landing system could consume substantial effort without improving the thruster model. The landing system should therefore be treated as a defined payload mass and in-

terface unless entry and landing are explicitly part of the project.

B. The rover's surface power system is not the same as the spacecraft's electric-propulsion power system

A rover needs power after landing. A Hall thruster requires substantial electrical power during operation, along with power-processing equipment and associated thermal management.

These requirements occur in different mission phases. The interplanetary spacecraft may use a nuclear-electric power system, while the rover may use a separate surface power architecture. The team must decide whether the nuclear power system is used only for transport, remains with the lander, or powers a larger integrated architecture.

Each choice changes the mass budget and mission design. For example, carrying a large power system through landing may introduce additional mass and complexity, while separating the cruise stage from the landed system changes the spacecraft interfaces and operational concept.

The group should not simply add a rover payload to a Hall thruster and assume that the complete vehicle is a consistent design.

C. The mission may become a landing-system project rather than a thruster project

A rover is a compelling exploration platform, but its requirements may be dominated by surface mobility, scientific instruments, thermal survival, landing risk, and communications.

If the team tries to design all these elements in detail, the Hall-thruster study may become a relatively small part of the overall work. This is a scope-management problem, not a scientific argument against rover missions.

A reasonable boundary is to define the rover and landing system through a reference mass, power requirement, and interface specification. The propulsion team can then evaluate the

transfer stage required to deliver that payload.

D. The rover does not inherently require an atomic clock

A rover mission may benefit from accurate timing and navigation, but the presence of a rover does not establish the need for a dedicated atomic-clock spacecraft.

The navigation architecture could rely on orbiters, ground tracking, onboard sensors, local mapping, and other techniques. Whether a precision clock adds useful capability depends on the task and the measurement architecture.

If the mission includes an atomic clock, the proposal should explain whether it supports interplanetary navigation, Mars-orbit navigation, rover localization, or a broader network. These are distinct functions and require different performance analyses.

A clock-equipped Mars orbiter might be a useful addition to a rover-support architecture, but it should be justified as a separate subsystem rather than treated as an automatic consequence of selecting a rover.

E. The mass benefit of NEP must be evaluated across the complete vehicle

A rover mission can make high specific impulse attractive because the payload and landing system may be substantial. However, the power system and electric-propulsion hardware also contribute mass.

A fair comparison should include at least

- *thruster and power-processing-unit mass;*
- *nuclear-electric power-system mass;*
- *propellant and tank mass;*
- *structural and thermal mass;*
- *cruise-stage mass;*

- *rover and scientific payload mass;*
- *landing-system mass;*
- *any additional hardware needed for the selected transfer and arrival architecture.*

The result may favor NEP, chemical propulsion, or a hybrid architecture. The answer depends on the mission and cannot be inferred from specific impulse alone.

G.1.2 Assessment

A rover is the strongest candidate if the group's central objective is an exploration mission with a concrete surface payload. It offers a clear science case and a meaningful spacecraft mass budget.

It is less suitable if the goal is to isolate Hall-thruster design and computational validation. The landing system and surface operations create a large set of requirements that must either be handled explicitly or bounded as external interfaces.

For a propulsion-focused project, the rover should be treated as a payload-delivery mission. The team should avoid expanding the scope to include full rover, entry, descent, landing, and surface-power design unless those are explicit project deliverables.

G.1.2.0.1 Cross-Mission Comparison

G.1.2.0.1.1 Qualitative comparison *The following table is a qualitative engineering assessment. It is not a numerical ranking derived from a trajectory simulation. The terms "high" and "low" refer to the relative clarity or complexity of the mission concept for the stated project.*

Criterion	*Mars-orbit clock*	*L1 clock*	*Mars flyby*	*Mars rover*
Immediate mission purpose	*Strong if tied to navigation or timing*	*Conditional on a defined network role*	*Strong if a specific encounter or science objective is selected*	*Strong and easy to communicate*
Connection to existing Mars operations	*Direct*	*Indirect unless integrated into a network*	*Limited after the encounter*	*Direct through surface exploration and relay*
Trajectory complexity	*Moderate to high; depends on arrival orbit*	*High; target-region dynamics and bounded orbit must be specified*	*Moderate to high; encounter conditions are critical*	*High when the complete arrival and landing architecture is included*
Need for long-term Mars operations	*Yes, if the clock is intended as a persistent Mars node*	*No Mars orbit is required, but sustained operations may still be intended*	*No*	*Yes, for surface exploration*
Navigation-system research potential	*Strong and directly motivated*	*Potentially strong but conditional*	*Limited for persistent Mars navigation*	*Depends on whether navigation is a central payload objective*

Criterion	*Mars-orbit clock*	*L1 clock*	*Mars flyby*	*Mars rover*
Landing-system complexity	*None*	*None*	*None*	*Major additional subsystem*
Scope risk for a thruster project	*Moderate*	*High if navigation and three-body dynamics become dominant*	*Low to moderate for a bounded flyby study*	*High if landing and surface systems are included*
Clarity of NEP justification	*Must be shown with trajectory and mass comparison*	*Must be shown with trajectory and mass comparison*	*Must be shown; chemical baseline may be competitive*	*Potentially strong for a large payload, but system mass must be counted*
Best project emphasis	*Navigation and propulsion co-design*	*Distributed navigation and three-body dynamics*	*Low-thrust trajectory and encounter design*	*Payload delivery and multidisciplinary systems engineering*

G.1.2.0.1.2 The four missions answer different questions *The missions should not be ranked as if they were interchangeable payloads on the same trajectory.*

The Mars-orbit clock asks whether an onboard clock and a Mars-based spacecraft can improve navigation or timing in the Martian environment.

The L1 clock asks whether a spacecraft near a different dy-

namical location can contribute useful information to a distributed navigation or timing system.

The flyby asks whether the propulsion system can meet a finite-duration interplanetary encounter requirement, possibly including a useful gravity assist.

The rover asks whether the propulsion and transportation architecture can deliver a comparatively complex surface payload while meeting the required arrival and landing conditions.

The preferred mission depends on which of these questions the group actually wants to answer.

G.1.2.0.2 Recommended Quantitative Evaluation

G.1.2.0.2.1 Step 1: Define the mission requirements
Before selecting a thruster operating point, establish the requirements that all four architectures must satisfy or explicitly identify which requirements differ.

At minimum, specify

- *launch date or launch-window assumptions;*
- *initial spacecraft mass and payload mass;*
- *destination or encounter conditions;*
- *allowable transfer duration;*
- *required operational lifetime;*
- *available electrical power;*
- *arrival or encounter velocity constraints;*
- *communication and navigation objectives;*
- *propellant and thermal constraints.*

A mission concept without these parameters cannot support a defensible thruster selection. If exact values are not yet available, use a documented range and conduct sensitivity analysis.

G.1.2.0.2.2 Step 2: Build a baseline mass budget *For each mission, separate the payload, spacecraft bus, power system, propulsion system, and propellant. Avoid double-counting the cruise stage or landing system.*

A useful initial accounting identity is

$$m_0 = m_{\text{payload}} + m_{\text{bus}} + m_{\text{power}} + m_{\text{thruster}} + m_{\text{PPU}} + m_{\text{tank}} + m_p + m_{\text{other}},$$

where m_{other} *represents any additional subsystems not listed separately. The definition of each term must be consistent across the comparison.*

For the rover case, the rover and landing system should be represented separately. For the clock missions, the clock, radio hardware, and any communications or navigation equipment should be included in the payload or bus budget.

The team should avoid assigning a highly precise value to a subsystem before the architecture is defined. Early mass budgets should use credible estimates and uncertainty ranges.

G.1.2.0.2.3 Step 3: Compute first-order propulsion requirements *For a chosen specific impulse and efficiency, estimate the thrust available at the selected electrical power:*

$$T \simeq \frac{2\eta_t P_{\text{in}}}{g_0 I_{\text{sp}}}.$$

The corresponding mass-flow rate is

$$\dot{m} = \frac{T}{g_0 I_{\mathrm{sp}}}.$$

The approximate propellant consumed during a period of continuous operation is

$$m_p \simeq \dot{m} t_{\mathrm{on}}.$$

This last expression is only a first-order estimate when the operating point is fixed. A trajectory with variable thrust, variable power, or changing specific impulse requires integration over time:

$$m_p = \int_{t_0}^{t_f} \dot{m}(t)\, dt.$$

The group should calculate several operating points rather than select a single arbitrary value. A useful trade study would vary specific impulse, power, and efficiency, then determine which combinations are compatible with the trajectory and lifetime requirements.

G.1.2.0.2.4 Step 4: Model the trajectory at an appropriate fidelity *A first-order impulsive transfer estimate can help establish whether a mission is plausible, but a Hall-thruster mission ultimately requires low-thrust trajectory analysis.*

The trajectory model should include the relevant gravitational environment, spacecraft mass variation, thrust magnitude, thrust direction, and mission boundary conditions.

For Mars orbit, the arrival condition should specify the target orbit or a defined orbit-insertion objective.

For L1, the target should be a specified bounded orbit or other explicitly defined operational trajectory near the Mars–Sun L1

region.

For a flyby, the model should specify encounter time, closest-approach distance, incoming and outgoing states, and any post-encounter objective.

For a rover, the transfer should deliver the defined lander and rover mass to the required Mars arrival state. The entry, descent, and landing system can remain outside the detailed thruster model if its mass and interface requirements are explicitly represented.

G.1.2.0.2.5 Step 5: Compare against a credible alternative *Each mission should be evaluated against a baseline that does not assume the proposed thruster is automatically the best option.*

Possible comparisons include

- *chemical transfer versus electric transfer;*
- *chemical orbit insertion versus electric orbit acquisition;*
- *electric propulsion with different power levels;*
- *different specific-impulse operating points;*
- *different mission durations;*
- *different payload masses;*
- *for the clock concepts, Mars orbit versus L1 using the same navigation-performance assumptions.*

The team should compare complete system outcomes, including power-system mass and transfer time, rather than compare only propellant mass or specific impulse.

G.1.2.0.2.6 Step 6: Tie the mission to the thruster design *Once the mission operating envelope is defined, it can inform the thruster's preliminary design.*

Relevant design variables include

- *discharge voltage and current;*
- *channel inner and outer radii;*
- *channel length;*
- *magnetic-field topology and field strength;*
- *propellant injection and distribution;*
- *cathode placement and coupling;*
- *thermal loads and material limits;*
- *expected operating duration and erosion exposure.*

These variables should not be selected independently of the required operating point. For example, the mission-level power and thrust target may constrain the discharge power, while the chosen propellant and desired specific impulse influence the voltage and mass-flow regime.

A simulation of the entire thruster should be treated as a separate validation hierarchy. A spacecraft trajectory model does not validate the plasma physics, and a plasma simulation does not establish mission feasibility by itself.

A defensible workflow is

mission requirements → trajectory → thruster operating point → thruster g

The chain must also be iterated in the reverse direction: if the thruster cannot deliver the assumed performance, the trajectory and mission mass budget must be updated.

G.1.2.0.3 Final Recommendation

G.1.2.0.3.1 If the priority is a clear mission justification *Choose the **atomic clock in Mars orbit** as the baseline concept, provided that the mission is defined as a navigation or timing demonstrator rather than merely as a clock placed near Mars.*

It has a direct operational context, an identifiable relationship to deep-space navigation research, and a straightforward way to define a bounded mission architecture. The team can compare conventional and electric-propulsion transfer options, then assess whether the resulting spacecraft can meet the required orbit and operational lifetime.

The principal unresolved question is whether a dedicated NEP spacecraft offers enough advantage over hosting the clock on another platform. That question should be addressed through the mass, power, and trajectory comparison.

G.1.2.0.3.2 If the priority is a more novel navigation study *Choose the **Mars–Sun L1 clock** only if the group is willing to investigate the distributed navigation system in detail.*

The proposal should identify the other nodes, the measurements made by the L1 spacecraft, the target bounded orbit, and the metric by which the new node improves system performance. The analysis should compare the L1 architecture against a Mars-orbit alternative using consistent clock and radio assumptions.

Without that analysis, L1 is a less defensible choice because the location itself does not establish a useful navigation service.

G.1.2.0.3.3 If the priority is a focused trajectory project *Choose the **Mars flyby** if the primary deliverable is a well-bounded low-thrust trajectory and encounter study.*

It avoids landing-system design and long-term Mars-orbit operations. It can support a clear comparison between propulsion architectures, but the team must show that the electric-propulsion option provides a meaningful benefit for the chosen encounter and post-encounter objective.

G.1.2.0.3.4 If the priority is an exploration mission *Choose the **Mars rover** if the group wants a multidisciplinary payload-delivery mission with an obvious scientific purpose.*

For a thruster-focused project, however, define the rover and landing system as an externally specified payload and interface. Do not allow full entry, descent, landing, and surface-system design to dominate the propulsion work unless those subjects are part of the intended scope.

G.1.2.0.3.5 Overall conclusion *For a group whose main deliverable is the design and computational evaluation of a Hall thruster, the recommended starting point is a Mars-orbit atomic-clock mission with a clearly stated navigation objective. It offers a useful balance between mission relevance, technical specificity, and manageable scope.*

The L1 clock is a stronger choice only if a distributed navigation analysis can demonstrate a measurable benefit from its location. The flyby is the cleanest alternative for a propulsion-and-trajectory-focused study. The rover is the strongest surface-exploration concept, but it carries the greatest risk of expanding the project beyond thruster design.

The final choice should be made only after a preliminary mass budget and trajectory comparison. In particular, the team should establish whether NEP improves payload fraction, transfer feasibility, or mission flexibility relative to a credible alternative. That result, rather than the apparent sophistication of the destination, should determine which mission becomes the

baseline.

Bibliography

[1] A. S. Ely et al., Advancing Navigation, Timing, and Science with the Deep Space Atomic Clock, NASA Technical Reports Server, 2016. https://ntrs.nasa.gov/citations/20160008206

[2] The Deep Space Atomic Clock: Ushering in a New Paradigm for Radio Navigation and Science, NASA Technical Reports Server. https://ntrs.nasa.gov/citations/20150007332

[3] NASA Science, Mars Relay Network. https://science.nasa.gov/mars/mars-relay-network/

[4] NASA Science, Basics of Space Flight: Trajectories. https://science.nasa.gov/learn/basics-of-space-flight/chapter4-1/

[5] NASA Technical Reports Server, A Martian Telecommunications Network: UHF Relay Support of the Mars Exploration Rovers by the Mars Global Surveyor, Mars Odyssey, and Mars Express Orbiters. https://ntrs.nasa.gov/search.jsp?R=20070023754

[6] NASA Technical Reports Server, Preliminary Investigation of Onboard Orbit Determination Using Deep Space Atomic Clock Based Radio Tracking. https://ntrs.nasa.gov/citations/20190025513

Outline of plan

1. *Find parameters*
2. *Baseline operating point(discharge voltage, flow rate, etc)*
3. *Find secondary performance via calculations*
4. *Design vague physical architecture*
5. *Optimize for size of channel and magnetic circut*
6. *Design propellant delivery*
7. *Simulate and further optimize*
8. *Integrate models*
9. *Write report*

Once we have the basic goal and such

- *Plasma physics and discharge*
- *Electromagnetic coil*
- *Propellant system*
- *Simulate and optimize*

G.2 Design

G.3 Optimization

G.4 Teachings

G.4.1 Numerics

Preliminary Workshop

I didn't use this for planning but I will probably write notes on how it went and next steps.

CFL/finite Difference

Didn't use this for planning but may not evaluation and next steps

****MHD***

G.4.2 Plasma Physics

First Go

- *What is plasma*
- *Lorentz Force*
- *Parameters*
- *Types of representation's*
- *MHD in more detail*
- *How plasma Produces Thrust*
- *Plasma in relation to our Thruster*

Appendix H

Story

H.1 Scope and Concept

H.2 Characters

H.2.1 POV Characters

H.2.1.1 Idea for POV characters

Each character is meant to be a subsection of who I am. Meant as an extreme version of who I might have been should I have gone for one of the jobs that I have thought of in the past. Somewhat simplified, taking things away, etc as meant to allow for conflicts, generally conflict that I myself have with myself.

H.2.1.2 Dr. John Jackson, 32

Occupation: Relativistic Theoretical Physicist at Princeton Institute of Advanced Study

Physical Presence: Plain looking, moderately tall, same plain

clothing,

Intellectual character: Thinks in structure, in a very abstract way, tendency towards the abstract and fascinating, does physics for a love of it, arrogant, extremely rigorous, first-principle thinking, deductive, philosophical, argues with himself, thinks all of the time through all things, enjoys other abstractions as well, and similar things.

Emotional: Condensed personality, feels little but the joy of discovery, forgets about people and general emotions, few close friends that he does more intellectual talk with than general.

Drive: To think

Voice: Precise, caught up in definitions, makes new words or redefines others, and direct

Arc: ??

Character Connections:

H.2.1.3 Dr. Marcus Venn, 46

Occupation: Plasma Computational Physicist at Princeton's Plasma Physics Laboratory

Physical Pretense: Homely, stereotypical professor look

Intellectual: Mathematical modeling, objective, empericalist, careful, and middle ground between each of them

Emotional: Has a loving family, cares extremely deeply for this family, large laugh, funny, makes himself the butt of jokes, and close mentor to graduate students

Drives: Create something meaningful

Voice: Conversational, analogies, loves to talk about physics and explain it, precise, and charasmatic

Arc: ??

Character connections:

H.2.1.4 Commander John Cram

Occupation: Astronaut, leader with a study of stars

Physical Presence: Large, muscular, commanding presence

Intellectual: Systems, contingencies, thinks quickly, pragmatically, procedural, patterns

Emotional: Stoic, rigid, leader as an arctype, separate, designed, understands people in an abstracted way, works a lot, few to any real friends

Drives: Mission

Voice: stoic, operational, economical, loud, authoritative

Arc: ??

H.2.1.5 Secretary

H.3 Plot

Appendix I

Random Small stuff

I.1 Singularity Formulation in 3D Euler and Nonuniqueness in unforced 3D Navier-Stokes

Part IV

Essays

Appendix J

Introduction

This part is used as a collection of essays on whatever topic. Whether it be philsophy, physics, economic, whatever.

Appendix K

Post-Surgery

K.1 Overview

K.2 1/9/2026 Essay: Cultivation in Degregation

As mentioned in previous chapters, the development of virtue stands at the cornerstone of the existence of man. Thus, how to develop it is key.

The question of how to develop virtue, what virtues to develop, etc; differ from person to person. I will focus on myself for this essay, on a practical development, rather than in the abstract general way I would traditionally go for.

Specifically, I will focus on my recent 'degregation' in many fields. What I mean is I have been struggling with health problems that caused constant pain, nausea, and brain fog. To combate this I used "quick" and "automatic" gratification. Watched lots and lots, of Instagram videos and YouTube shorts. With quick and simple comedy TV in the background. While it is difficult to divide what is caused by using these simple causes of

gratification and what is caused by my still recovering body, but I am having trouble focusing, get tired easily, less intrinsically motivated, and other similar malignities.

I have a belief that a person can create themselves based upon their habits. Thus, that is what I design to do, create myself in the image of my highest values. I desire that I am someone who enjoys challege for it own sake(especially intellectual challenge), persistent high energy, optimistic outlook, pro-social tendencies, high focus, and the continual push towards greater virtue in all fields of life.

To reach these goals I have decided to pick up "The Karamazov Brothers," a book I had abandoned earlier due to my declining condition. I did this not only for enjoyment, but to accomplish my goal. To slowly focus my mind on something that lasts longer than a minute, to remind myself of my enjoyment of intellectual type activities and engage my mind in something that could challenge it, and to reconnect myself with my faith.

I am also reading "Numerical Hamiltonian Problems," trying to learn to code a website, code personal pet projects, watch educational and long-form youtube videos rather than shorts, exercise each day, and write each day. I do all of these things to re-develop my attention, enjoyment of effort, regain my energy, and to develop my skills.

Developing myself on the social side of things is a bit more challenging. Currently as of writing this essay I am still isolated from my peers, but I have a plan. Teaching Sunday school, volunteering for church outreach groups, returning to my scout troop, participating in JROTC volunteer activities, teaching karate, and so much more. Things to redevelop my enjoyment and understanding of social responsibilities and natural altruism.

With this current plan in motion, I will start making purposeful and willful actions in order to develop in the way that I design.

K.3 3/../2026: New Beginnings and Failures

Appendix L

Metaphysics

The verbal interpretation, on the other hand, i.e. the metaphysics of quantum physics, is on far less solid ground. In fact, in more than forty years physicists have not been able to provide a clear metaphysical model. — Erwin Schrodinger

L.1 Scientific Materialism and God

Despite the fact that there exist no formal term for such philosophies(I should come up with a name for it, Christian Scientific Materialism?)it in a broad stroke is most likely the most common view of meta-physics of all, though I plan to expand upon the concepts more deeply than that which is common. In essence this view can be summarized by the combination of two seemingly contradictory concepts:

First, is scientific materialism, this is the idea that there is nothing further than the material, that reality can be derived empirically through the laws of science and nature.

Second, the idea of the Christian God. That there exist an external omnibenevolent, omnificent, omnipotence, omnipresence,

and omniscience as described by the Christian Bible. Later I will explain my faith in more detail, but for now this is sufficient.

You may be wondering how these concepts can be combined due to their seemingly contradictory nature. This is a simple enough question, if the material is all that exist how can a non-material God exist? The answer in all reality is very complex, but it can be simplified to the fact that I view the material world to follow the philosophy of scientific materialism but there to also be a non-material "world" that God exists within. The reality of such a non-material world is sadly beyond my grasps due to my life within the material, I make no assertion that I understand such things. Partly because by definition such things are not possible to understand, like a 4th spacial dimension. While we can understand it through analogies and abstraction based upon prior more concrete idea, but sense we have no clear or objective priors we have no capacity to do such.
You can simplify such thoughts through formal logic

Let:
X=Material
Y=Non-Material
C=Laws of nature and reality
C(X)=Matter/Energy/Both interactions(all material action)
C(X,Y)= Observable phenomena A=Scientifically explainable phenomena

$$X \Longrightarrow C$$

$$C(X) \Longleftrightarrow C(X) \models A(X) \top$$

$$\forall C(X,Y) \exists C$$

$$\nexists Y(C) \therefore \nexists A(Y)$$

L.2 Scientific Realism vs Anti-realism

What is scientific realism and anti-realism? It is the debate between whether our theories of science are ontologically real, or simple predictive models.

My view is that it depends, when it comes to macroscopic and easily observable and prove able; these theories aren't true. You may be thinking the opposite, that it is those that are most definitively true and the rest that aren't. Though I would argue these are the models; a ball don't fall; a collection of subatomic particles follow their geodistic, which is restricted by other particles and forces, and so much more. This can be applied to all things other than the most fundamental of physics. It can especially true for the softer sciences; chemist, biology, psychology, and so many more.

While I think science does study the metaphysical truth through fundamental physics and abstract mathematics, on a broad stroke it does not.

L.3 Causality

Now I am sure you are confused, I mentioned I would dive into deeper questions related to free will, determinism, what is real, divine voluntarism, occasionalism, how quantum mechanics plays within this and other 'deeper' concepts, but we must take things one at a time.

So, let us dive in. What is causality, it is very simple. It is the concept that things happen because other things had happened first. That all things have a 'cause' and that that thus causes the effect(then they later become the cause of a later thing)

I can hear you screeching that this is non-sense, who in their right mind would question something as basic as cause and ef-

fect? Why even look into it. First, that is extremely intellectually lazy, all things have debates against it, you should always look into it. Though, more importantly, there actually are debates against it.

The easiest to dismiss is Hume's skepticism, that causality is simply a human made concept. That things doesn't really happen because of another action but simply that both actions are just sequences of events. Though, I would argue that this pedantic argument of definitions doesn't truly work. On an epistemological basis, causality is easier for humans to understand, and though breaking things into chunks we may better understand the world. That either concept is equally true, but one concept as a greater pragmatic 'truth' and you can argue that that correlates to a better empirical truth. That there is even a speed of causality written in the nature of reality.(The speed of light) A bit more deeply, once you quantize to the plank time, things truly do divide. Something happens, then a next thing happens. The complexity and abnormal reality of quantum mechanics makes this argument a little fuzzy, and thus leads me to my next paragraph.

Now those unfamiliar with quantum mechanics may be confused, how does quantum mechanics may be a little confused. I will attempt to explain it to the best of my ability, but the reality of quantum mechanics is extremely complex and without a clear understanding of differential equations, discrete mathematics, linear algebra, and much more it is impossible to understand. I know many science communicators claim to teach "quantum physics and mechanics" to they layman, but this is a lie. These half backed analogies are not quantum mechanics. It does not come close, but luckily for me I am the only one reading this book I will be able to understand my attempt of analogizing(?) the philosophical question.

In essence, in quantum mechanics. Things are not local, for many reasons. The most basic is the idea that particles do not

exist in the way we can imagine. They exist in multiple places at the same time[not really but useful analogy], simply with different probabilities(don't pretend to understand, nobody does, they just prove the math and accept as is.) You know what, that analogy was not enough. I will explain in more detail. Particles do not exist(not really). Simply quantum fields exist. The electron field for electrons, the photon for photons, and so on. The particles are not particles as we imagine, but excitations within the field. It is a bit more complex than that, they can act as particles when observed but that is a whole other can of worms. Now back to the matter at hand, now waves act in certain ways that is hard to explain. Due to their quantum nature they must be decomposed first through Fourier modes to under stand more clearly. This is an example of a very simple form:

$$\psi(x) = \frac{1}{2\pi} \int \phi(p)^{ipx} dp$$

Now I won't explain the equation in its entirety but I will say this. That you cannot solve it for an absolute certainty of position(not that you could get a "true" answer even if you could given that is is a Taylor Mode and not an analytical equation). Now you may think this is just a quirk of the math, but it is not. It is real for reason that could take up the entirety of this book if I truly understood them.

Next, quantum mechanics goes against causality through entanglement. Now what do I mean, when people refer to the speed of light, they are actually referring to the speed of causality. To explain this is relatively simple with a good understanding of Special Relativity. It is obvious that the universe is invariant to position and velocity(this has been previously proven thoroughly through experiments in high speeds, specifically through electromagnetism but others can apply), but for this to be true there must be a finite constant cosmic speed(because it requires a Lorenz transformation[there is more explained why this si the

only answer but I will leave this hear because I have much much more physics to talk about in a philosophy book.]) This is the speed of causality, the speed of information. If something were to go above this speed it would in essence go back in time. This can be proven rather simply through simple algebra:

$$t' = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}(t - \frac{vx}{c^2})$$

From here you can look at the square root and easily find that if $v > c$ then γ(the value of the first part with the $\frac{1}{\sqrt{1-\frac{v^2}{C^2}}}$ would be imaginary. Thus the t value(of time would then become negative, this obviously breaks causality.

Finally, back to quantum mechanics, as you know entanglement transfers information at faster than the speed. This seems to break causality.

Now I could keep on talking about arguments between quantum physics and causality, but I will only talk about one more. Virtual particles. Now first of all, what are virtual particles. Virtual particles are non-existent but existent particles that exist within the math of quantum field theory(and thus above my pay grade) but I will attempt to explain it. The idea in essence is that due to the uncertainty principle there exist small amounts of excitements within their fields. These excitements are not enough for an actual particle to exist in the way we imagine, but occasionally a particle can appear, primary through the use of "borrowing" energy from another particle or through creating a negative particle(not to be confused with anti-particles those are negatively charged, these have negative energy/mass) These virtual particles are especially important to particle interactions, what I mean is the fact that when two particles are "close" enough together they excite the fields around them to create bosons(force carriers) to cause the actual interaction. As an analogy, imagine

two electrons, when close enough together they excite the fields around them to create a virtual photon, the photon then interacts with one electron(and also effects the other electron negatively[it is almost impossible to explain]) and cause the electromagnetic repulsion between the two. Now, back to philosophy, what in the world does this ramble have to do with causality. That is a great question. It is that all of our interactions are based upon something that isn't real. Electrons don't actually create virtual photons, these photons don't create negative photons. Despite having a tangible effect, they don't actually exist. How can something cause another thing if when you break it down to its component parts, particles are behaving 'irrationally' being effected by mathematical objects.

Now after I have listed the critiques, you are probably very aware about how you assumption of the obvious state of causality was intellectually lazy, but you probably also realize I wouldn't have written all of this unless I have an answer to these critiques, or at the very least a possible answer. In all reality I could be wrong about this and causality could be false. Though I am never wrong, so that must not be true.

The uncertainty of particles doesn't necessarily falsify causality, it simply evolves it. We previously see causality as something that is caused by a specific thing, but it can be caused by a probability of things(if that make sense).

Next I will talk about virtual particles, and yes I hear you saying that I should go in the order of which I introduced the ideas, but I do not care. Now, virtual particles. How does causality work with things that are not 'real'? Now the most astute among you might claim, why would causality not apply, and you are exactly right! You all may feel a sense of betrayal, that it was simply a trick of the light. But now, listen carefully, this is how philosophy works. Though asking questions that are hard to understand, questions that play tricks on our language, on our mind. Though these questions we refine our thoughts. This

is the problem with modern philosophy, they confuse critiques with criticism. While criticism has a place within philosophy, it should not be as big as it is now. Now before I continue I will define why I consider a critique and a criticism as different things. A criticism takes a part an idea, concept, action, or really anything for the sake of taking it down. This can be justified if such ideas have no worth and the criticism is an attempt to prove it. A critique is a question used to refine a thought not destroy it.(though if the critique can not be resolved then it may.) This critique of virtual particles is not supposed destroy causality by any means, but like the earlier critiques, it is meant to refine our thoughts. It asks us to question what a cause is, what can be a cause. Can virtual particles be causes, modern science suggests yes, but this requires a refinement of our concept of causality. Of what is 'real.' In a later section I will delve into this in a future section, but now I will leave you to ponder.

Now finally, to entanglement. Does this refine our definition or break it? It breaks it, but due to the evidence surrounding causality, including the fact that the special and general relativity are simply logical derivatives of the idea of a finite special to causality suggests that our idea of entanglement is flawed not our idea of causality. Now what is the answer? In short, nobody knows. There are many theories trying to come up with a solution. These include Einstein-rosen bridge, super-determinism, and local hidden variables theory. Though since these ideas are both extremely complex and offer little beyond the problem at hand I have elected to move on. Now you may say that my Faustian arrogance has lead me astray to ask questions I can't answer, and to that I say... No dummy.

Now you may think we are done, we are not. We haven't even mentioned the nature of God in causality and the teleological nature of Hamiltonian mechanics. There is also further refinements of time which must be discussed to even further define causality.

L.4 Divine Law Theory of Causation

You thought I was done with with causality. You thought wrong.

Now, what is the Divine Law Theory of Causation. It is very simple, it is that God predefines the laws of physics and nature. This differs from the ideas of occasionalism. Occasionalism is that all actions are the conscious decision of God, that the uniformity of the laws of nature is simply a coincidence.

Now, why do I choice Divine Law Theory of Causation? While I do not attempt to venture into the idea that I can even begin to comprehend the machinations of the God most high. Though in my mind the idea of predetermined laws make more sense, a more efficient method for God. Though this is not my primary reason for this belief. As I will mention latter in this chapter I will explain several beliefs of my predicated upon divine law theory of causation. Also, it is more in-line with scientific materialism

L.5 Material, Immaterial, and Reality itself

Now we are getting to the meat of things. What makes something real? What makes something itself? How do we define such things?

These are all great questions that I will go through one by one.

I will begin with what makes something real. While it is easy to look at things in the materialistic view, that of which that can be observed and have an observable effect is "real." No this has some implications, what about what about God(this is based on my assumption that he is 'real') but more importantly what about things that have an effect but of the concept of virtual particles.

First, the immaterial. Given the fact that by definition, these things are beyond us I will ignore them because we cannot gain any intuition from it. Just remember that it must be included within the concept of real, simply we are unable to define it accurately.
M= Material
I=Immaterial
$\mathbb{R} = Real$

$$M \wedge I \subset \mathbb{R}$$

Finally, back to virtual particles, I have kept you in suspense enough. How do we satisfy the existence of virtual particles? Just as we have several times before, say it with me, we must refine the definition. Now, how to do this. We must first realize that at the subatomic level, no particles are particles in the way we describe it, as I mentioned earlier they are simply excitations within a field. They are energy in a specific manner. rom here we can divide the material into two types; true material and partial material. The true material refers to what we generally consider matter, and the the partial(photons, gluons, and other massless but still interacting objects.) Which leads me to my final definition; material is that with observable effect[observable does not just simply mean seen, it can be any form of observation].

L.6 Identity

While many have abstract ideas on the identity of object, I will simple say that all subatomic particles are identical and thus it is not hard to define, and the others are simple as humans(another word defined by humans) define it. Good day gentlemen

L.7 SpaceTime

Space and time, the two things most clearly and least clearly understood at the same time. Things that we know do to our interactions with it every second(ha-ha) of our lives but still don't truly understand.

Many philosophers have questioned the reality of space and time. Are they simply human made? I would suggest not, the mathematical empiricalism and importance of them in higher level physics points to their real nature.

You can see this clearly through special relativity, there is an implied "proper" time at which things like electromagnetism operate correctly and without it our modern view of physics collapses.

Now you may ask about quantum mechanics, where such ideas of concrete space and time become fuzzy. Where particles exist within multiple place, some theories suggests 'time travel"(not real time travel) and other. Though, as I have mentioned before, I think his more evolves our ideas matter than our ideas of space and time. Also, there are examples of the objectiveness of space and time through things like quantum field theory(combines special relativity) and the Plank constants. You can't have constants of space and time without an objective reality of space and time. Now, what is the Plank length and time. They are the units of which distance and time become quantized, and some suggest the smallest units possible. Now, that seems like an odd idea, especially considering how much smaller it is that anything real and the random constants within it $c, G, \hbar$*. Speed of light(or causality), gravitational constant, and reduced Plank constant. This is because as you get smaller and smaller there is a new uncertainty, a 'plank' uncertainty. Beyond this the uncertainty becomes absolute. That the realities of a new quantum gravity take hold The easiest way to look at this is*

through finding at what point the wave-length of light(our primary method of observation) becomes a black hole.

L.8 Interpretation of Quantum Mechanics

The long awaited further explanation of my views on quantum mechanics. Throughout this chapter, I have hinted at these ideas of interpretations of quantum mechanics. Also, throughout this I have primary relied on the Copenhagen interpretation of quantum mechanics. Which may lead several into seeing this as my view of quantum mechanics, it is not(I will elaborate further later.) Though let us first go through the different interpretations of quantum mechanics.

The Copenhagen interpretation. In essence the Copenhagen, the most popular view among scientist, is that quantum physics is impossible to understand through our classical views of metaphysics. That our theories of quantum mechanics works, and to ponder further is a waste of time given its impossibility. It takes our ideas of wave functions and entanglement as reality, not because it is real reality, but because the model works. That at the end of the day physics is not the study of reality in the way metaphysics is, but the study of models of the universe.

Though this explanation isn't really fair, so I will expand upon it. The key concepts are things like wave-particle duality, that particles are both particles and waves and that things like observation can effect how they interact with itself(yes particles interact with themselves) and with other things.[Though observation is never truly defined] Quantum superposition, something we all know fairly well and requires no further explanation. That quantum physics is probabilistic by nature, and truly probabilistic not governed by hidden laws berried beneath. There is a lot more to

it but given that I have already explain part of it previously and how popular it is, you can fill in the gaps on your own. While this theory has been extremely important for our modern views, and important to explaining how quantum mechanics works without esoteric 'nonsense' it has a couple flaws like observer dependence and the fact that it quite literately states that it is not the ontological truth but rather a model.

Next, the many worlds theory. One of the popular interpretations by the general public. Basically, it is that when probabilities collapse; whether it is an interaction, wave-particle duality, superposition, or many of the several other things; the universe splits into two(or however many are used to describe the 'whatever'). That one universe has one and the other has the other. This is quite outlandish idea; for its reliance on the observer and the fact that there is no mechanism of 'universe splitting.' One thing I would like to add, is that I have always hated how people use the idea of a multiverse in this context, it is multiple verses it is one with multiple observable components. A better term, generated by Sir Roger Penrose and a Latin Professor that he was friends with(which I think is absolutely amazing, he made up a new word because he didn't like the word being used) he now calls is an omnium(meaing of all)

An even more outlandish theory is many-minds... which is exaclty what it sounds like... yeah

The Pilot wave interpretation is that there are hidden variables; that just as things like coin flips, brownain motion, plasma instabilities, and so much more seem random from the outside, if you really break them down they aren't. Now what are these hidden variables. Now what is this hidden variable, you might ask. Well lets take a step back and explain a bit deeper. First, lets define a quantum wave function, this is different to the excitations of quantum fields mentioned earlier. The wave-function is a complex variable(I mean complex as in in the complex field not challenging to understand) that depends on the particles po-

sition and can be used for several things. It can de described:

$$i\hbar\frac{\partial\psi(r,t)}{\partial t} = (-\frac{\hbar^2}{2m}\nabla^2 + V(r))\psi(r,t)$$

where $\hbar$ is obviously the reduced Plank constant. r is the position vector. ∇^2 is the Laplacian operator. m is mass. v is the potential energy. t is time.

Once you have the wave function you can do several things with it, square it for probability distribution, collapse the wave function, etc. But I am getting away from myself, you want philosophy so I will get back on that.

Basically, pilot wave theory suggest that this wave function influences this particle more than we think, and thus can lead to the particle's behaviors being deterministic through equations like this for velocity, v:

$$v = \frac{j}{\psi}$$

Now, in all reality the equation is obviously bigger; must expand j and ψ, but I didn't want to write it out(also there are expanded versions for relativistic and whatnot.) though back to what we had in mind, what did we have in mind, oh yeah superdeterminism. Basically, quantum mechanics is deterministically not probabilistic.

One final thing I will say about pilot wave theory, is that some people suggest that the many worlds interpretation and pilot wave theory are one in the same just describing from different points.

Next, we have environmental decoherence theory. Sadly, I know very little about this one at the moment, I will attempt to learn more to better understand it better(for its own sake and given that it could convince me) But it is basically that like how a coin isn't truly random, neither is the quantum world. That there are density matrices made by interactions that effect fu-

ture actions leading to a theoretically deterministic world.At the moment I don't really understand the mechanisms so I can't prove, disprove, critique, or criticize.

Lastly, Objective collapse theory. The objective collapse theory is similar to Copenhagen, with it's more literal view of quantum mechanics. Though, it has two differences. It has a true literal and ontological belief that goes further into everything within it. Also, it does not rely upon the observer. It relies upon 'random' collapses of wave-functions, such as gravity, scholastic field, or the interactions of other particles.

This is all well and good, and I do truly suggest you try to figure it out yourself(even though you don't exist), but this is a book MY philosophy. So what do I believe you might ask. I find the objective collapse theory to be best with the pilot function being a close second. Now why do I believe this?

First, for reasons I will explain later; the idea of non-determinism is important to other theories of mine and thus I need it for them to function. Though I am sure you all don't think that is enough, so I will go further. Second, in all reality given we cannot empirically prove any of this it doesn't really matter. Three, given that it doesn't really matter, objective collapse theory makes the most intuitive and philosophical sense to me intuitively. Four, you know what, I give up(for know) these justifications are kind of weak, even in my eyes. I plan to do more research and come up with further justifications. Right now it is basically which sounds right.

L.9 *Other Questions in the MetaPhysics of Quantum Mechanics

...(will update later)

L.10 Quantum Consciousness, Arminianism, Super-Gödel thinking, OR Orch, why I even decided to write this book, and how long can I make the title for a section within a chapter

As much as I love how hilarious the title is; with its length, random buzz word, and joke on the end. I also kind of hate it; quantum consciousness seems like some esoteric nonsense. It feels like I am admitting to believing in a specific theory just to justify my religious beliefs, but this chapter will attempt to prove that that isn't the case. That it is rigorous, while maybe not on a true empirical scale, at least on a formal logic one.

While there are many reasons for the creation of this book, the main ones is to evolve this very idea. It was that I had this idea at one point and it challenged a lot of previous thoughts, thus I wanted to evaluate my philosophical thoughts from the ground up, then apply that.(While it was the original reason, I plan to continue this book much further and continue it.

Let me first explain what my theory is in broad strokes. It is the belief(more so a belief than a full theory given its unprovable nature, unscientific background, and basis on the belief in an omni-potent God unconstrained by the confines of physics, formal logic, or even abstract-mathematics.) It is that our consciousness and free will is preserved by God through the use of quantum mechanics. This theory in it-of-itself in not groundbreaking, original, or even scientific in nature, but I intend to go about explaining some possible mechanisms of this(though sadly, by definition this theory will be unprovable, which I will explain why in a second.)

L.10 Quantum Consciousness, Arminianism, Super-Gödel thinking, QR Orch, why I even decided to write this book, and how long can I make the title for a section within a chapter

First, is there any justification for this idea of quantum consciousness? There is some, it is weak but existent nonetheless. It is related to Gödel's incompleteness theorem. Gödel incompleteness theorem has two part. First, "For any consistent formal system F that is capable of expressing elementary arithmetic, there exists a statement G in the language of F such that if F is consistent, then G is true, but G is not provable within F." The second is, "For any consistent formal system F that is capable of expressing elementary arithmetic, the consistency of F cannot be proven within F." This claims that formal logic has its own limits; you cannot prove an axiom with itself and you cannot prove a derivative with an axiom alone, you must prove such axioms with other axioms(because you cannot prove an axiom with itself.) Now what does this have to do with free will, well to have free will you must first have some form of consciousness, something outside of our ideas of formal logic. We have some proof of this, the fact that we are able to derive pure math at all seems to prove that we are not algorithmic based, something beyond. Now there are a lot of arguments against this, namely that our ability to do mathematics comes from our informal thinking rather than beyond-formal logic thinking. In all reality, I like this explanation better, but I place Penrose's argument of consciousness here as a sample of somewhat scientific rigure rather than simply religious thought.

You may now be asking how this super-Gödel thinking proves free will. It doesn't, but it implies as deeper idea of consciousness allows us to theoretically side-step determinism in order to 'allow' free will. Now, before we get into how God can protect free will, and why I think he would, let us first explain some possible mechanics of explaining this seemingly unexplainable phenomena. Though before that, let me say the fact that by definition it goes beyond Gödel's theorems, we by definition can't prove it so take all of this with a grain of salt. It is an idea, not a law, not even a true theory, and idea.

Let us first go through the most popular explanation, the reason I even came up with this idea. Sir Roger Penrose's OR Orch(I seem to be mentioning him a lot) Essentially it is that microtubals, a potion of the brains neurons, and theoretically a part of consciousness, are highly subjective to quantum 'strangeness' in a way that most other complex structures aren't. That the interactions of quantum mechanics can somehow safe guard conciseness and allow super-Gödel thinks and even possibly free will.

While this idea is specifically for objective wave collapse theory, it also works for things like environmental decoherence and kind of work for many-worlds, but not for Copenhagen and pilot-wave.

For Copenhagen, no extra theories are required, by definition the idea of consciousness in 'beyond science' and thus no extra loop-holes are needed.(they don't specify what an observer is so therefore other explanations are possible, but they hold no further ground than this one.)

Now how does quantum 'brains' lead to free will. It doesn't, not in the most literal meaning, but it does give us extra wiggle room. While most modern views see our view of determinism as falsifying free will, but these theories give us room to say that there are other possible answers; dualism, God's choice, beyond material, and others.

I would say that God, by some manner, protects free-will. Whether it through some hidden ideas of quantum mechanics or it is that our consciousness is beyond the material and effects the material through the quantum world. I don't know, as I mentioned earlier, I am not even confident it is correct. It is an idea.

Now you may or may not be asking, why would God use quantum mechanics to protect consciousness, if by definition he is beyond formal logic and thus wouldn't need such things.

While I cannot comprehend the machinations of God, I would suggest that a possible explanation would be to create our logically, self-fulfilling world that we are able to comprehend(for the most part) and live in a true free will.

Now to end, I want to reiterate how theoretical and disconnected from regular science this is, most of my ideas are at least respected by a decent portion of the scientific community, this is not. It is an idea, nothing more. This is an exercise to see if it holds weight, it currently does not seem to.

L.11 Compatibilist Free Will

Even though I wrote a whole lot about quantum libertarian free will. I don't necessarily believe in that. I view the world as either semi or super-deterministically and that that doesn't actually effect free will. Though I still wanted to examine the idea in more detail nonetheless.

L.12 *Super-determinism, semi-determinism, Retro-causality, and Chaos theory

L.13 Role of God

I have already gone over the role of God in many ways; creator of the universe, its laws of nature, protecting free will, but now I wil go further.

The first question is the relation of miracles, in the Christian faith, we believe that our God is a personal God that creates miracles for us. Now how does he do that if he predetermined the laws of nature?

Well that is fairly simple, God is omnipresent, meaning he not only knows what you are going to pray for before he created the universe, but he also knows how to create such miracles from the creation of the universe. Thus he 'created' the miracles when he created the universe itself; with knowledge of all that will ever transpire.

How does the divine foreshadow effect our idea of free will. Well, God is beyond our free will, he is beyond whatever quantum mechanisms protect our free will, so he can know without interfering with our free will.

Is something good because God says it is, or because it is objectively. It is because it God says it is, beyond God there is not mechanism for the creation of truly objective morals, there are logical models based upon subjective values, but they cannot stand upon themselves. God makes them right because he is beyond formal logic.

L.14 Post-Axiomatic Nature of God

I guess pre, would be the better prefix given what I am about to say, but Post-axiomatic sounds better. Now back to the actual concept.

This idea is simple. That our axioms, as later examined are derived by God rather than God being subservient to the nature of these axiomatic principles, "A line is the shortest distance between two points," "Existence exist," "consciousness in its totality," "A is A," and many more. He, God, derives them. Now this twists our mind given that axioms by definition cannot be imagined without their existence. It is like inventing a new dimension for us to see, it doesn't exist.

This concept enforces many ideas already present; God omni-(something that cannot exist with current axioms), God being

beyond comprehension, that God is self defining(see non-infinite definition of words and axiomatic semantics), and being outside of causality, outside of formal logic. He creates existence(something that makes no sense in our current axiomatic understanding of 'existence'), he is three in one(against A is A), and much more.

While they don't 'solve' these questions in the traditional sense, they open the door in a more esoteric sense. While by definition we can never understand these concepts intuitively they do give rise to much of our understanding of God.

On to our esoteric identification(I know must of what I write is intended to be rational and examined this is one of the exceptions.) This idea of pre/post-axiomatic identification leads to our idea of faith. God, isn't just beyond us, he is ineffable and unprovable though our axiomatic thought processes. Beyond that, this further goes into my idea of the "non-material." We can further define this as that of which is based upon our known axioms and that of which is not.

L.15 Metaphysics of cosmology

While I could go on and on about physics of things like the big bang, entropy, arrow of time, anthropic principle and how they could theoretically effect our views of metaphysics. I don't need to, my belief in God nullifies such arguments. That all possible incites have already been satisfied by God and thus have little need for exploration.

L.16 Ontology of Information and Entropy

Once again, we must first define what our concepts are rigorously.

A classical perspective. Within this concept information is the physical quantities that en essence give you traditional information. That allow you to decode past events and even future ones. Entropy is the inverse of this affect. Geometric instabilities and 'randomness' that hide 'information' from observers.

From a quantum perspective; quantum information even includes superpositions, wavefunctions, density matrices, and more. The quantum bits, qubits, can exist and superpositions and exist within the same quantum state(position)

In this view, entropy then becomes a byproduct of entanglement, a process that 'hides' the information from view.

There is much more on the semantics of this. Going into Geometric identities, black hole information paradox, and quantum thermodynamics. Though, for now I will move on to the more metaphysical parts.

One common debate is if information is truly fundamental or simply a human made connection. That the universe has a 'computational structure' where information is the substrate and entropy the measure of the complexity. With modern understanding of the geometric identities and the fundamentalness of information, the theory of fundamentalist seems to take the lead.

Next is the arrow of time. The second law of thermodynamics/entropy states that in isolation, states drive towards entropy. This necessitates an arrow of time. It breaks the symmetry of time.

Now many question whether or not entropy is a by-product of

the arrow of time or the other way around.

The way I see it, the direction of entropy is the consequence of the already time symmetry breaking existence of dark energy, black holes, and other similar objects in this preview. That the direction towards entropy precludes low-entropy existancess.

Though this is just one theory. Holographic principle, Gravitational Entropy, Quantum Mechanics and Unitary Evolution, and more. In-fact the quantum mechanics and unitary evolution makes more sense and less problems.

The symmetric theory ignores that some forms of entropy are consequences of statistical physics that doesn't deny time symmetry

L.17 Dualism

I have already mentioned this in interpretations of quantum mechanics, but in physics many possible mathematical(and thus metaphysical) interpretation exist. Different ways to look at the world. They can't both be true, can they?

L.18 The Problem of the one-way speed of light

Quantum Mechanics is not the only physical model with inherit uncertainty. Special relativity has a similar principle. Essentially, the problem is that it is impossible to find the one way speed of light. Basically, while we can use a clock to find the two way speed of light, to find the one way would require two synchronize clocks. To synchronize clocks along a distance, the speed of light is a required number. Thus, the speed of light could theoretically be different in different directions. In one

way it could be 0.75c and the other could be 1.25c or 0.5c and ∞. There is no possible way to actually prove it. Not only does this theory not prove it, but by definition changing the directional speed of light would change the transformations, effectively making everything equal. Just like changing the coordinate system(there are some versions where it does change things, it will get to that)

Now why does it matter if the observables stay the same? There are several reasons I will get into.

One is the question of scientific realism and instrumentalism. A kind of epistemic humility about how one of our singular most verified theories is simply convention. Not ontological truth, simply convention. That there could be hidden mechanisms causing these changes and speed and we could have no way of knowing.

Beyond that, the different theories based upon this fact offer wildly different views upon the ontological nature of space and time. Einsteinian, space and time are relative and this relativity causes the physical views we see. Lorentzian, there is simultaneity and objective time. Relationalism, time is a network of relations and the speed of light is a consequence of that relation not the opposite. Block, everything happens at the same because simultaneity is frame dependent and so is the speed of light. Presentism, there is an objective present, but the nature of light diseludes human observation by its nature. There are many other theories beyond this, but I think it envelopes the idea.

Lastly, there are some theories that while equivalent in most situations(situations that we have empirical data for) but is different. Different in not just ontological but physical differences. Differences that could lead to a unified field theory.

L.18.1 *Phase-space and the Euclidean Mind

L.19 *Ontology of Numbers

L.20 Reality and the Abstract

When reading this chapter, one can easily get a esoteric/gnostic sense of reality. Of pure abstraction with little regards to reality. Though this is not the case. In fact I find the abstract "non-material/pre-axiomatic" world as inconsequential. The only true consequence being God, but he matter because he influences the material. The focus on this abstraction being the fact that the ideas of reality are largely explored and are easily understood and so large amounts of writing and exploration on the topic is relativity inconsequential.

The focus on abstract, structuralism, post-axiomatic is simply to satisfy my need for cognition(as later explored) rather than being core parts of who I am. These theoretical ideas that I am not even sure if they are right or not(post-axiomatic God, Consciousness, etc) are explored not because they are core to who I am and what I believe(maybe the God one) but because it is fun.

L.21 Dialects' Critique of Relativity

I have recently been watching a YouTuber Dialect, he is quite intelligent and well spoken, but one thing that quite surprised me was the fact that he, and fellow researchers, had an intense disdain of Einstein's Relativity. While I think his extreme disdain and inflammatory writing it untoward, the essential idea is worth exploring. While there is much more that he goes on, but I will focus on two things that he mentions; critiquing physical positivism and Platonism in modern physics.

Let us examine positism first. Positivism, essentially says that science should be relegated to only observable phenomena. That physics does not and should not concern itself with ontology and metaphysics beyond what is strictly observable. Things like the one-way speed of light or the nature of simultaneity are strictly convention and should not be taken seriously. This is seen in a much harsher way in the field of interpretations of quantum mechanics. The idea of positism is obviously closely linked with instrumentalism as mentioned in other sections.

Opponents argue that while we should have epistemic humility and understand that from a view of instrumentalism this is as far as science can realistically go, we should nonetheless continue to find ontological arguments that posit real underlying entities. That science should not renegade itself to measurement bookkeeping and remain at its heart natural philosophy that looks into explanatory mechanisms. Looking for something semi-physical like ether that it a medium for a field rather than the nature of the field itself. That viewing gravity as a nature of how we measure things within spacetime is overly simplistic and lacks realism.

Another common argument being as mentioned that relying on the nature of time and simultaneity as simply convention as a fact given the mathematical nature of time is similarly quick to judge. As mentioned, other theories like modern ether, matrix, etc. explain these things in a much more natural physicallist way rather that give further causal explanation than Relativity does.

For the first argument, I do agree in the abstract. While I recognize that as physics becomes more and more advanced, and further and further from Galilean intuition, connecting abstract physics with ontology is hard if not impossible. Especially given how even these most basic ideas are open problems in the field. Though, modern physics has become too divorced, in many cases ignoring physicallism entirely. In the case of Rela-

tivity, I would not say this is the case. While some of the more mechanistic theories like ether can technically be considered to be "more mechanicstic," it isn't truly. Not in a truly paradigm shifting way. Relativity does offer a true mechanism, the curvature of SpaceTime. A more intuitive medium that comfines better with the quantum fields would, in my mind, be better fit, but that is far from saying that their are true misgiving in the nature of Relativity that must be rectified. It is just as likely that quantum field theories nature is what is misplaced.

As for the second point, I once again agree in the abstract. I do not like relying on such a fundamental nature to be at best convention and at worst be convention; though, we cannot rely on what we cannot prove. To take one of the other equivalent theories as fundamental would once again be convention. Thus, the problem is not of Einstienian Relativity but of the basic uncertainty principles of nature.

One final note, it that in the instance of many sets of Ether theories, it can be taken to be that Relativity and Ether are equivalent and to simply have relativity be a descriptive theory and ether as an explanatory theory. While I do like the deeper explanatory power of ether theory, given its unprovable nature it is simply a deeper more intuitive explanation of the same thing but with no backing. One can argue that the Einstienian view has no deeper backing and thus no better. Thus choosing one above the other is simply from an aesthetic choice, mathematical elegance or clear ontology. In this case, there is no answer. From a personal stance, while I generally find clearer ontology to be superior to mathematical elegance, given that the majority of the physics community is on the opposite side, it is more pragmatic to build upon Einstenian Relativity to avoid confusion and to have the tools set forth by those that have come before. For instance, a general Relativity equivalent Ether theory is still an open problem with it being quite underdeveloped, far from the clearly developed theory that has set up itself in dozens of

different formalism's, metrics, and tools like numerical solvers.

While in this writing I have also addressed much of the arguments related to Platoism, I will address that here as well.

In many interpretations of Relativity, namely Minkowski, the mathematical four-dimensional geometric object known as SpaceTime is considered to be real. Critics claim that it is not testable and thus not scientific and that this over emphasis on mathematical elegance is foolhardy. That to take a literalist approach to the consequences of mathematical symmetry and reduction is going too far.

This has quite some merit, for instance to take a literalist view of Hamiltonian reduction of fluid dynamics would give quite a different view. The way scientist avoided this was the fact that reduction through inductive observation came first, this is obviously not possible for the nature of SpaceTime. Thus, having trying to look for deeper mechanism, like how particle kinetic theory or molecular theory does, is likely a worthwhile investigation. Two problems arise though. Once, given the nature of SpaceTime, modern science sees no way to study it in such a way that it is possible to go beyond simply mathematical interpretation. The other being that mathematical exploration is argueably a better way to explore the nature of reality. In other chapters(namely the next one) I explore a more mathematically abstract view of ontology than traditional newtonian view.

L.22 Structure as Fundamental

Appendix M

Epistemology

"Man is neither infallible nor omniscient; if he were, a discipline such as epistemology—the theory of knowledge—would not be necessary nor possible: his knowledge would be automatic, unquestionable and total." — Ayn Rand

M.1 Limited Objectivism

Before I begin upon explaining this, I must first specify the influence of Ayn Rand is limited to Epistemology. It has not influenced my politics nor ethics. Now lets begin with the philosophy.

In a short simplified sense, objectivistic Epistemology is the inverse of transcendental idealism(though despite what many say, not the opposite). Both philosophies take both reason thought and empirical data as useful tools in the accumulation of knowledge.

Though they differ in importance and direction. Transcendental idealism takes reason as the most fundamental and such

abstract thought can then be applied to the outside world through empericalism. Objectivism takes the inverse, that we create our abstract models based upon our interactions with the world. Now these may seem like small differences, but the expand themselves through their logical conclusions. I could explore how they differ in more detail, but I will focus on objectivism.

Now, where did the limited part come from. As I have mentioned before, I am a religious person. I believe there are something beyond our sense and possible our mortal comprehension, therefore we are limited by such things when acquiring knowledge. I do believe what we are capable of discovering is 'objective' and just as metaphysically true as what we can't, but that does not take away the fact that some abstractions are beyond us.

M.2 Consciousness and Reality

A common continental question, is the question upon whether our consciousness truly models objective reality. While these questions can be useful when directed in a purposeful and specific way; though most people don't direct it in a useful way, but rather in a simply abstract way with little refinement of knowledge than the ability to trick others through strange and useless question.

Now, why do I think this is a useless question? Well, for several reasons I will address categorically. Now I will prefis this by saying this is a criticism, not a critique, I may later may later add a more refining critique but for now it is an acknowledgment of the stupidity of this, and to show that all derived concepts have no true analytical framework.

The most basic argument against this is the pragmatic thesis. In essence, it doesn't matter that reality is consistent with the observations of the mind, the mere fact that our observa-

tions closely associate enough for practical purposes is reality enough. There is no purer form of reality than this functions.

Though, this isn't enough for me, I think it could be enough if it was all there was but I want more. I do believe in truly objective reality and not just our semi-subjective observations of reality.

A refinement of the pragmatic argument to include the existence of an objective world would be a question of statistics. If a further world with separate physical laws existed it wouldn't work with our vast and extreme observations. Though, this wouldn't work with some more extreme theories wouldn't work with more radical and complete ideas, like Recreant's dream theory. Though, this implies that whatever objective reality is it must not be capable of interacting and effecting our reality. This can be attributed to nothing that could interact with our world, or our world is independent of the interactions of reality by its own definition. This may seem like a minor and obvious reduction but nevertheless it such reductions are the essence of philosophical discussion.

Next, such dream theories are useless. Derive no further discussion, only exist as a way to trick people up and thus serve no purpose. For these, the original pragmatic argument is all that can be derived secularly.

Now, you may say this is a good way to question our scientific premises. An argument against scientific realism, but in reality it doesn't change anything. Scientific realism stands or doesn't regardless of the 'validity' of such theories.

My final secular argument is that such philosophical arguments are harmful. That philosophy should not start with doubt but rather with perception. Such perception must be doubted, but doubt existing in its own self-containerized munition is useless. It defines nothing and leads to strange and possibly dangerous ideas. For instance Desecrate lead to Spinoza who lead

to Hegel, who lead to Italian idealism, who lead to Giovanni Gentile the philosophical founder of fascism. Now, you may criticize that this was one complex branch, you could also derive similar lines through Aristotle or just any philosopher far back enough, and this is a good argument. Though, there is a clear line, also to other dangerous ideas like communism, NAZIsm, and so much more. These things are predicated on these strange cartesian/continental questions that derive no true thought, and thus should be understood in this way. Simply as pathetic and dirty tricks and shouldn't be see as true paths to knowledge. While yes, we should still engage sometimes, truly placing it on equal footing(or higher as we have been) is dangerous for our collective understanding reality and derived sense based upon it.

Though, many of you may have noticed I specify secular. There is a religious argument as well. If God created the reality as reality, then it is by definition reality. The only argument otherwise is that God doesn't exist(either entirely or he is not God)

M.3 Nature and Validation of Axioms

First, like always, we must define axioms thoroughly. The dictionary definition is "a statement accepted as true as the basis for argument or inference." For instance, our understanding of scientific realism is the understanding of the fact that our observations of reality are accurate(there are some observations that can be morphed due to other things that are observable.)

A simplified and more observable definition can then be defined to say that an axiom is a statement that either cannot be proved or cannot be proved currently that is used within a greater statement that can be proved upon its edifice. These axioms, while don't require 'proof' generally desire some type of

reasoning even if it is not objective or purely analytical in nature. For instance, the above arguments for the objective nature of reality aren't true objective proof, they still exist as semi-arguments. While we cannot prove the objectivity of reality with true and objective reasoning, some reasoning can be applied to get some kind of 'proof' of the statement so that further studies can be applied based upon that said axiom.

M.4 Non-infinite Definition of words and Axiomatic Semantics

In the field of axioms, there becomes a problem. Part of the reason that 'some' words cannot be further defined. They are defined by themselves, they can only be intuitively understood, and even there is a limit.

A great example is existence, because to define this thesis, you need an antithesis. That doesn't exist(non-existence by definition needs to exist to be something and thus cannot exist because it would need to exist to be in existence.), so therefor existence is defined by itself.

This brings up a great deal of problems, paradoxes, and much more that will be explored.

M.5 Induction & Deduction

In the expansion of human knowledge there are two main ways in which they can be reached; induction and deduction.

Deduction is by taking premises/axioms and derive conclusions based upon this. Induction is the opposite, from observing 'effects,' define whatever knowledge is desired.

This can be seen in physics, deduction is taking a funda-

mental theory and applying it in a new way to find novel uses, predict, or whatever may be needed. Induction would be taking data to come up with a way to model reality or come up with an entirely new theory.

Both of these are extremely important. Without induction we could never go beyond basic logic, but without deduction we can never find objective truth. The way I have found to be the most useful is to use induction to attempt to find our 'axioms' and then use these to deduct the reality around us, though this is only really works for my fields of interest. In other fields, like psychology this strategy would fail.

M.6 The Problem of Induction

Nothing can be proven true. As controversial of an idea as this may seem, it isn't. This basic axiom is the foundation of modern scientific epistemology. That theories can only be proven wrong, never correct. That is why they will always remain theories.

Now, you might think something as foundational as something like Newton is true. Though, there are questions through MOND(Modified Newtonian Mechanics) and others. Given, Newtons empirical evidence, there would need to be a lot of evidence for MOND for it to be taken anywhere near the same level

Appendix N

Ethics

"Finally, brothers and sisters, whatever is true, whatever is noble, whatever is right, whatever is pure, whatever is lovely, whatever is admirable—if anything is excellent or praiseworthy—think about such things." — Philippians 4:8

N.1 Introduction to ethics

Many that know me, know the question I love to ask, "what is your moral philosophy?" I love this question for many reasons. First, it intrigues people, they love to talk about their own morals and beliefs, but engages their curiosity with these newish words. Second, Morality is something that is both simple and fundamental enough that everyone has thoughts of it but still complex enough that it can have engaging discourse. Three, it get to explain stuff, most people haven't heard of virtue ethics, deontology, egoism, etc.

Despite this fundamentalness to ethics, many questions are still unanswered, which fundamental theory? Many questions are raised for each theory that must be addressed? Even people question if Morality is even objective or not?

N.2 Virtue Ethics

My fundamental theory is virtue ethics

Virtue ethics is based upon the idea that Morality is derived from living a virtuous life rather than the adherence to rules or a logical stance. That you must pick a set of virtues and live by those principles.

Now why do I choice this over the other types of theories of morality and ethics. To understand this I will first criticize(yes criticize not critique) other theories then I will defend virtue ethics.

First, deontology. Deontology comes from the Greek word deon meaning duty. That you have a duty to follow certain 'rules.' That morality is based upon a set of rules and concepts rules and concepts to follow. Say, don't kill, don't steal, give to others.

My problem with this is not that it is too ridged, but that it is impossible. It would be impossible. You can't make rules for all possible interactions so must theories of deontology tend to just be contorted virtue ethics. Also, that both the action and consequences should be included when making an ethical decision but generally rules can lose this complexity.

Next, egoism. Yeah no

OK, fine, I'll do a little more. Most theories of egoism fall into either three camps; nonsense hedonism(which needs no criticism), logical consequentiality based concepts, or altruistic egoism. First, the consequentialism type eventually leads to normal consequentialism but with more steps to find the conclusion so I will focus on that when I get to it.

Next, consequencialism. Frist, for consequencialism to work a set of values must first be prescribed(so it isn't fundamental) there are others like utilitarianism but they aren't self-justifying nor truly justified by other means. There are some that exist

but I will show that to analyses all in-depth is a waste of time. My reasoning is that both actions and consequences should be analyzed. Why, good question... Well, first there is religion obviously. Though a more secular reasoning is that ethics is used as a binding agent, people emotionally don't bind well to 'immoral' actions, and those that do tend not to bode well with other things.

Now for justification of virtue ethics. The first and most important is that I am a Christian, from my analysis most Christian ethics is highly tided in virtue ethics. The fruits are virtues. Most sins are vices. Most ethical advice is based upon abstract virtues rather than a simple value(consequentialism) or rule(deontology). While there are many rules, most of them are based upon the virtues later mentioned rather than justified by themselves.

Next, on a more secular level, virtue ethics are far more complete and level for people to analyze, create, and apply to your life. They can included anything you want to have in them without creating paradoxes and problems(as long at the two concepts are not are paradoxical in it of themselves)

I will show this through the future sections that go through a list of virtues and why I feel they should represent my ideas.

One final thing I will talk about, is something that may be a strength or weakness depending on how you look at it. Virtue ethics has a much more 'subjective' connentation. A lot of virtues that I value another may not, also these virtues can be analyzed in your own way.

The way I see it, some virtues are fundamental and should be applied to all given they are principles derived by God, others I feel as personal virtues and feel no sense that they should be forcefully applied to others(though I feel things would be better for all that way.)

In essence, the idea of virtue ethics is to pursue perfection of the mind, body, moral thought, etc in order to create a purity of

intention. This is the goal, not absence of sin, but the perfection of moral intentions. To create yourself in the image of your own ideal and to pursue it. That morality isn't just a responsibility but ontological alignment. To pursue the ideal of Jesus.

N.3 Stewardship

Throughout the rest of this chapter, many times I reflect on morality as my values. Values that are observed because of some 'internal promise.' This is because it is the way I have naturally thought, though I am trying to evolve past this.

I have always seen the internal promise as more pure than any other form of moral enforcement, though I have found this not to be true in the highest extent.

This obsession with dignity and following through with my own values alone is cornered in the vice of pride. It sits upon it, being thus controlled by it, and by proxy then so am I.

Now I have found, not responsibility, but existence in stewardship. In stewardship to the will of God, his grace, his mercy, his creation, and most importantly each other.

You see, God brought Adam and Eve into the world, as images of himself: builders, molders, and beings capable of reason. A creation to tend the garden, name and care for the animals, and to care for each other. This was the reason for humanities existence.

After the fall of mankind, humanities purpose shifted, but not much. We are still arbiters of his will and creation, by authority he has given us. Though, because of the imperfections made by us, our responsibility is to fix them. To perfect the world around us through the words that God has authored for us.

For God himself has given us this divine commission; though he hasn't left us to do it alone. He gives us gifts, hope, and

healing. He feeds us and nurtures us so we made be sanctified in his love for us.

Now in practice, this becomes our purpose of moral perfection. Both for ourselves and for those around us. To create, not only pure intentions for ourselves but for the society and people around us. I will discuss this more throughout this book, especially this chapter, but here I will leave with this.

Though our actions, we may be able to co-create with God to bring Heaven on Earth. Bring about social holiness caused by our own actions with others. This is the purpose of Christianity, of ourselves. Only secondary to a relationship with God.

N.4 Categorical Good

Based upon my religious beliefs I propose this axiom, "that all of reality is designed in such a way in that "good" can be logically inferred and the actions to find this good is what we call virtue."

$$\forall x(R(x) \rightarrow \exists P(I(P, G(x))) \wedge \forall A(V(A) \Longleftrightarrow \exists y(L(A, y) \wedge G(y))))$$

In essence, there exists a good for all categories(ex. communication, get information delivered clearly). Thus, there are good ways to do this, virtue(ex. use precise language when needed but don't use overtly complex language when not). Finally, not only can iterations of this build up an idea for how to "be a good person" based upon a complex model of different circumstances throughout life, but that what virtues exist reveal what makes a person "good" by what God has consciously divined, giving the actor "wisdom". This is all centered around the Stewardship given to us by God as mentioned in the previous section.

Obviously, this isn't a purely religious view. Aristotle came through with an idea very similar idea, but even his idea had a pantheistic view, and other secular versions generally make

it an unquestionable axiom without further backing other than the fact it creates a non-paradoxical system. Also, this system allows further development, like founding it in stewardship or having all things reveal "wisdom" to further develop virtue.

N.5 Cardinal Virtues

Now, while my cardinal virtues have basis in the traditional cardinal virtues of the catholic faith they are not the same. The name comes from the fact that cardinal comes from the Latin word 'cardo' which means hinge. In essence all other virtues hinge on these concepts.

Here is a short list of the main ones

- *Temperance: Excess in most matters proves detrimental; self-control is paramount in one's life.*
- *Prudence: Deliberation is essential before action in all circumstances.*
- *Fortitude: Mastery over one's emotions is a paramount virtue, as the inability to do so inflicts harm upon oneself and others.*
- *Faith: Maintain strong belief not only in God but also in oneself and one's values and beliefs; these should remain immutable unless confronted with supreme evidence.*
- *Duty: One has obligations to oneself, family, others, and one's values, which should drive one's life. Duty to oneself includes ambition and adherence to personal values; duty to family involves support and politeness; duty to others is similar but to a lesser extent.*
- *Individualism: Embrace self-reliance, self-respect, and ambition, and adhering to personal values.*

As you can see all of these virtues are simple and widely acknowledged and don't require much additional justification. The only true exception being individualism....

N.6 *Salient Virtues

N.7 The Philotimic Virtue

Now Philotimic is a word I have made up, I will go into more detail about what exactly it means later, for now assume it means pride.

Now, I hear your vapid and lost mind screaming, "isn't pride a vice, what are you doing" Though I will examine why I think pride in some forms is a virtue not a vice(at least certain types of pride)

First to examine we must first define pride and what it is. One definition is ,"feeling of deep pleasure or satisfaction derived from one's own achievements, the achievements of those with whom one is closely associated, or from qualities or possessions that are widely admired." or "consciousness of one's own dignity." Now what does this mean and how do they fit together.

Let us start with the first one, a deep feeling of pleasure in ones own achievements. This can manifest itself in many ways; work, morality, intellectual achievement, and so many others. Beyond that it can effect people's actions in both positive and negative ways. Let's start with negative, it can lead to obsession with ones own capacity and the product of such. On the flip side, a healthy obsession will lead to personal growth on whatever they feel pride in. Also, it can give them confidence in their own capacity. An example would be with morality, if they feel pride in their own sense of morality it can lead to self-assurance in their own senses leading to them being able to apply and do

them, but it can lead to close-mindedness and looking at just your own thoughts and nothing more.

Now lets look at the second definition, consciousness of one's own dignity. Meaning that they are aware and influenced by concepts of their own dignity. The belief that somethings are within their ability and others are beneath them.

Now what I mean by philotimic, as essentially the positive aspects of this. That to have philotimic virtue is to take your own life seriously. To be principled, to be absolute. Not just in behaving moral in a moral and dignified manner, though this is definitely a part of it. In fact, even within this manner this virtue is partly about fanatical pursuing such moral perfection. Though, outside of this there are many things. Most notably is having such fanatic belief in things beyond morals, other values like being early, being prepared, working hard, being mature, and being other similar concepts. That while there are values that God doesn't arbitrate, but you do, and thus you should keep them to almost the same esteem.

This concept, as mentioned earlier is about taking your own dignity, self-esteem, self-respect, virtue, and above all life seriously. Because that is what you should do, take it seriously in a dignified and ridged manner. To never compromise upon even the most minimalist value, because to compromise on such things is far worse than any other conceivable interaction.

Another concept within it is control. Control over your innermind. Your thoughts, emotions, temperament, and even personality being bent to the will of your own 'ego.' Not ego in the regular sense, but as in the inner consciousness of your own self. The self of morality, values, reason, and more(more closely related to the idea of 'superego')

In essence, recursively creating your own self based your higher values rather than letting your environment impact your supposed values.

Philotimia in its highest sense, is not merely pride, but the love of what is worthy of love. To find what is worthy of such esteem and pursue it with fanaticism and create your soul around such values. To take responsibility for your own existence, morality, self-esteem, and so much more.

An even further simplification is it is taking upon yourself the responsibility to be human. The responsibility to think, to fear God, to live accordance with your values, to have values, and to live in the image of your highest self. Not only that but bend the world into the image of your highest self.

One thing I will add, is in the conflict of universal morality(derived by God) and personal values. This conflict doesn't really exist. Both of these values exist in my mind. While universal morality is obviously stronger, this doesn't dissipate the importance of personal values. The largest differentiator is others; for universal morality, others disobeying. is an infringement of morality, while for personal values only you 'break them,' no one else.

N.8 *The Internal Promise and Dignity

N.9 Good Life and Eternal Struggle

There are two parts to this. Obviously the good life and the eternal struggle as defined in the title. This are both extremely connected, as I will show momentary.

First, the good life. To live a virtuous life is to live a good life, but there is challenge. One, you should train yourself to desire this good and enjoy it. as defined in the philotimic virtue, take this training upon yourself rather than simply passively let

culture and external figures train for you. While yes, culture can do a good job, training yourself creates a better feedback loop and consequence; beyond that, training yourself is your own moral responsibility as a human being.

Find this good to be your highest desire, so that you not only can but will naturally follow it and ignore sin and vice around you. To desire it fanatically. While many consider fanaticism to be a bad thing, in the pursuit of virtue, it isn't just a good thing but a lack of it is evil.

Beyond that, this struggle will be enteral. Virtue by its very nature is never perfected. Therefore, you should always struggle. To take this further, you should love the struggle against it.

N.10 *Personal Virtues, Values, and God's Morality

N.11 Intentions

In my theory of morality, intentions matter just as much as the action in it of itself. "The LORD does not see as man does. For man sees the outward appearance, but the LORD sees the heart." 1 Samuel 16:7.

I could go on and on, but this idea is relatively intuitive and while many don't consciously think this way it isn't far from our minds.

N.12 *Morality and Religion

N.13 *The Ethics of Knowledge

N.14 Recursive Connection of Civilization & Virtue

Our capacity towards civilization and virtue are strictly linked in a depth rarely acknowledged.

Let's begin with the start, for civilization to happen, we must be able to do it well. This is the heart of virtue, while virtue can be connected with any doing any good thing well, the ability to be civilized and do civilization "well" is the very heart of it.

Now, how do we do civilization well? To make sure people do not become distrustful, honesty should be brought about. To bring about any positive outcome, discipline becomes a need. To expect pro-social activities, empathy is not only an ideal but a requirement. While I won't explore every angle, it is clear where this leads. Our modern ideas of culture and morality is primary based upon how to build a civilization, a society.

Though, how to develop virtue on a mass scale? Society. Through community virtue is developed, the most obvious being how we enforce guilt and pride into those that exhibit virtues and vices. To teach them young, how ethical theory works.

A bit deeper, through habits. Virtues are all about who we are, our attributes, our character, presence, intellect, and much more. We become who we are through our habits; habits of action, ideas, thoughts, and much more. Society creates these habits, even when we don't realize this. To explore this idea, I will present a case study: a commonly criticized culture rule is to not curse. It may seem illogical, they are words, sometimes

not even directed at the person offended. Now, before I begin on this analysis, I will say that this particular analysis will focus on social rules as habit forming implies virtue forming. It will ignore the dozens of other reasons and arguments.

From this idea of it being illogical, you can clearly see the reason by what the person is actually doing. They are tempering their own impulses for the sake of social harmony and empathy towards others. Having interpersonal tact and learning from moral authority and teachers without rebelling because you didn't see the big picture. These habits are both skill and virtue building so we may be civilized adults. This is also why it is so extremely emphasized during childhood.

Now, while these parts are extremely important, they aren't simply circular self-defining concepts. Virtues exist beyond civilization and their primary justification being to mimic Christ and fulfill our duty to steward his creation.

N.14.1 *Social Rules and Morality

N.14.2 *Culture as Moral Education

N.14.3 Intensity of Duty*

While on a strictly moral view, ethics and their virtues are extremely important. I have gone over again and again, how these ideas should be pursued fanatically, but individually. While they should be taught to others, expecting purely moral actions from other is not something to be expected from them.

Virtues of civilization are another story. Some lenience is required, but to much leads to a denegation, a slippery slope. Things like duty, respect, loyalty,

N.15 Morality in the End of the World

Appendix O

Personal Philosophy

"Only through the radical examination of one's own life and the reality surrounding may one gain authority over it. Through such authority their own happiness becomes clear. To have the freedom to be responsible for themselves and the integrity of their own mind. This is the principle and highest form of human action, to divine their own morality and values then shape themselves and reality into the image of their highest values."
— AJ Cason

O.1 *God as Foundation

O.2 Will as Identity

All throughout history, the question of identify has been constant. Who we are: are we are personality, outward actions, inward thoughts, natural impulses, cognitive style?

I propose a different method, we are our will. That is our method of identity. This proposition has many consequences.

Before I get to that, I must define your will. Your will is your

highest self within you, that which pushes you to your highest ideals. The image that burns within you of what you could become, that whispers how you may achieve it. That is your will, your will to your own power over yourself.

First, it allows acting through purpose to be acting authentically. Many people find paradox between our need to live 'authentically'(a moronic ideal) and fulfilling their purpose and duty to others and themselves. Though, from my view these are the same. To act and bend your personality to your personal will, to fulfill your duty, isn't unauthentic, but rather the purest form of authenticity.

To 'act' and author yourself is your responsibility as a human, a creation of God the most high that has been made in his image as a being with a conscious mind. Don't live passively letting your whims and childhood define you, but rather your will is your definition.

Second, a devotion to your will is your responsibility. To 'falter' is to kill yourself and let a husk(personality, impulses, etc take the place of the human mind that God has put within you. To let the flesh control you is dishonorable, evil, disgusting, and never to be considered. It is not understandable, it is not acceptable, never surrender, never compromise against anything beyond your will.

This can be observed through giving up, compromising, complaining, etc. These are evils. People seem to accept complaining as a natural reaction to 'unfairness' and that you are nothing more than your "childhood traumas," don't become this subhuman husk. Become more, become a thinking human.

Four, to let others to your thinking for you is to outsource your soul. To degenerate God's creation. Think for yourself, think logically and analytically, derive your 'first principles' clearly.

O.3 *Fanatical Ethics

O.4 The Duty as an Actor

As mentioned in "Will as Identity" you must in a sense become an "actor" in your life. What I mean by this is that you must control your reactions(be civilized and dignified) to fulfill your duty.

Be cheerful within the misk of irksome tasks and weighty responsibilities, hold your head up high, never complain, don't profane what is sacred, and so much more.

One, through "acting" you will become that in which you pursue, virtuous. You will become happy, become moral, become dignified, become who your will decides you must be. It is through this "act" you will become that image of your highest self within you.

O.5 *The Aesthetic Pursuit of Truth

Appendix P

Personal-Analysis

"The unexamined life is not worth living" — Socrates

P.1 Introduction

I will use this journaling of sorts to examine myself and my own personality.

P.2 Disgrace and Pride

(if anyone else reads this, don't pay too much mind. I use words differently than their actual meaning. I don't word things very well. I was simply trying to capture my own indescribable and esoteric and possibly failable thoughts in this moment on this topic, my actual feelings are very different than what can be interpreted through the word choice.)(Additional update: upon later analysis, I have found that these ideas of pride and disgrace are for the most part the idea of seeing my earlier mentioned "personal values" in a light very similar to traditional morality. This doesn't fully explain it and I will continue to work

on refining these ideas beyond the esoteric concentration currently presented.)

Why is it that you are so accustomed to the use of disgrace in your speech and why does it effect you in such a manner. The idea of 'disgracing' yourself is so vital to your worldview, it effects everything; morals, interactions with other, and so much more. It isn't even like you are all that effected by the thoughts of others. In fact the most abnormal and extreme feelings of self-disgust and disgrace are related to a refection of the thoughts of other. Why is it that I feel like to see other, value their opinions, and let them influence men to disgrace myself. It is an odd thing, yet it is so very natural. It influences everything, and it is so much more extreme when I am in isolation as I am now. What is it about the concept of being influenced by others to be so repulsive, so disgusting that I won't just let it influence my actions but I will try to persuade others my way is best with pride. With this eternal pride that locks me in, says that I am always all of the way in all things. There are so many things to think about in this discussion so I will try to take it piece by piece and hopefully add some things.(also, what it with the strange switching from dialogue with an external, internal, and this explanation style?)"

Well, disgust is a natural motivating force. It motivates far better than most, it is consuming. It involves fear, pride, righteous anger, annoyance, and so many more extreme and strong emotions. It motives far more than most, and it comes so easily, especially for someone like me. There is no greater fear than falling in ones own eyes. My disgust of others is just who I am, it is what makes me who I am. It disconnects me from the petty emotions around me. I mean, can people really say that those are better; to be insecure, jealous, vapid, empty, etc? Is it really better? Take insecurity, it is such a strong emotion that consumes people so very easily, making them bicker and fight, making them claw at each other with their broken paws that do

very little besides hurt themselves and hurt others by their own volition of nonsensical blight. My 'pride' can rise me above that. If I do not see them as anything, then for what reason do I have to be insecure about this vapid collection of dust. Now I am not truly some raging narcissist who sees no one beyond and object for my own use. I see people as they are, I care for others, value them. I just simply do not value them in the way others prescribe them. I don't see them as threats, competition, just broken monkeys in need of assistance(and that I am the same way and can occasionally use assistance myself, just less than most.) I still see them, I make friends just as easy, if not easier than most due to this.

This disgust is a strengthening endeavor that assistance me in many ways. My self-disgust pushes me further. It is what lead my to teach myself calculus in elementary, and has brought me here know with graduate level knowledge, research, and so much more. It has disconnected my from the vapid desires of others in their looks and whatever other nonsense.(though I will add, this disgust is not some obscene emotional self-distrain but rather a more abstract understanding that doesn't truly make me feel bad, just concous of 'wrong doing.' More so like a passion against, I feel no negative emotions about myself)

Though I must acknowledge that this does not make me better, it makes me better within my own eyes, my own values, not others. My values have no greater truth than that they are my own. So my disgust should always remain abstract, looking down at the ideas not the people. I have done great at this, never prescribing habits or concepts to people. Though as of late I have had trouble with this, seeing people beyond the moment. Being conscious of their deniers and thoughts, it can create some discontent. I should remain as I was, seeing people only in the instant. Nothing more. I am the only conscious being within the confines of my own mind, for it is better this way. You cannot accuse a rock of moral failure, only a man. For which I am the

only man that I see, the only man that I know, I am to constituent for the moral blame. The disgust should be within me and me alone seeing only me. For I am the only one held to my values and the only one who could suffer the breaking of them. For they are my values alone, that is the purpose of the disconnect. Why my values must be mine alone, for I am the only one to be connected to them, the only man in my eyes.

For this I also hold the burden of thought and reason. The thought behind my morals, my values, my beliefs, my religion, my reality. I see the world as my land for the conquest of my own knowledge to build up it all. That is the whole purpose of this book, to fully disconnect, to find myself fully and fully alone. Because, in the end that is all that matters in the abstract. For sure, I love my family, my friends, and my fellow man, but I love myself in the way that one can only love oneself, the expectation of my own measurement, the pride in my achievements, the disgust in my faults, the understanding of my beliefs. For one cannot love others without loving oneself, for love is derived from ones understanding of their own values, who they are in all of reality.

Now why are these feelings and thoughts all the more present in isolation.For when with others, those are my real feelings. In isolation I attempt to derive and find, deriving and finding the strange and unwieldy emotions of the mind does not come truly and with the same accuracy of that of physics. It comes in strange botches of thought that don't mean what they literally mean but can be described through thoughtful examination of the words, other words, and the actions of the person.

When did I become this, so thoughtful behind it all. Seeing the world beyond the material, seeing my thoughts beyond the exact. Maybe I really am chaining, in a way that must lead to the changing of even my most basic assumptions.(this last paragraph is really stupid)

P.3 On Ayn Rand

It would be obvious to admit that this section would be an analysis and critique of Ayn Rand's ideas, but like the more erudite among you will notice that this would go against the structure of this chapter. This is exactly correct, this will rather be an analysis about my propensity towards Ayn Rand's ideas.

While this may seem useless and without purpose, it isn't. It has been a strange psychological question about my enjoyment of Ayn Rand's novels even though I starkly disagree with actual philosophy and the fact that her books lack the many of the general characteristics that generally allow me to enjoy such novels. So what is it?

There are several reasons, though I will start with the most fundamental. Simply put, I see myself within the characters. Most particularly Howard Roark. Many of the descriptions of his won emotions and others description of me mirror is a strange way. While not absolutely similar, it builds off in a way more closely connected than any other fictional character. How he exists as an independent entity, not noticing others but living by their moral code not out of other means but as its own mean. Because of integrity above all. The way he is an act of moral striving rather than a disgusting abstraction for those even more disgusting to connect to. He lacks those pathetic neurotic tendencies that those around me let control and give authority over them.

Beyond that, he gives me ways to articulate my own personal feelings in a way that I have never seen before. Being "to proud to boast" by seeing both the criticism and complements of the world to be equally insignificant because it does not come from myself. To find pride, value, truth, within myself not within the pathetic world beneath me. Seeing another person see how pathetic the social validation games are, not is disdain for others

but as seeing it beneath myself.

The way happiness is the his natural order rather than some far flung ideal that is beyond, that negative emotions seem dulled by their pathetic attempt. That he is truly happy at all times, just as I am.

The way compromise, even in the slightest way seems to be evil. That this concept has been a driving force in my life thoroughly. Because it is a self-betrayal, a betrayal that can't even be thought about. An evil beyond belief and idea. Even a white lie, a broken ideal without real backing, a principle made as a child, and so much more seem evil and I can't figure out how others live with it. Though these compromises are never entertained long enough to feel anything beyond the knowledge of it.

The way others see him as cold and arrogant despite him clearly not being, due to there misunderstanding of him. Because he is independent, because he doesn't care about friends and what insignificant interactions they had, what their friend's did to annoy them. That they don't care about philosophy, physics, mathematics.

How he is both happy in isolation and with others. Equally, because the existence of others doesn't have that effect on him.

The way his creative and logical thoughts of Philosophy, physics, mathematics, and other intellectual topics are all that matter, well beyond the existence so often people confine themselves to.

Finally, he lacks all neurotic tendencies. No desire for complements, praise, people to soften their words, bend down. These neurotic tendencies are found everywhere, in everything. People, fictional characters, and others. Though I feel such thoughts so rarely. It is a impeded idea in almost all of fictional by its own virtue, but I never see it within my own mind. He like me is truly beyond these petty neuroticism, no capacity for them at all. No vulnerabilities, no anxieties, no insecurities, no of it.

P.4 On the Pursuit of Thought

NFC, or Need for Cognition is a psychological concept seen clearly in this book. Though it many not be obvious the the extreme extent it is true.

Much of my ideas of philosophy, ethics, physics, and more may seem like a desire to know reality in its greatest extent(and this is certainly true), but in its most basic and primate way, it is my need to think.

I love thinking, it is my favorite thing. This is what drive me, my desire to satisfy that part of me. I think about anything complex enough; philosophy, ethics, physics, mathematic, coding, economics, political philosophy, literary analysis, world building, international relations theory, geopolitics, formal analysis, psychology, meta-cognition, and so much more. The more thinking required the better.

In fact, I love it the most when it is beyond me. When it takes me weeks to not even fully understand what questions to ask, when it feels beyond my comprehension, when I think for days and go no-where. I love it, the scavenger to knowledge, then to actually get it. For it to all fall into place just as it should.

Now very little does this, in fact most things just come. They are understood intuitively. Seem to basic to even consider. Even some of my other hobbies like psychology, politics, and such seem to basic, and most of the other things beyond my intellectual hobbies seem so basic as to not even give it time at all.

Another funny consequence, is despite my almost compulsion to efficiency, I still am drawn to complexity. Despite my normally physicalistic and literal tendencies I am also drawn to the abstract. I have found this most easily seen in my obsession with using higher order math. I use tensor calculus, hamiltonain mechanics, geometric identities, when similar methods can do it. I obsess over these ideas when there are more practical matters.

P.5 Nietzsche's Sovereign Man and Morality

Upon further reflection, my chapter on disgrace and pride has much connection with Nietzsche's "sovereign man." Here I will first explain what that is, its connection, and divergence.

Before I begin, one thing with Nietzsche's writing is that it is up to controversial interpretation. Some say the sovereign man is sincere, other ironic, others meta-ironic, others see it simply as a literary device, and there are still other interpretations out there. Luckily though, for the purposes of this exercise it doesn't matter Nietzsche's intentions, only my intentions and ideas. For instance, Nietzsche's reasoning for presenting the sovereign individual differ from mine; which I will present later.

Now to actually begin with the analysis. Who is the sovereign individual? The sovereign individual is the archetype of Nietzsche's will to power(the Ubermensch later takes its place as a further extreme but I care little for this idea.) It is an implied ideal in which a person creates mastery over their own life. To live their life in accordance with their values, that they themselves had created rather than inherited. They do this with their mastery over their own impulses; a mastery so extreme that they eventually shape their impulses into however they consciously wish. Further more, these values and 'morals' are created though anesthetic ideas, creativity, future-bound, and affirmation to the power of life rather than traditional fear. Finally another key part in is surprisingly forgiveness, though not in the traditional sense. Rather than forgiveness caused by God or some other values it is a combination of the acknowledgment of the fact that most people are incapable of true moral thought, self-mastery, and to let go of their resentments and impulses; another key idea is simply the fact that not forgiving hurts the sovereign individual, Nietzsche's suggests that it is better to sim-

ply forget about the trespasses rather than hold on to resentment and call it responsibility to forgive when it is hidden resentment. Essentially the sovereign individual is the archetype of the promise made flesh(or word)

Now how does this relate? Well before I get toe pride and disgrace, lets explore the philotimic virtue. The philotimic virtue highly mirrors the concept of creating your own values, setting your life and soul in pursuit of them, and finally holding yourself to them in a way that can be seen as fanatic to the outside. Also, creating these values individual of the existence of others(though the philotimic virtue includes the will and thought of God to yet be higher than the will to power of myself.) To live your life as you will it so. T

Now disgrace, the idea that my personal values that I have willed hold the same good and evil. That to betray them is evil. But both ideas have a similar them, by not hiding from it but rather carrying the burden is guilt erased. By taking the conscious action of purifying oneself one becomes worthy of forgiveness(not God's forgiveness but my personal one) and because I have achieved this, while I carry momentary understandings of mistakes unlike the sovereign man, I hold no returning 'guilt' or resentment of any kind. No neurotic tendencies, no projections onto other, none.

Pride, this as mentioned earlier is the same as the earlier mentioned philotimic virtue. The idea of creation and happiness as a norm that has been willed by me as an action of my pursuit of virtue and creation. This is where the disgrace truly comes in, it is reabsorbed, not as pain but as energy for the recreation of the self in the form of higher values.

Another key idea is that of forgiveness. Once again there is the connection between my idea and his. The fact that for the self; moral, ethical, value-driven, etc is not a same thing. It is a rapture of the very existence, but rather than waste energy on being sorry for oneself, one must use that energy on self-

perfection. Though, forgiveness of others is another story. Other people are not capable of moral action, they are slaves to impulses, temperament, social constraints and much more. While people claim to be moral, much of there actions disgrace their ideals, their intentions align with physiological temperaments rather than values, and they abandon so much so easily. Their transgressions should be forgotten, or better yet not noticed at all.

Though, I am sure you are thinking about the differences(I mean I have been sprinkling them about these section) and while they are important to some extent. I don't want to go though a point by point refutation of Nietzsche ideas, especially when this analogy of the sovereign man is meant more an a psychological analogy to help explain intuitively my own personality rather than philosophical affirmation.(I disagree with Nietzsche on much)

One thing I will say though is the difference in end goal. For me, the sovereign man in a more refined sense(axiomatically driven, recursively made, God fearing, etc) is the final form. The Ubermensch is useless. The chaotic form of disparaging logic, reason, and God in forming morality disbanded the entire project and is just as disgusting as the slave morality of others.

P.6 Half-Growth and Half-Death

While many proclaim that my ideas of seeing people not as thinking being, not as moral agents is misanthropic and anti-social. That I should have more respect for others, that this lack of respect can lead to later personal problems with others. Though I disagree, first I will go over how I have seen others interact that makes me disagree, and my own personal evidence from my falterations with these values.(one thing I will add, is once again these are not true literal beliefs but analogies to explore esoteric

identities.)

Think of how people treat children, then people they deem as equals, then themselves, and then those they deem as higher than them when they perceive a slight.

The child's slight is either ignored or the person takes conscious action to help them, not by expecting moral action as they would with others but by understanding the child's abilities and inabilities and working around them in a kind manner.(Bar extreme causes or grotesque individuals)

With others there exist a range of reactions to slights, but their is a clear differentiator. They expect others that are 'equals' to exist in a semi-moral manner(this is because most people hardly understand moral manner to begin with). They try to act in kindness and understand others faults and temperaments but when push comes to shove they resent others, they expect from others. This can come in a wide variety of ranges: resentment, anger, fear, annoyance, instability, and so much more. These reactions are almost always unproductive and hurt relationships, people and such. They also use the perceived moral responsibility to negate their own moral responsibility in many cases(under my analysis, this is where most pain received by the affliction of others comes from, either consciously or unconsciously)

Then from here it is easy to see that the extremes of these go up and up.

These are where a majority of the petty, vapid, and pathetic emotions I wrote about earlier come from.

Now what I suggest isn't as radical as it seems, it is just shifting the average person closer to where others see children, no one other than God as above me. That I intentionally analysis and 'handle' other people. I understand them both to better see them as pathetic not in a disgusting way but like a child. Also, to better handle them for interactions with them.

Though, one thing I didn't include in my earlier writing is that

some people do come closer. Select people I know that I am close to enjoy(or don't depending on how you look at it) the responsibility of a human and conscious life within my own ideas.

Now, where does my growth and death come from? Well in recent time I have faltered, I have subconsciously found that I do expect things from others. These is so terrible in my eyes fro several reasons. One, as I have mentioned earlier, this 'equality' is the source of much of our troubles in the modern age void of true horrors. Second, for my personality and moral theory this effects me in a much more extreme sense than it does other. From a shallow perspective I am an extreme puritanical person who is obviously very prone to disdain. Beyond that I hold some values in high regards that many don't hold at all. Though, more truly than that, my in my moral theory intentions mean everything and so very few people have truly pure intentions. Now I don't mean everyone is a selfish jerk(though many are) but beyond that many are husks of people that only follow moral theory because they don't comprehend any other way, others hedonists that find minor moral action as easier, others do it because of insecurities, and so on and so forth. The concept of doing things for the mere fact they are moral/logical is lost on almost every person you will ever meet.

Here I will give an illustration. Last week(as of writing this section) I had an interaction. Someone had very condescendingly given me 'advice.' Telling me in a clearly condescending and antagonistic tone to remember to unroll my selves I had rolled up to wash my hands.(this is also after of many other similar interactions with this person) Now because I knew that to explain the ethical ideas of attempting to 'assert' fake authority on another in such efforts was immoral due to the fact that on a psychological basis that such things could very easily be comprehended as attacks on their own self-determination and competency. Also, that such actions could easily be interpreted(and likely truly) as things like need for superiority, low self-esteem,

extreme lack of social awareness, control issues, projection of their own incompetence, desire for a reaction, or malignant/-grandiose/competitive narcissism. Though I instead let it slide and said thank you, I didn't feel it would be worth it or would give any results on the matter. Now I didn't expect them to realize their fault or anything of that extreme. I expected them to say your welcome either as a empty social connection or an understanding of the fact that I understood their game and they would either give up or hopefully be filled with self-disgust over the conscious acknowledgment of why they did what they did. Instead they didn't make eye contact and instead "hhmf" at me. I still had the rational ability to understand that this person was either too stupid or immoral to understand whatever I wanted to say on the matter, so I decided not to. Though regardless I was filled with disdain. I was disgusted by such extreme evil. I know for many this doesn't seem like evil, but in the eyes of intentions it can only be logically assumed as. Those of lacking ability of intelligence to react in moral manners wouldn't make such an extreme mistake, those powered by resentment or anger would likely be filled with shame of such actions, and so on and so forth. Likely the only remaining interpretations would be that of malignant/competitive narcissism, desire for the reaction/pain of others, and other similar grotesque ideas. Now while I see the earlier stated ideas as immoral, these take special places, especially for someone like her who has the capacity of living in dignity.

Now this isn't some one shot, lately I have been more and more likely to to feel disdain for others. I don't know if it is puberty, social connection, increases in closeness with others, or whatever. All I know is that while this is something that so many other have proclaimed as what would be the greatest thing to my self-perfection is rather the worst thing.

P.7 Lack of Aesthetic

When I am referring to aesthetics in this section, I do not mean philosophical aesthetics, as in valuing something over another; I mean traditional ideas of beauty in music, art, and natural affairs.

In this sense of the word, I lack almost all aesthetic values. I have no favorite color, no concept on beauty in most things, and no care for musics, visual art, or other 'artistic' concepts.

This doesn't mean I have none. I have some interest in the artistic understanding in complex and intriguing literature, I can also be temporary incapsed by complex art/music(though only as long as it takes for me to understand it, and if it is too abstract as to be useless I have little or no care for it); beyond that I can find beauty in mathematics, logic, efficiency, ideas, plans.

In relation with that I have no feelings of sentimentality, meek emotions, and other similar emotions.

Now, I have always naturally found all of these things as beneath me and childish(and still do to some extent) though I have been taking time to try to expand my horizons.

As I have been doing this, I have noticed some changes. I have thought in more lofty ways than before, been more open to some experiences, and even engage with emotions in a way previously unheard of. Now here I will leave with this, but later I will explore this in more detail. What is actually causing this change, Explore what the change actually is rigorously, and whether is it actually a good or bad thing?

P.8 Boredom

Just as NFC is a primary driver in my life, so is aversion to boredom.

The most obvious connection is simply for my NFC is grown and in a feedback loop with my aversion to boredom. I satisfy my boredom with challenging cognitive tasks. This is one of if not the largest driver in my self-education, research projects, writing this book, and so much more are driven by aversion to boredom.

A little beyond that is ambition. Just as I choice challenging cognitive tasks to satisfy my boredom, so do I choice other challenging tasks. Leading, planning, mentorship, responsibility, work, physical activity, and more. This is what pushes me in SVA, Slack, JROTC, Scouts, OA, and much more. These activities satisfy my boredom.

I will go even further and say this desire for challenge is primary from aversion to boredom, not fear, perfectionism, and other traditional psychological reasons for people pushing themselves further than most.

Now, my aversion to boredom isn't just in ambition and other productive means. The most obvious is in watching TV or non-educational videos. Though this is largely to simple to take any time analyzing.

Though there are other things to analysis, relations. First I will go over casual relations.

For some reason, as I will discuss in more detail later on, I switch between 'extroversion' and 'introversion' in a sense. When I am away from others I hold no desire to be with them, I satisfy my boredom through abstract thought, arguments with myself, physics, and other similar things. That this feels the most natural thing. Though when I am with others this switches. Getting lost in thought no longer feels natural but rather challenging and hard to focus in the same way. I naturally feel the way to satisfy my boredom is through conversation with others. It is an interesting development that I will likely look into further.

P.9 Why Physics

While I have many interest; math, coding, economics, psychology, philosophy, logic, and many more. Though, one interest stands above and beyond the others. Physics. Ever since I was a kid I have been obsessed with physics.

Now I am sure I can come up with some lofty reason on how physics is the nature of the universe in its purest form, or the most fundamental science of them all. While I am sure that it is part of the reason I like physics so much, it is not one of the primary.

The most easily observed is its balance of abstract thinking and practicality. It is one of the most abstract and purest forms of logic other than pure maths and some forms of philosophy, but unlike the others it has a more direct connection to real-world application. For instance, my work is working towards fusion reactors, something I think once working will be one of the most influential and greatest creations of the modern age. To combine both reduction to axioms and analytics and contructionism of creating something practical.

Next is the challenge. Not only is physics itself extremely challenging, with some aspects of it taking months of studying to even comprehend it, it also combines many other hard disciplines; math, coding, engineering, and even metaphysics as times. This challenge is exhilarating and as mentioned elsewhere is one of the things that i most desire.

On combining other things, physics is a multi-disciplinary subject. Combining and using many of my other hobbies.

There are many more reasons but these are the main ones.

P.10 Lack of Resistance

As developed throughout this book, especially in "On the Pursuit of Thought" the idea of challenge as a goal.

The joy of an intellectual challenge specifically. To mull over a topic for hours. To have something beyond current comprehension. Something that doesn't make sense. Then all of a sudden explodes, not only to explain itself, but making connections all over. Like a flood.

This is my greatest joy, what I live for. Though, even as a teenager I am coming to limits. I will examine this in parts, first through my self-education and then through external world interactions.

What is going on personally is I fear that I won't feel this feeling. Many topics like economics, psychology, geopolitics, history, philosophy, literature, and more don't give this rush anymore.

As an example I will examine literature. In the past year I have found interest in literature, a topic I had previously missed. It was enticing, learning about the usages of symbolic characters(and figuring out who was and what was), setting as a character, philosophy through stories, and so much more. I had that excitement, though lately I don't. While there is still more to learn, it isn't the same. Everything new, is obvious. No requirements of complex thought when the workings are already there, only new information to be feed into the pre-made algorithm.

Now, I still have some exiting things in the above fields and much, much more in abstract mathematics and physics, but I am only 16 at the moment. If I am already this far along now, what is to say in 10 years there will be much left, what about 40? I have already surpassed all traditional classes and all that is left is research articles and manuscripts, these will leave me for quite some time, but who knows how long that will be.

This also continues in my personal life, school has the intellectual engagement of watching paint dry, talking to people is a burden, leadership roles still have excitement but have less and less return,

P.11 Am I Misanthropic?*

My misanthropic like tendencies are clear throughout this book, but am I really misanthropic? No, well... maybe a little.

I do have many friends, friends that I enjoy. I like talking to people. I am truly an extrovert in nature.

Though, beneath that there is a disconnect. I don't truly enjoy talking but rather staving off boredom. While in isolation I generate distrust and disgust of others. Many people I dislike more than I dislike boredom and everyday the percentage of people within that camp grows.

P.12 Depth of Feeling

All of this discussion on emotions and psychology can leave a reader with the thoughts of a sensitive and emotional boy, but this is far from the case.

While I use extreme words to convey my ideas, the extremity is not there. All my life my feelings have been dulled. While others are overwhelmed by feelings, for me they are hard to observe. Like they don't fully touch me.

This isn't true for all emotions; my passion, excitement, fanaticism, and need for cognition are all real. This doesn't take away from my thesis.

Beyond these, I have always required conscious effort to hold on to emotions. While this may sound like lamentations, it is

not. It is a great thing to forget anger and pain because of there insignificance. To never feel overwhelmed or anxious, to have control over my thoughts and actions, I am in control. I love that I am this way.

There was once a time that this made me feel inhuman. While others talked about there emotions, though that feeling like all has passed. Though, don't take this too far, I am by no means some robot devoid of all emotions, just simply that for any population with variance some people will be above or below average at things. For me I am simply far enough that observing others like me is a rare enough occurrence to think it truly is rare(but it really isn't.)

P.13 Language Usage*

Before I actually begin upon the main idea of this section I feel the best way to introduce this is to explain the inspiration of this section.

The inspiration is the oddest place place possible; Ted Kaczynski, or the Unabomber.

I remember when I was a kid I always hated the idiom "Have your cake and eat it too." The reason for this is due to the fact that the have your cake is placed first, and this fit(in a way that doesn't fit with the idiom.), because you must have your cake to eat it. So, I started to switch it around to "Eat your cake and have it to" though never really liked that. Until I read that part of the reason the Unabomber was caught was due to his strange phrase "Eat your cake, and have it all the same." When I had read this my first thought was insult that I hadn't thought of that myself.

Once I thought of that I realized more connection. Some more 'normal.' Starting with a thoughtfully researched and observed

effect being the fact that highly educated members of fields like physics, mathematics, logisticians, computer science, and philosophy use mathematical and logic terms in regular speech. This is because their thorough and clear definitions have clear applications. Axioms, manifolds, prior, bias(in a mathematical sense), fallacy, paradox, induction, inference, deduction abduction, ambiguity, equivocation, disjunction, mapping, inversion, duality, convergence, divergence, topology, entropy, singularity, resonance, phase transition, field, fractal, lattice, span, gradient, symmetry breaking, null, second order, and more. This isn't that odd considering that it is more wide spread, I mean a couple of those terms described above have even introduced themselves into colloquial speech(though used slightly differently.)

Another is using words that were created in academia, then introduced into regular speech that then changed it, in their original usage.

Though some are a bit stranger, mainly the fact that I simply create my own definitions for words or even create my own words.

So lets go through these one by one.

The first is very simple, these words are very useful in regular speech, convey abstract ideas easily, and can be applied easily. The mere fact that I understand them makes it only logical that I use it.

Next, the usage of academic definitions has a similar requirement. The fact that these academic definitions are generally more completely and logically defined than generally used word.

Finally, the changing of words and creation has a similar efficiency and logic.

Though, this brings new questions, why do I feel so comfortable with this, what criteria do I use for greater 'logic,' what is with this obsession with logic, why don't I just pick either change

current words or create new ones instead of do a bit of both, and much more.

...

P.14 Euphesus*

As I mentioned earlier, there has been change in me as of recent, and it has recently come to head with the advent of 'dating'(not actually dating the girl formally|will explain more later|Also not completely sure it is 'coming to head' just more extreme than before).

What I mean, is for the first time in a long time I feel conflicted. As strange as it may sound, I have been complete for a while now. I never feel internal conflict, neuroticism, second guessing myself, or anything of that such. I have for a very long time clearly defined my morals and values and thus never needed to second guess myself. While I am not perfect, the few times I make mistakes, I clearly identify them after the fact and rectify them. There is no lingering guilt nor do I go back and forth between what I should have done or should do for the future.

Now, in the last year or so, I have had some rare conflicts; many of whom I have talked about in earlier sections. Though with this ambiguous relationship, it has become an almost daily thing.

I went back and forth between whether or not I should ask her out in the first place. When I did I had a lingering regret about doing it over text rather than in-person. Later when texting her after the fact, I felt the inexplicable desire to talk to her, without any real end-desire other than the conversation. First, I felt conflicted because I have never felt the such a desire, in fact I often was disgusted by others for this desire. It always seemed beneath me, while it wasn't against my values, for someone who

has always been so 'complete' and whose entire personality and temperaments are recursively defined by myself the mere fact of something new is insane upon the face of it. It is like a self-betrayal which it took me a while to figure out what the betrayal was in the first place. Furthermore, this caused me to start questioning thing, why I didn't desire to talk to my friends about nothing in particular, should I start doing it, etc. It even lead me to try it. Then this started to make me think about how some random girl I am not even formally dating, has caused me to alter my own behavior. Just slightly, but nobody alters me even slightly.

Once I had finally gone through all of that and moved on, I decided I would talk to her over text. During that time I neurotically tried to when would be a good time and what to say. Once again, I don't do that. I don't care what other people think of me, I don't care if the message wasn't perfect, I don't feel neurotic over anything, certainly not a person. Yet I did, I thought about it. Finally, I texted her, it went nice.

I decided I should keep in touch, because I wouldn't see her in person for a month. So I decided to reduce ambiguity I would simply have a schedule of when to initiate a conversation. The number of days between initiations was meaningless and I knew I just needed some arbitrary number, yet I thought about what would be an embarrassingly long amount of time.

Later, came the day for my next initiated conversation. and once again I spent way to long figuring out what to say, but when I did, she hadn't responded. I didn't think much of it, due to the fact there were dozens of possible explanations for the lack of communication that were fine. That was until the next day, when she responded on a group text, but continued to ignore the individual one. Now even still, hours later I am still thinking of it. Still conflicted, wondering if asking her out was the right choice, or my number, or what I said. Wondering what my next steps should be.

While yes, I understand these emotions are normal for an adolescent boy of my age, but they aren't for me. Not only am I rarely conflicted about anything. I am much less about people. I am the kind of person who interacts with people for amusement or out of responsibility; deep down I don't care other people in the way required to feel these neurotic thoughts and emotions.

It is so strange, having the last couple years of my identity and foundation so clearly defined for everything. Only to bring out something new, something that doesn't fit. Now I don't know what to do; stay rigid or change(but change what)

Over the last month or so I have gotten over the visceral reaction. Now that I am back from my trip we can see each other in person, but haven't very much.

Appendix Q

Self-Authorship

insert here

Q.1 Scope and Understanding

It has always been a clear belief of mine that it is a persons duty and responsibility to define themselves though conscious thought and action. It has been something on the forfront of my mind. As mentioned in an earlier chapter on post-surgery and sicknesses, I haven't been the person I used to be.

Regardless it is a very important thing to be done. To clearly define the virtues that define how I should live my life and then make sure that I live in accordance to them.

I will begin defining who I want to be, much of it has already been defined in ethics, but personal ethics will be defined as well. Then, I will look into actions I can take.

Q.2 Ideals of Becoming

Q.2.1 Overview of Virtues

In becoming I define a set of categories of virtues that will be further defined.

Virtues Intellectual

Virtues Volitional

Virtues Executive

Virtues Stability

Virtues Adaptive

Virtues Social

While these are not equal categories nor exhaustive, they work as a good way to organize these essays for now.

Q.2.1.0.1 Intellectual Virtues *To be able to create oneself, one must be conscious of how to do so and who one wants to be. Some underlying things must be true.*

I believe that someone(me) should have a world-view. One that is completely consestaint, that brokers all areas, and is useful for further creation. A person should not simply listen vapidly and repeat the words of others without clear thought, but must have a formalized system to be able to not simply add information but be capable of sythesising it and using it in a broader context. To have a rigorous view on everything and logic behind it. This is the only true way to actually learn and function, without it you are simply repeating others with the variations of subconscious bias.

One must also have clear meta-cognitive skills, to question oneself it the only way to clearly understand something. While this can become natural through rigour, developing the natural tendency may speed of the process without formal rigour at every turn.

Knowledge and understanding is incredibly important. To use, develop oneself in the abstract sense, and to be able to abstract for future knowledge.

The ability to abstract things to develop clearer understanding of it or relating systems. Always look for the underlying mechanism beneath, not simply the begining and top and leave.

Humility to understand that which is beyond.

I want to be someone in constant though, of ethics, social responisbility, understanding the world around me. While maybe not the best desicion, it is mine.

Also, have your ideas be your own, fit them within your formal world-view and build upon them.

Appendix R

Philotimic

R.1 Intro

As I am moving on to college, one thing I note, beyond traditional internal self-authorship, there also exists the aspect of image. Image enforces behavior through wanting to be consistent, consistency builds virtue, image brings you together with like minded people, etc.

To continue to develop myself both internally and externally through consciously practiced drills and habitsis my goal. The highest human good in my world view. This area is going to be used both to explore the virtues and habits I wish to do and my way of practicing them in order to work on it. Then, log my activity.

R.2 Overview

R.2.0.0.1 Self-Perfection *In this proccess, I will make mistakes. The purpose is to grow. I tend to forget about mistakes as soon as resolved, thus, a plan would be to write down daily*

errors, mistakes, way to improve, etc. I will likely put them in the drill log later in this chapter. Another option would be in the journal. I will figure out soon.

Some other drills would be to argue against my own position, Steelman other, and write about when my views change.

R.2.0.0.2 Outward image *Before entering any professional room, take three seconds to check posture, breath, and warm neutral face.*

Try to repeat people's names in order to help develop memory of them.

Whenever walking into a new room, come with a question or striking statement prepared.

I will also try to record 60 second introductions and work to improve them.

R.2.0.0.3 Speech, Vocabulary, Diction, and Rhetoric *To improve my speech, I should work on trying to remove both filler words and vague quantifiers.*

Three ways to practice this:

1: as I am developing my YouTube Videos, I will naturally develop these skills when working on the scripts reading them.

2: I can develop things that I wish to describe, explain, or define. Then, record myself speaking on them in order to see ways to naturally improve.

3: Lastly, take something that I said to another person and take some time each day to rewrite in a better form.

R.2.0.0.4 Academic Communication *After stating a result, explain the scope and research.*

when explaining or talking about a result. State the result,

evidence for said statement, and finally end with limitations of my claim.

Write down paragraph summeries of research papers.

R.2.0.0.5 Assumptions and Cognative Habits *Many people, including myself, rest ideas on assumptions we don't even know we have. I would like to work on this.*

One way that I have already started is by writing my axiom book.

Another would be randomly taking random assumptions and either questioning them or proving them.

R.2.0.0.6 Bayes Reasoning *Using Bayes reasoning to think more logically it Paramount. Getting to the habit is the problem.*

One way would be to continuously be making predictions, guesses, conclusions, etc. Evaluate them using Bayes.

An example would be through every new room, evaluating people around me. Look at what they are doing, guess why, guess their job, age, height, and many more. When meeting people truly, try to come to a psychological profile of them of sorts.

R.2.0.0.7 Meta-analysis *When evaluating my thinking, actions, plans, etc. Don't be vague, come up with an number.*

R.2.0.0.8 Attention *Evaluate wandering and conclude I am doing it. Don't feel shame and such, just fix and move on.*

Try to be attentive even without deep work. Ponder, think on things, observe, to constantly be in a state of thought.

R.2.0.0.9 Observational awarness *Many examples have already been said. Conclusions on people, observe things like exits, direction, fire extengushers.*

R.2.0.0.10 Leadership *The most obvious thing is to get into it and figure out ways to obtain leadership positions.*

Some practice subjects could be mock meetings,

R.2.0.0.11 Dealines *Every deadline or commitment, write it down to not forget. Get into the habit of using outlook and not relying on memory alone.*

R.2.0.0.12 Body Language *Straight posture, long gait, eyes straight, weight even, feet shoulder width apart, hands unclasped and seen, use hands rationally, and no stress responses.*

Videos of me explaining things can be helpful in this regard.

R.2.0.0.13 Memory *Practice memory skills. For instance, now I want to work on LaTeX snippets and Physics library, so I will work on that.*

R.2.0.0.14 Desicion making *Evaluate desicions. The information used to make them, my confidence of the results, actual results, how good/bad of a disusing it was, etc.*

R.2.0.0.15 Teaching *Practice teaching as much as possible; YouTube, personal videos, or writing.*

R.2.0.0.16 Schedule *I have now written schedules, I want to follow them and by my nature develop them for the future.*

Once a week I will work on refining them.

R.2.0.0.17 Humor *Avoid too much self-depricating humor as I have done in the past. Also, avoid making fun of others. Work on continuing my general ironic humor delivered in an ironical dry tone. I think that works well.*

R.2.0.0.18 Writing *Write more in this book. Even when I go to college, try to write something beyond lecture notes once a day.*

I also plan to carry pocket note books with me.

R.2.0.0.19 Socratic Dialog *In working on speaking, teaching, and debate. I would like to incorporate a kind of Socratic dialog with others. An obvious way to work on this is continuing to write essays. Another would be to actually practice it with others. How to slowly do that remains a challenge.*

R.3 Doing it

R.3.1 9/21/26

R.3.1.1 Morning Outlook

As mentioned many times in previously, life is a project to develop ones self in virtue and understanding. I will be working on this project of mine in more detail today.

I plan to create a schedule based upon my earlier writing, I also wish to write a dialog on the "paradox of altruism" as I have been refering to it, essentially how can one be altruistic in a value system that puts altruism at its forefront.

I have been inspired by a program called the week of stoicism based upon early stoic teachings and thus am taking some ideas from them.

R.3.1.2 Notes

For my future actions, of the mentioned actions above I wish to prioritize

1. *Taking time to think without stimulation*
2. *Socratic Dialog*
3. *Beyond my traditional Socratic dialog where I wish to interrograte my thoughts not yet developed, I wish to interrogate my thoughts as they have been developed*
4. *Evaluating ideas, actions, and mistakes at the end of the day*

I have written my Socratic dialog in a partially formulated way. Next I want to work on my dialectic section but I am not sure how to decide what to have a dialectic on.

An idea I wish to examine further is dual intentions, and how to apply that to myself and my actions.

R.3.1.3 Review of the Day

Appendix S

The Structure of Motion: A Philosophical Journey Through Hamiltonian Mechanics

"The Hamiltonian formalism of mechanics, especially in its canonical form, has been an inexhaustible source of inspiration for modern theoretical physics." — Albert Einstein

S.1 Introduction

This chapter is planned to be an extreme examination of Hamiltonian mechanics, especially from a philosophical, logic, and abstract way. Eventually, it will go from the basic logic and axioms underlying it, to the metaphysics and epistemology that are derived, and finally to advanced and modern usages. The primary purpose of this exercise is for me to find the gaps in my knowledge and gain a more intuitive and deep understanding of these concepts.(It was also written in 10th grade before the rest of the

book so may not have the same standards.)

The reasoning for this is due to the extremely complex and abstract nature of such mechanics. To see the world through 'flows' instead of forces.

S.1.1 Physical Reality and Frameworks

Now, there are many theories, models, descriptions, framework, and so much more. Though, what really differentiates them? What makes one thing one and the other the other. While I won't truly go into extreme linguistic detail for every single one, I will go into moderate detail within the art to get the general idea down.

Let's focus on frameworks, because that is what a Hamiltonian is(others define it as schemes which is another great word and arguably more accurate, but I will refer to it as a framework because it has a more intuitive grasp). It is an entirely different framework than Newtonian. It comes from an entirely different perspective, looking into geometric identities rather than the causal relation of forces and such interactions. Also, other theories and identities are derived through it, independent of the actual reality of it. For instance, the flow of electromagnetic interactions, relativistic, or even gravitational are all derived from this concept of the flow of energies through sympathetic geometry.

Even beyond that, Hamiltonian mechanics can represent any function that changes; statistics, viruses, and many others. This is due, to that Hamiltonian's is a pure mathematic relational concept rather than based upon physical axioms. For instance, Newtonians is based upon the axiom of inertia. Though, I will add one interesting involvement is the idea of privileging Hamiltonian over Lagrangian and stating that Lagrangian is a subset of Hamiltonian rather than its own framework.

Then this begins to ask our questions of scientific realism vs

anti-realism(I found out the more neutral language is instrumentalism). Basically, what makes such theories more fundamentalist, predictive modeling or metaphysical truth. Further, this now engages the complexities of mathematical axiomatic reduction or physical.

Let's start with realism vs instrumentalism. I have already made my stance clear, having metaphysical truth in important but not in the way that non-metaphysical schemes should be thrown away. Though, this then brings about the question of what frameworks have greater metaphysical truth, but I will later look into this in later sections.

This now leads to the question of mathematical abstraction vs physical realism. Though, one thing I will add is that the physical axioms of Newton's laws are based upon falsified ideas like absolute space and time(but you can get around such semantics by employing Whewell's axioms of Mechanics[1st: "Every change is produced by a cause." 2nd says that: "Causes are measured by their effects." Finally, the third remain unchanged from Newton's formulation.] Now back to the main event, I feel that the structuralism of Hamiltonian is better due to the fact that when you break down physics to the quantum the classical forces, inertia, and such break down and you can see that the structure is what remains. That while forces still interact as abstract 'causes of change" through bosons, their classical and 'physical' realism fails to encapsulate their extreme complexity.

S.1.2 Philosophical Context*

The Kantian influence on Sir Hamilton is clear....

S.1.3 Notion of Time, States, and Trajectories

Let's start with time. I have already explained my thoughts on time, but here I will deepen my explanation. Time, is the factor on which states evolve. It is then evolution of entropy. This axiom is very simple and intuitive. While objective empirical proving the nature of time is extremely challenging due to its abstract nature. For a more in-depth analysis of the nature of time visit Chapter 1, section 3&7.

One thing I will discuss is time reversibility. Now, Hamiltonian mechanics works best for time reversal symmetries(it can still work in some use cases but rarely.). In fact, time reversal symmetry is a corner stone of much of modern physics, especially quantum mechanics, but this interferes with our intuitive grasp of reality, the second law of thermal dynamics, and even dark energy; in essence the emergent properties of reality seem to interfere with our 'fundamental mathematical' derivations. So I will examen these questions; both for their own state and because it continues our question of the fundamental nature of reality. I will also attempt to discuss imaginary time with regards to quantum mechanics.

Actually, upon further thought the nature of time will be further developed through after the derivation of analytical mechanics.

Next will the on trajectories; now like many other ideas presented within this book, that may seem simple but the derivation of the ideas will be thorough and show its true complexity. Within Hamiltonian mechanics, the trajectories hold special geometric identities due to the nature of motion, but what can we derive from this simple observations

S.1.4 Introduction to Phase Space

Another question that arises from a Hamiltonian view of metaphysics is Phase Space. The fact that, according to our theories, phase space almost seems real, but yet it is so very different than what we observer. Or, we think so.

Obviously, there is the idea of the conflict between the teleology that comes as a consequence and our view of causality. I will address this in a second.

A simpler idea is that, from a perspective view, change becomes fundamental and any identity becomes simply emergent. This isn't truly new, simply more emergent in phase space, so I will move on.

One new thing, is that Phase Space allows for entropy irreversibility. Traditional theories find time reversibility to be a corner stone, this going against it.

This is due to the fact that entropy increases

$$S = -k_B \Sigma_i p_i log p_i$$

Where the volume of the phase space stays the same

$$Vol(\phi_t(\Omega_M)) = Vol(\Omega_M)$$

Lastly, the 6D phase space seems very different to ours. Having generalized position and momentum coordinates is odd. An instrumentalist view makes this simple, but a scientific realism makes this difficult. Sadly, I don't have the knowledge to come up with.

S.1.5 Teleology vs Causality

My privileging of causal relations has already become clear, but now I will justify my thoughts through a Hamiltonian view point.

While many claim that the nature of Hamiltonian mechanics necessitates a teleology, and my stated view on structuralism seems to contradict my views on causality, but I will address these claims categorically.

First, the nature of Hamiltonian mechanics necessitates a teleological view of the natural world. This statement will seemingly logical, lacks the tautological strength required of such an extreme statement. For one, Hamilton himself had a clear belief in causality(while this obviously isn't enough, I will explain why he and I see it that way.) First, the concepts of causal forces(forces are by definition causation principles) can be derived through the inverse gradient of potential energy. Second, the connection of the 'principle of least' action and teleology is flimsy at best. This is because their main argument is based upon intuition of mathematic alone not physical realism(I know this goes against stated structuralism but I will come back to that), the reason I state this against due to the fact that, mathematical intuition in the face of metaphysical truth. For instance, as stated earlier the teleological 'energy' is found as a derivation of causal force. Finally, returning to our arguments placed in the first chapter.

S.1.6 Role of Variational Principles in Mechanics

S.2 Axiomatic Mechanics

S.2.1 Axiomatic Derivation of ...

S.2.1.1 Introduction

For any mathematical physical axiomatic derivation, a mathematical derivation is first required. I won't go super in depth but

I will still go over it for rigors sake. I may go back and do a deeper dive but for now I will simply put these things.

S.2.1.2 Axiom of Real numbers

Why needed: Physical quantities like position, velocity, time, energy, and action are represented by real numbers. The calculus used in mechanics relies on the properties of real numbers.
Key axioms:

- *Commutative, associative, and distributive properties for addition and multiplication.*
- *Existence of additive and multiplicative identities (0 and 1).*
- *Existence of additive and multiplicative inverses (for non-zero numbers).*
- *Completeness axiom: Every non-empty set of real numbers with an upper bound has a least upper bound (ensures continuity, critical for calculus).*

Role: These axioms enable arithmetic operations, inequalities, and the construction of functions like the Lagrangian $L(q, \dot{q}, t)$.

S.2.1.3 Axiom of Euclidean Geometry

Why needed: If the system involves spatial coordinates (e.g., particles moving in 3D space), Euclidean geometry provides the framework for defining positions and distances.
Key axioms(informal):

- *Points, lines, and planes exist.*
- *A straight line can be drawn between any two points.*

- *Distance between points is defined (e.g., via the Pythagorean theorem).*

Role: Defines generalized coordinates q (e.g., Cartesian or polar coordinates) and kinetic energy terms like $\frac{1}{2}m\dot{x}^2$. Note: For abstract systems (e.g., in generalized coordinates), geometry may be less critical, but it's foundational for physical intuition.

S.2.1.4 Axiom of Set Theory (Basic)

Why needed: Sets are used to define the domain of variables (e.g., time $t \in \mathbb{R}$, coordinates $q \in \mathbb{R}^n$) and functions.
Key axioms(informal):

- *Existence of sets: Sets can be formed to represent collections of objects (e.g., possible paths of a system).*
- *Union, intersection, and Cartesian product: Allow combining and manipulating sets.*
- *Axiom of choice (implicitly): Ensures a choice of path exists in variational problems.*

Role: Provides the language for defining functions, spaces, and the configuration space of a system.

S.2.1.5 Axiom of Calculus (Differential and Integral Calculus)

Why needed: The principle of stationary action involves integrals (action $S = \int L\,dt$) and derivatives (in Euler-Lagrange equations).
Key concepts (built on axioms of real numbers):

- *Limits: Define continuity and differentiability of functions like $L(q, \dot{q}, t)$.*

- *Derivatives: Partial derivatives (e.g., $\frac{\partial L}{\partial q}$) are used to describe rates of change.*
- *Integrals: The Riemann integral defines the action S.*
- *Fundamental theorem of calculus: Links derivatives and integrals, essential for variational calculus.*

Role: Enables the formulation of the action and the variation $\delta S = 0$.

S.2.1.6 Axioms of Variational Calculus

Why needed: The principle of stationary action requires finding the path that makes the action stationary, which is a problem in variational calculus.
Key axioms(informal):

- *Functional: The action S is a functional (a function of functions, e.g., paths $q(t)$).*
- *Variation: Small changes in the path $\delta q(t)$ are used to compute δS.*
- *Stationary condition: The path satisfies $\delta S = 0$, leading to the Euler-Lagrange equations.*

Role: Provides the mathematical machinery to derive the equations of motion from the action.

S.2.1.7 Axioms of Linear Algebra (Basic)

Why needed: For systems with multiple coordinates or degrees of freedom, vectors and matrices describe generalized coordinates q_i and momenta p_i.
Key axioms:

- *Vector space axioms: Addition and scalar multiplication of vectors (e.g., coordinates in $\mathbb{R}^n$).*
- *Linearity: Operations like partial derivatives in Hamilton's equations are linear.*

Role: Supports the formulation of the Hamiltonian $H(q, p, t)$ and phase space (the space of (q, p)).

S.2.1.8 Axioms of Variational Calculus

Why needed: The principle of stationary action requires finding the path that makes the action stationary, which is a problem in variational calculus.
Key axioms(informal):

- *Functional: The action S is a functional (a function of functions, e.g., paths $q(t)$).*
- *Variation: Small changes in the path $\delta q(t)$ are used to compute δS.*
- *Stationary condition: The path satisfies $\delta S = 0$, leading to the Euler-Lagrange equations.*

Role: Provides the mathematical machinery to derive the equations of motion from the action.

S.2.1.9 Physical Assumptions (Not Strictly Mathematical Axioms)

While not mathematical axioms, certain physical assumptions are mathematically formalized:

- *Time is continuous and one-dimensional ($t \in \mathbb{R}$).(Though Hamiltonains can work in other forms)*

- *Energy is well-defined: Kinetic energy T and potential energy V are functions of coordinates and velocities.*
- *Differentiability: The Lagrangian L is sufficiently smooth (at least twice differentiable) to allow partial derivatives*
-

Role: These ensure the mathematical framework applies to physical systems.

S.2.2 Axiomatic Derivation of Hamiltonian Mechanics

Two hundred years ago, William Rowan Hamilton reformulated classical mechanics using the ***principle of stationary action****, a single axiom that unifies the dynamics of physical systems. This principle states that the path taken by a system between two times makes the action S stationary.*

The ***action*** *is defined as:*

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt,$$

where $L = T - V$ is the ***Lagrangian****, with T as kinetic energy, V as potential energy, q as generalized coordinates, and $\dot{q}$ as their time derivatives.*

The system follows the path where $\delta S = 0$. Varying the action:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) dt = 0.$$

Integrating by parts on the second term:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \delta \dot{q}\, dt = \left. \frac{\partial L}{\partial \dot{q}} \delta q \right|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q\, dt.$$

*Since $\delta q = 0$ at t_1, t_2, we get the **Euler-Lagrange equations**:*

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0.$$

*Hamilton defined the **Hamiltonian** as:*

$$H(q, p, t) = \sum_i \dot{q}_i p_i - L(q, \dot{q}, t),$$

where $p_i = \frac{\partial L}{\partial \dot{q}_i}$ are generalized momenta. Typically, $H = T + V$.

*Using the Legendre transform, the dynamics are governed by **Hamilton's equations**:*

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

***Example**: For a particle in gravity, $L = \frac{1}{2}m\dot{h}^2 - mgh$. The momentum is $p = \frac{\partial L}{\partial \dot{h}} = m\dot{h}$. The Hamiltonian is:*

$$H = \frac{p^2}{2m} + mgh.$$

Hamilton's equations yield:

$$\dot{h} = \frac{p}{m}, \quad \dot{p} = -mg,$$

reproducing the equation of motion $m\ddot{h} = -mg$.

The principle of stationary action simplifies dynamics, applies to diverse systems, and underpins modern physics, including quantum mechanics and relativity. Hamilton's work remains a cornerstone of theoretical physics.

S.2.3 *Hamiltonian Mechanics ad a Geometric Theory

S.2.4 *Poisson Brackets and the Algebra of Dynamics

S.2.5 *Canonical Transformations: Symmetry, Simplicity, an Structure

S.3 Philosophical Implications*

S.3.1 *Time in Essence

S.3.2 *Hamiltonian Constraints and the Problem with Time

S.3.3 Noether's Theorem

Noether's Theorem is a brilliant application of formal mathematics to show that conservation quantaties define conserved quantaties. Such as time symmetry defines energy conservation.

S.3.3.1 *Ontolological Priority of Symmetry

As I keep coming back to, is symmetry more fundamental than more traditional physicallist things like energy or momentum.

S.3.3.2 Laws as Autonomous Structures

S.3.3.3 The Continual Definition of Energy

S.3.3.4 Nature of Time

S.4 Modern Hamiltonian

S.4.1 Modern Research

S.4.1.1 *Hamiltonian Formulation in Plasma Physics

S.4.1.2 *Hamiltonian and Metriplectic Mechanics

S.4.1.3 *Symplectic Integrators: Preserving the Structure of Nature

S.4.1.4 *HNNN(Hamiltonian Neural Networks)

S.4.2 Fun Mess Around

S.4.2.1 Non-Canonical Poisson Brackets

A Poisson bracket is a bilinear, antisymmetric operation $[F, G]$ *that defines the time evolution of a functional* F *via* $\dot{F} = [F, H]$*, where* H *is the Hamiltonian. Non-canonical brackets arise when the phase space variables (e.g., density, velocity, magnetic field in MHD) do not follow the standard canonical structure* $\{q_i, p_j\} = \delta_{ij}$*.*
Properties: The bracket must satisfy:

- *Antisymmetry:* $[F, G] = -[G, F]$
- *Leibniz Rule:* $[F, GH] = [F, G]H + G[F, H]$

- *Jacobi Identity:* $[F,[G,H]]+[G,[H,F]]+[H,[F,G]]=0$

For this I will be defining my own Poisson bracket specifically for plasma thrusters.

For this I will obviously need to define the variables. Things like:

- *Mass density:* $\rho(\mathbf{r},t)$
- *Velocity field:* $\mathbf{v}(\mathbf{r},t)$
- *Magnetic field:* $\mathbf{B}(\mathbf{r},t)$
- *Entropy or internal energy:* $s(\mathbf{r},t)$ *or* $\epsilon(\rho,s)$

Now define the Hamiltonian, for this I will use the pre-established ideal-MHD, but in all reality for plasma thrusters I should use a more expanded model.

$$H[\rho,\mathbf{v},\mathbf{B},s]=\int\left(\frac{1}{2}\rho v^2+\rho\epsilon(\rho,s)+\frac{B^2}{2\mu_0}\right)d^3x$$

Next, I must define the phase spaces and constraints. Incompressibility or magnetic field divergence.

Next I propose a bracket system, with many of them being

$$[F,G]=\int\sum_{i,j}\frac{\delta F}{\delta\xi_i}J_{ij}\frac{\delta G}{\delta\xi_j}d^3x$$

Such that Ideal MHD is:

$$[F,G]=-\int\left\{\rho\left[\frac{\delta F}{\delta\rho},\frac{\delta G}{\delta\mathbf{v}}\right]+\left[\frac{\delta F}{\delta\mathbf{v}},\frac{\delta G}{\delta\mathbf{v}}\right]\cdot\left(\frac{\mathbf{B}}{\rho}\times\nabla\times\frac{\delta G}{\delta\mathbf{B}}\right)+\frac{1}{\rho}\left(\nabla\times\frac{\delta F}{\delta\mathbf{B}}\right)\cdot\left(\frac{\delta G}{\delta\mathbf{v}}\times\mathbf{B}\right)+s$$

Another approach is using lie-groups. The Lie-Poisson approach is a method to construct non-canonical Poisson brackets

by reducing a canonical Hamiltonian system on a large phase space (e.g., particle coordinates) to a smaller phase space of collective variables (e.g., fluid fields in MHD). It is rooted in the symmetry properties of the system's configuration space, described by a Lie group.

...

The other way is through Casmir invariants. Casimir invariants are functionals C that commute with all functionals F under the Poisson bracket: $[C, F] = 0$. They are conserved quantities that arise from the degeneracy of the non-canonical bracket and provide constraints on the system's dynamics.

Degeneracy: Non-canonical Poisson brackets are degenerate, meaning their Poisson tensor J_{ij} has a non-trivial kernel. Casimirs live in this kernel, satisfying:

$$J_{ij} \frac{\delta C}{\delta \xi_j} = 0$$

Physical Role: Casimirs represent invariants tied to the system's topology or symmetries, such as helicity in MHD, which constrain the evolution of plasma in thrusters.

Identification: Casimirs are found by solving the functional equation $[C, F] = 0$ for all F, often using the Lie algebra's cohomology.

Appendix T

Dialog

T.1 Introduction and Scope

T.2 Politics

T.2.0.0.1 AI is a Net Positive*

T.3 Personal

T.3.0.0.1 Who Should I Become(part 1, Broad) *Q: Before we continue, let us narrow the question down to a pieces. Come up with various categories and sub categories.*

R: ok

- *Presence*
  - *Physical Presentation*
  - *Facial Expressions*
  - *Tone of Voice*

  - *Body Language*
  - *Tone of "warmth"*
  - *Humor and Levity*
  - *Professionalism*

- *Speech*
  - *Vocabulary and Precision*
  - *Technical Explanation*
  - *Academic Writing Structure*
  - *Cross-audience communication*
  - *Socratic Dialogue*

- *Conversation and Social Interactions*
  - *Core skills*
  - *Networking*
  - *Situation scripts*

- *Mind and Cognition*
  - *Cognative Habits*
  - *mathematical and physical thinking*
  - *Problem solving*
  - *Desicion making*
  - *Sustained and Consistent focus*
  - *Observation*
  - *Bayes thinking*
  - *Emotional Regulation*

- *Character...*

Q: Ok, that is enough. We don't need more than that. What would you say is the most fundemental on that list?

R: Character, our morals dictate our actions, how we interprate actions, and much more. It influences everything else while being independed of everything else.

Q: Good, we have already discussed that in detail in this book, what would you say the next most fundemental? For the purposes of this exersices we will use fundemental to mean building blocks towards our conversation, not any other meaning you could describe.

R: For that, I would have to say that presence. As your goal of presence effects sets your goals for speech and conversation. Mind is largely independent, but it is also independent of the others. More orthogonal than fundemental.

Q: I accept your argument. To further constrain this conversation, let us define what you mean by presence.

R: I mean who I wish to be processed as. In a general sense.

Q: Does that fit with your above categories

R: Sadly not, but it can be useful in other situations. I would go back an reorder our conversation, but let us move ahead.

Q: Wouldn't that challenge the exercise?

R: No, the point of the beginning was simply to define what we should talk about, now we know. Furthering it could be of use, but we can do it later without such hidden truths coming to trouble in this conversation.

Q: Ah, so, now define our conversation.

R: While I find the whole purpose is to change internally for the sake of overcoming myself. Something I have over looked is others perception of me. Thus I should examine that. Here we will define broadly who I want to be perceived to be.

Q: Why?

R: Why I haven't thought of it before or why it matters?

Q: The latter.

R: Very simply, modern society is... well a society. To succeed and achieve my broader goals I rely on others to see me, to help, to teach, and to let me lead when the time comes. It is also as I have mentioned a human's duty to be pro-social within their society. Having a defining oneself so others can see allows other to allow them to do those things much more easily.

Q: I could certainly examine that short statement further, but let us move one. for an experiment, let us start with writing some general ideas who you want to be seen as.

R: Rigorous, consiouncious(in all of its definitions), conscious, questions assumptions, actively helpful, kind, helpful, friendly, obedient, reverent, independent but still a leader, curious, broad minded, disciplined, consistent, deep, "always", precise, high standards, active, confident, purposefull, makes others more capable, composed, long-term vision, and principled.

Q: Alright, we will reconvine in the morning.

R: Alright, I have returned.

Q: Looking at the above, in broad strokes what would you say the image you want to convey to be?

R: Competent, in control, and helpful

Q: Do any constraints arise?

R: Being seen as competent, in control, and all of the related properties tends to lean towards stoic and general neutral line. While being helpful and a friendly would obviously lean towards being warm and "friendly."

Q:Remember, whenever paradoxes arise, test you assumptions, as contradictions do not exist.

R: Yes, there is no true paradox, only the question of picking a point on the spectrum and knowing when to move about that spectrum.

Q: Why is this?

T.3.0.0.2 My use of AI *Q: to question the use of AI, let us first constrain ourselves to ourselves.*

T.4 Philosophy

T.4.0.0.1 The Paradox of Absolute Altruism *Q: As you have mentioned clearly, altruism in an effect remains at the center of your moral framework, but if everyone should be altruistic, in that effect who are they being altruistic for and how so?*

R: My view gives a privilege to the person in themselves, but a privilege alone cannot justify egoism. They remain in their duties and development of who they are in society as a perfection of the telos of who they could become.

Q: that answers who, but not how. At what goal do they hope to achieve?

R: To a large part, a purpose, a purpose in society for a happiness in challenge, development, virtue, and accomplishment.

Q: Why do these things matter at all? If one can choice to be happy as you often suggest why go for these specific forms of happiness, and if it is an ease, why not other forms of achieving happiness.

R: On a purely pragmatic and secular light, these are the most effective long term functions of happiness that can support themselves. On a more religious note, I firmly believe that God has created us in such an image to have a specific telos and purpose that we must achieve, part of that being to become more like his begotten son Jesus.

Q: How does one find their purpose?

R: Careful thought and reflection as God has written it upon

our hearts.

Q: How does one distinguish between the telos written upon themselves and the pleasures of the flesh?

R: One can look at how altruistic these values are in the face. If they match a framework that works to develop society into the image of their highest values.

Q: Once again, we find ourselves back to the beginning, if one doesn't value pleasure or happiness, how is one altruistic to others.

R: I would say I do value pleasure, just that it can be chosen when it happens and why, so that having it as a deciding factor is a null point. Purpose is the foundation, this purpose can be found through examining emotions such as pleasure, but that does not mean it has its foundations in pleasure.

T.5 Physics

T.5.0.0.1 The Laws of Physics are the same everywhere

Q: To examine the claim, we must first define out terms. So, if you would please explain what you mean by the Laws of Physics and "are the same."

R: Well, that is quite simple. The Laws of physics are the underlying mechanisms and devices that govern the Universe. Whether that be symmetries, quantum fields, structure, or something else has little barring on this topic. As for are the same, simply that there exists a spacial invariants across the universe.

Q: Why does the nature of the laws of physics have little baring on the conversation at hand?

R: All of the listed above have the same observable nature by definition. It is not like how the electron behaves differently than the atom, but rather more like the quantum field fluctuations and electron. If they had easily distinguishable differences that can

be examined scientifically, such discussions would not be had.

Q: Ok, I accept your premises, now on to the claim itself. How exactly would one be able to verify that the laws of physics are spatially invariant if we use our current programme to continue research into the new?

R: While I consider that it is quite difficult, and while my axiom should still be held, a general skepticism is perfered. Now how to look out for it, modern advancements allow us to see areas separated by great distances. Inconsistencies accross different regions of space would be quite easy to observe. Allowing one to question held assumptions.

Q: What about areas beyond our observation?

R: No objective way to prove it, but science is not about unprovable "what-ifs," it examines observables. As mentioned, it is taken as an axiom unless falsified.

Q: Yes, until falsified. You make it sound just like another theory. Yet, is it not common that another criteria for a theory would be to adhere to this axiom?

R: Yes, just as any theory must adhere to modern scientific consencous, until reason to doubt such underlying theories.

Q: I realize something as we talk. When referring to spacial changes in physics, you take a more conservative view than I. Focusing on the idea of variable pockets of contiuum with slight changes. Alas, there could be massive changes. Changes in dimensions, symmetries, or even fundamental properties like momentum transfer.

R: As for fundemental properties, they are simply consequences of more fundemental properties like dimensions and symmetries. Possible changes have been thoroughly examined so one would be able to observe them and theoretically figure it out.

Q: What if the changes are to logic and mathematics? Rather than symmetry.

R: That is absured.

Q: Why is that?

R: No possible mechanism could possibly do that, as logic cannot be effected by mechanisms.

Q: How could you know that? Do you have omniscient knowledge?

R: No, but deductive reasoning can show that math is divorced from physical reality and thus cannot be effected by physical mechanisms. The idea of logic being capable of changing by external forces is paradoxical. If logic changes, that would simply be a consequence of other logic, thus consistent with logic. Only new.

Q: So there could be new logic that we have no knowledge of?

R: Yes, but, such idyllic daydreaming is unrealistic.

Q: Why, what makes it any less unlikely than anything else?

R: There is no possible evidence of it.

Q: Would there be evidence of it?

R: Yes, changes in the laws of physics over time or space.

Q: You assume you understand this unknown, how could you? How could you know you could identify it with any for of accuracy? What makes you think the current mysteries aren't that our current understanding it reliant of completely false axioms?

R: Such avenues are being explored.

Q: In very limited forms. Not in the forms I have been suggesting.

R: Yes, but your forms are paradoxical.

Q: How so?

R: Your forms throw out induction. This is simply Hume's problem dressed up, but the solution stays the same. One can-

not move forward without some basic axiom of the invariant nature of reality over some configuration space. Primary space and time.

Q: Why can't we move forward?

R: If one questions basic logic, how could one possibly come to any sort of understanding of observations.

Q: As you most certainly know, a lack of an idea is hardly evidence that such an idea is impossible.

R: Ok, ok... If one where to doubt logic, deduction goes out the window. If deduction goes out abduction goes out. If we have no abduction we cannot trust our senses nor our insurements. Thus no evidence, ways to evaluate evidence, nor any theories to understand any evidence. Science goes down with it.

Q: Such a fragile thing science is.

R: I know where you are going. This calls into question the validity of science, hanging on by an unprovable axiom. Yet it matters not. If science works, then it helps, if it doesn't, it has helped in the past and abandoning it would not do us any better.

Q: So pragmatism becomes the answer?

R: In the question of axioms, yes.

Q: Why not everything else?

R: Once we have the basic axioms, such hand waving is unessecary as more rigorous derivations are possible.

Q: Why are these more rigerous methods preferable?

R: We have gone all of the way from spacial invariants to the foundations of epistemology, why don't we give it a rest.

Q: Isn't this what you wanted?

R: What are you going on about, you are me. This meta-humor makes no sense, especially as no one will ever read this.

Q: Do you want someone to read this and think you clever?

R: No, at least I had not thought of it until you(I) wrote that

down... I still hold it that it be a personal choice rather than the desire for external validation.

Q: Ok... I won't push it, back to where we where. Why are more rigerous methods perfered in epistemology. If you already concede that absolute truth is unattainable(I know you haven't already outwardly have, but you will get there as you are me and this is quite long[what in the world am I doing, writing to myself?Yes, that is the whole point((Golly I sound like I had back when I started Logos[[Is that a bad thing, isn't that what you are trying to do?Alright, that is enough]])]]) Anyways, wouldn't a pragmatic view be more consistent and make more sense?

R: In many cases, a pragmatic world view can lead to idyllic views where idealism leads to pragmatic outcome. In science this is obviously true.

Q: That is hardly a view in the philosophical sense, but more in the aesthetic.

R: Then I take that on. Pragmatic in epistemology, instramentalism in pragmatic thought, and scientific realist in ideals.

Q: That is quite complicated.

R: It may be, but so is the universe.

T.5.0.0.2 Computational Models and their Accuracy*

T.5.0.0.3 Further Development in Structural preserving methods

T.5.0.0.4 Explaining Gravitational Waves *Q: Explain to me gravitational waves*

R: Essentially, gravitational waves, also refereed to as gravitational radiation, are small perturbations in spacetime. Mathematically equivalent to waves.

Q: What causes them?

R: It is something called quadropole moments, where quadropole refers to motion where the distribution of the stress-energy density changes while the "center of mass" remains the same. Examples would be binary orbits or CCSN.

Q: Why quadropole?

R: Quadropole moment, refers to distribution, it is a higher dimension extension than monopoles and dipoles. Monopoles referring to changes in the cumulative, such as total mass. While dipoles would be change in center of mass. Obviously, mass is conserved so any change in it would be by its nature a higher dimensional pole. A dipole movement would be uniform and thus unable to preterb spacetime in wave form. While quadropoles would be non-uniform by definition.

Q: What about higher dimensional?

R: I am not sure what higher dimensional poles would be so I would have to look that up.

Q: Before we do that, let us try to examine it without that. Think about an abstraction of what we are looking at.

R: Well, we go from the integral, to the $1/n$ *of the integral, to the function itself. The general idea is on the tip of my tounge but I keep think aout it going nowhere.*

Q: Ok, look it up, but then tell me about what that unearthed.

R: Multipoles generalize "shape descriptors." Going from a scalar that tells you the total but nothing of movement, symmetry breaking, etc. Then adding movement, then flattening or elongating(quadropoles), then three-fold. They look at the asymmetry of the asymmetry of an earlier order. Taking that all quadpoles plus define the distribution changes while quadropoles alone look at compression and elongating the structure.

Q: Great, what does this say about gravitational waves?

R: Quaropole plus tend to have the heaviest effect while later ones add correction terms.

Q: How do the above examples give these?

R: Take a binary orbit, the distribution of mass obviously changes, the matter appears in different areas. Taking the example of flattening or elongating, it effectively does that, imagine the curvature of spacetime rather than the point particles. This makes the earlier analogy of the sphere much more clear.

T.6 Mathematics

T.7 Research Culture

T.8 Broader Culture

T.9 General Conversing

Appendix U

Clarification

U.1 Scope and Purpose

Essentially, to write down clarifications on things I had forgoten or realized I didn't understand like I thought I had.

U.2 Physics

U.3 Math

U.3.0.0.1 Multilinear *A tensor is a multilinear map of vectors and covectors. What is a multilinear mapping?*

It means it takes several variables(or vectors), takes the rest as constant and maps only one, and does that for all of them.

Appendix V

Random Stuff

V.1 Euphesus, Continuation

V.1.1 Anaysis of the Person

First things first, I must define my scope and what I am going to do. Let's start with interests, values, motivations, writing and speaking style, her friends, personality patterns, context, how these influence her interactions with me, my interactions with her, and finally moving forward.

As for interest; politics, philosophy, theology, law(spesifically criminal and constiutional), European history, art(mostly realism with charcoal), legos, cooking, and such. I have also learned recently she is suprisingly into Lego NinjaGo. Excited to ramble about character development, meta-analysis of the show, and plot points. She generally enjoys card games, conversations, simple and easy games like catch, and surreal or absurdist humor. The humor often time juxtoposes childish tone or manner of speaking with witty creation and topic.

Career, she has a bit of a somewhat range of ideas for career. Ranging from constitutional lawyer to homicide detective, with

constitutional lawyer being at the top. Beyond that I am not sure. A general enjoyment of law is very obvious but why is less so. There are ome obvious ideas such as an interest in social order and moral responsibility.

For Values, she is a devote Christian with a quite conservative outlook on the world. Generally valuing the ten commandments, respect, protestant work ethic, and liberal arts. She is quite consistent, having a structured theory of value. Seeing sin in a conservative manner, everyone is doing it constantly and that even thoughts are a sin, but people are not slaves to such sin. Through action they can control even their thoughts, even if imperfactly. Also, when discussing morality she tends to lean towards the side of virtue ethics.

General personality patterns, she is generally introverted. As mentioned earlier has a surrealist sense of humor. Often times focused on other people desite this introvertedness. Slightly more on the side of spontanaity. Pretty shy and tend to downplay her own abilities and exagerate others. Generally focuses on being needed to the point in there is little self-introspection in wants. Enjoys my questions on morality, philosophy, science, ect; saying that it makes her think in new ways about new things.

Speaking/Writing style: starting with writing, as in texting. Tends to uses short grammatically correct sentences, quite often uses emojis to convey tone, generally sends multiple short messages rather than one longer one, uses a relatively suprising amount of em and en dashes, uses exclamation points, very direct and to the point, and writes quickly. As mentioned multiple times, generally uses sureal and witty humor during most of the conversation in rapid fire. On more in-depth conversations uses condensed psuedo-paragraphs.

Motivations, we haven't spesifically spoken on motivations but we have on fears. She has spesifically mentioned a fear of losing inner child/imagination/creativity and forgetting what it

is like to be a kid. Citing that she feels she is already doing this in her life.

Context, she grew up in a quite conservative home in Chicago, before moving to Tennessee. Her mother suffers from cancer that debilitates her in an extreme way. Much of what Keira does is to help her mother. Her mother is very sweet but with her disabilities is mostly focused on her daughter. I don't know much about her father, he is introveted, polite, and hard-working.

Part V

Journaling

Appendix W

*Introduction

Appendix X

Plans

X.1 Now

X.1.1 Self-Education

elf-education has always been a great value of mine. As mentioned earlier, I became obsessed with learning Einstein's field equation in 3-4th grade. Beyond that I have spent much of my time reading books, watching YouTube videos, and more about topics that interest me. Though, late in 8th grade, this desire to teach myself grew a great deal. Before I can go about my future goals for teaching myself I must first go over what I have already down.(This list is not full, mainly lectures finished and books from lectures, this does not include books not finished, non-lecture based video education, research papers read, projects where I learned things for the project alone, and other similar programs)
Mathematics:

- *AP Calculus AB and BC - Khan Academy*
- *Multivariable Calculus*

- *Khan Academy*
- *18.03 Differential Equations - MIT OpenCourseWare*
- *Gilbert Strang on linear algebra - MIT OpenCourseWare*
- *Vector Calculus - Trevor Bazett*
- *Tensor Analysis - eigenchris*
- *Tensor Calculus - eigenchris*
- *Crash Course in Complex Analysis - Steve Burton*
- *Introduction to Applied Numerical Analysis - Richard W. Hamming*
- *Symplectic geometry & classical mechanics - Tobias Osborne*
- *Lie groups, algebras, brackets - Mathemaniac (more conceptual than mathematical)*
- *Differential geometry - Robert Davie*
- *Calculus of variations - Faculty of Khan*
- *Working on*
  - *Advanced Analytic Methods in Continuum Mathematics: Fundamentals for Science and Engineering - Hung Cheng*
  - *Discrete Differential Geometry - CMU*
  - *Differential Geometry for students of Numerical Electrodynamics - Alain Bossavit*

Physics:

- *8.02 Physics II - MIT OpenCourseWare*

- *8.03 Physics III - MIT OpenCourseWare*
- *8.033 Relativity - MIT OpenCourseWare*
- *General Relativity - Stanford Online*
- *Introduction to plasma physics - USYD senior plasma physics lectures*
- *Introduction to electromagnetism - Griffiths*
- *Computational physics - Mark Newman*
- *8.224 Exploring Black Holes: General Relativity and astrophysics - MIT OpenCourseWare*
- *Classical Dynamics of Particles and Systems - Thornton Marion*
- *Introduction to cosmology - Stanford Online*
- *Introduction to fusion energy and plasma physics course - PPPL*
- *Seminar: Fusion and plasma physics - MIT OCW*
- *Statistical Mechanics - Stanford Online*
- *Hamiltonian description for magnetic field lines in fusion plasmas: A tutorial - AIP*
- *Fusion economics: power density, materials and maintenance - PPPL Frontiers Colloquia*
- *Flash-X code tutorial, a users perspective - University of Chicago(skimmed)*
- *Flash-X user guide - Flash-x*
- *Flash4 User support(skimmed)*

- *Radiative Procuresses in Astrophysical Phenomena- George B. Rybicki, Alan P. Lightman*
- *Goldstien Classical Mechanics Lectures - Prof. Jacob Linder*
- *PiTP 2016 - Institute of Advanced study*
- *2024 PPPL Graduate Summer School*
- *Hamiltonian description of the ideal Fuid - P.J.Morrison*
- *Quantum Mechanics with Applications - David B. Beard & George B. Beard*
- *"What IS Structure, How Do You Create or Recognize It, and How Can You Use It? or Metriplectic Dynamics: Using the 4-Bracket for Constructing Thermodynamically Consistent Models." - workshop Geometric Mechanics Formulations for Continuum Mechanics*
- *"On an Inclusive Curvature-Like Framework for Describing Dissipation: Metriplectic 4-Bracket Dynamics." - workshop Infinite Dimensional Geometry and Fluids*
- *Haven't finished*
  - *Theoretical Physics - Georg Joos*
  - *Advanced MHD with Applications to Laboratory and Astrophysical Plasmas - Cambridge*
  - *Classical Mechanics: A Modern Perspective - Sudarshan and N. Mukunda*
  - *The Interpretation of Structure From Motion - Ullman, S.*
  - *Radiative Processes in Astrophysics - George B. Rybicki & Alan P. Kightman*

  - *Symplectic geometry & classical mechanics - Tobias Osborne*

Coding:

- *Computational physics - Mark Newman*
- *Python Numerical Methods - Berkeley*
- *Applied Numerical Methods - Crice Carnahan, H.A Luther, James O.Wilkes*

Other:

- *Management in engineering - MITOpenCourseware*
- *Principles of microeconomics - MITOpenCourseware*
- *Dynamic leadership: using improvisation in business - OpenCourseware*
- *Logic 1 - OpenCourseware*
- *Policy for science, technology, and innovation - MITX*
- *Reducing The Danger Of Nuclear Weapons And Proliferation-MITOpenCourseware*

X.1.2 Research

- *Egg drop simulation*
- *Numerical Methods for 3D compressed Plasma using Lattice Boltzmann(paper written)*
- *Simulated Annealing for plasma thrusters*
- *High Altitude EMP for primordial black holes(never finished)*

- *Flash-x simulations:*
  - *Basic Sedov*
  - *Running simple MHD simulation of my own design*
- *A Hamiltonian Framework on ICF Implosions Rocket Equation Based on Rayleigh-Taylor Instabilities*
- *Contributed to PlasmaPY*
- *Relativistic Rocket Equation(Newtonian)*
- *Relativistic rocket equation(Hamiltonian)*
- *Finite difference MHD Model in Fortran*
- *Computes velocity based upon Navier Stokes equation adapted to magnetic and electric fields*
- *Outputs graphical color field of velocity*
- *Asteroid game*
  - *Classical*
  - *Time*
  - *Survival*
  - *Newtonian(player and asteroids have gravity)*
  - *Dark Matter(Invisiable gravity fields)*
  - *Relativistic(time dilation, spacial contraction, black holes, etc)*
  - *Proximal Policy Optimization "AI" played game*
- *Hamiltonian based plasma thruster Fortran Numerical Code*
- *N body simulation*
- *Fluid based N-body simulation*

- *Death Star*
  - *Basic gravitational 'planet'*
  - *MHD Plasma beam hitting*
  - *Basic evaporation and kinetic energy transfer*
- *OpenMHD usage*
- *Usage Gkyl*
- *Omega-X project*
  - *Inspired by FLASH-X, attempting to build a multi-physics plasma physics codebase using Metriplectic forumism rather than Lagrandian and Newtonian*
  - *I have successfully implemented the 1D thermofluid metriplectic discretization onto a discontinuous Galerkin subspace using an L^2 projection. Integrated using implicit midpoint rule.*
  - *Working on module registry & driver pattern*

X.1.3 Other activities

I also had some other activities that I have done in high school, I won't go too much in them.

- *SVA*
- *Boy Scouts*
- *Church activities*
- *JROTC*
- *FCA*
- *OA*

- *Etc*

The OA, Order of the Arrow, I will continue into college.

X.2 Pre-University

X.2.1 Independent Learning

One thing I need to do is nail calculus completely. I am going to be taking the AP calc BC test without taking the course.

Beyond that, I am completing differential geometry. Once done I will finish Symplectic geometry. Then lie (brackets, algebra, group, and theory)

I also want to get into functional analysis.

I should also learn C++ beyond what I have done in the past.

X.3 Undergrad

X.3.1 Classes

X.3.1.1 Requirments

X.3.1.1.1 General

I have tested out of many of the general education requirements, so here is the remaining bit:

- *Written communication(pick 1)*
  - *Honors ethical theory*
  - *Haslam knowledge*
- *Oral communication*

  - *A survey in Contemporary Physics*

- *Arts and humanities(pick 2)*
  - *Haslam Knowledge*
  - *honors intro to philosophy*
  - *Professional responsibility*
- *Civilization(pick 2)*
  - *Honors development of Western civilization*
  - *Honors history of world civilizations*
  - *University Honors*
  - *World Religions*

X.3.1.1.2 Academic Physics Major

- *Prerequisites*
  - *Intro Computer Science*
  - *Differential Equations 1*
  - *Matrix Algebra*
- *Core*
  - *Fundamentals of modern physics*
  - *Mechanics 1*
  - *Thermal Physics*
  - *Electronics Lab*
  - *Intro to quantum mechanics*
  - *Modern optics*
  - *Electricity and Magnetism*
  - *Survey in modern physics*

  - *Modern Lab*

- *Academic*
  - *Mechanics 2*
  - *Intro to quantum 2*
  - *Electricity and magnetism 2*
- *Recommended*
  - *Mathematical Methods for scientist*
  - *Partial Differential Equations*
  - *Complex Variables*
- *Thesis in physics*

X.3.1.1.3 Applied Maths Major

- *Computer Science*
- *Core*
  - *Diff equations*
  - *Matrix Algebra*
  - *Intro to abstract mathematics*
  - *Math proficiency*
- *Breath*
  - *Honors abstract algebra 1*
  - *Honors Analysis 1*
  - *Numerical Analysis*
  - *Stochastic processes*
- *Applied breath(physics courses)*
- *Depth*
  - *Numerical Algebra*

X.3.1.1.4 Computer Science Minor

- *Intro comp sci*
- *data structures*
- *Logic design*
- *Numerical Algorithms*
- *Numerical Analysis*
- *Numerical Algebra*
- *Parallel Programming*

X.3.1.1.5 Electives

- *PHYS 441 - Introduction to Computational Physics*
- *PHYS 494 - Special Topics in Physics*
- *2nd PHYS 493 - Research and Independent Study*
- *PHYS 531 - Mechanics(Maybe, graduate)*
- *PHYS 541 - Electromagnetic Theory(Maybe)*
- *PHYS 571 - Mathematical Methods in Physics I(maybe)*
- *PHYS 573 Numerical methods in physics(maybe)*
- *PHYS 615 Astrophysics(maybe)*
- *PHYS 643 Computational Physics(maybe)*
- *MATH 462 - Differential Geometry*
- *MATH 467 - Honors: Topology*
- *MATH 536 - Partial Differential Equations II*

- *MATH 567 - Riemannian Geometry I*
- *MATH 568 - Riemannian Geometry 2*
- *MATH 572 - Numerical Analysis II*
- *MATH 577 - Optimization*
- *MATH 579 - Seminar in Numerical Mathematics*

X.3.1.1.6 Total

- *Honors ethical theory(x)*
- *honors intro to philosophy(x)*
- *Professional responsibility(x)*
- *Honors development of Western civilization(x)*
- *Honors history of world civilizations(x)*
- *Intro Computer Science(x)*
- *Differential Equations 1(x)*
- *Matrix Algebra(x)*
- *Fundamentals of modern physics(x)*
- *Mechanics 1(x)*
- *Thermal Physics(x)*
- *Electronics Lab(x)*
- *Intro to quantum mechanics(x)*
- *Modern optics(x)*
- *Electricity and Magnetism(x)*

- *Survey in modern physics(x)*
- *Modern Lab(x)*
- *Mechanics 2(x)*
- *Intro to quantum 2 x*
- *Electricity and magnetism 2 x*
- *Mathematical Methods for scientist(x)*
- *Partial Differential Equations x*
- *Complex Variables x*
- *Thesis in physics (x)*
- *Intro to abstract mathematics x*
- *Honors abstract algebra 1(x)*
- *Honors Analysis 1 (x)*
- *Numerical Analysis x*
- *Stochastic processes (x)*
- *Numerical Algebra(x)*
- *data structures(x)*
- *Logic design(x)*
- *Parallel Computing (x)*
- *35(-1, -4, maybe) main(13-15 electives)*
- *MATH 462 - Differential Geometry(1)(x)*
- *MATH 467 - Honors: Topology(2)(x)*

- *PHYS 441 - Introduction to Computational Physics(15)*
- *PHYS 493 - Research and Independent Study(3)(x)*
- *PHYS 531 - Classical Mechanics(4)(x)*
- *PHYS 541 - Electromagnetic Theory(5)(x)*
- *PHYS 571 - Mathematical Methods in Physics I(6)(x)*
- *PHYS 573 - Numerical Methods in Physics(7)(x)*
- *MATH 515 - Analytical Applied Mathematics I(8)(x)*
- *MATH 567 - Riemannian Geometry I(9) x*
- *MATH 568 - Riemannian Geometry II(10) x*
- *MATH 569 - Seminar in Topology and Geometry(11)*
- *MATH 574 - Finite Element Methods(12)*
- *MATH 578 - Introduction to Scientific Computing(13) (x)*
- *MATH 585 - Optimal Control Theory(14)*
- *PHYS 599 - Seminars(x)*
- *PHYS 593 - Independent Study(x)*

X.3.1.2 Schedule

physics *general* *math* *comp sci*

X.3.1.3 First Year

1. *Honors ethical theory*
2. *Honors intro to philosophy*
3. *Professional Responsibility*
4. *HR: Western Civilization*
5. *HR: World civilization*
6. *Intro computer science*
7. *Diff equations 1*
8. *Matrix Algebra*
9. *Fundamentals in modern physics*
10. *mechanics 1*
11. *Intro to abstract mathematics*
12. *PHYS 503 - Physics Colloquium*

X.3.1.4 Second Year

1. *Thermal Physics*
2. *Electronics Lab*
3. *Intro to quantum mechanics*
4. *Modern optics*
5. *Electricity and Magnetism*

6. *Survey in modern physics*
7. *Modern Lab*
8. *Mechanics 2*
9. *Intro to quantum 2*
10. *Electricity and magnetism 2*
11. *Partial Differential Equations*
12. *Numerical Analysis*

X.3.1.5 Third year

1. *Honors abstract algebra 1*
2. *Honors Analysis 1*
3. *Stochastic Processes*
4. *Numerical Algebra*
5. *Mathematical Methods in physics*
6. *Research & independent study*
7. *Differential Geometry*
8. *Honors: Topology*
9. *Classical Mechanics, grad*
10. *PHYS 541 - Electromagnetic Theory, grad*

11. *MATH 515 - Analytical Applied Mathematics I*
12. *Independent study and research*

X.3.1.6 Fourth year

1. *Thesis in physics*
2. *MATH 567 - Riemannian Geometry I*
3. *MATH 568 - Riemannian Geometry II*
4. *Numerical methods in physics*
5. *data structures*
6. *Logic design*
7. *Parallel Computing*
8. *PHYS 599 - Seminars*
9. *PHYS 513 - Problems in Theoretical Physics I*
10. *MATH 578 - Introduction to Scientific Computing*
11. *MATH 534 - Calculus of Variations*
12. *MATH 537 - Mathematical Principles of Continuum Mechanics I*

X.3.1.7 Others

1. *PHYS 514 - Problems in Theoretical Physics II*
2. *MATH 585 - Optimal Control Theory*

3. *MATH 574 - Finite Element Methods*
4. *PHYS 615 - Astrophysics and Cosmology(grad required)*
5. *PHYS 643 - Computational Physics(grad required)*
6. *MATH 515 - Analytical Applied Mathematics I*
7. *MATH 571 - Numerical Analysis I*
8. *MATH 572 - Numerical Analysis II*

X.3.1.8 Possible summer

1. *Graduate Reading in Mathematics*
2. *Research and independent study*
3. *graduate reading in physics*
4. *Special Problems*

X.3.2 Goals for the first three weeks

1. *Talk to Dr. Mezz*
2. *Create LaTeX libraries for class*
3. *Organize schedule template*

as for organizations I am interested in

1. *Society of Physics Students()*
2. *Book to build*
3. *Students for Philanthropy*
4. *Academic quizbowl*

5. *Alpha Omega Campus Ministry*
6. *American Nuclear Society*
7. *Canoe and Hiking club()*
8. *Christian Student Fellowship*
9. *Christian Volunteers()*
10. *Climbing club()*
11. *Cycling club*
12. *Data science*
13. *Forum on science, ethics, and policy*
14. *Hema at UT*
15. *Nuclear Managment*
16. *Martial Arts club*
17. *Model UN*
18. *Naturalist club*
19. *Out in STEM*
20. *Pickle ball*
21. *Astronomy*
22. *Caving*

X.3.3 First Week

I want to walk to and time walk between and time from

- *Physics*
- *Humanities*
- *Business*
- *Health and Recreation*
- *Recreational fitness*
- *Rocky top dining*
- *Stokely Hall*
- *Volenteer Hall*

I would also like to explore the library, Weslyan house,

I want to try biking around the area

I am required to go to Authoring your story, rocky top road map, and torch night

I would consider going to dining scavenger hunt and brunch

I will work on my schedule to come up with something more detailed to my current model.

X.4 Second Semester

X.4.1 Classes

Possible

- *Physics 251(Optics and modern physics)*

- *Physics 361(electronic lab)*
- *Math 251/PHYS 301(linear algebra)*
- *Eng 102*
- *Math 300(abstract)*
- *MATH 435(partial diff)*
- *MATH 462(diff geometry)*
- *ASTR 421(general relativity)(Not sure, has a recommended background but no prerequisites)*

X.5 *Grad

X.6 *Early career

X.7 *Late Career

Appendix Y

Schedule Templates

Y.1 Preliminary Semester 1

Time	*Monday*	*Tuesday*	*Wednesday*	*Thursday*
6:00				
6:30				
7:00				
7:30				
8:00				
8:30				
9:00				
9:30				
10:00				
10:30				
11:00				
11:30				
12:00				
12:30				
1:00				
1:30				
2:00				
2:30				
3:00				
3:30				
4:00				
4:30				
5:00				
5:30				
6:00				
6:30				
7:00				
7:30				
8:00				
8:30				
9:00				
9:30				
10:00				

Time	*Friday*	*Saturday*	*Sunday*
6:00			
6:30			
7:00			
7:30			
8:00			
8:30			
9:00			
9:30			
10:00			
10:30			
11:00			
11:30			
12:00			
12:30			
1:00			
1:30			
2:00			
2:30			
3:00			
3:30			
4:00			
4:30			
5:00			
5:30			
6:00			
6:30			
7:00			
7:30			
8:00			
8:30			
9:00			
9:30			
10:00			

Appendix Z

Journal

This section will hold daily journaling. Essentially it will be day-to-day notes and plans. These will then be synthesized into the previous chapters in this book.

Z.1 2025-11-27

Z.2 Emacs

This is my first journal write using EMACS, for that has been my project at the moment. Creating a doom Emacs with Org-mode enhancements, AUCTeX/pdf tools. Org-roam(graph+server), Org-bable for multiple languages(FORTRAN, python, Julia, Jupyter, and shell), LSP for coding, Magit(for git), projectile/treemacs(for using projects), a daily journal that auto inserts into my LaTeX book, and create Org nodes and graphs for chapters and journals.

For the most part I have this set up. I need to edit AUCTeX so that it uses latexmc rather than latex, so that it will run multiple times for the book-like structure. I think I have more dependen-

cies I need for the languages. I still have more to learn to get comfortable with this program. Also, I am not sure that the program auto inserts my journals.

```
print("2+2 =", str(2+2))
```

Z.3 2025-11-29

Z.4 Axiom

As is very obvious, I am making a transfer from now on. I will have this section for day-to-day journaling and notes, while the previous sections for more concise and developed thought. I won't change the already made stuff. This new This takes me to Emacs, I have figured out how to incorporate Org files into my LaTeX book.

I do need to work on attaching it to the table of context. I believe I should have this completed by the end of the day.

Z.5 Emacs

I am learning much about how to use Emacs, as I have mentioned above I have been working to better understand and use Emacs. I have created two new function's. One to make IDs for Org files written outside of Emacs and also to turn Org files into LaTeX to use.

Z.6 Media

Z.6.1 Zettelkasten

I was originally having this section be a Zettelkasten-like notes taking. Though after more consideration, I have reconsidered. While a similar linking will be useful, I am unlikely going to have the same extreme modularity that is accustomed to the system. Thus, I am unlikely to use it fully, but some use is likely.

Z.7 2025-12-03

Z.8 Links to Book Chapters

- *coding Axiom* `roam:Omega-X roam:plans roam:personal_analysis`

Z.9 Emacs

Z.9.1 Emacs updates

For this journal, remember that the command is SPC n r d t. Beyond that I have fully completed the Org → LaTeX file. Though, as useful as these programs can be it does raise the question of if I should use it. I have begun to question its usefulness.

Z.9.2 Emacs vs Vim

I have found that as I have created these things, I have realized I can substitute a lot of the programs by simple scripting and vim. While Vim's speed and ease of use has its use. I do think

the scripting of Emacs it also quite fast, ease of moving between files, and even the spell check. Beyond that the integration of things coding languages is potentially very useful once I gain the hang of it.

Z.9.3 Org vs Tex

One question I have been looking into is whether I should write my notes directly in LaTeX or continue to write in an Org file and export to LaTeX after the fact. ***Org***

Pros	*Cons*
Org babel to run commands	*Easy errors*
Org-capture to quickly make notes	*whitespace inconsistency*
Graph view	*Hard to write math with*
Easy connections	*Difficulty with more complex LaTeX co*
Easy to use syntax	*Degregation of performace over*
Easy to use Hierarchy	

Tex

Pros	*Cons*
Absolute control	*more verbose*
none of the easy errors	*Have to script new stuff*
Can be scripted to simulate Org-capture-like	*No babel*
More freedom of use in LaTeX commands	

Z.9.4 Future

Currently my plan is to have continue to use Org for journaling, though I may revisit this idea later on.

With this I will continue to learn how to use Org/LaTeX integration and use. Also some of syntax of Org. I also need to learn more about running code and using it in Emacs. I currently only use it for notes.

- *note-book enviorment(like jupiter)*
- *build-up the graph view*
- *Agenda*
- *linking*
- *task automation*

Z.10 Omega-X

Z.10.1 Arguments

Currently, I have the poor practice of setting variables in a module and ignoring using arguments or such things. I plan to rectify this. This is for several reasons; to protect from errors and to allow for MPI(my next step).

Z.10.2 Updates

I need to continue on a couple of sets. Create a IO files, finish the evolution driver, and have ways to analyze the data.

Z.11 Future Ideas

I have a couple ideas for future projects:

Z.11.1 Categorical Metriplectic

Using category theory to set up a fully and complete formal theory of the Metriplectic system. This is highly ambitious considering my limited knowledge in category theory, but it is a field that is thoroughly explored and thus I believe it is within my reach to accomplish.

Z.11.2 Cason Brackets Dynamics(name pending)

I believe I can integrate metriplectic systems to Nambu mechanics. As far as I know this is a completely novel goal and thus would be my own. Let's see with the names: Cason Formalism, Cason Mechanics, Cason dynamics, Cason framework, Cason Analysis, Casonian Dynamics, Casonian Mechanics, Cason's Method, Cason's Dynamics, Casonic Dynamics, The Cason Principle, Cason Bracket Dynamics, Cason system, Cason Framework for hybrid Dynamics. I know this seems overtly narcissistic, trying hard to come up with a name based upon my own; I don't really care.

Z.11.3 Metriplectic PINN

I have read several papers using Metriplectic dynamics to create PINN(Physics-informed-neural-networks).

Z.12 Media

Z.12.1 Star Trek

I forgot how much I like Star Trek next Gen, especially Picard.

Z.13 Self-analysis

Z.13.1 Stress & TV

For some reason, watching TV where the show lets you in on how stupid an idea is and yet the character does it stresses me out. Especially with lying and illicit activity. When they lie and it spirals out, then they lie more. I don't like it. Messes with me.

Z.14 2025-12-04

Z.15 Links to books

- *plans*

Z.16 Projects

I have recently come into more free time and would like to use it to my advantage to increase my abilities and capacity; though, I am having trouble considering what to do, in what order, and how to block my time.

Z.16.1 Formal projects

The actual projects I want to work on are continuing:

- *FORTRAN Omega-X project([Omega-X]finish it, develop parallel computing, develop TUI, continue documentation, profile it, add higher order methods, more time stepping methods, add modules for working with additional systems, data analysis, visualization, GPU acceleration, etc)*

- *Cason Bracket dynamics: synthesis Nambu and Metriplectic and incorporate into Omega-X*
- *Categoric Theory of Metriplectic Dynamics: Self explanatory*
- *Metriplectic PINN: PINN based upon PINNs*
- *Emacs Building: I want to build workflows and other things in Emacs to make stuff easier for me(coding workflow, see data, writing book stuff)*

Z.16.2 Skills

There are several skills I want to continue to develop:

- *Julia coding: right now in the basics*
- *C++: similar to Julia*
- *Emacs: I want to get more comfortable with the key-bindings and creating commands to make things simple*
- *Parallel computing: most clearly can gain this through my Omega-X project*
- *Org/LaTeX: Learing how to use both org and tex so they work together.*

Z.16.3 Plan

I am thinking schedule it:

1. *develop a plan for the day in Org, experiment with Org/TeX integration to gain that experience in that regard.*
    - *Incorporate Julia whenever possible*

2. *Theoretic study - work on one a day*
    - *Categoric Theory of Metriplectic Dynamics*
    - *Cason Bracket Dynamics*
3. *Programming Project - work on one maybe two a day*
    - *Develop Omega-X project*
        - *finish it*
        - *develop parallel computing*
        - *develop TUI*
        - *continue documentation*
        - *profile it*
        - *add higher order methods*
        - *more time stepping methods*
        - *add modules for working with additional systems*
        - *data analysis, visualization*
        - *GPU acceleration*
    - *Develop Emacs tools*
        - *Create Julia REPL in Emacs and keybinding to make it simple*
        - *custom LaTeX keys*
        - *Set FORTRAN build and run*
4. *Skill Development - work on one or two a day*
    - *Julia*
        - *symbolic Poisson/Nambu brackets solver*
        - *ODE solver using the SciML interface*
        - *GPU-accelerated metriplectic evolution*
        - *Category theory for dynamics*
    - *C++*

  - *Simple library*
  - *SIMD-accelerated*
  - *learn expression templates*
  - *GPU version using CUDA or HIP*
  - *Plug-in architecture*
  - *simple MHD solver*
- *FORTRAN*
  - *Coarrays*
  - *Polymorphic OOP with type-bound procedures*
  - *Generics (Fortran 2023)*
  - *MPI*
  - *GPU Offloading*
  - *Memory alignment + SIMD-aware design*

Z.17 2025-12-05

Z.18 Links to Book Chapters

-

Z.19 Plans for Today:

I have been currently working on my school work; getting caught up with that. I have also conntacted NSCC about the dual enrollment program to make sure I am not missing anything.

For my next steps:

- *Programming Project: Omega-X*
  - *First: I am going to build a type binding for Omega-X's ininitilize subdriver.*

  - *Second: formalize $declare_{vars}$ module to work for this new update*
  - *Third: Move C++ binding to a safe place(C++/FORTRAN?[either fits])*

- *Theoretic study: Catogoric Metriplectic:*
  - *Create LaTeX notes sections*
  - *Write list of possible papers to read*
  - *Read one and write notes*

- *Programming/skill: EMACS building/Julia*
  - *Develop Julia REPL in EMACS*
  - *Play around with it once in*

Z.19.1 Omega-X Notes

As I mentioned in the above section my first task is to develop a type system for $driver_{init.F90}$. To create the types:

```
module types_make
     !individual types
      type :: name_of_type_1
        real :: variable_in_type_1
      end type name_of_type_1

      type :: name_of_type_2
        real :: variable_in_type_2
      end type name_of_type_2

      ! higher level type
      type :: higher_level_type
        type(name_of_type_1)
```

```
        type(name_of_type_2)
      end type higher_level_type
end module types_make
```

The to use it:

```
program use_types
        use types_make

        type(higher_level_type) :: H_T_L

        ! initialize fields
        H_T_L_of_type_2%variable_in_type_2 = 1.0
```

I ended up doing less than I imagined. I created the type system and corrected the arguments for many things. My next steps should be to develop the

1. *declare$_{vars}$*
2. *fix uses of modules*
3. *Change python code to work with new type arguments.*

This is what I can do tommorow.

Z.19.2 Emacs Build

While working on Omega-X I figured I should focus on FORTRAN integration before Julia, I also saw a cool calculator thing I will set up and shouldn't take long.

Calculator:

- *Get in M-x calc or C-x * or M-x full-calc*
- *It is in reverse polish notation: operators follow their operands*

- *t to get to trail*
- *i to enter*
- *u to undo*
- *DD to redo*
- *TAD to swithc*
- *DEL to delete one*

There is also a LaTeX embeded system:

- *C-x * e → Enter calc-embedded mode.*
- *C-x * r → Replace the marked expression with its result.*
- *C-x * q → Quick evaluation of inline expressions.*

*calc-embedded C-x * e Enter embedded mode on the current region (or insert delimiters if none). calc-embedded-word C-x * w Evaluate the word at point as a Calc expression. calc-embedded-activate C-x * a Activate embedded formulas in the current buffer. calc-embedded-new-formula C-x * n Insert a new embedded formula at point. calc-embedded-select C-x * s Select the next embedded formula in the buffer. calc-embedded-edit C-x * e (when inside formula) Edit the embedded formula in Calc. calc-embedded-update C-x * r Replace/update the formula with its evaluated result. calc-embedded-update-all C-x * u Update all embedded formulas in the buffer. calc-embedded-clear C-x * c Clear (remove) the result part of an embedded formula. calc-embedded-clear-all C-x * C Clear results for all formulas in the buffer. calc-embedded-duplicate C-x * d Duplicate the current formula. calc-embedded-next C-x * n Jump to the next embedded formula. calc-embedded-previous C-x * p Jump to the previous embedded formula. calc-embedded-reload C-x * l Reload Calc settings for embedded mode.*

Now I have created the FORTRAN stuff:

1. *Fortran-lang LSP*
2. *Fortran-indent improvements and compile.el*
3. *custom build commands*
4. *Org babel*

Z.19.3 Categoric Metreplectic

Z.20 2026-01-03

Z.21 Links to Book Chapters

-

Z.22 Emacs

Z.22.1 TABS

I have recently added the tab like system. Here are the commands: New tab tab-new C-x t 2 Close tab tab-close C-x t 0 Switch tab tab-next / tab-previous C-x t o / C-x t O Rename tab tab-rename M-x tab-rename Duplicate tab tab-duplicate M-x tab-duplicate

So that's that.

Z.23 2026-02-05

Z.24 Links to Book Chapters

-

Z.25 Plans

As I mention in "Self-development" at this time I am working to return to normal. For this I have to re-establish and create habits to be the person I want to be.

Z.26 Schedule

Z.26.1 9:00 - 11:00

- *Wake-up*
- *Prepare for the day*
- *Hair-cut*

Z.26.2 11:00 - 1:00

- *Exercise*
- *Eat Lunch*

Z.26.3 1:00 - 4:00

- *Do Calculus review exam*

Z.26.4 4:00 - 5:00

- *Read*
  - *The Karamazov Brothers*
  - *Numerical Hamiltonian*

Z.26.5 5:00 - 5:45

- *Eat*

Z.26.6 5:45 - 6:00

- *Math discussion post*

Z.26.7 6:00 - 7:00

- *Project*
  - *Numerical Hamiltonian*
  - *Website*
    - *One idea is to create a kinda to do website based upon my schedule of that day.*
  - *Writing*
  - *Other*

Z.26.8 7:30 - 8:30

- *Blank period for extra time to push things back*

Z.26.9 8:30 - 10:00

- *Watch TV*

Z.27 2026-02-16

Z.28 Bible Reading

Z.28.1 Genesis 41:17-42:17

God protects Joseph through establishing gifts within him. He does the same for us.

Why does Joseph test his brothers?

Z.28.2 Matthew 13:24-46

The parables of Jesus, establish stories and myth as the methodology of developing the abstract concepts of God's will.

It refers to the weeds as everything that causes sin and all who do evil, what exactly does this mean? Everyone does evil and everyone has within that which tempts them. Taking out the "weeds" prematurely will destroy the wheat.

The final stories is about how the kingdom of God gives everything for the grace of those who are lost.

Z.28.3 Psalm 18:1-15

Love of God

Z.28.4 Proverbs 4:1-6

Community, family, and God fosters wisdom. Wisdom that protects you.

Z.29 Today

Today was a good day. While I wasn't feeling my absolute best, I was productive. Not as in I did much work, but I was able to actively engage in things I enjoy. I wrote a large section on the fundamentals of metaphysics, some other things, got back into reading the bible, and read from Numerical Hamiltonian Problems.

The main thing was the writing, it has been a while sense I have been able to have sustained effort on such a complex and in-depth subject. Much of my stuff is rederived, but that is kinda the point.

Z.30 2026-02-17

Z.31 Bible reading

Z.31.1 Old testament

Why is Joseph testing with them even after hearing what they had to say while also giving them money

Z.31.2 New Testament

The kingdom of God takes in all, but not all remain.

Interesting how such a radical take on ones word is used.

Z.31.3 Psalm

God rescues you and protects you from the enemies that are greater than you

Z.31.4 Proverbs

Pursue wisdom over all other virtues

Z.32 Ideas for projects

While I am still not in the headspace for in-depth research like before. I can still do things.

Z.33 Schedule

Z.33.1 Coding

- *Simple Julia*
- *Play around with parallel computing*
- *Learn some optimization*

Z.33.2 Writing

- *Continue Axiom Project*
- *More on the metaphysical implications of duelist frame works*
- *More physics writing outside of metaphysics*
  - *Deriving basic equations in other formalism's(Nambu, metriplectic, etc)*
  - *Classifications of poisson manifolds*
  - *Understanding of Lie-Poisson reductions*
  - *Chaos theory and integradability*

  - *Numerical theory*
  - *Infinite dimensional Hamiltonian systems*
  - *Noether's theorem*
    * *Understanding on metaphysical lens*
    * *understanding mathematically*
    * *General relativity breaking it*
  - *Constructing 4-bracket metriplectic*
  - *Metriplectic VS Generic*
  - *Contact Geometry*

- *Review papers*

Z.33.3 Math/Physics

- *continue reading my books*

Z.34 April 14

Today has been somewhat successful. What I need to do is focus on simply things that are productive and enjoyable. I will break it into a couple of parts here.

Things I need to do:

- *School work*
- *Fall Fellowship*

Simple projects

- *LaTeX sty file to make writing easier*
- *Abbreviation in emacs to make writing easier*

Reading/videos

- *Bible*
- *Relativistic Quantum Field Theory*
- *Numerical Hamiltonian Problems*
- *Handbook on the Philosophy of physics*
- *Advanced GR by Wandu*

Z.35 7/24/26